SWAFL Desktop 1.1.0 — User Guide and Technical Reference

Volume 2 of 3: screenshot atlas at the default settings — every screen, every menu and dialog, and every layer from top to bottom with the equations it draws on

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Version 1.1.0

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About this volume

This volume shows every screen of SWAFL Desktop 1.1.0 at the default settings (Coarse Loam, conventional tillage, the cereal/oilseed rotation, wheat-straw mulch of strength 1.0, surface irrigation, trigger −300 cm at 20 cm, the built-in weather record, the working date of the capture). Each layer is shown first in the window, then drawn from top to bottom and cut into page-size parts at its section headings; the caption of each part names the sections it holds. After the drawing, every equation that the layer draws on is printed in full with its provenance. The three designs are shown after their run, and the virtual farm is shown for each design on an irrigation day.

Chapter 1

The window

1.1 The workbench

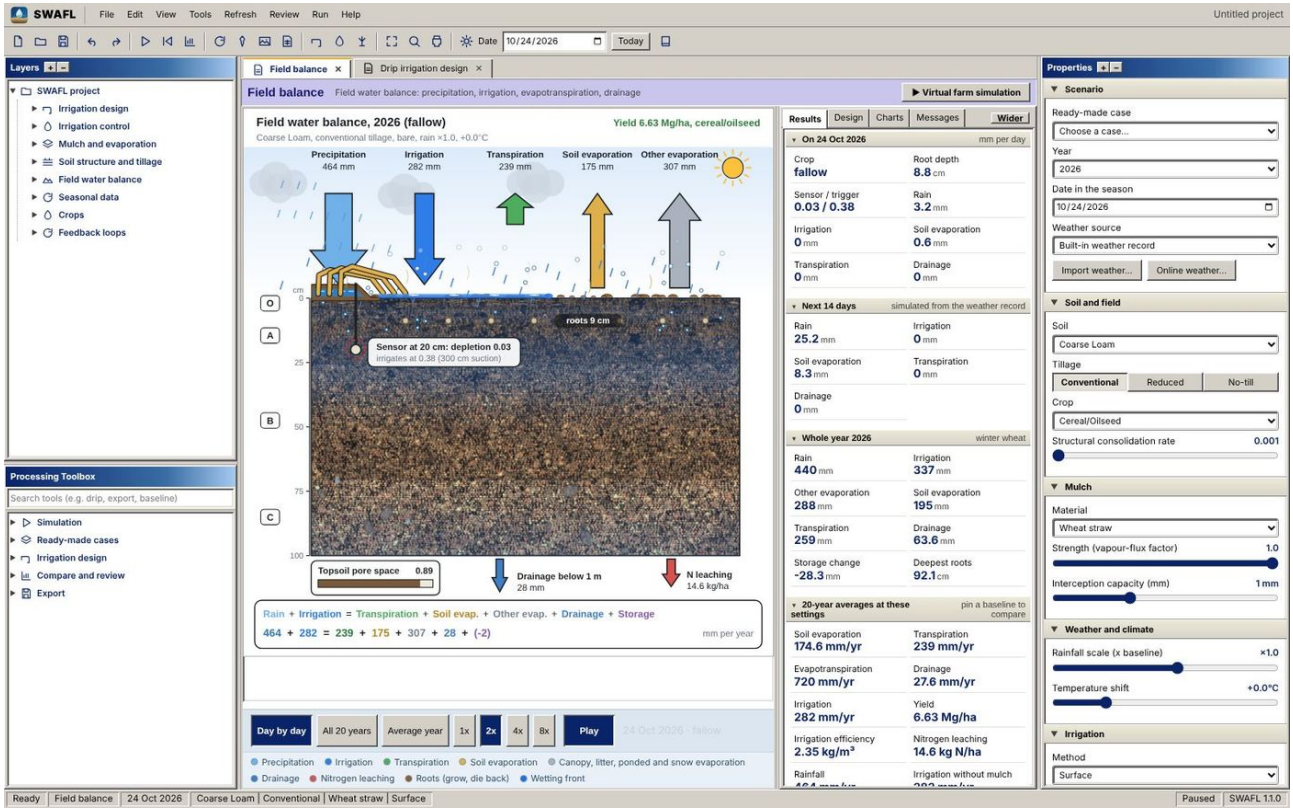


Figure 1.1: The workbench on start: Field balance for the working date.

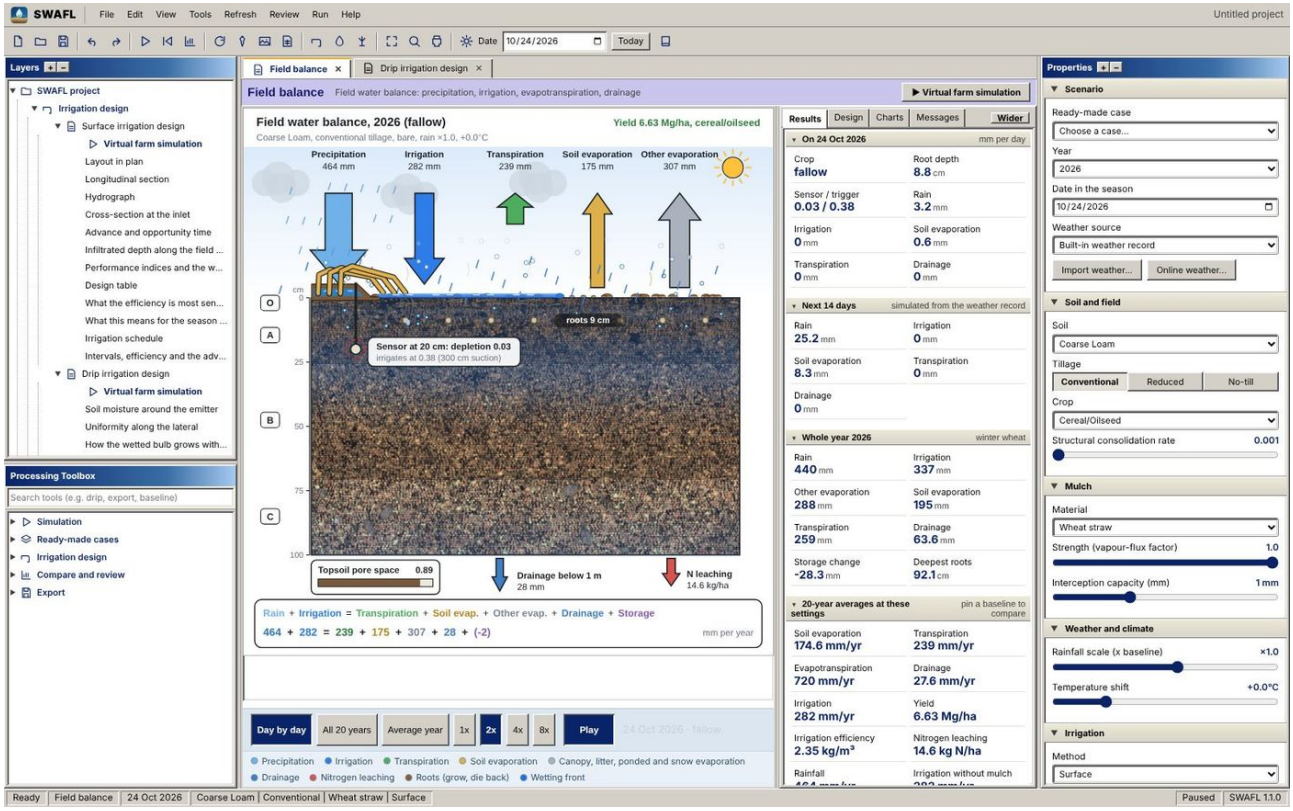


Figure 1.2: The Layers panel with every group and layer expanded; each design lists its sections and its virtual farm.

1.2 The menus

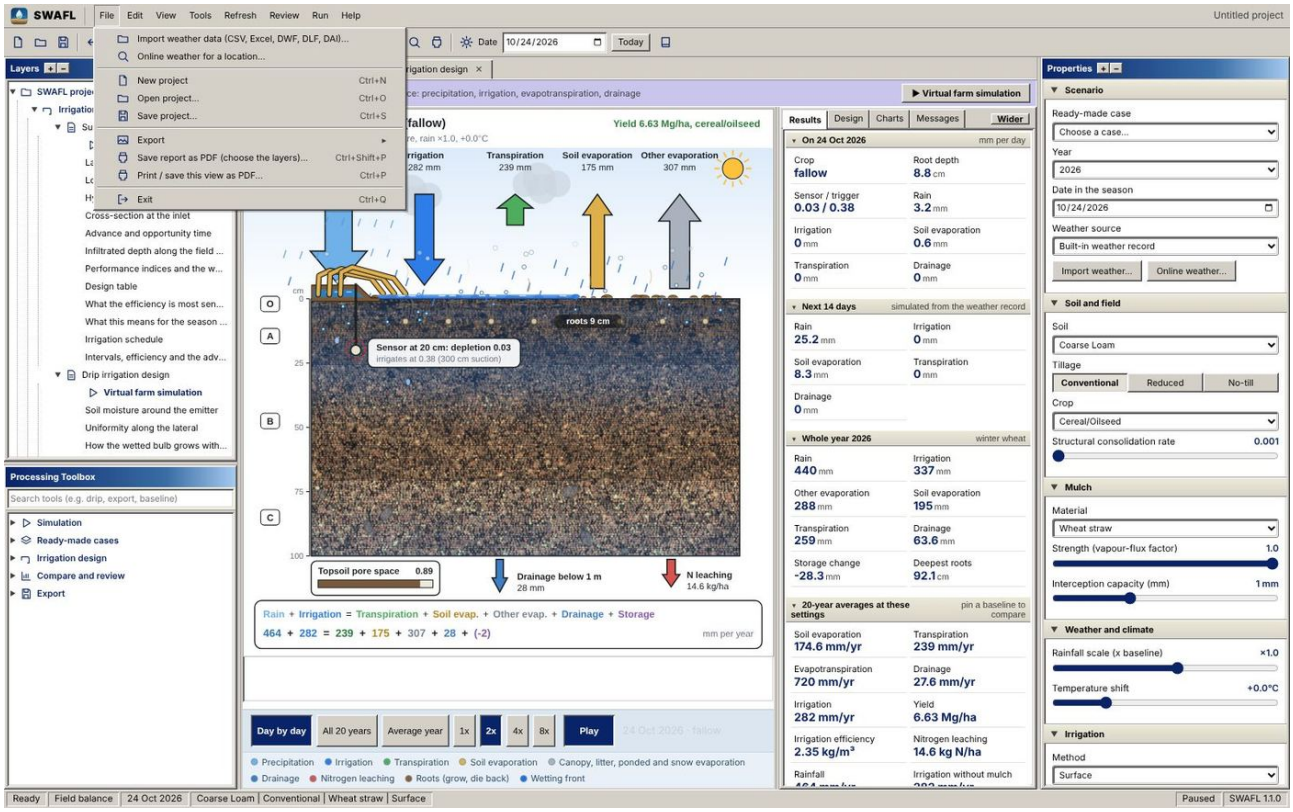


Figure 1.3: The File menu.

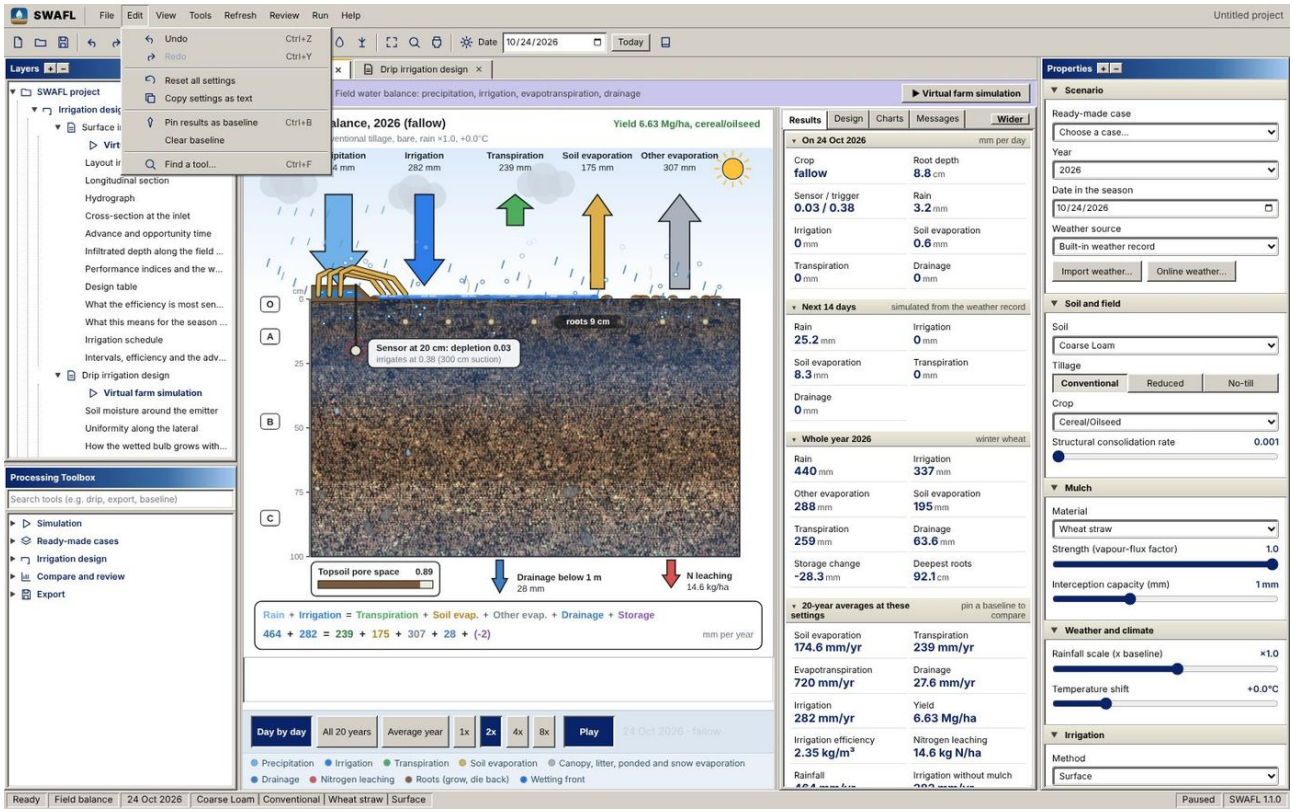


Figure 1.4: The Edit menu.

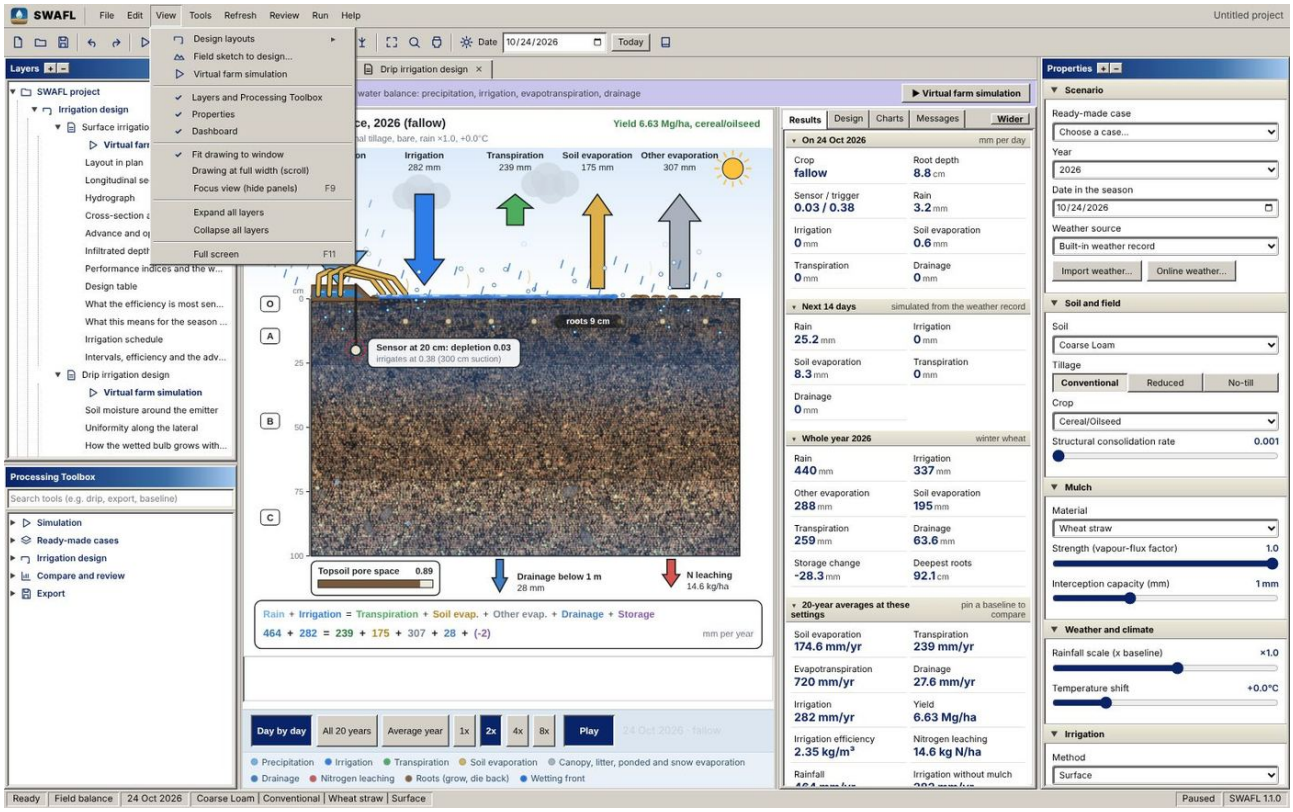


Figure 1.5: The View menu.

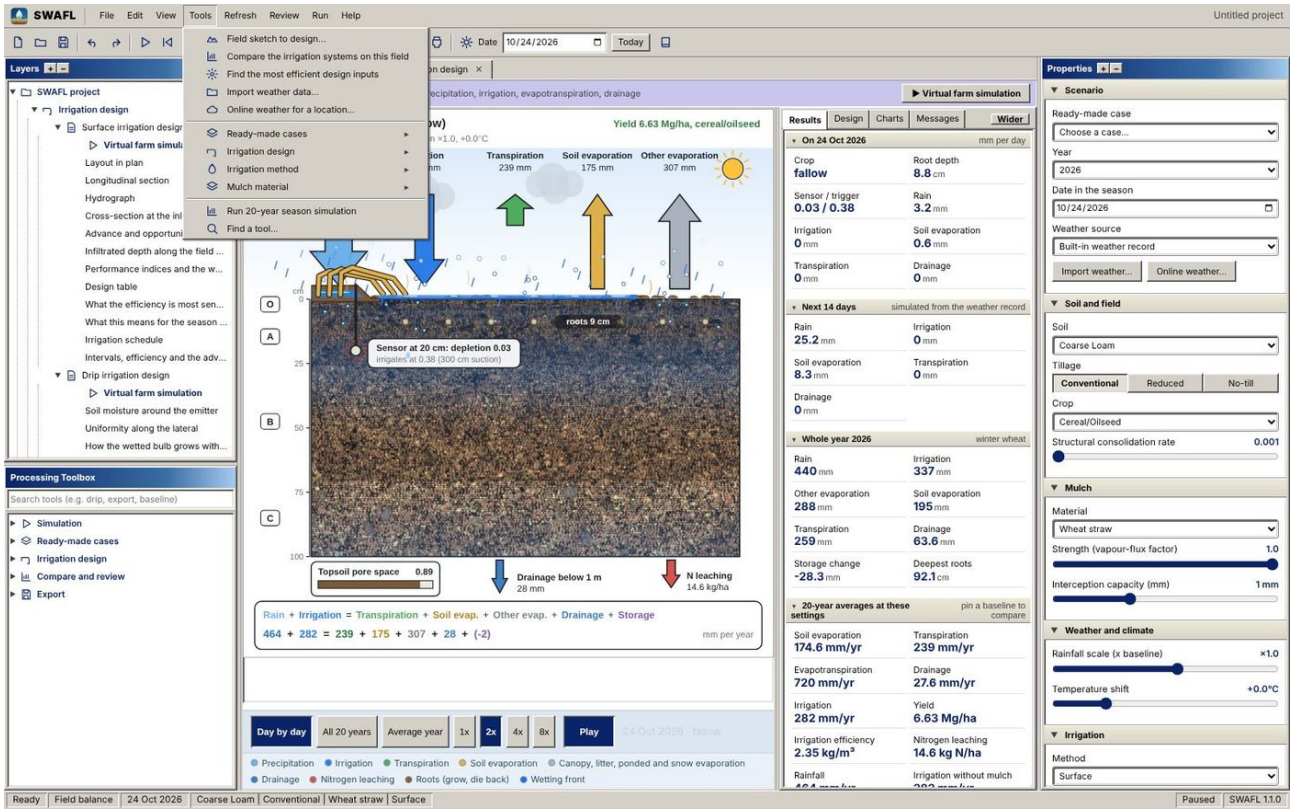


Figure 1.6: The Tools menu.

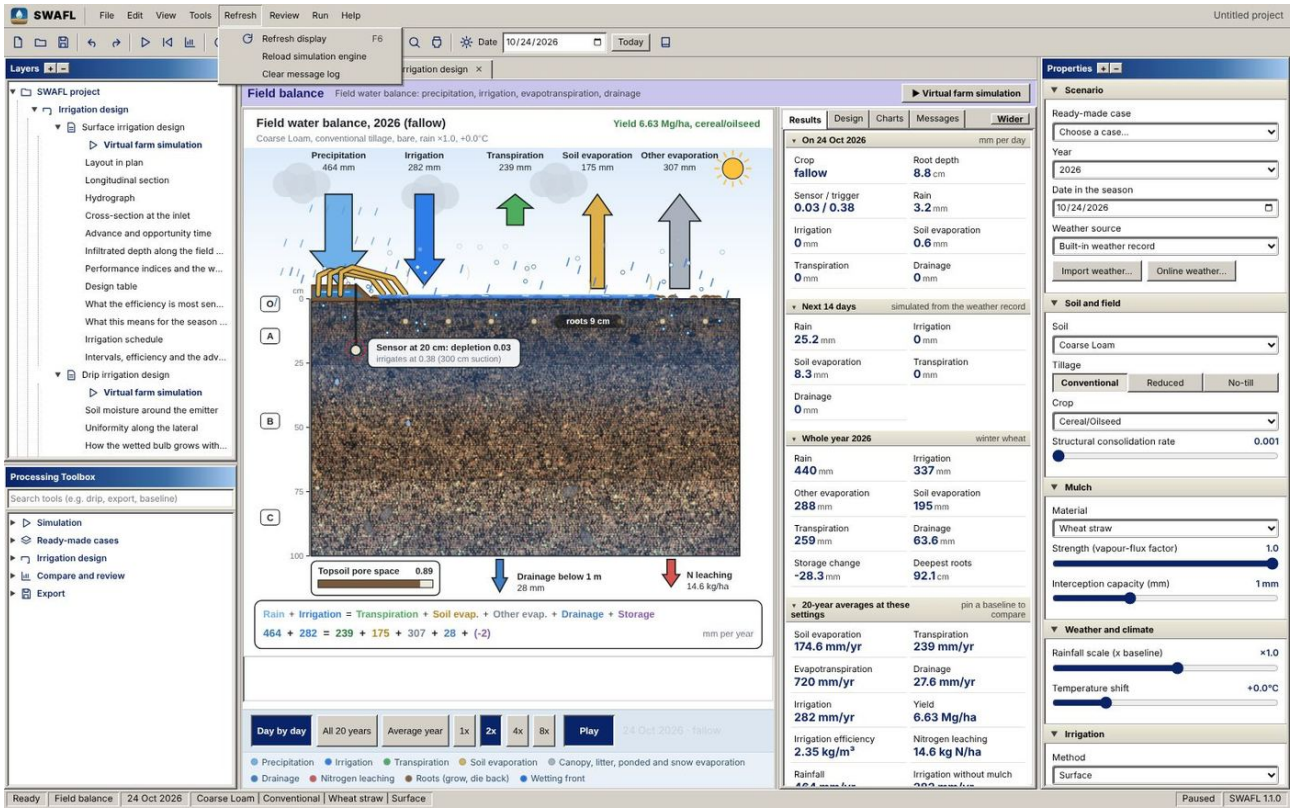


Figure 1.7: The Refresh menu.

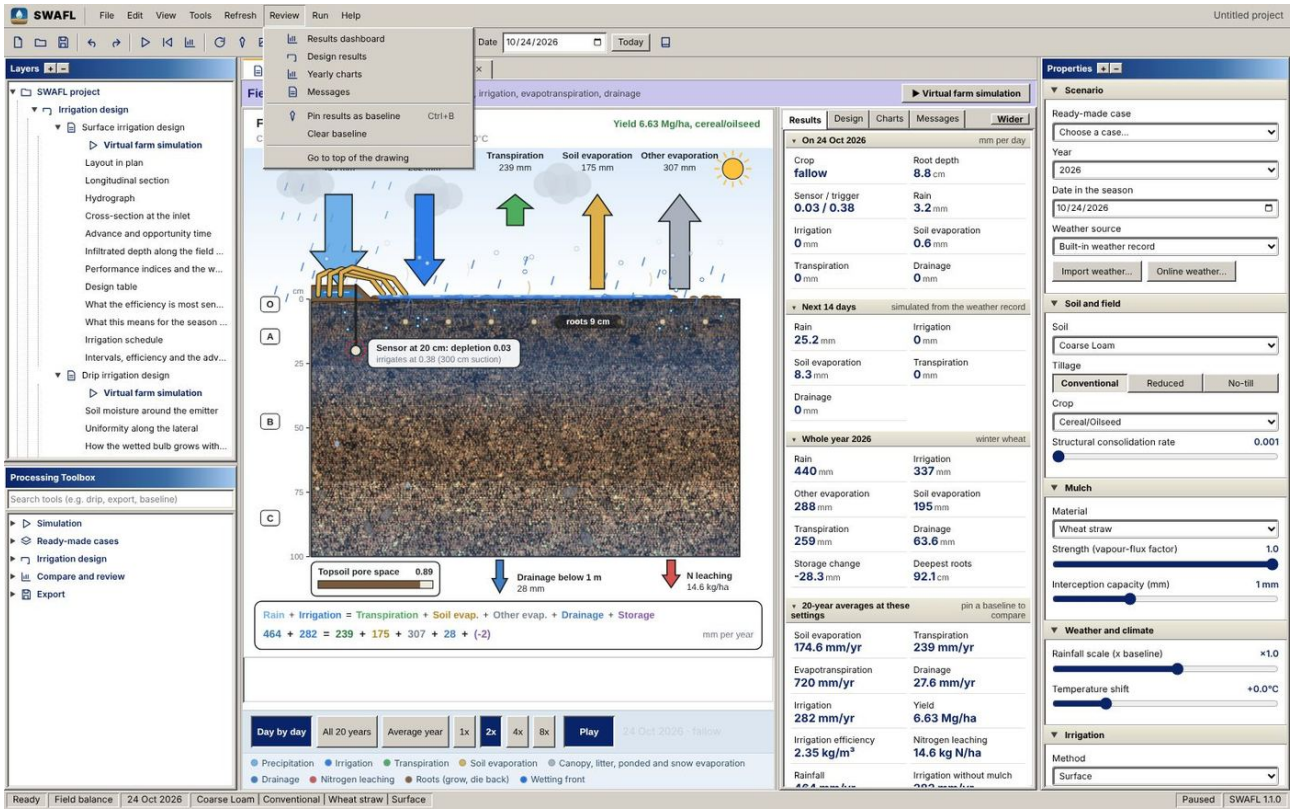


Figure 1.8: The Review menu.

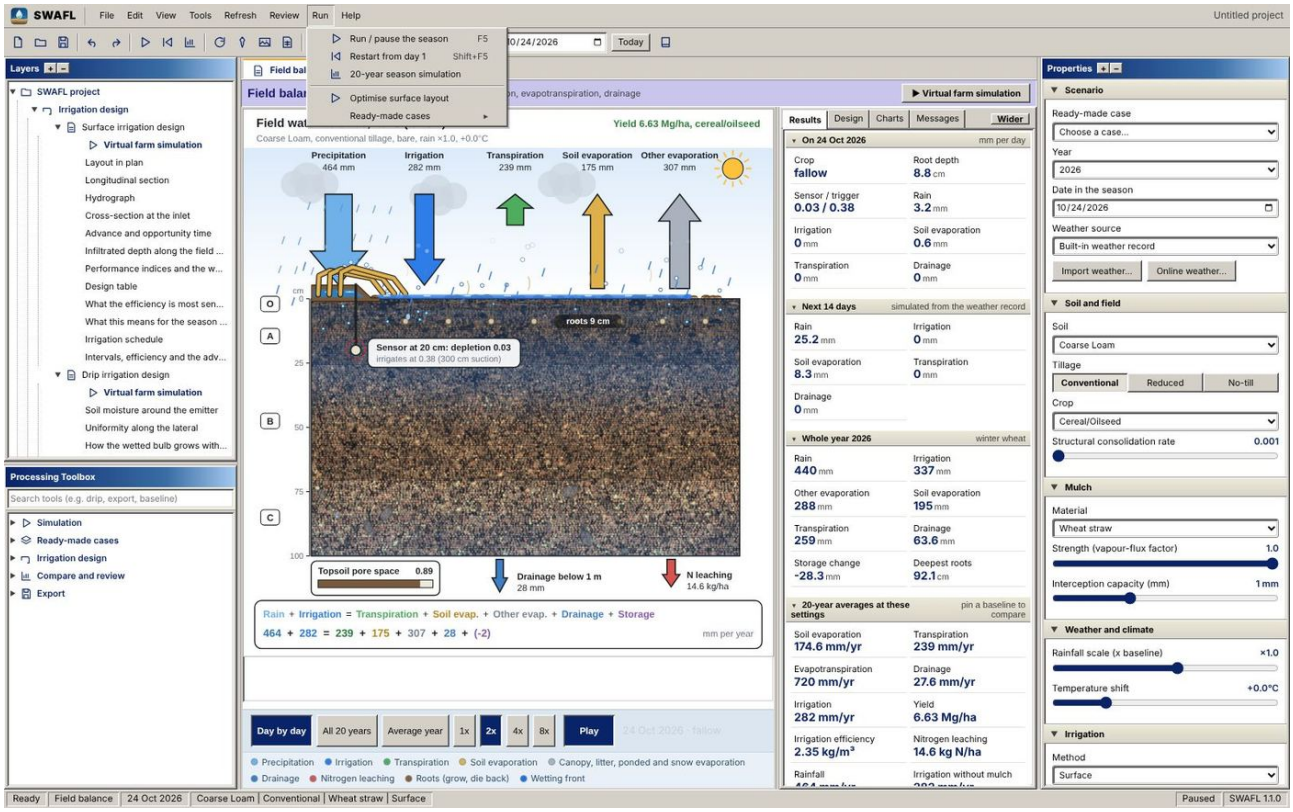


Figure 1.9: The Run menu.

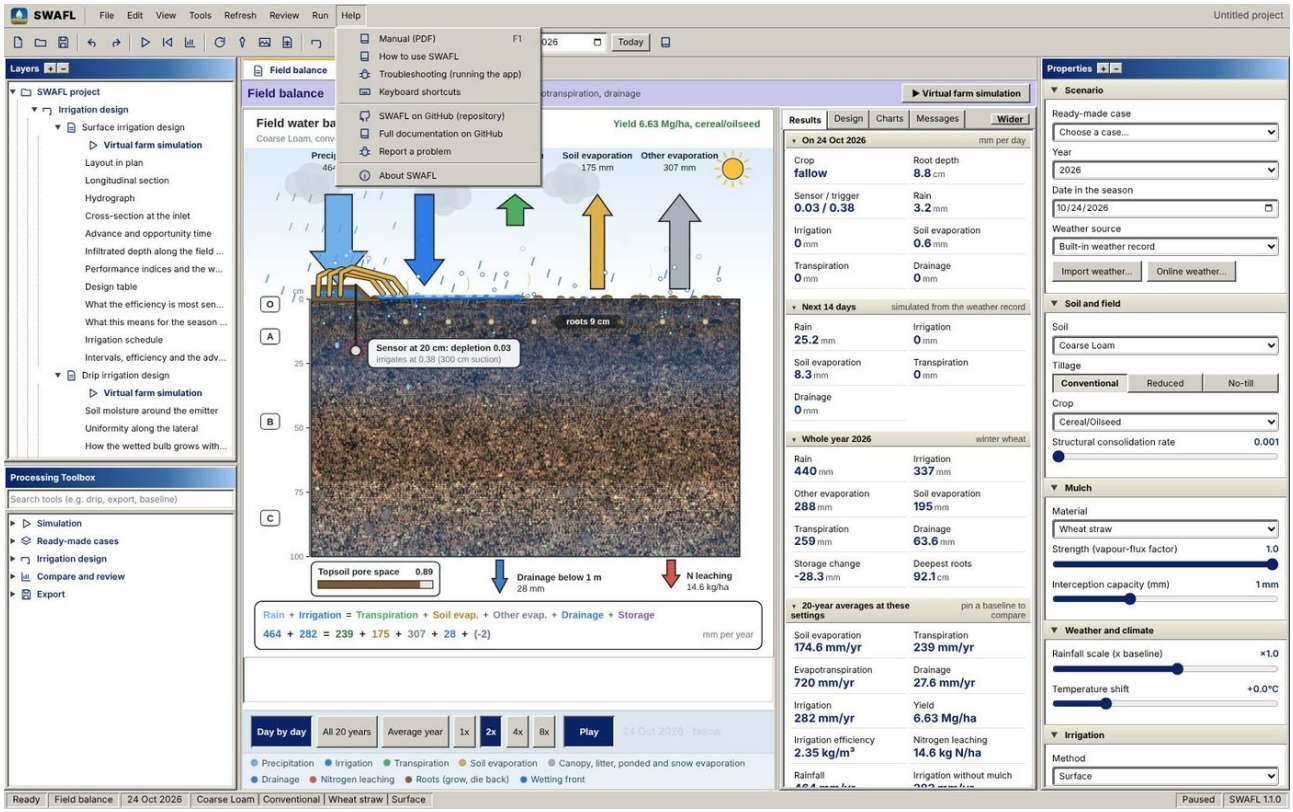


Figure 1.10: The Help menu.

1.3 The dialogs

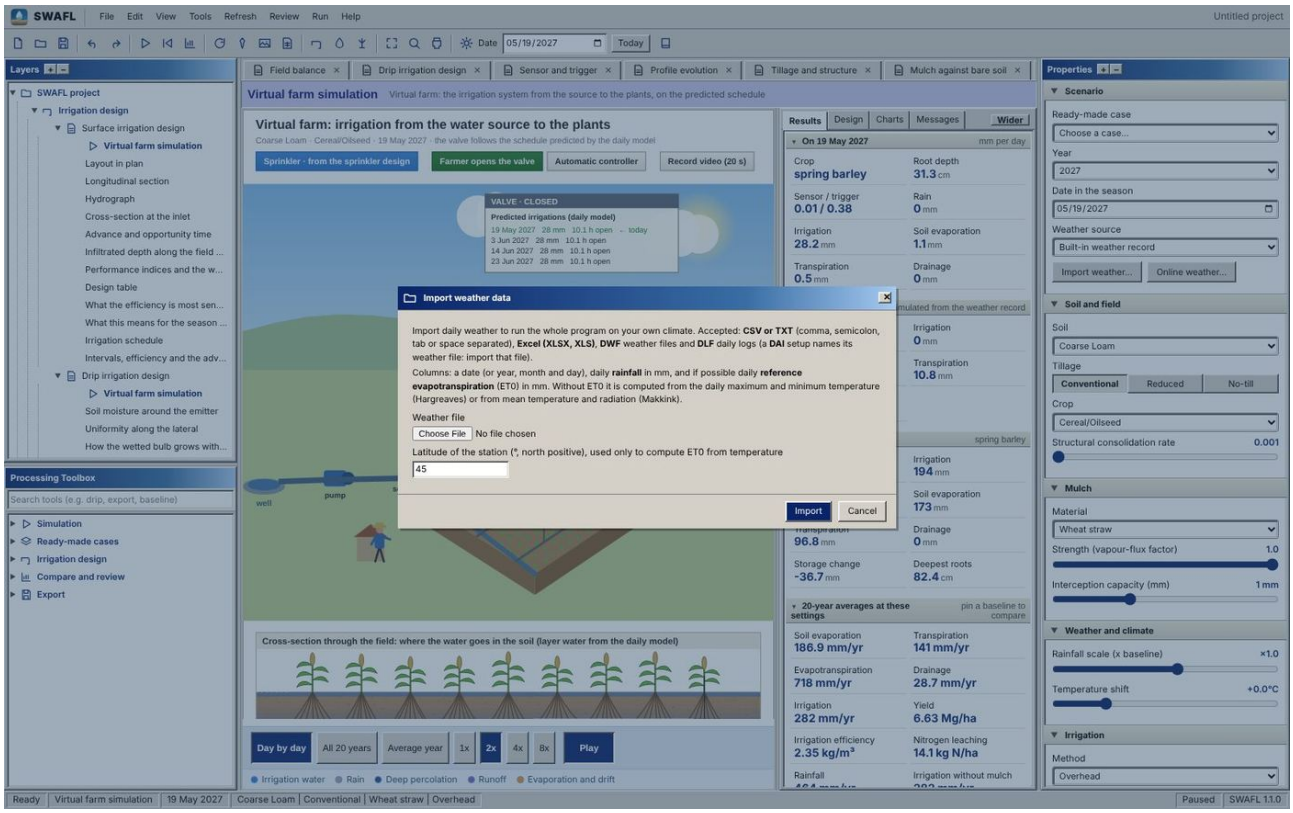


Figure 1.11: Import weather data.

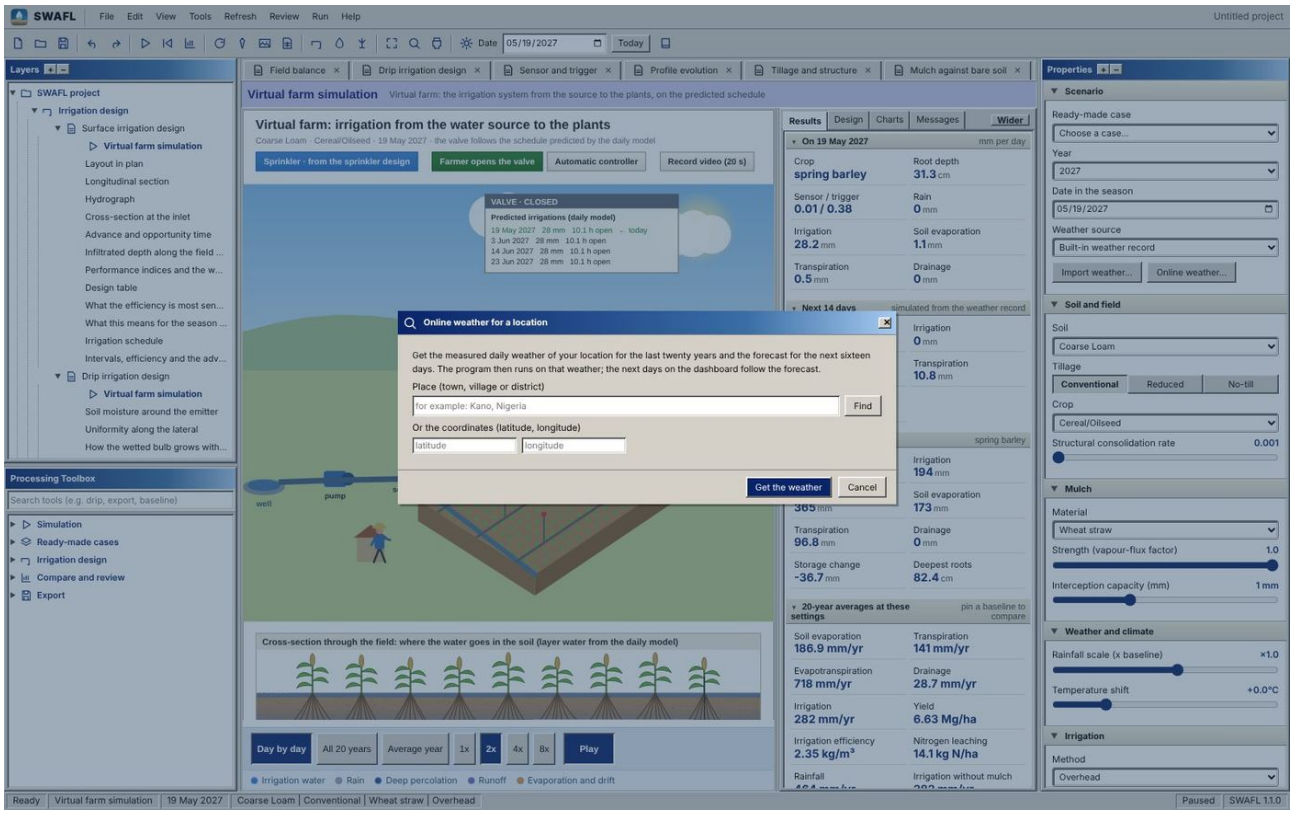


Figure 1.12: Online weather for a location.

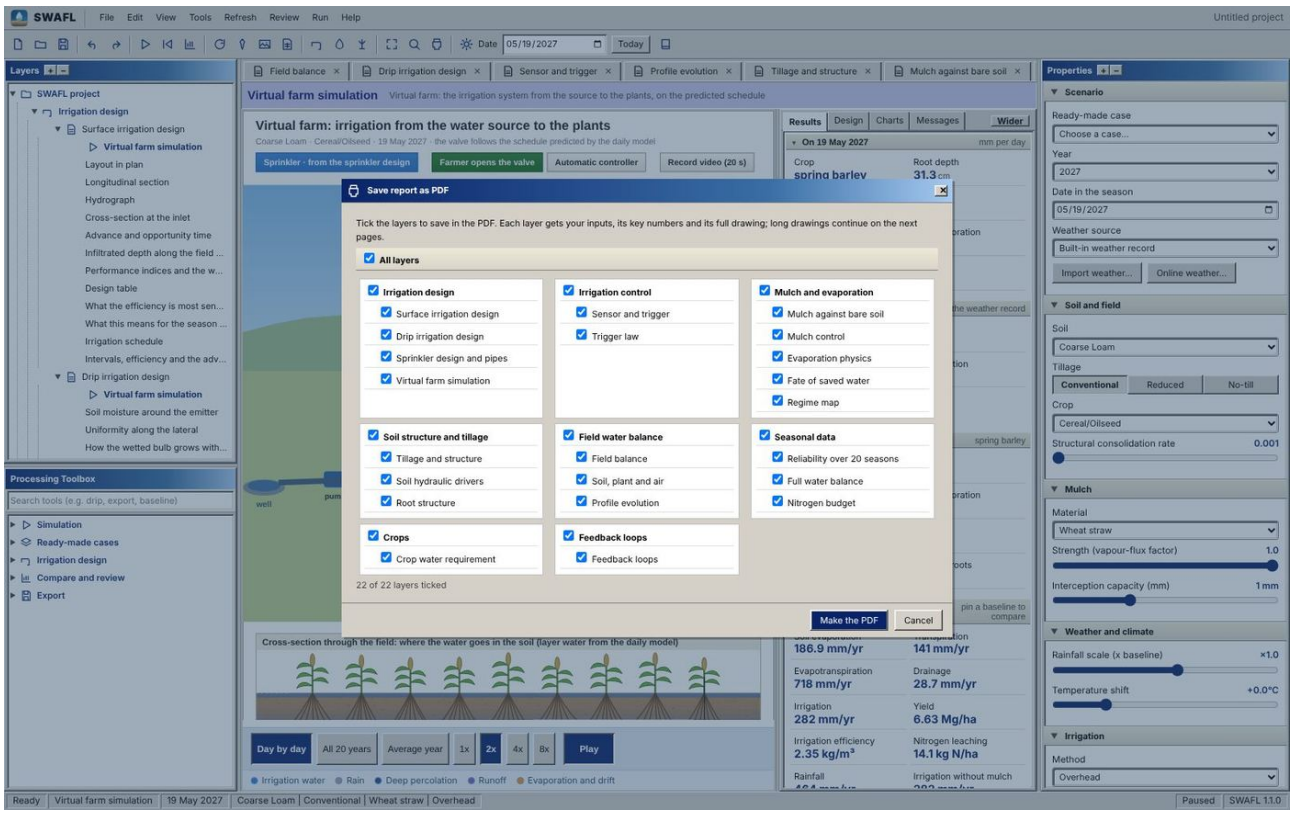


Figure 1.13: Save report as PDF: choose the layers (all ticked).

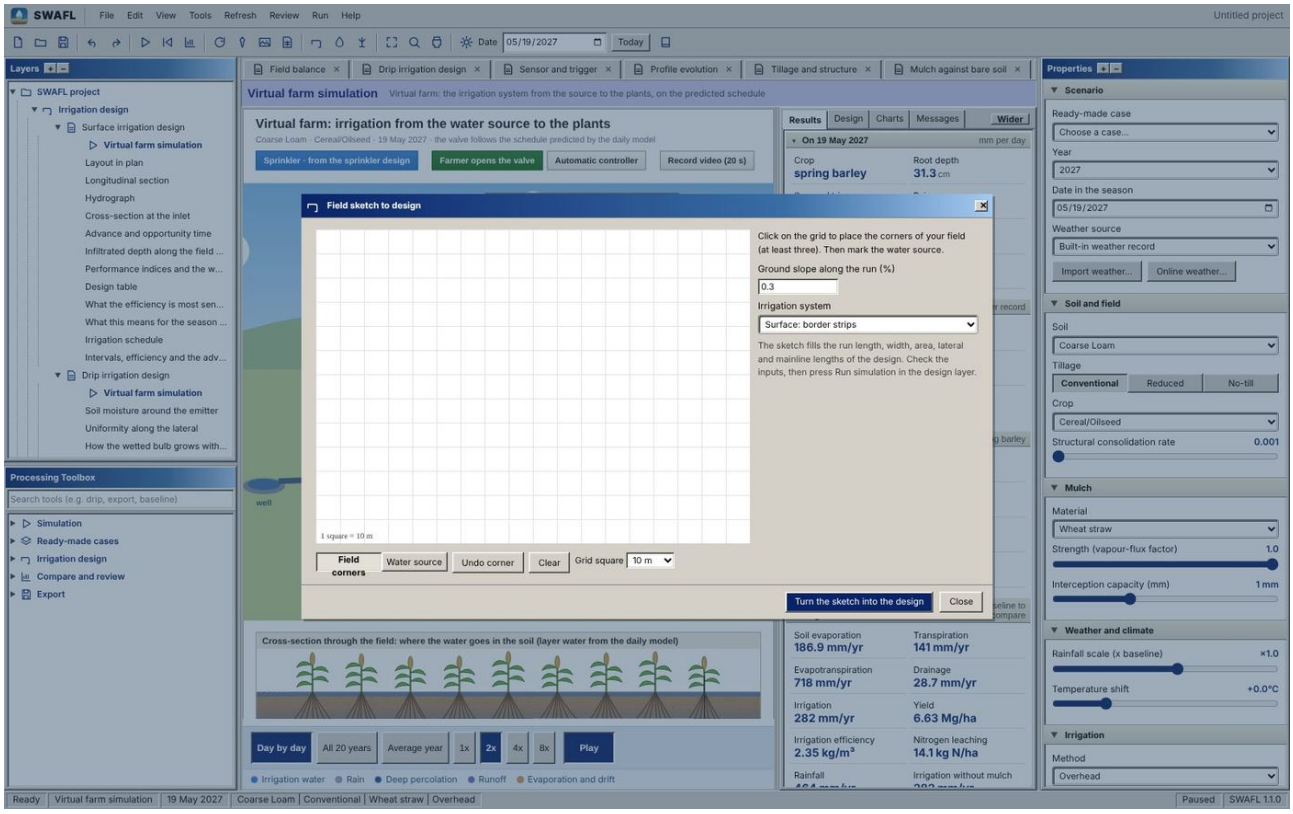


Figure 1.14: Field sketch to design.

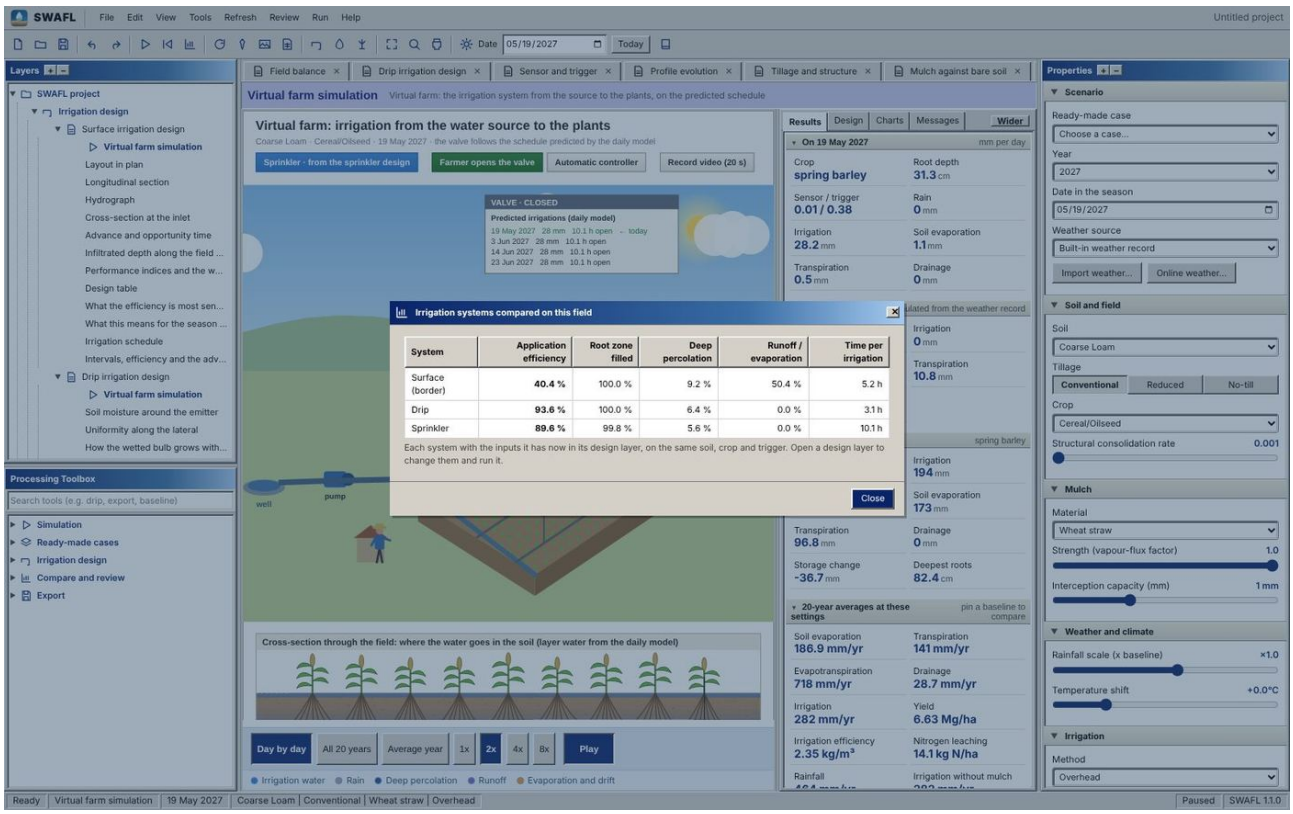


Figure 1.15: The irrigation systems compared on this field.

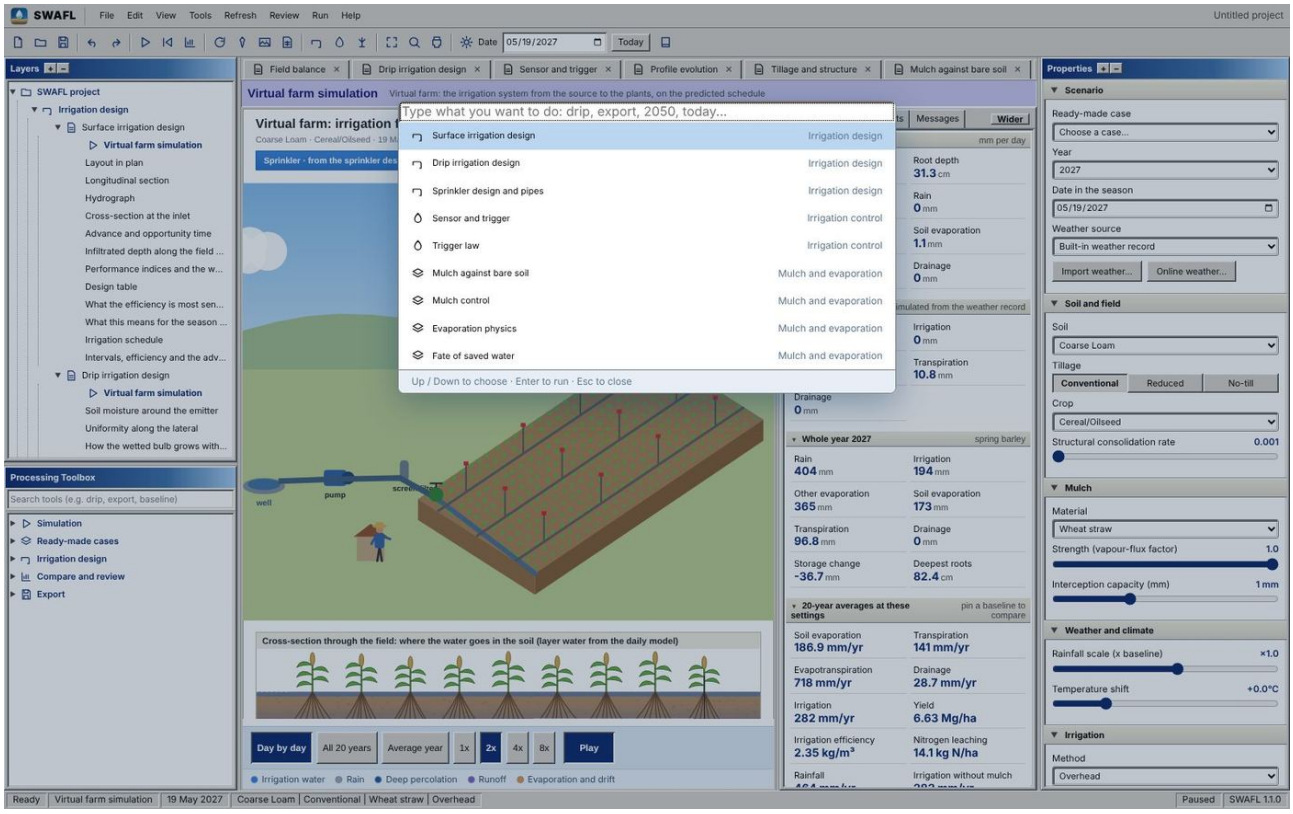


Figure 1.16: The command palette (Ctrl+K).

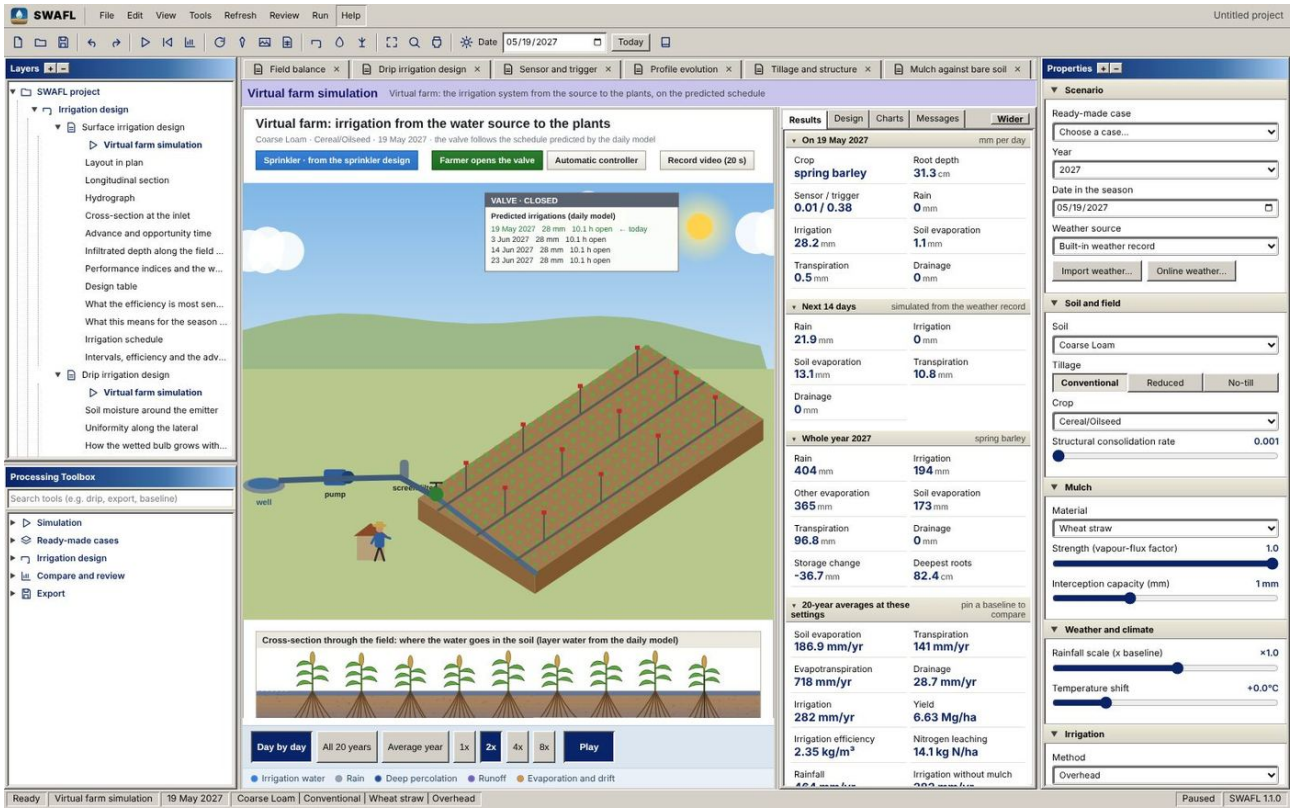


Figure 1.17: Help (F1).

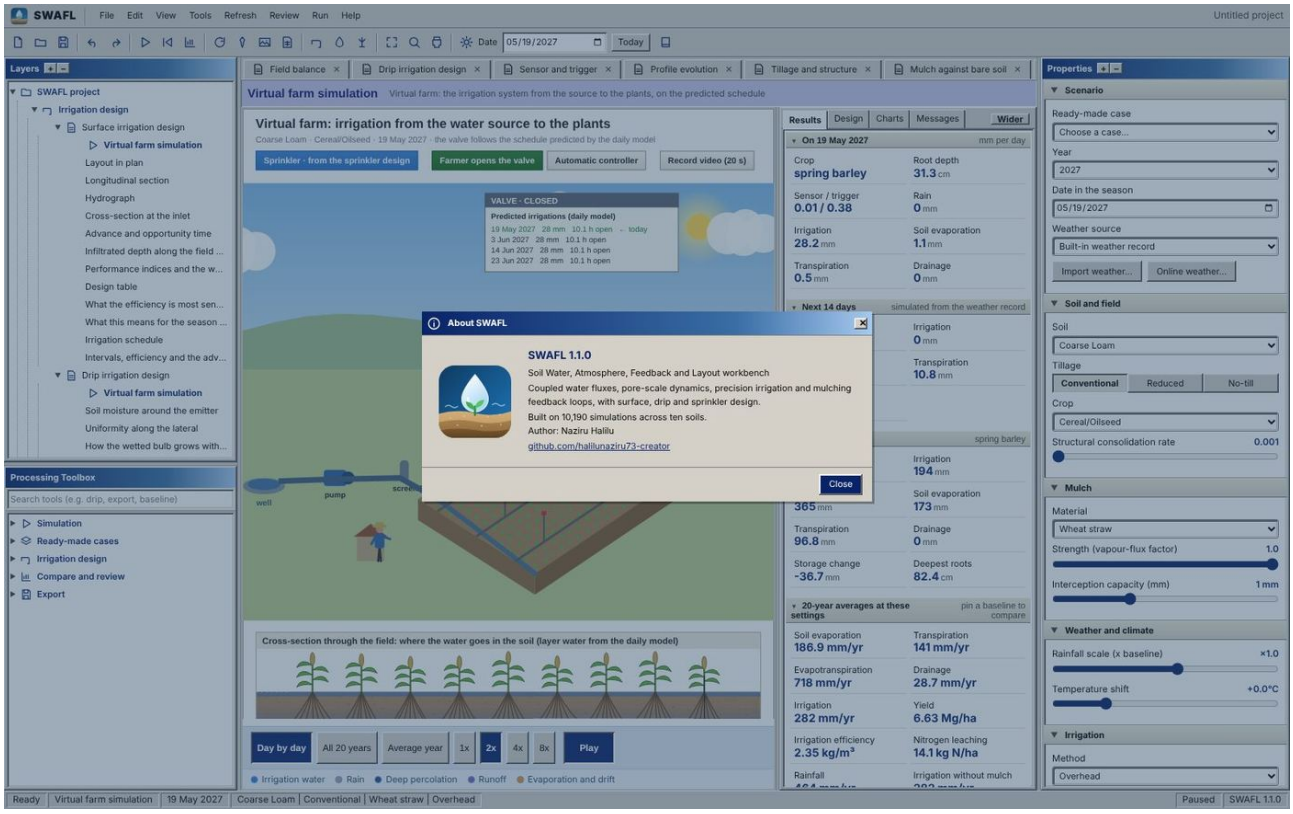


Figure 1.18: About SWAFL.

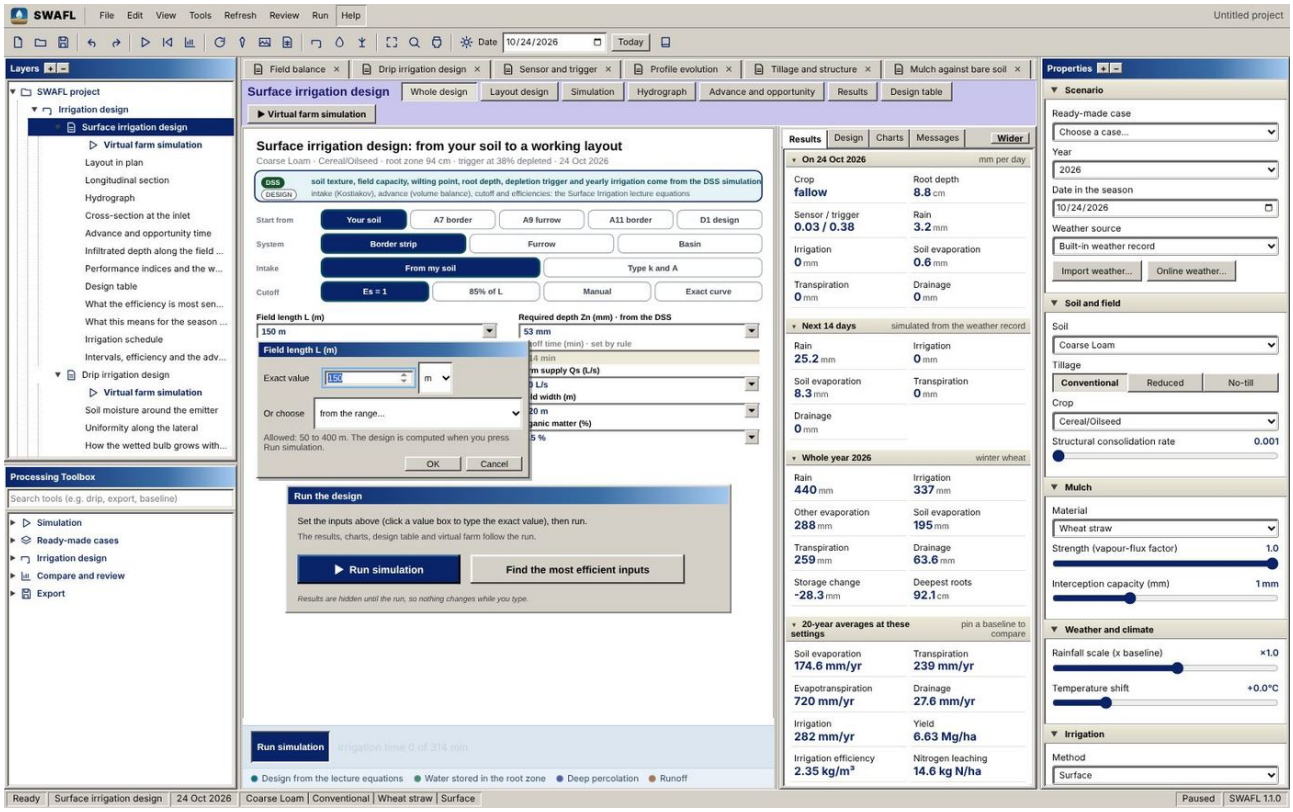


Figure 1.19: The value window of a design input: exact value, unit and the list of the allowed range.

1.4 The Properties panel

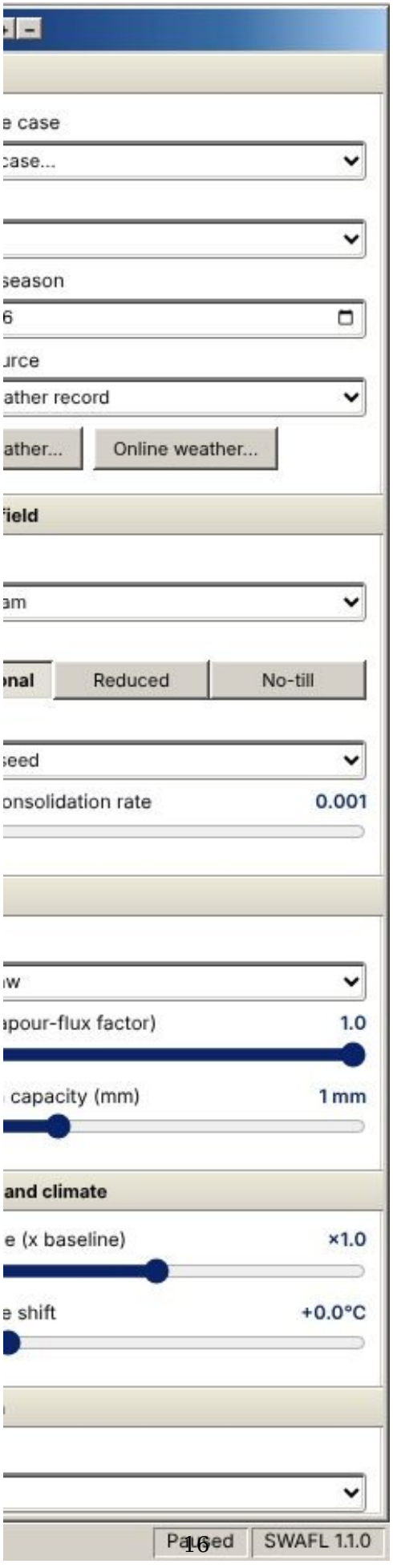


Figure 1.20: The Properties panel: Scenario, Soil and field, Mulch, Weather and climate, Irriga-

1.5 The dashboard

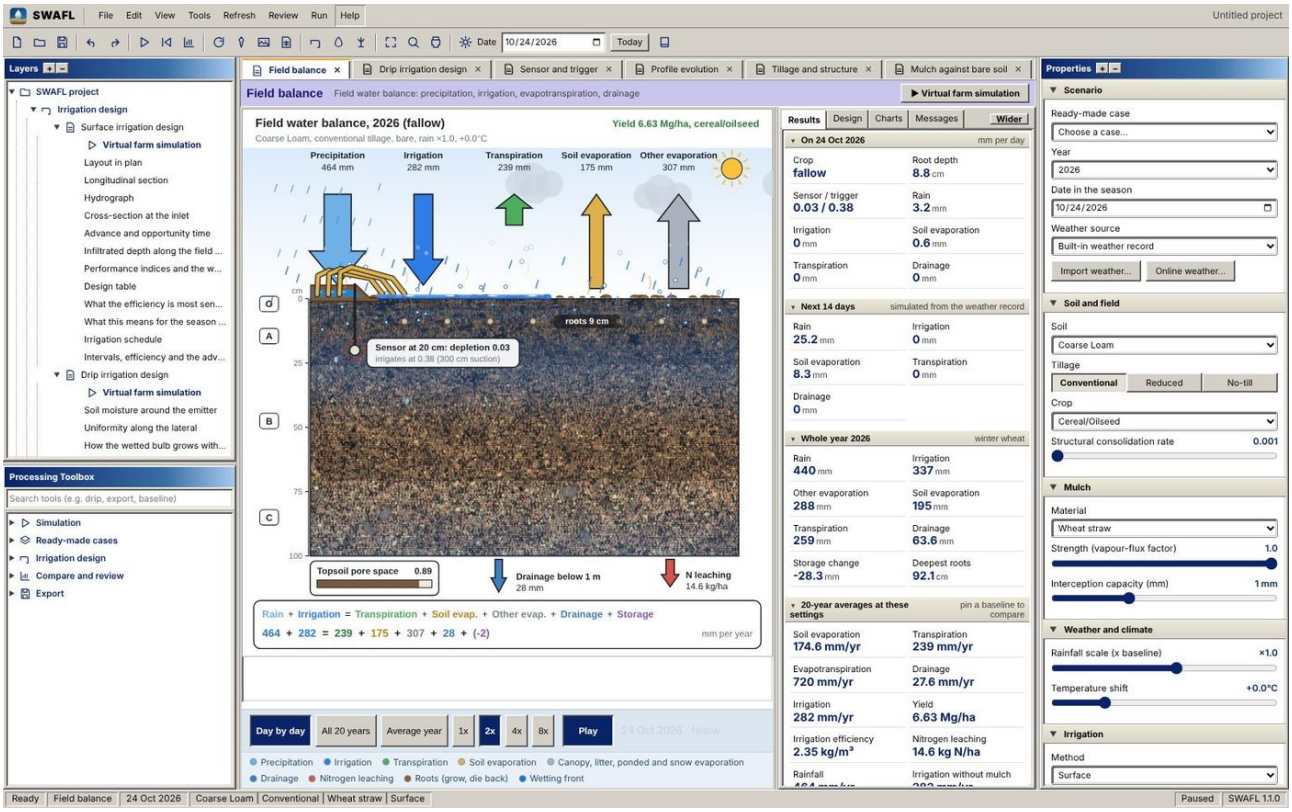


Figure 1.21: The Results tab of the dashboard with Field balance open.

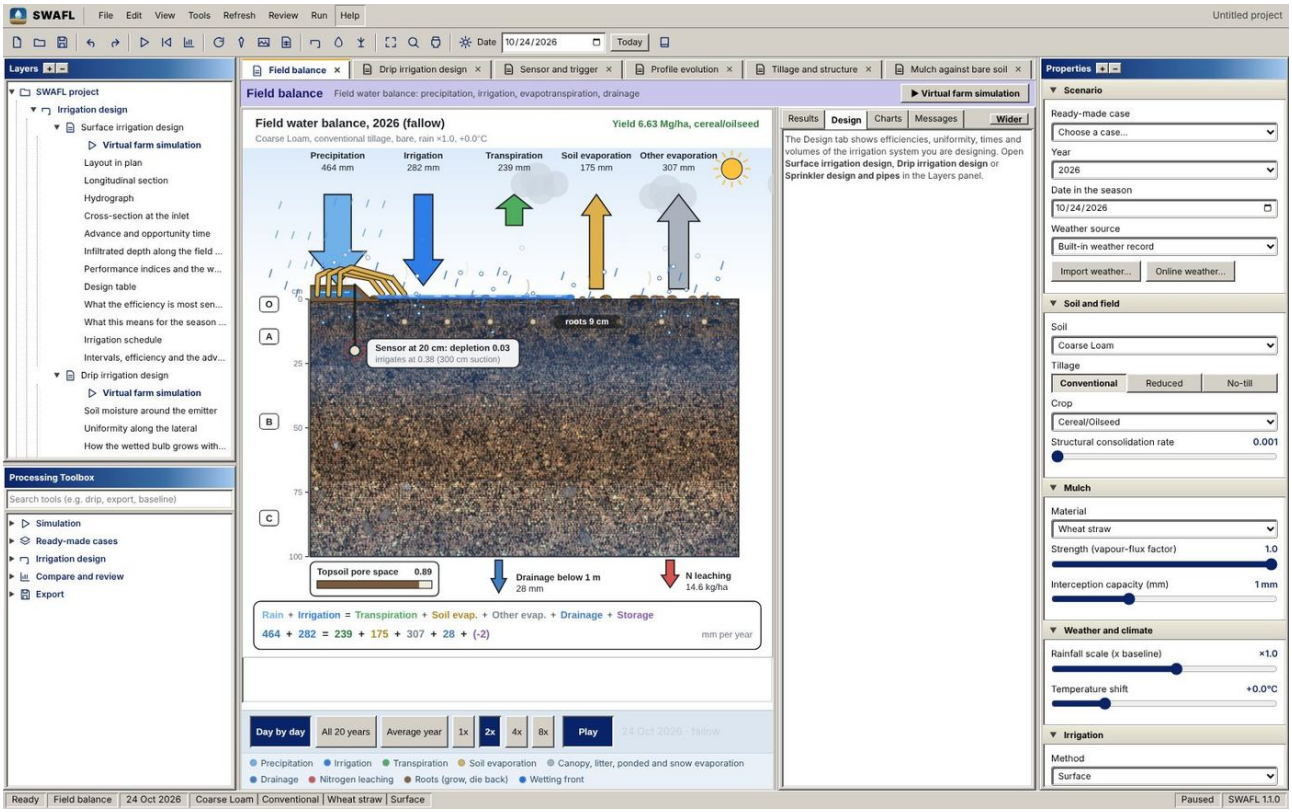


Figure 1.22: The Design tab of the dashboard with Field balance open.

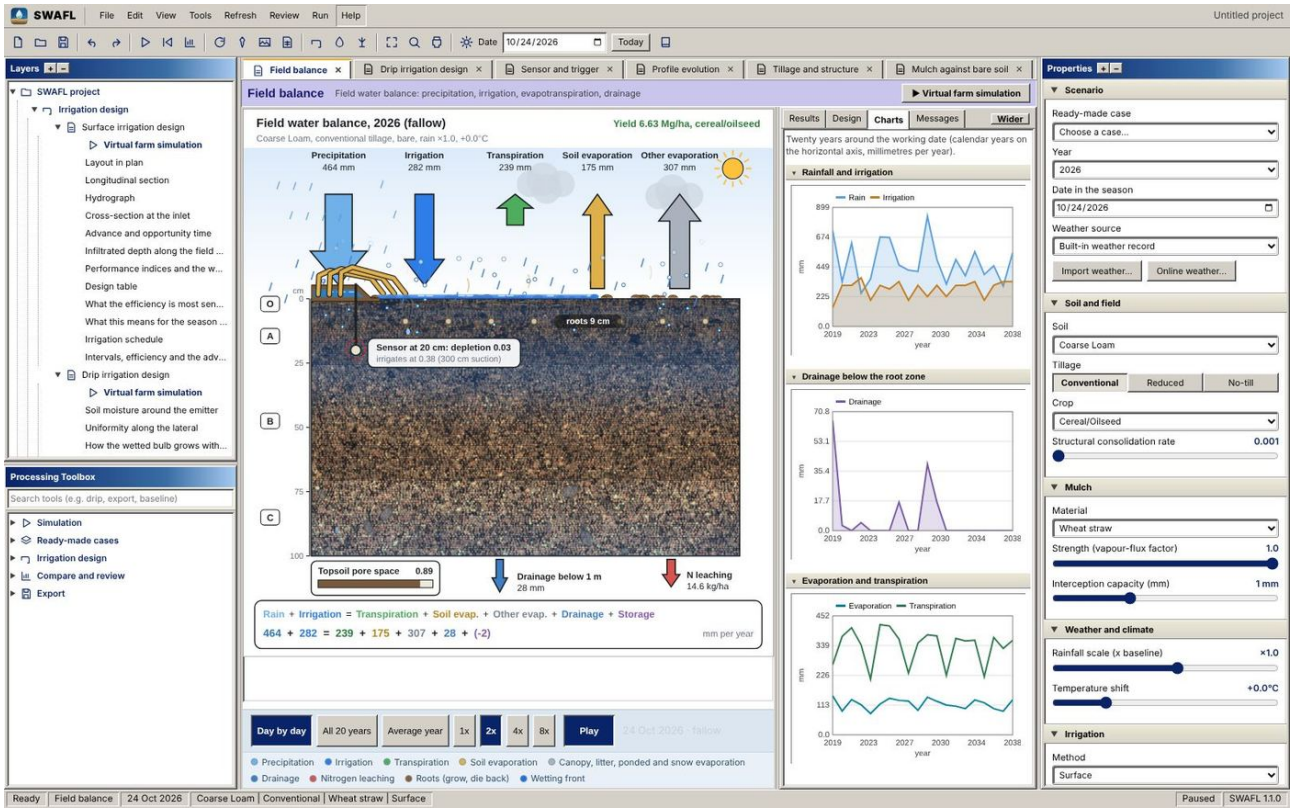


Figure 1.23: The Charts tab of the dashboard with Field balance open.

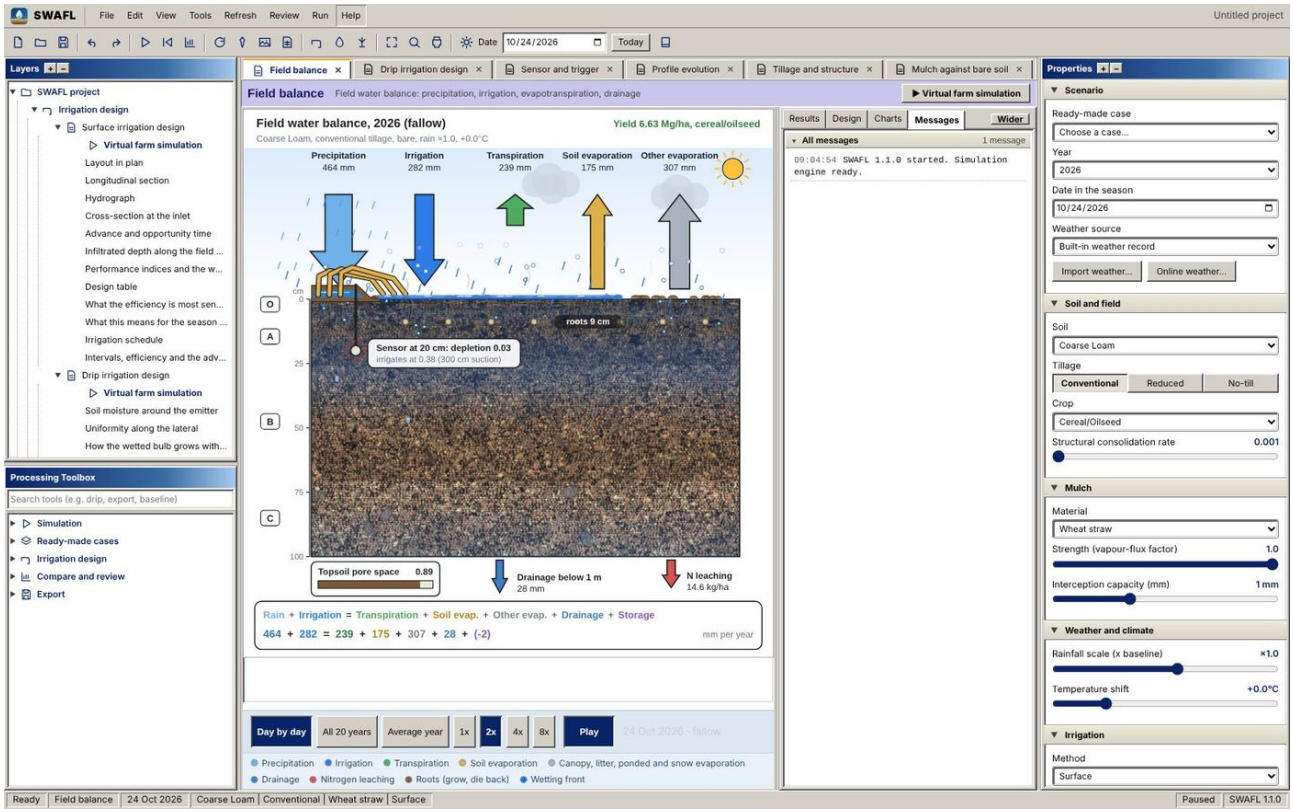


Figure 1.24: The Messages tab of the dashboard with Field balance open.

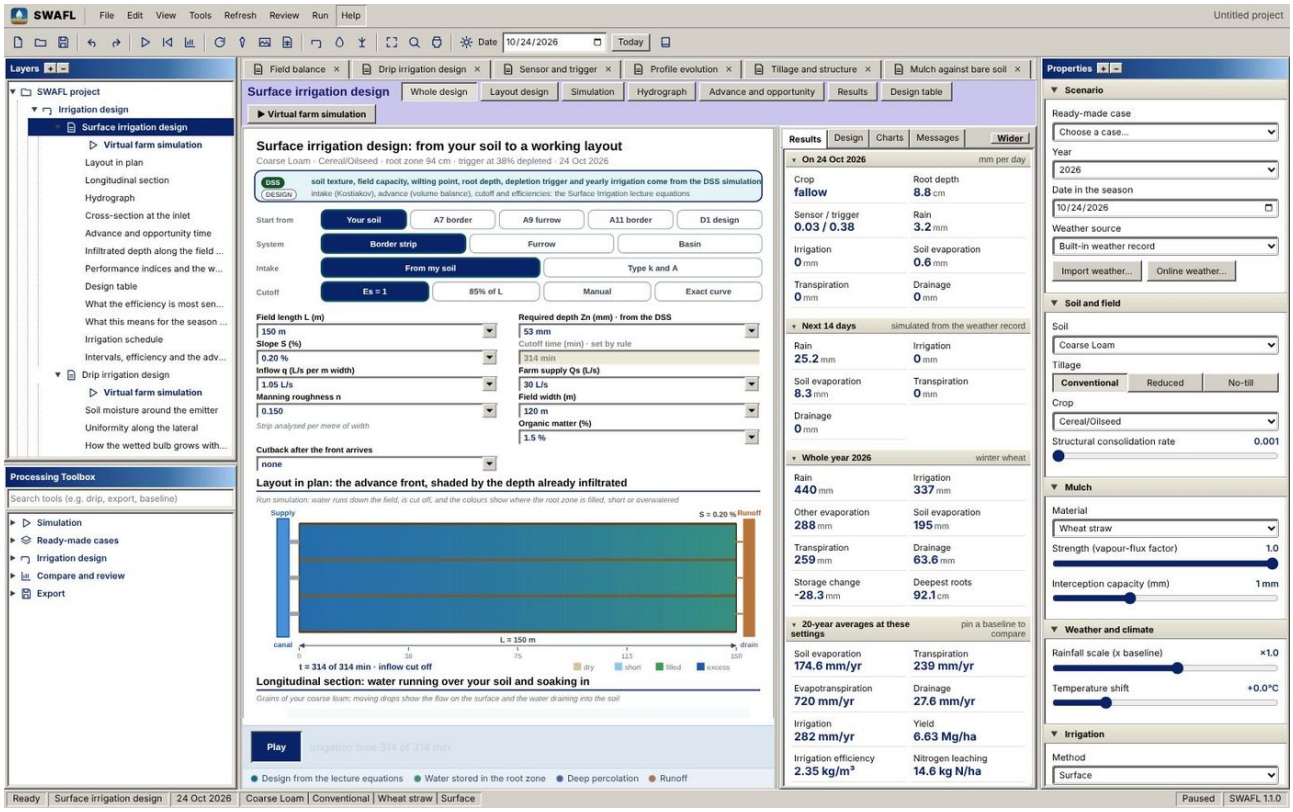


Figure 1.25: The Results tab of the dashboard with Surface irrigation design open.

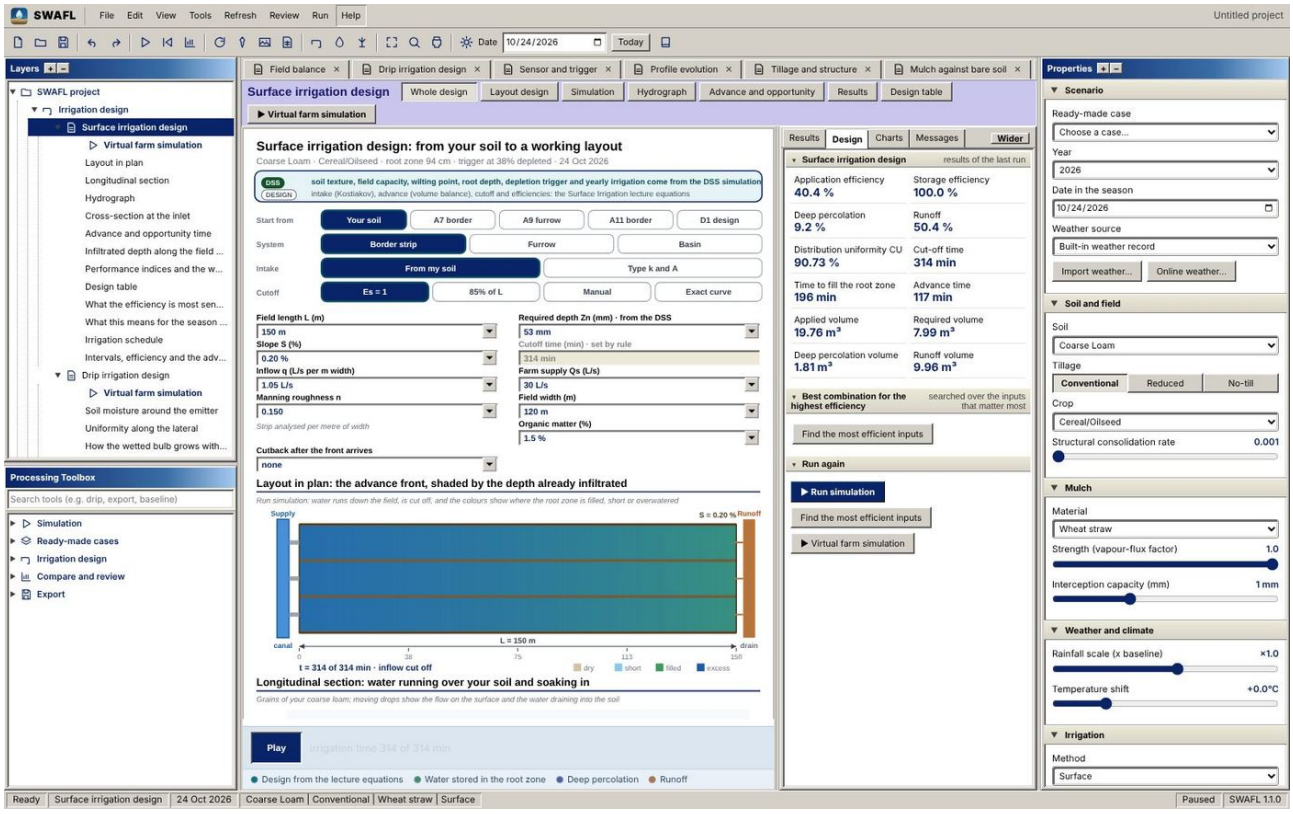


Figure 1.26: The Design tab of the dashboard with Surface irrigation design open.

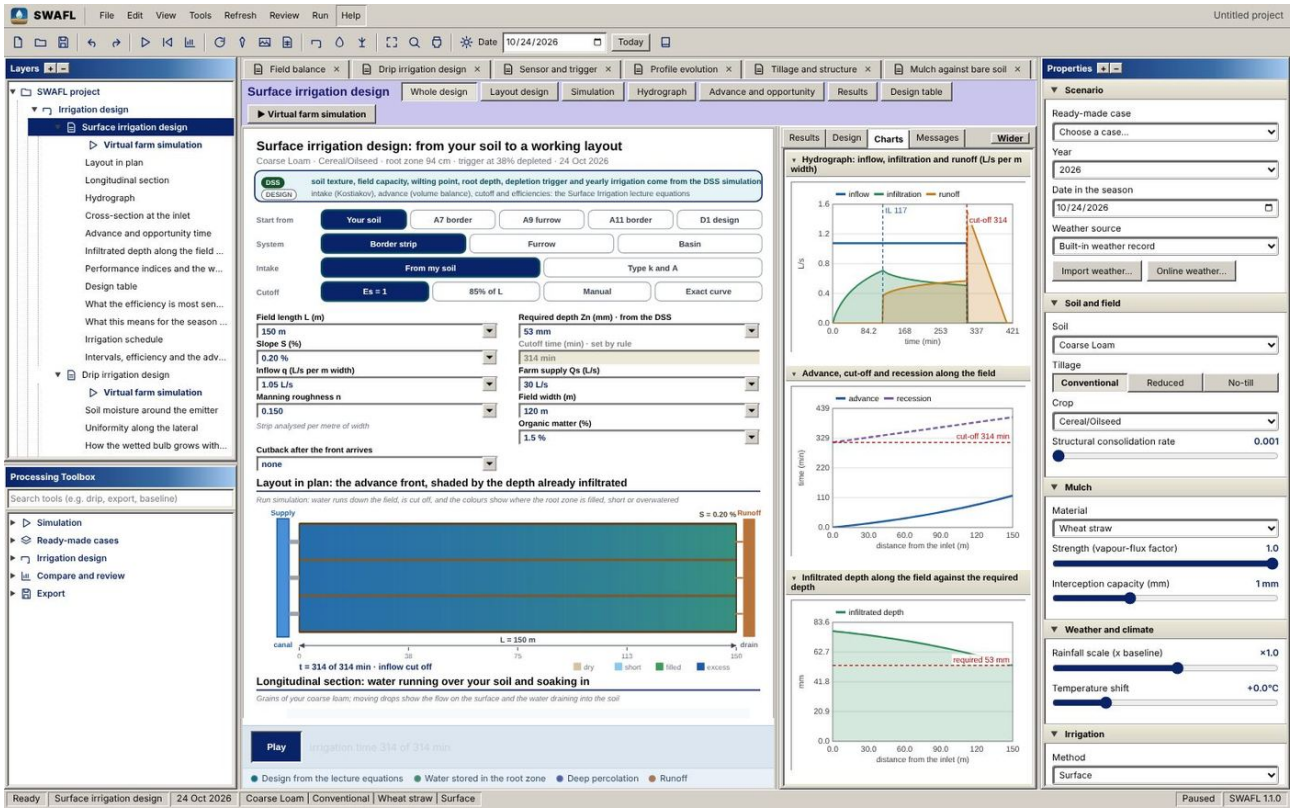


Figure 1.27: The Charts tab of the dashboard with Surface irrigation design open.

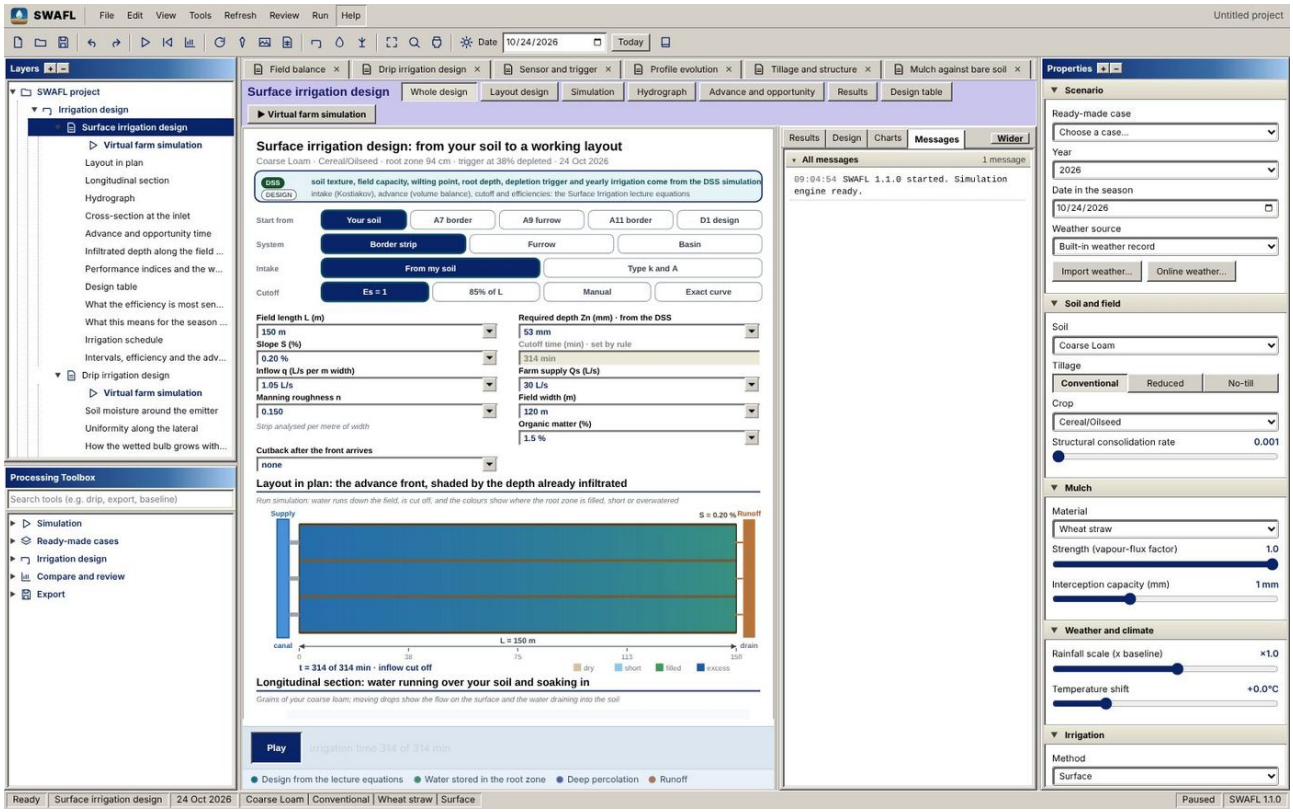


Figure 1.28: The Messages tab of the dashboard with Surface irrigation design open.

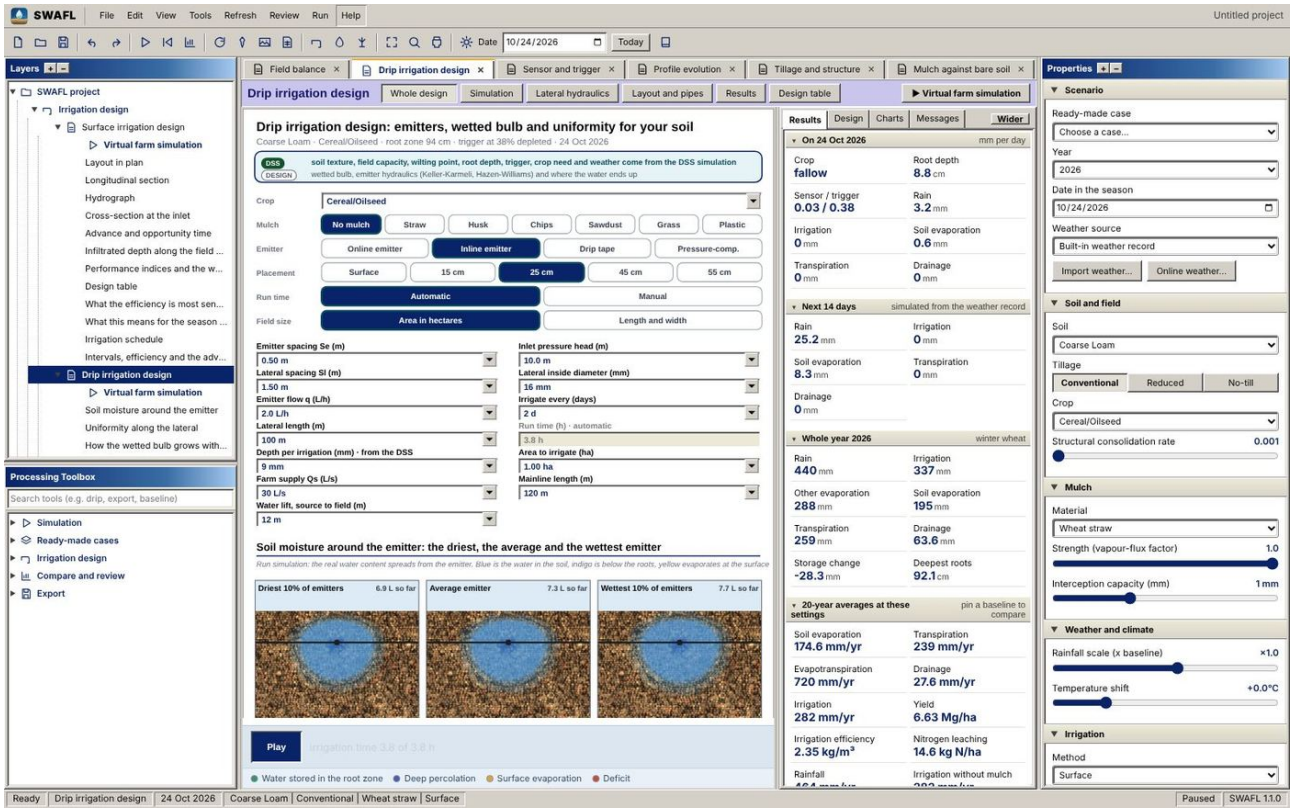


Figure 1.29: The Results tab of the dashboard with Drip irrigation design open.

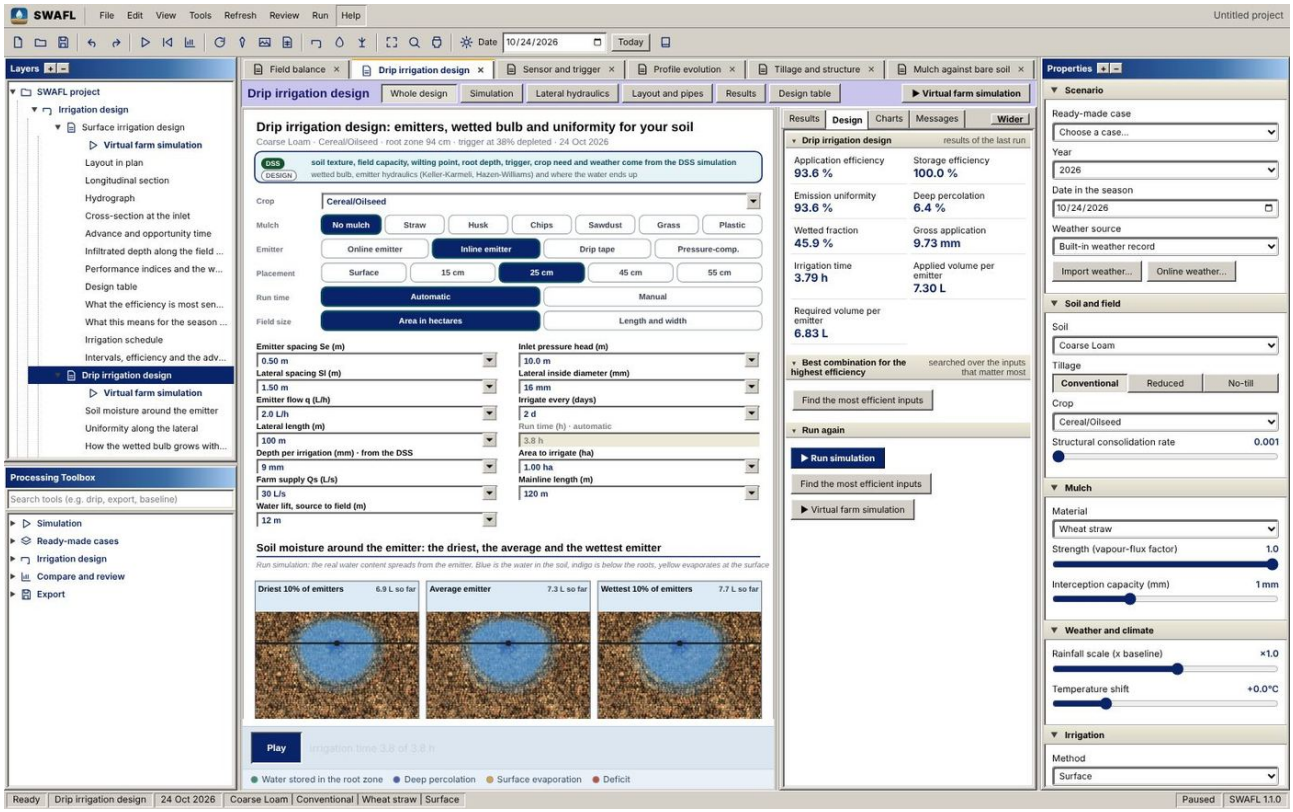


Figure 1.30: The Design tab of the dashboard with Drip irrigation design open.

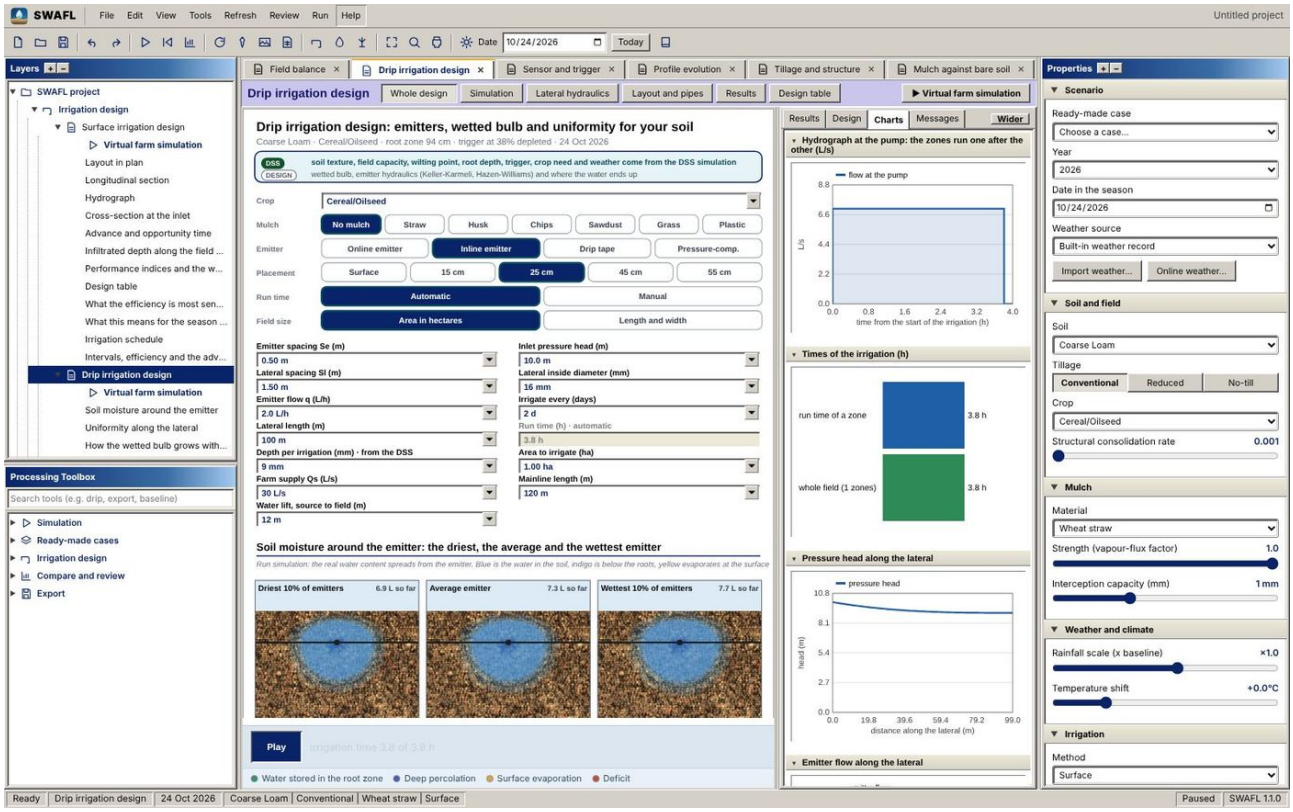


Figure 1.31: The Charts tab of the dashboard with Drip irrigation design open.

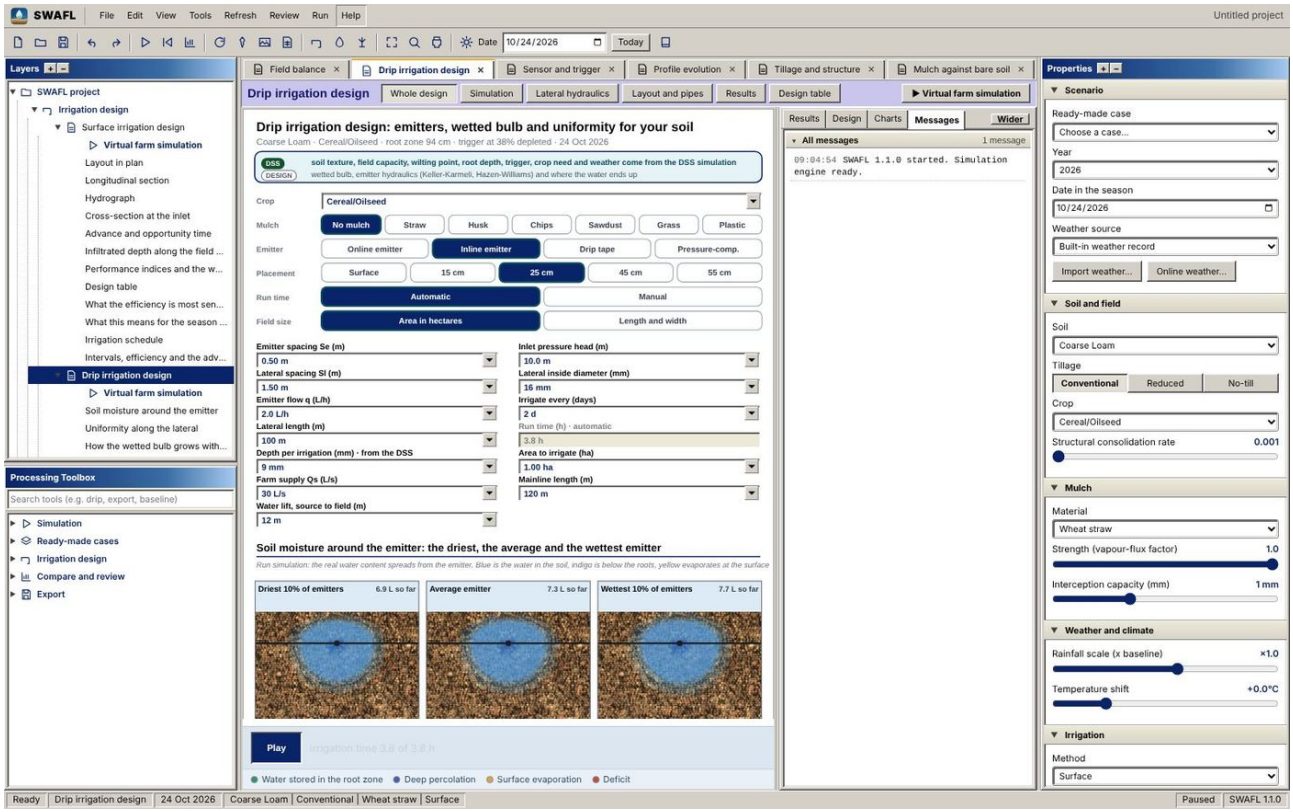


Figure 1.32: The Messages tab of the dashboard with Drip irrigation design open.

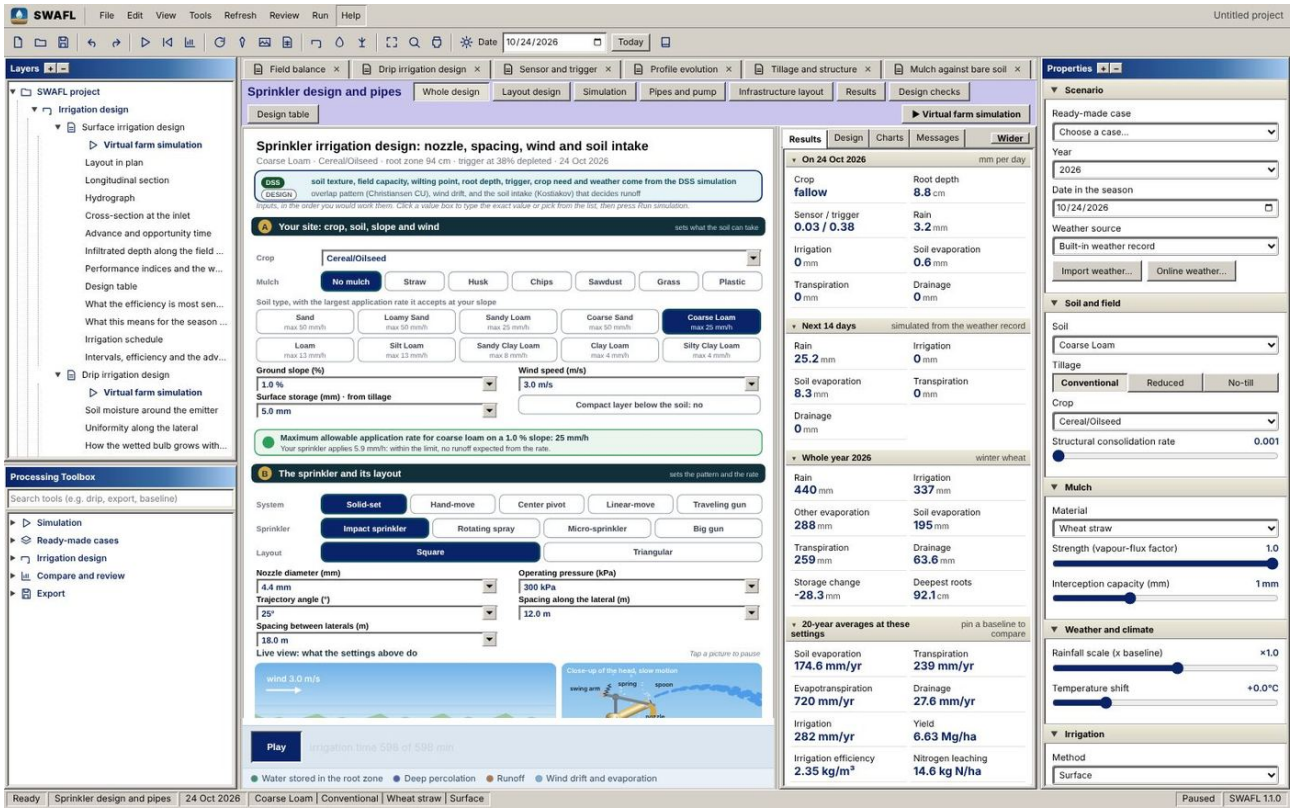


Figure 1.33: The Results tab of the dashboard with Sprinkler design and pipes open.

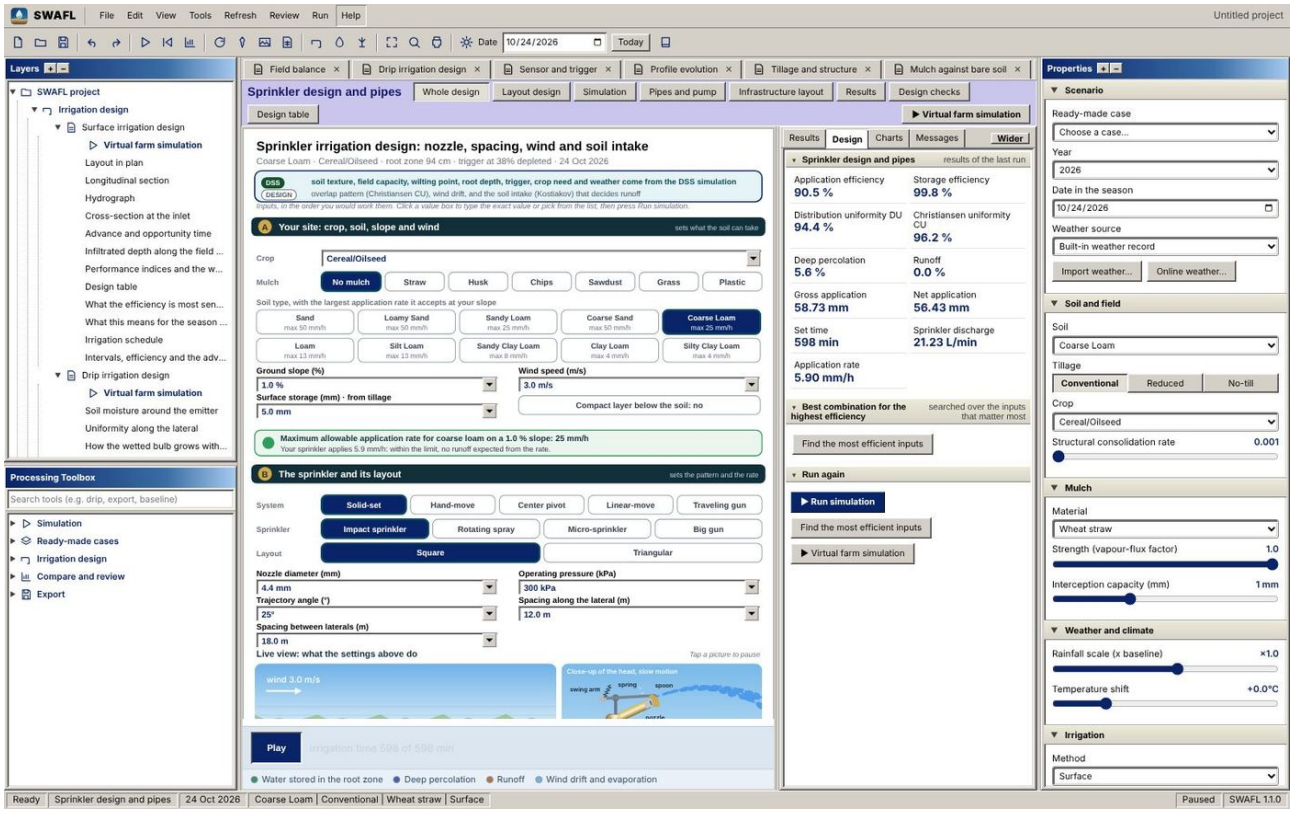


Figure 1.34: The Design tab of the dashboard with Sprinkler design and pipes open.

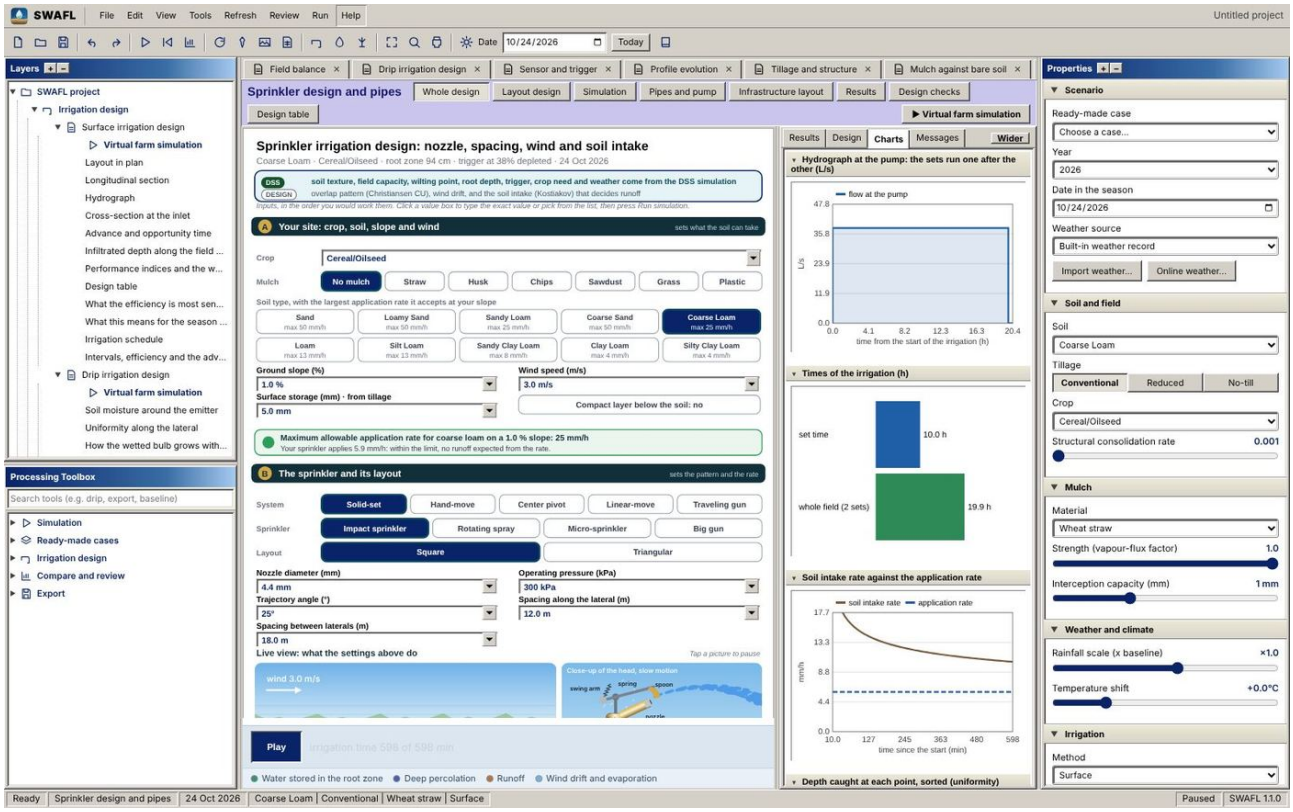


Figure 1.35: The Charts tab of the dashboard with Sprinkler design and pipes open.

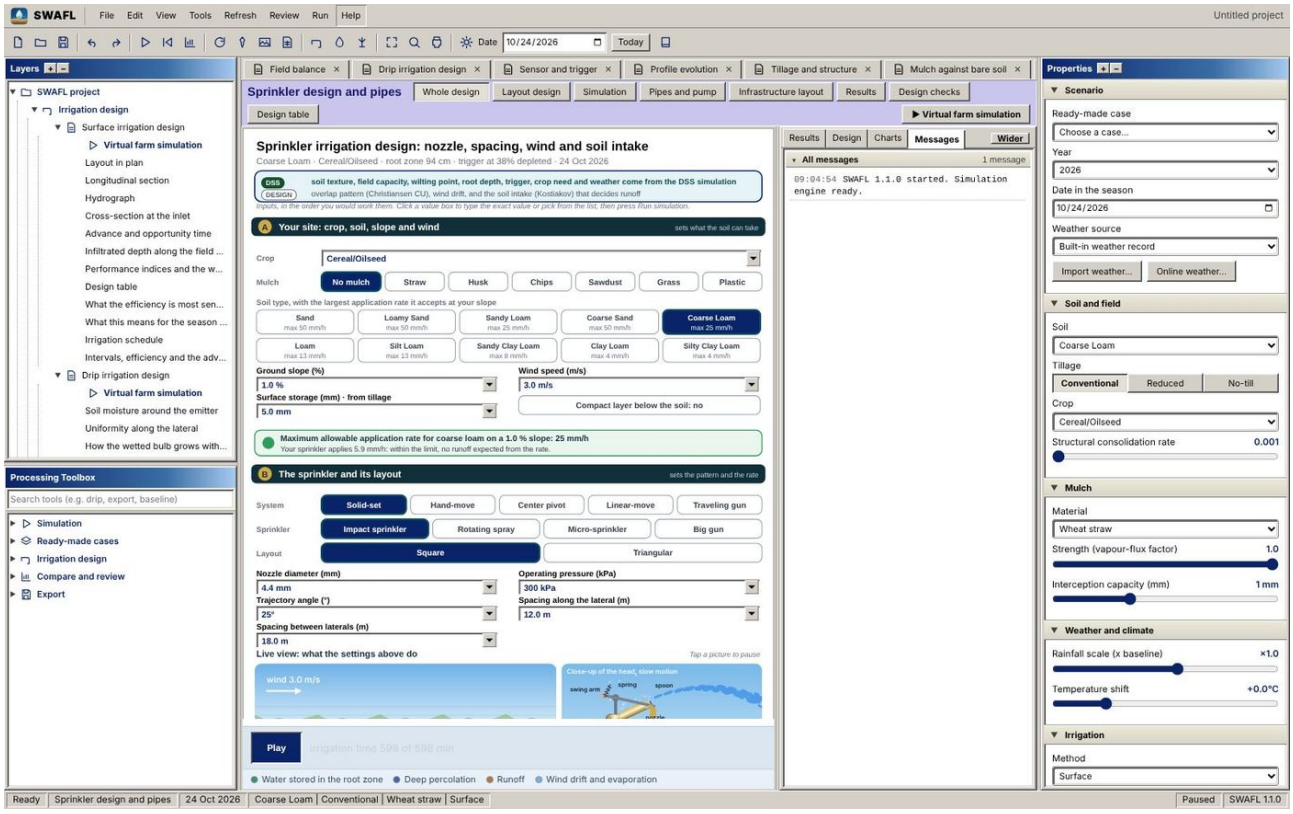


Figure 1.36: The Messages tab of the dashboard with Sprinkler design and pipes open.

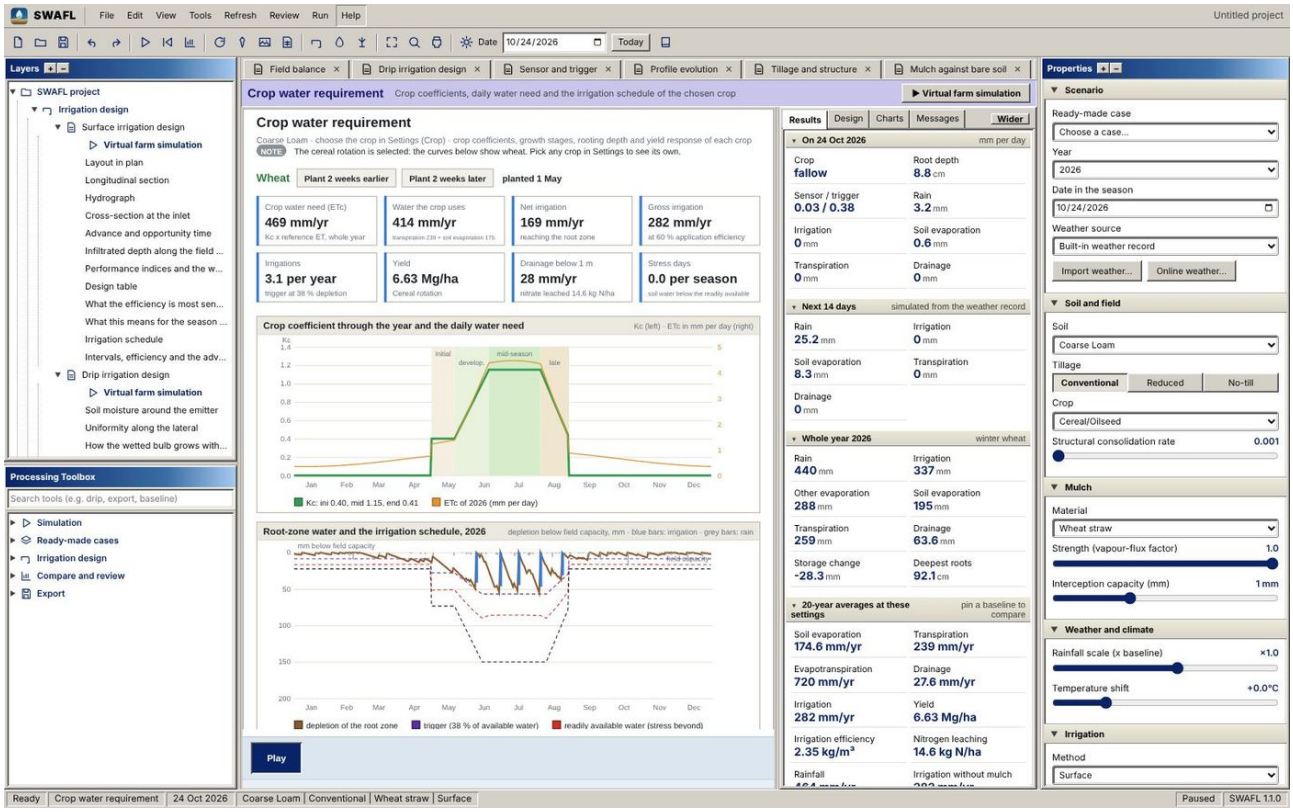


Figure 1.37: The Results tab of the dashboard with Crop water requirement open.

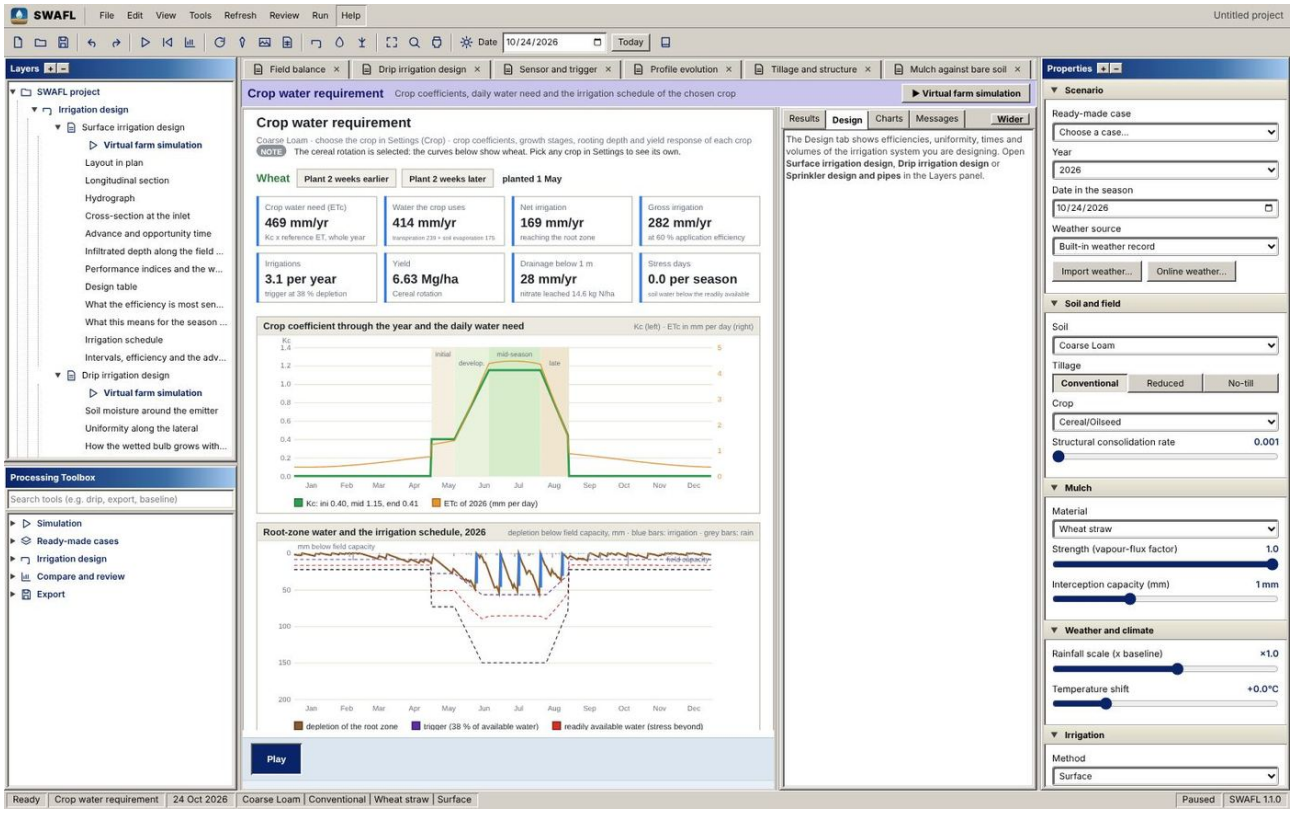


Figure 1.38: The Design tab of the dashboard with Crop water requirement open.

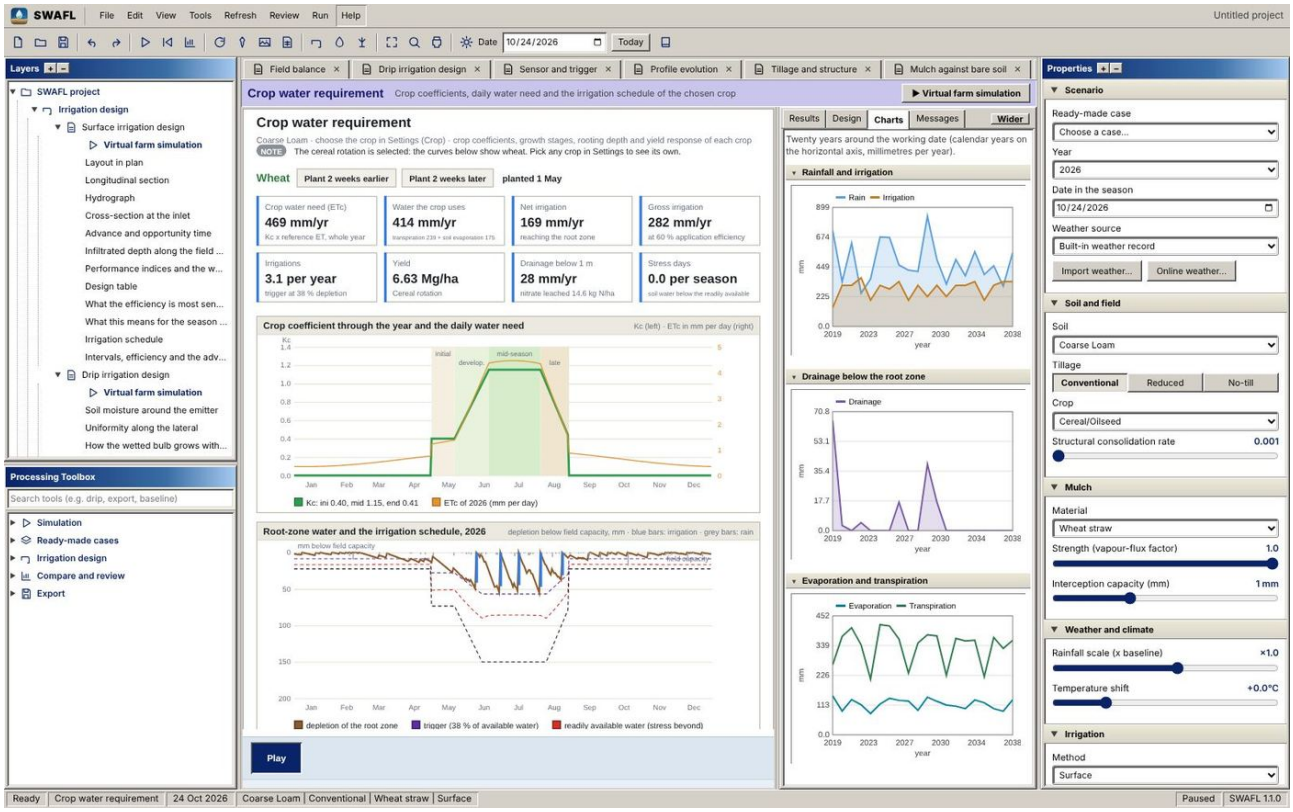


Figure 1.39: The Charts tab of the dashboard with Crop water requirement open.

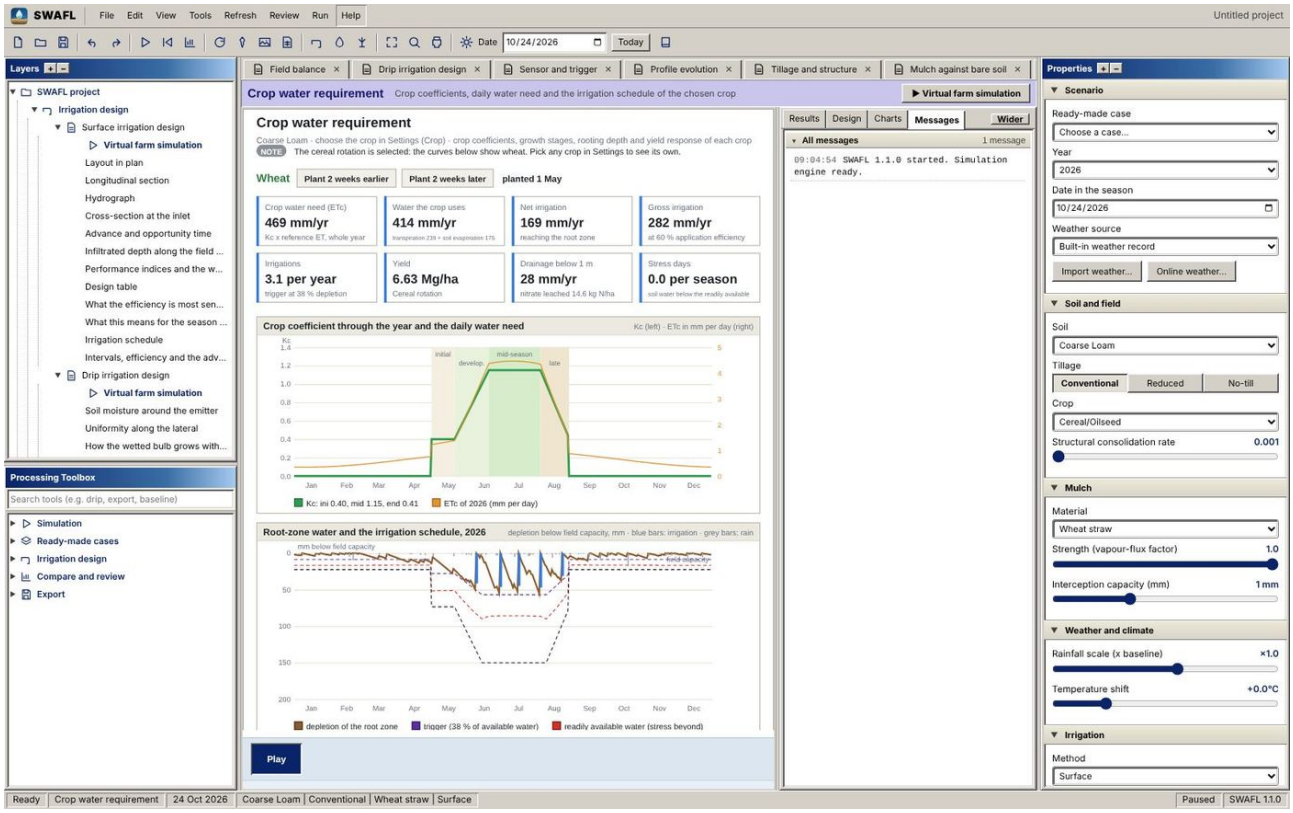


Figure 1.40: The Messages tab of the dashboard with Crop water requirement open.

1.6 Before a design runs

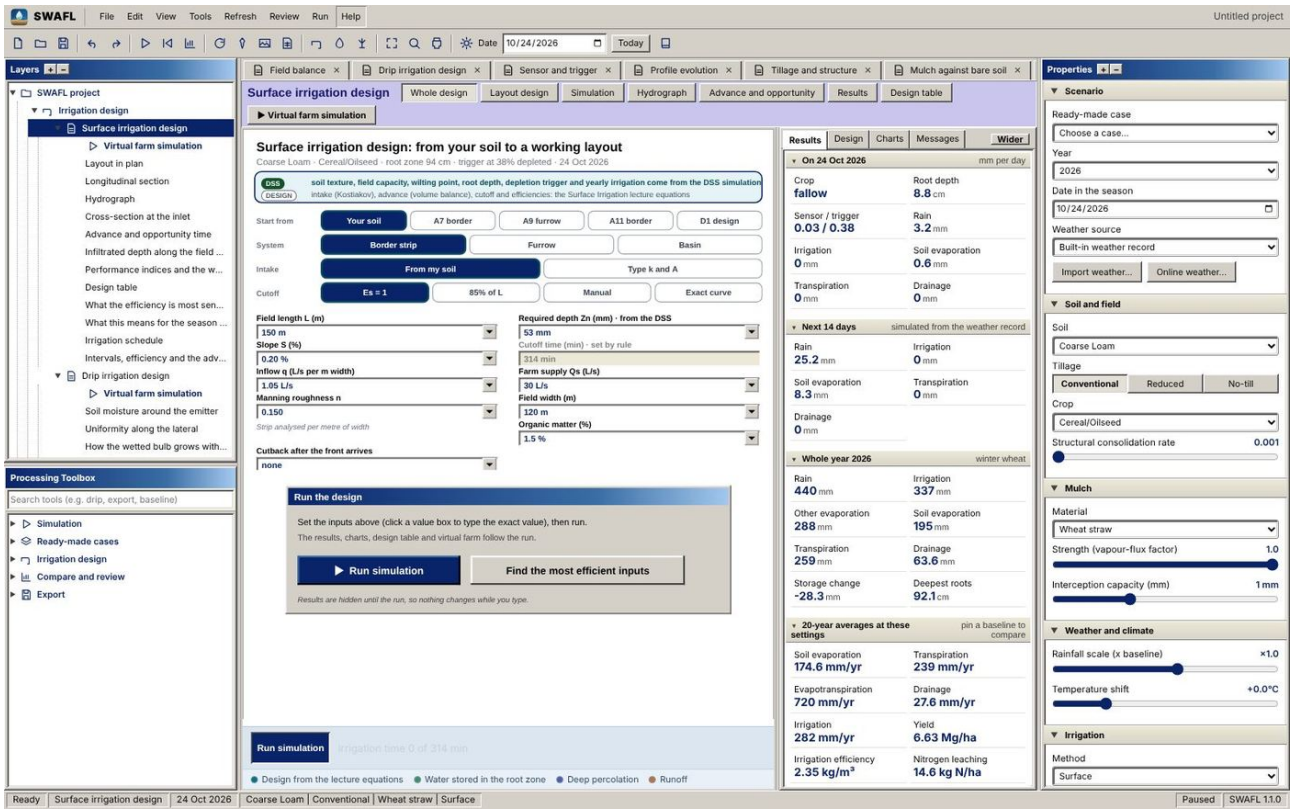


Figure 1.41: Surface irrigation design before its run: the inputs are set, the results wait for Run simulation.

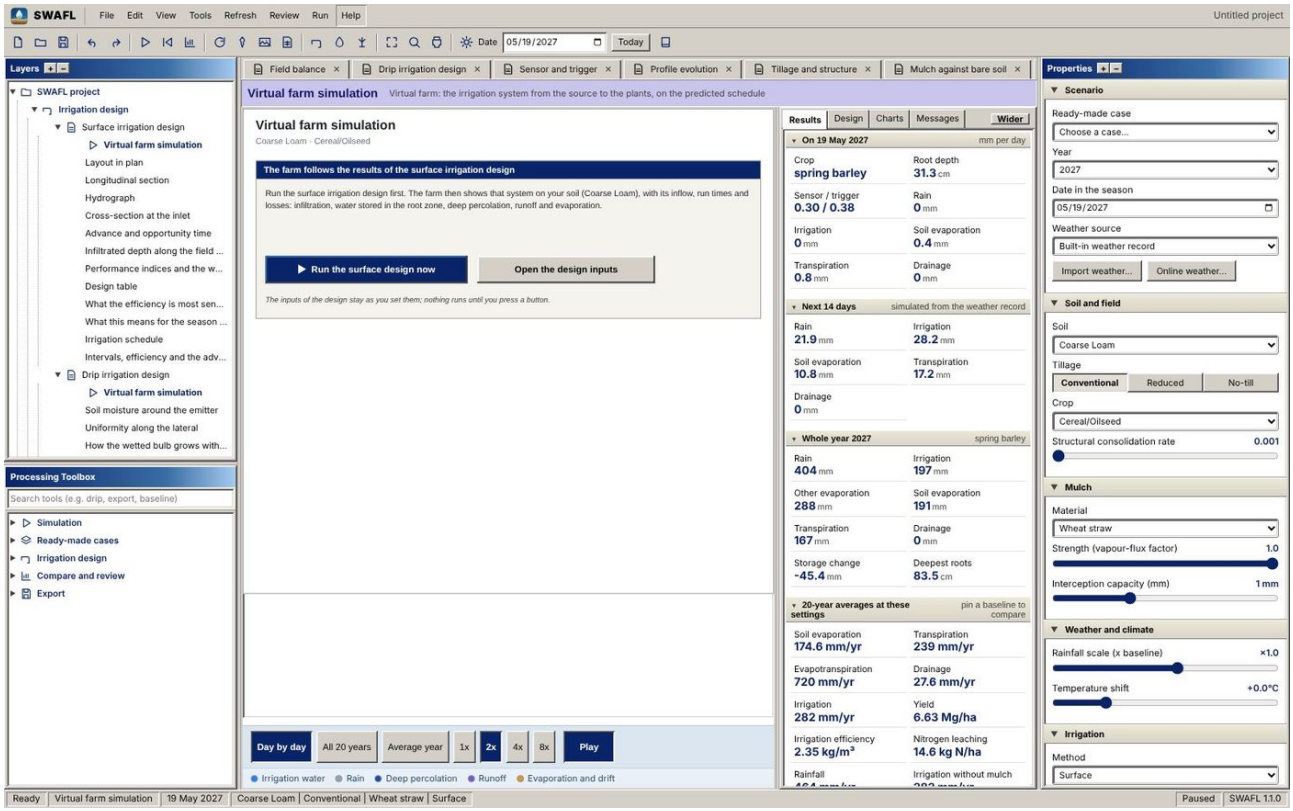


Figure 1.42: The virtual farm of the surface irrigation design before the design has run.

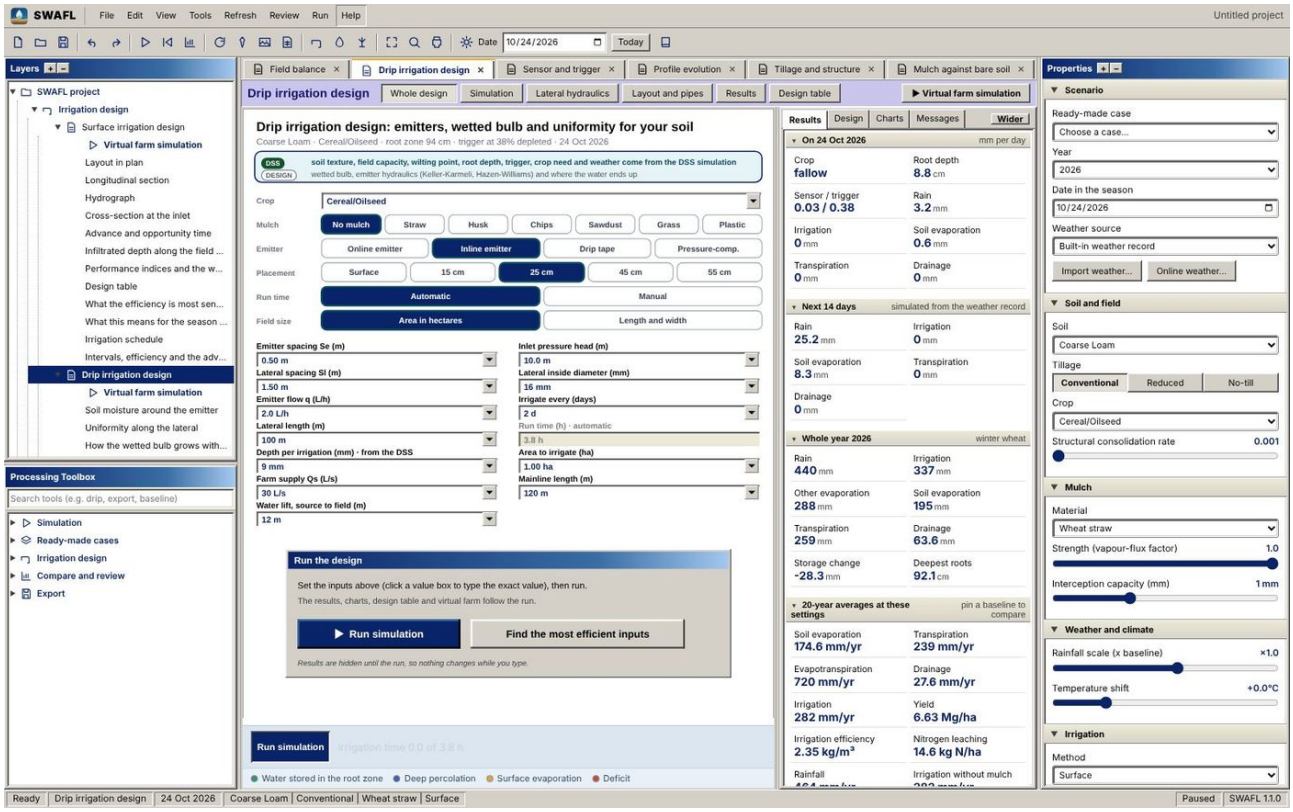


Figure 1.43: Drip irrigation design before its run: the inputs are set, the results wait for Run simulation.

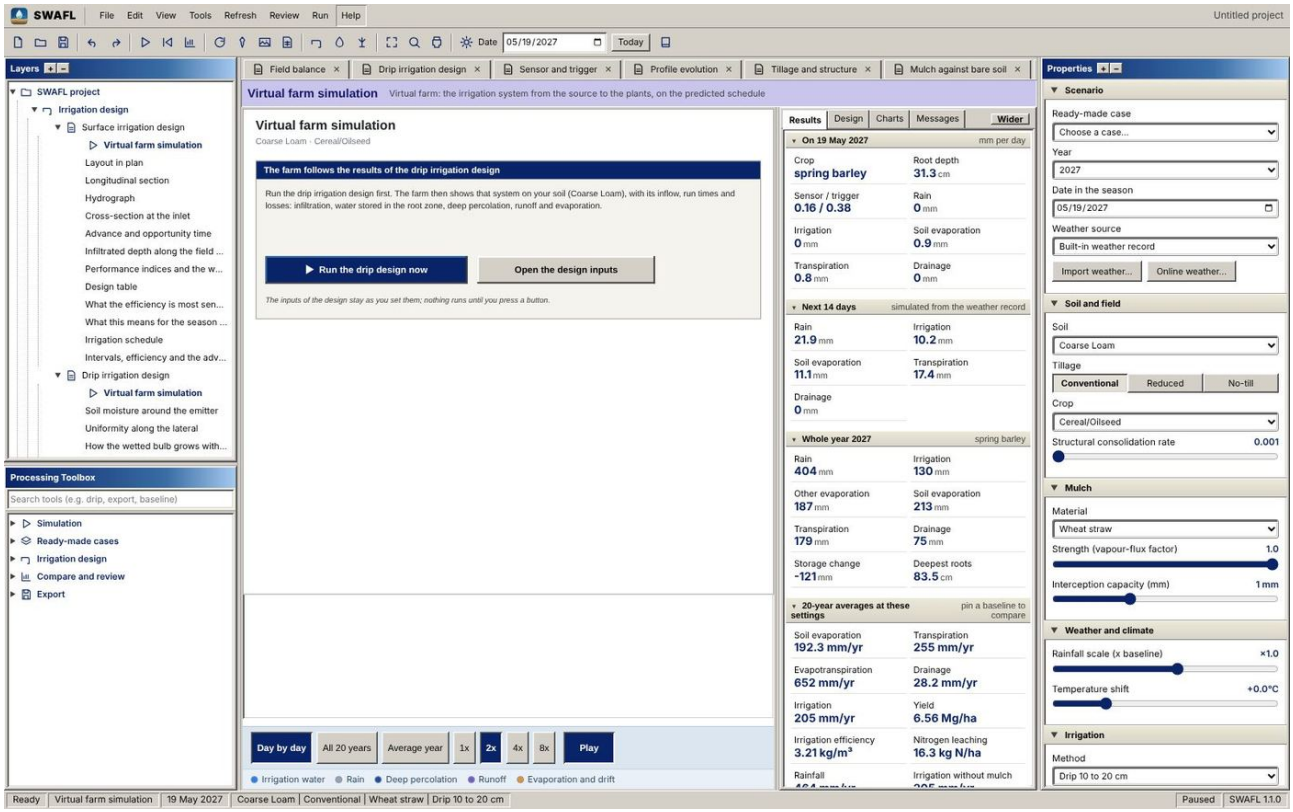


Figure 1.44: The virtual farm of the drip irrigation design before the design has run.

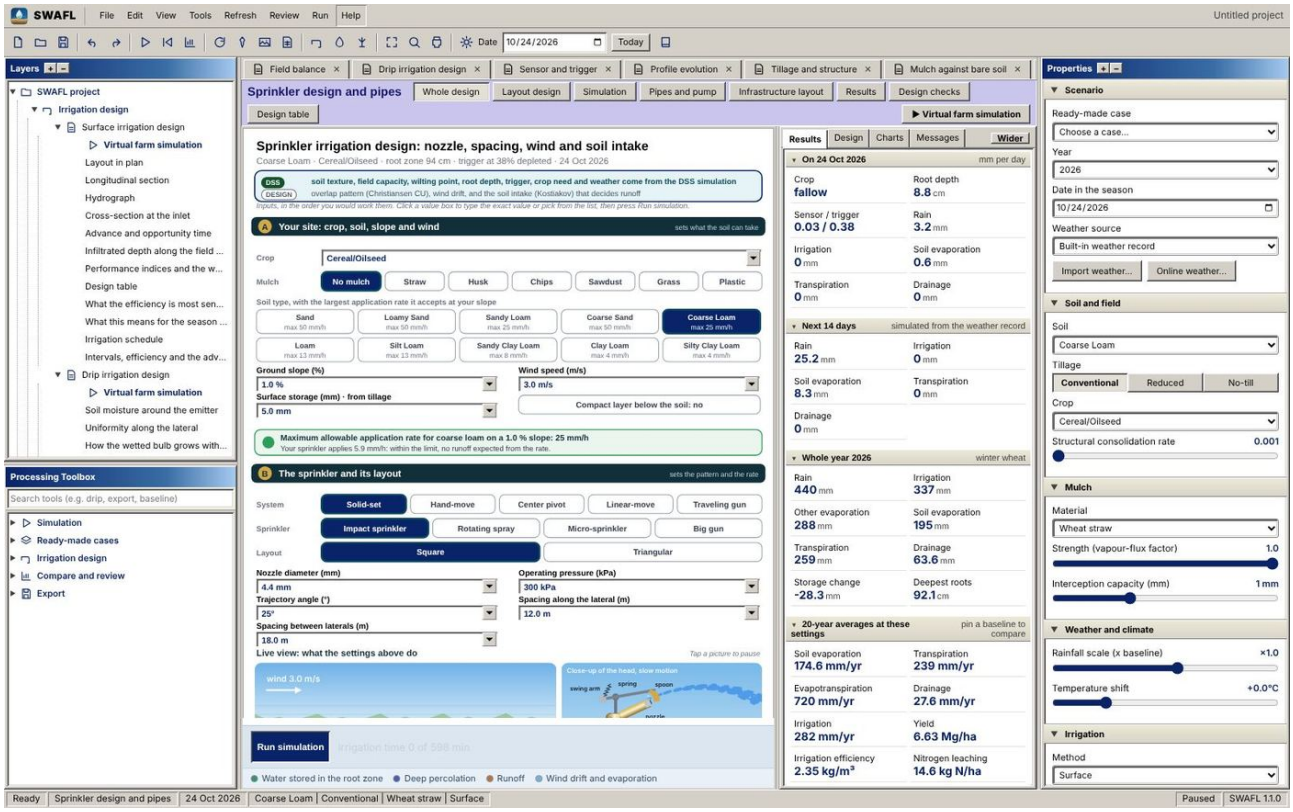


Figure 1.45: Sprinkler design and pipes before its run: the inputs are set, the results wait for Run simulation.

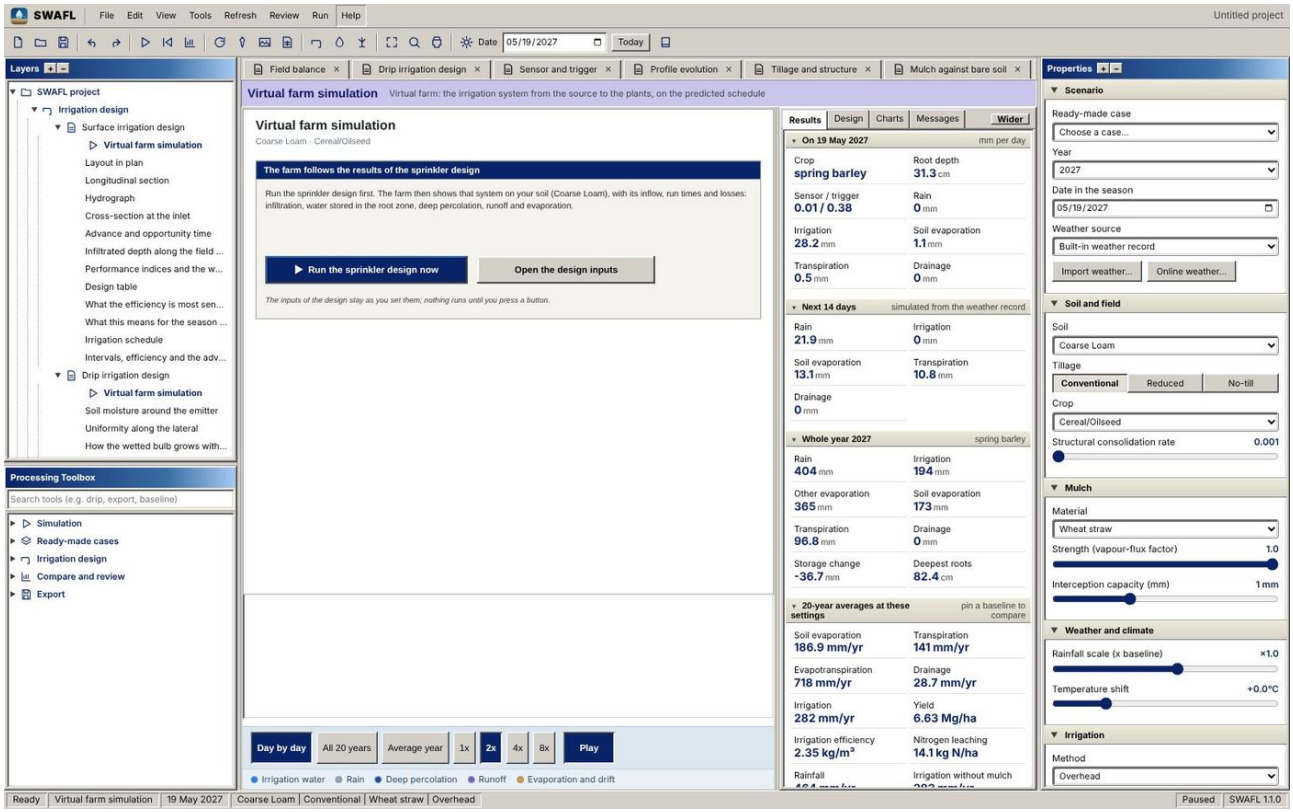


Figure 1.46: The virtual farm of the sprinkler design and pipes before the design has run.

Chapter 2

Irrigation design

2.1 Surface irrigation design

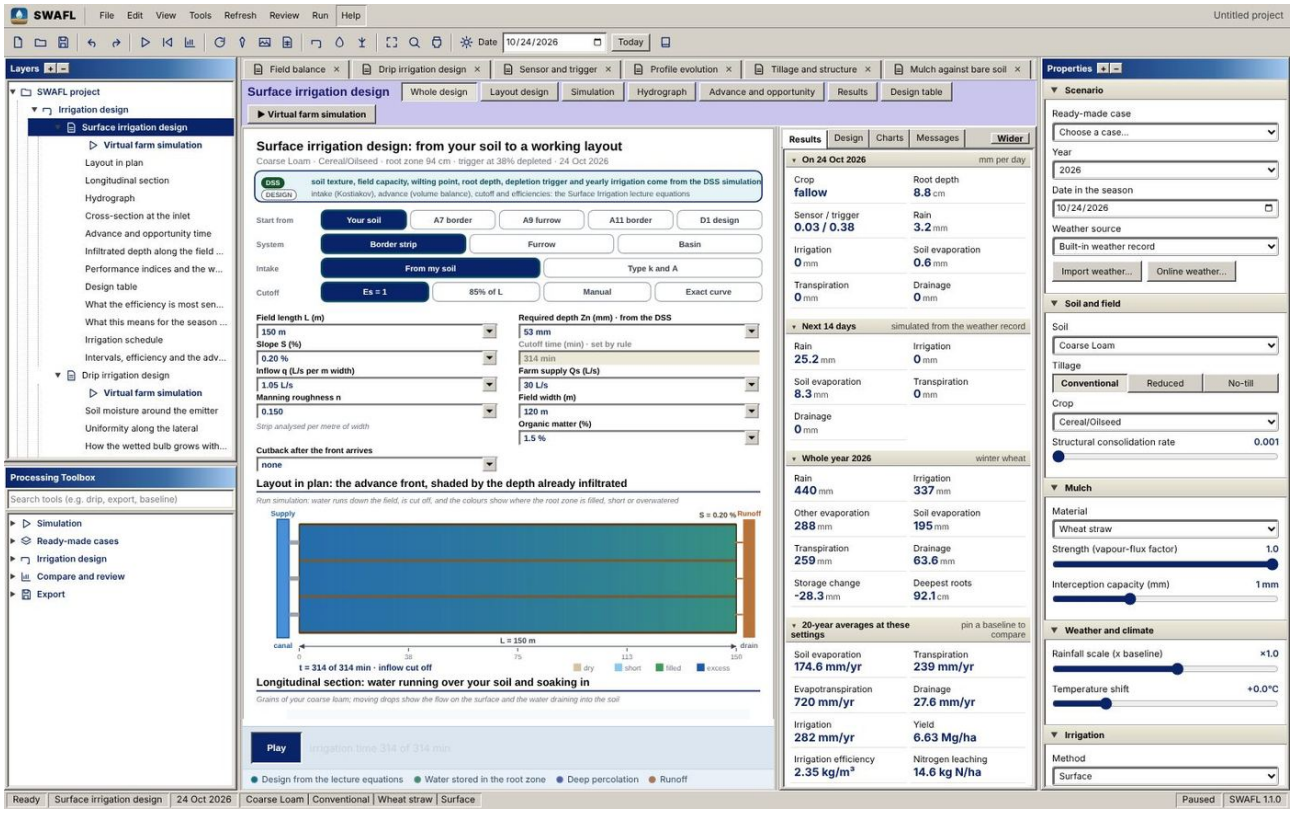


Figure 2.1: Surface irrigation design: the window with the layer open.

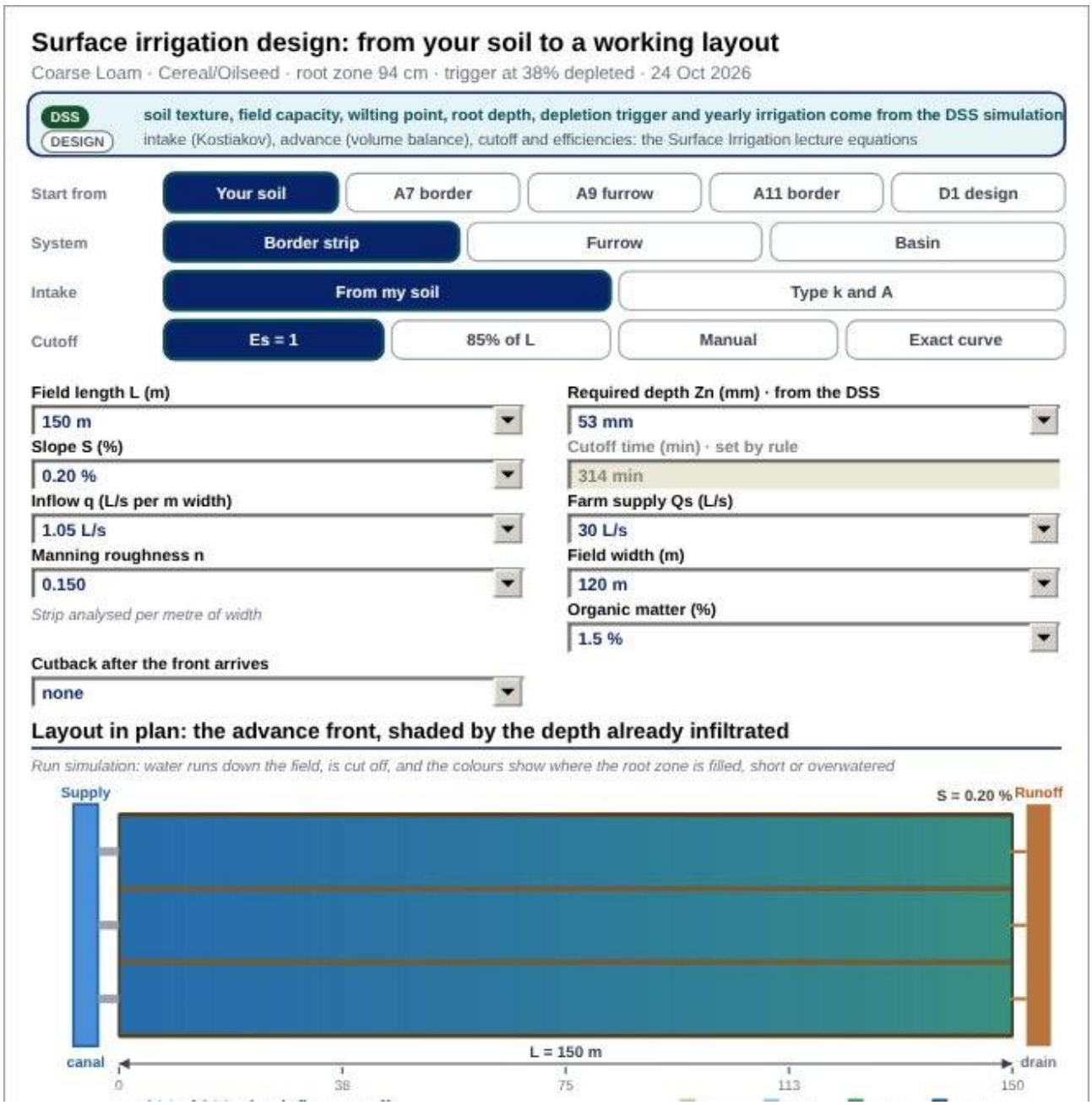


Figure 2.2: Surface irrigation design, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

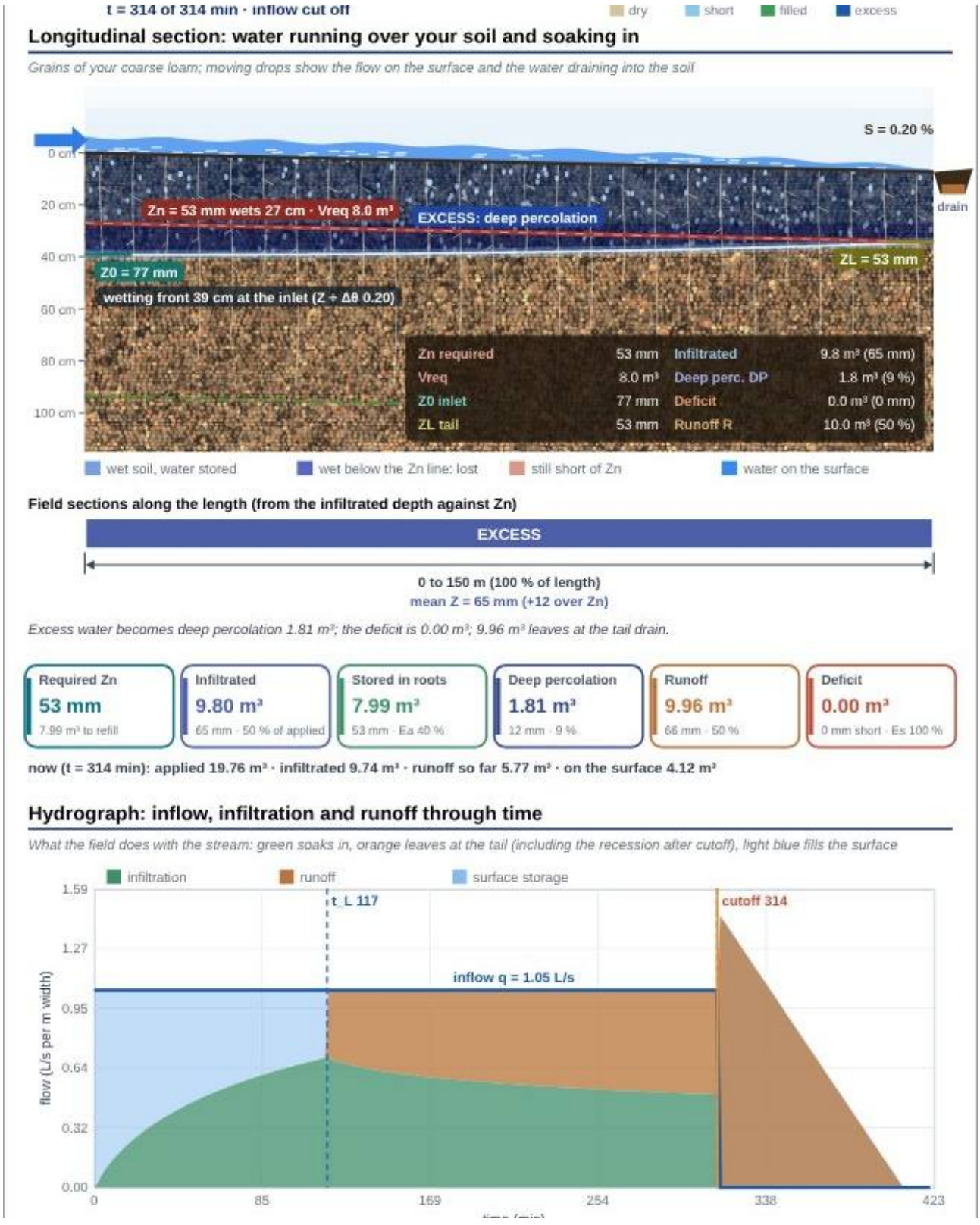


Figure 2.3: Surface irrigation design, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

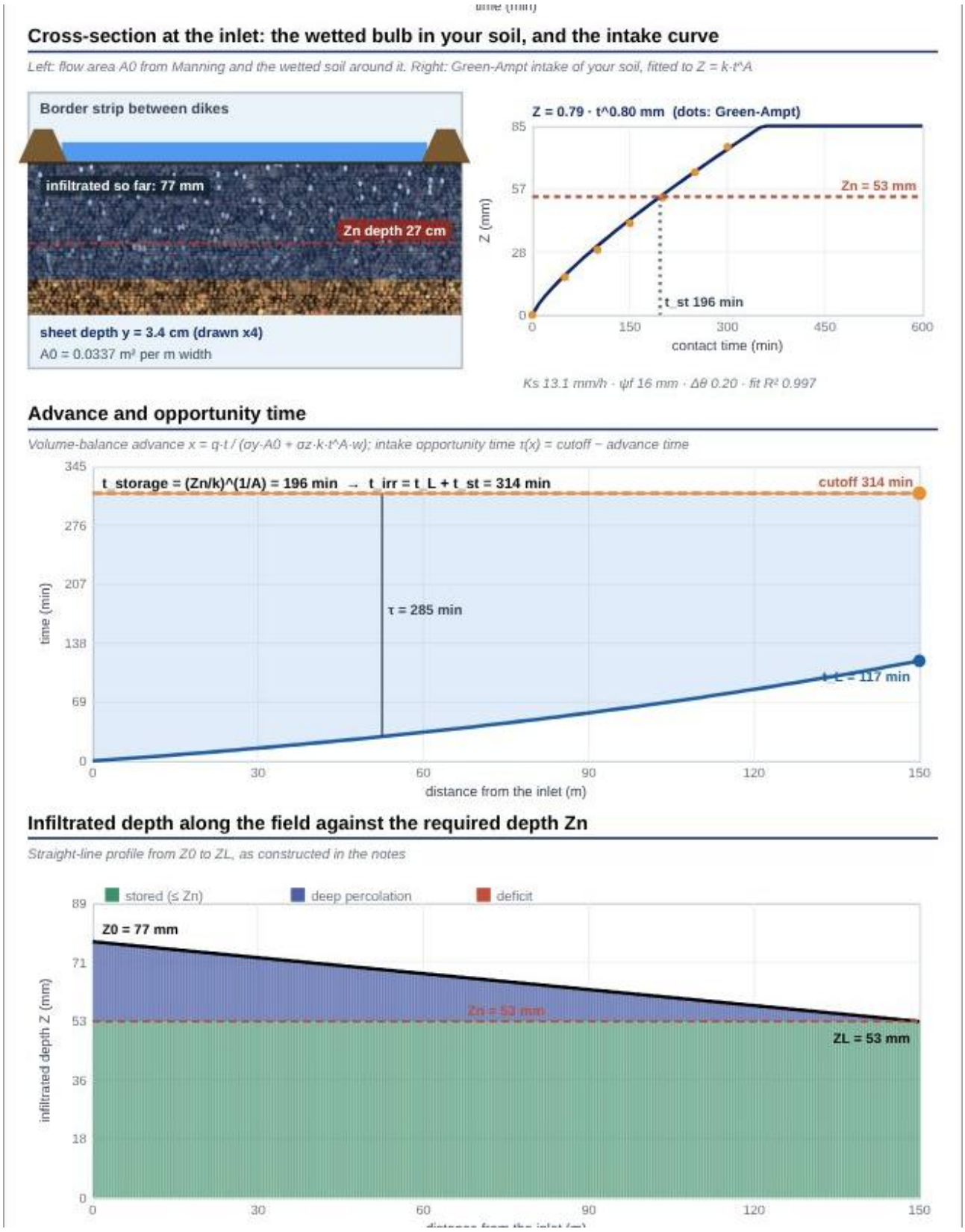


Figure 2.4: Surface irrigation design, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

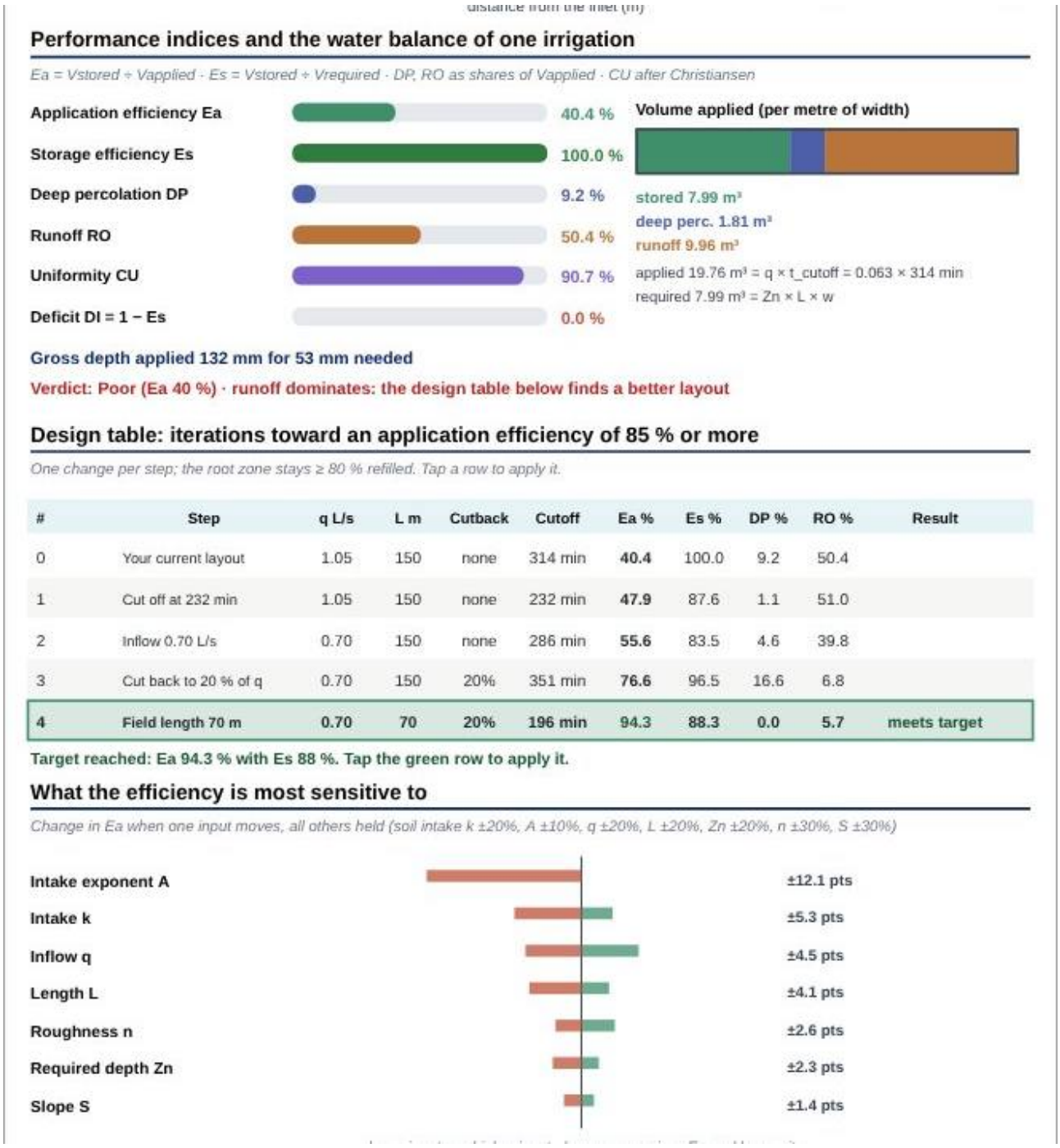


Figure 2.5: Surface irrigation design, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.



Figure 2.6: Surface irrigation design, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

2.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 4.1 (Volume 1, dates)

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 5.1 (Volume 1, engine)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\,S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 (Volume 1, engine)

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\Big[\tfrac{\text{now}.E}{\text{ref}.E}\Big]\Big[\tfrac{\text{tN}.E}{\text{tB}.E}\Big]\Big[\tfrac{E_b}{E_{b,\text{ref}}}\Big] \tag{5.4}$$

$$T = T_b\,\Big[\tfrac{\text{now}.T}{\text{bare}.T}\Big]\,\text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\Big[\tfrac{\text{now}.ET}{\text{bare}.ET}\Big]\,\text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\Big[\tfrac{\text{now}.I}{\text{bare}.I}\Big]\,\text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\Big[\tfrac{\text{tN}.Y}{\text{tR}.Y}\Big]\Big[\tfrac{\text{now}.Y}{\text{nm}.Y}\Big] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$$ET \in \left[\, P + I - D - 30,\ P + I - D + 15 \,\right] \text{ (if not, } ET \text{ is clipped and } E, T \text{ are scaled by the same factor)},$$
$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 5.27 (Volume 1, engine)

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \text{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.1 (Volume 1, surface)

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\text{OM} + 0.005\,S\,\text{OM} - 0.013\,C\,\text{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\text{OM} + 0.006\,S\,\text{OM} - 0.027\,C\,\text{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2 (Volume 1, surface)

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\text{OM} - 0.018\,S\,\text{OM} - 0.027\,C\,\text{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3 (Volume 1, surface)

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4 (Volume 1, surface)

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5 (Volume 1, surface)

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6 (Volume 1, surface)

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7 (Volume 1, surface)

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

Eq. 11.8 (Volume 1, surface)

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9 (Volume 1, surface)

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10 (Volume 1, surface)

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11 (Volume 1, surface)

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12 (Volume 1, surface)

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13 (Volume 1, surface)

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15 (Volume 1, surface)

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16 (Volume 1, surface)

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17 (Volume 1, surface)

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.18 (Volume 1, surface)

$$\begin{aligned} &V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)), \quad V_{\text{req}} = Z_n\,L\,w, \\ &V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx, \\ &V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx, \\ &V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \end{aligned} \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19 (Volume 1, surface)

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.20 (Volume 1, surface)

$$V_c(t) = w\sum_i k\left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.21 (Volume 1, surface)

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.22 (Volume 1, surface)

$$q_{\max} = \min\Big\{q_{\max}^{\text{sys}},\ \frac{0.04}{S}\cdot\frac{1000}{60}\Big\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

Eq. 11.23 (Volume 1, surface)

$$\text{score} = \min(E_a, 1) - 3\max(0,\ 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\ 0.05 - \text{RO}) - 5\max(0,\ E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

Eq. 11.24 (Volume 1, surface)

$$\begin{aligned} n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000\ \ (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\ N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60}\ \ (\text{h}), \end{aligned} \tag{11.24}$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

Eq. 18.0 (Volume 1, runs)

$$\kappa(\text{state now}) = \kappa_{\text{run}}, \qquad \kappa = \big(\text{every Properties control except the irrigation method},\ \text{weather mode}\big), \tag{18.0}$$

Provenance (Eq. 18.0): **[Engineered]**: bookkeeping of the run.

Eq. 18.2 (Volume 1, runs)

$$s = E_a - 3\ \max\!\big(0,\ 0.95 - E_s\big) + \begin{cases} 0 & \text{surface} \\ 0.05\,E_u & \text{drip} \\ 0.05\,\text{CU} - 2\,R_o & \text{sprinkler} \end{cases} \tag{18.2}$$

Provenance (Eq. 18.2): **[Engineered]**: the application efficiency, with a penalty when the root zone is refilled to less than 95 % (an efficient design that leaves the crop short is not efficient) and a small reward for uniformity.

Eq. 14.6 (Volume 1, dashboard)

$$Q_{\text{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\ n_z\, t_{\text{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

Provenance (Eq. 14.6): **[Engineered]**: zones of the layout (Eqs. 12.20–12.23 and the sprinkler analogue) run in turn.

2.2 Drip irrigation design

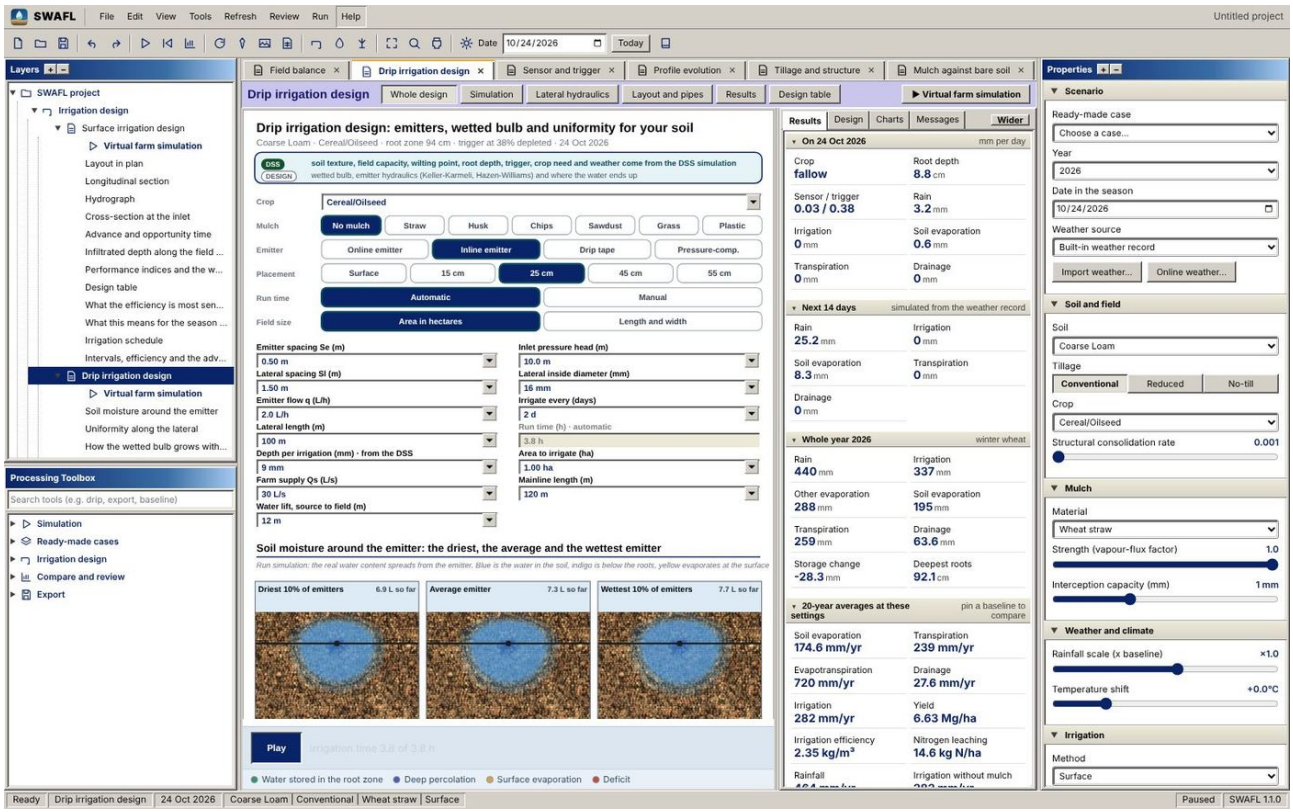


Figure 2.7: Drip irrigation design: the window with the layer open.

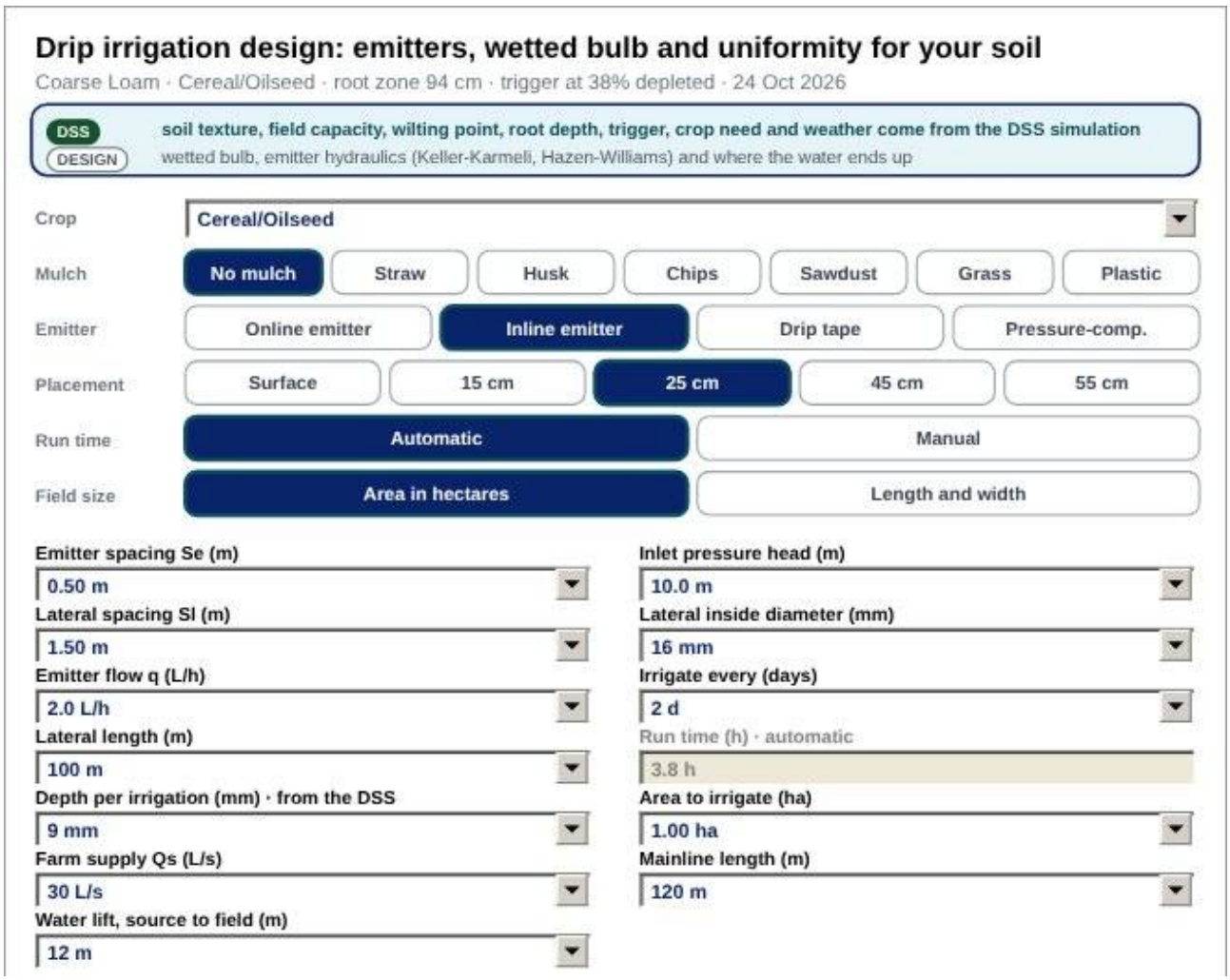


Figure 2.8: Drip irrigation design, part 1 of 6.

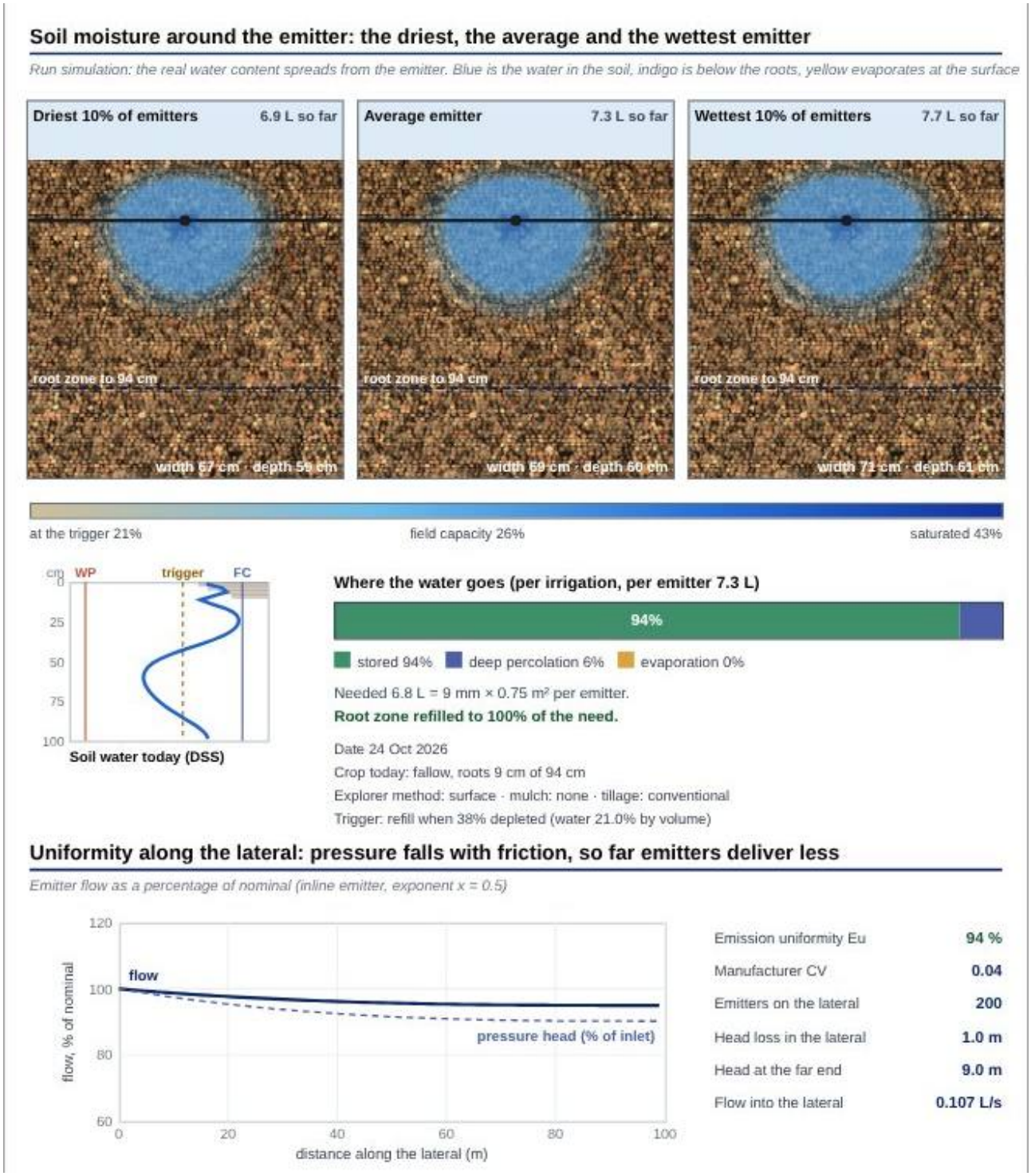


Figure 2.9: Drip irrigation design, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

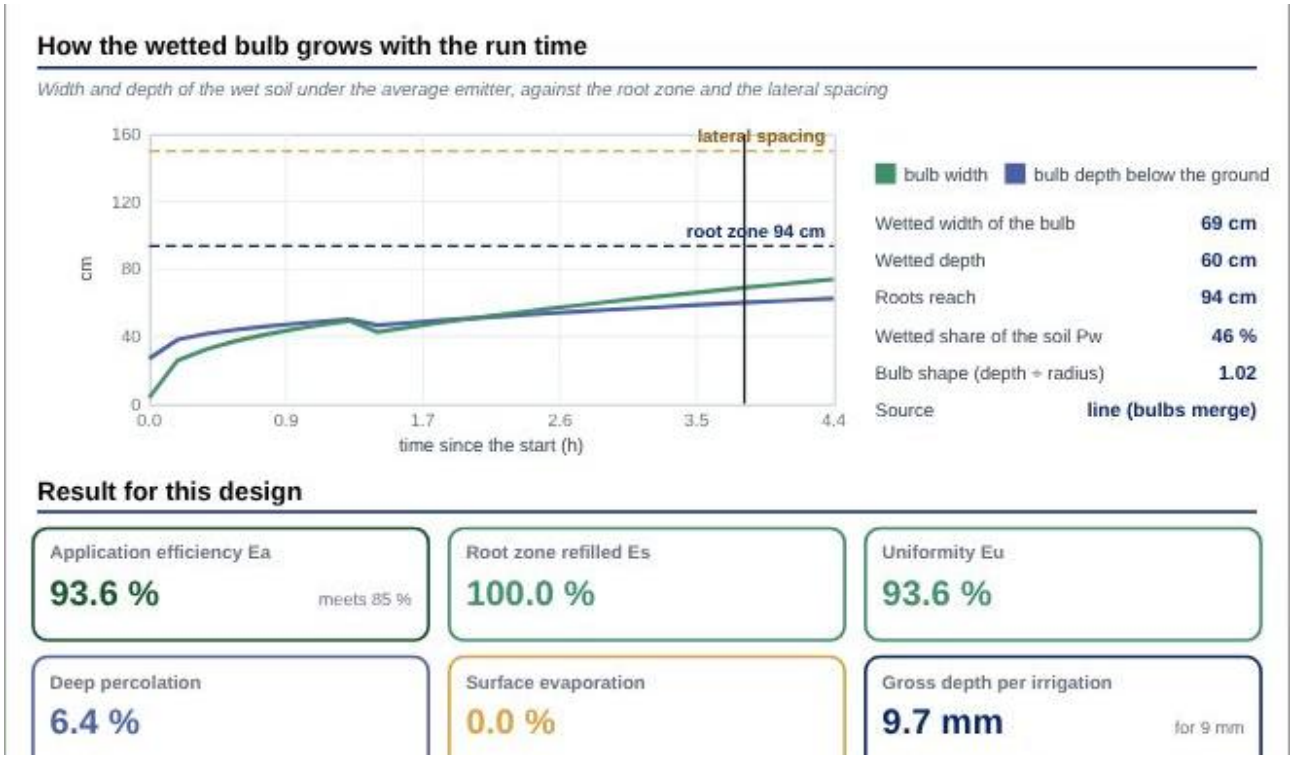
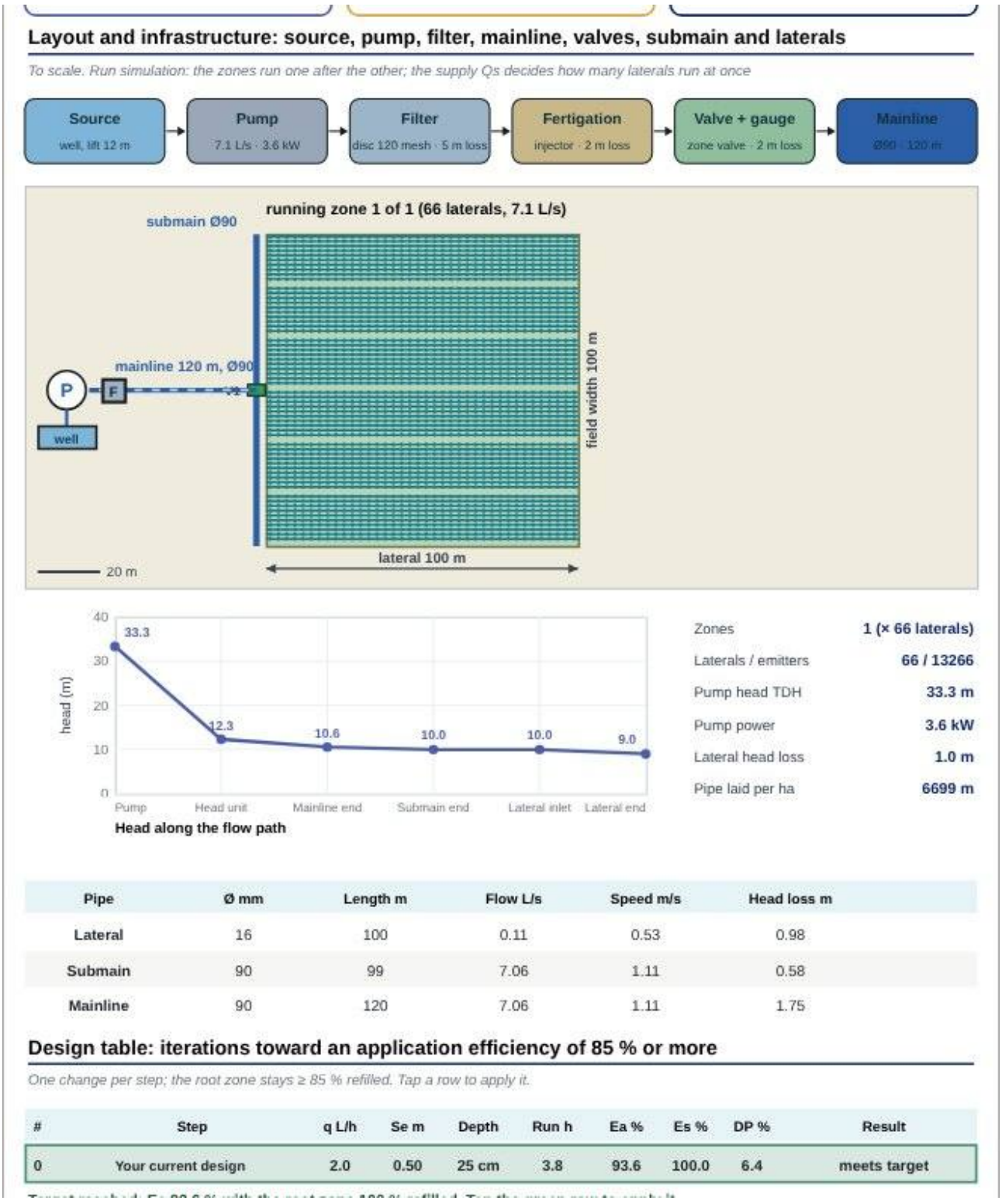


Figure 2.10: Drip irrigation design, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.8	93.6	100.0	6.4	meets target

Figure 2.11: Drip irrigation design, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

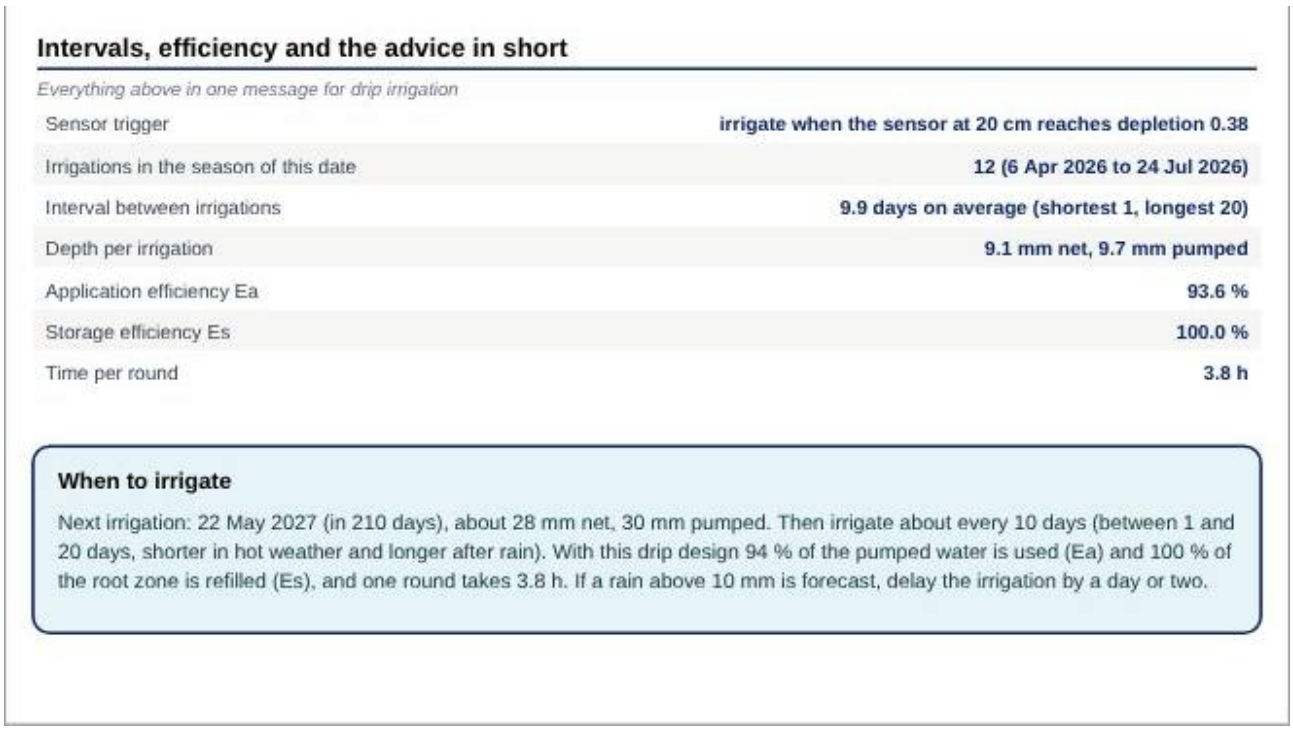
Net irrigation need in a season (DSS)	**339 mm → 37 irrigations of 9 mm**
Gross water per irrigation	**9.7 mm (Ea 94 %)**
Gross water per season	**362 mm = 3618 m³ per hectare**
Peak crop need / interval	**4.6 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.8 h = 3.8 h per round**
Pumping per season	**141 hours · 500 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**22 May 2027**	210 days	28	30	3.8 h	13 days
2	**4 Jun 2027**	223 days	28	30	3.8 h	11 days
3	**15 Jun 2027**	234 days	28	30	3.8 h	9 days
4	**24 Jun 2027**	243 days	28	30	3.8 h	9 days
5	**3 Jul 2027**	252 days	28	30	3.8 h	10 days
6	**13 Jul 2027**	262 days	28	30	3.8 h	16 days
7	**29 Jul 2027**	278 days	28	30	3.8 h	251 days
8	**5 Apr 2028**	529 days	28	30	3.8 h	-

Figure 2.12: Drip irrigation design, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.38**
Irrigations in the season of this date	**12 (6 Apr 2026 to 24 Jul 2026)**
Interval between irrigations	**9.9 days on average (shortest 1, longest 20)**
Depth per irrigation	**9.1 mm net, 9.7 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**3.8 h**

When to irrigate

Next irrigation: 22 May 2027 (in 210 days), about 28 mm net, 30 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 3.8 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 2.13: Drip irrigation design, part 6 of 6: Intervals, efficiency and the advice in short.

2.2.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.27 (Volume 1, engine)

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\; 25,\; 150\big)\ \text{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.3 (Volume 1, surface)

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.1 (Volume 1, drip)

$$Z_n = \min\big\{Z_n^{\text{DSS}},\; \max(4,\; \text{ET}_{\text{pk}}\,\Delta t)\big\}\ \text{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.2 (Volume 1, drip)

$$\text{ET}_{\text{pk}} = \text{clip}\Big\{\max_{d}\,\tfrac{1}{5}\sum_{j=0}^{4}(E_{d+j} + T_{d+j});\; 2.5,\; 9\Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.3 (Volume 1, drip)

$$h_{f,j} = \frac{10.67\,S_e\,Q_j^{1.852}}{C^{1.852}\,D^{4.871}}, \qquad Q_j = \sum_{i \ge j+1} q_i\ \ (\text{m}^3\,\text{s}^{-1}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4 (Volume 1, drip)

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\; h_j/H_0 \ge 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5 (Volume 1, drip)

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \ge H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6 (Volume 1, drip)

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7 (Volume 1, drip)

$$E_u = \operatorname{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8 (Volume 1, drip)

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.9 (Volume 1, drip)

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10 (Volume 1, drip)

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11 (Volume 1, drip)

$$a = \operatorname{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12 (Volume 1, drip)

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13 (Volume 1, drip)

$$P_w = \operatorname{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14 (Volume 1, drip)

$$V_{\text{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15 (Volume 1, drip)

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\}\quad (\mathrm{L},\ \text{area in m}^2,\ E_0\ \text{in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16 (Volume 1, drip)

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\},\qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17 (Volume 1, drip)

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}),\qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\},\qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18 (Volume 1, drip)

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19 (Volume 1, drip)

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle},\quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}},\quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle},\quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle},\quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.20 (Volume 1, drip)

$$W_f = \frac{10^4 A_f}{L_L},\quad N_L = \Big\lfloor \frac{W_f}{S_l}\Big\rfloor,\quad n_{\mathrm{zone}} = \Big\lceil \frac{N_L\,Q_{\mathrm{lat}}}{Q_s}\Big\rceil,\quad n_l = \Big\lceil \frac{N_L}{n_{\mathrm{zone}}}\Big\rceil,\quad Q_z = n_l\,Q_{\mathrm{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.21 (Volume 1, drip)

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \mathrm{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22 (Volume 1, drip)

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}},\qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23 (Volume 1, drip)

$$\mathrm{TDH} = z_{\mathrm{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\,\mathrm{m}, \qquad P = \frac{\rho g\, Q_z\, \mathrm{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\mathrm{TDH}}{0.65}\ \mathrm{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.24 (Volume 1, drip)

$$n_{\mathrm{ev}} = \frac{y_I}{Z_n}, \quad Z_{\mathrm{season}} = n_{\mathrm{ev}}\, Z_{\mathrm{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\mathrm{L\,s^{-1}\,ha^{-1}}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25 (Volume 1, drip)

$$n_{\mathrm{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\mathrm{round}} = n_{\mathrm{set}}\, t, \qquad H_{\mathrm{pump}} = n_{\mathrm{ev}}\, t_{\mathrm{round}}, \qquad E_{\mathrm{kWh}} = H_{\mathrm{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.26 (Volume 1, drip)

$$\mathrm{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 18.0 (Volume 1, runs)

$$\kappa(\text{state now}) = \kappa_{\mathrm{run}}, \qquad \kappa = \big(\text{every Properties control except the irrigation method, weather mode}\big), \tag{18.0}$$

Provenance (Eq. 18.0): **[Engineered]**: bookkeeping of the run.

Eq. 18.1 (Volume 1, runs)

$$A_f = \frac{L\,W}{10^4}\ \ [\mathrm{ha}], \qquad L_{\mathrm{lat}} = \mathrm{clip}\Big(\frac{L}{\lceil L/L_{\max}\rceil};\ L_{\min},\ L_{\max}\Big), \tag{18.1}$$

Provenance (Eq. 18.1): **[Engineered]**: geometry; laterals longer than $L_{\max}$ are split into equal runs from a central submain.

Eq. 18.2 (Volume 1, runs)

$$s = E_a - 3\ \max(0,\ 0.95 - E_s) + \begin{cases} 0 & \text{surface} \\ 0.05\,E_u & \text{drip} \\ 0.05\,\mathrm{CU} - 2\,R_o & \text{sprinkler} \end{cases} \tag{18.2}$$

Provenance (Eq. 18.2): **[Engineered]**: the application efficiency, with a penalty when the root zone is refilled to less than 95 % (an efficient design that leaves the crop short is not efficient) and a small reward for uniformity.

Eq. 14.6 (Volume 1, dashboard)

$$Q_{\mathrm{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\ n_z\, t_{\mathrm{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

Provenance (Eq. 14.6): **[Engineered]**: zones of the layout (Eqs. 12.20–12.23 and the sprinkler analogue) run in turn.

2.3 Sprinkler design and pipes

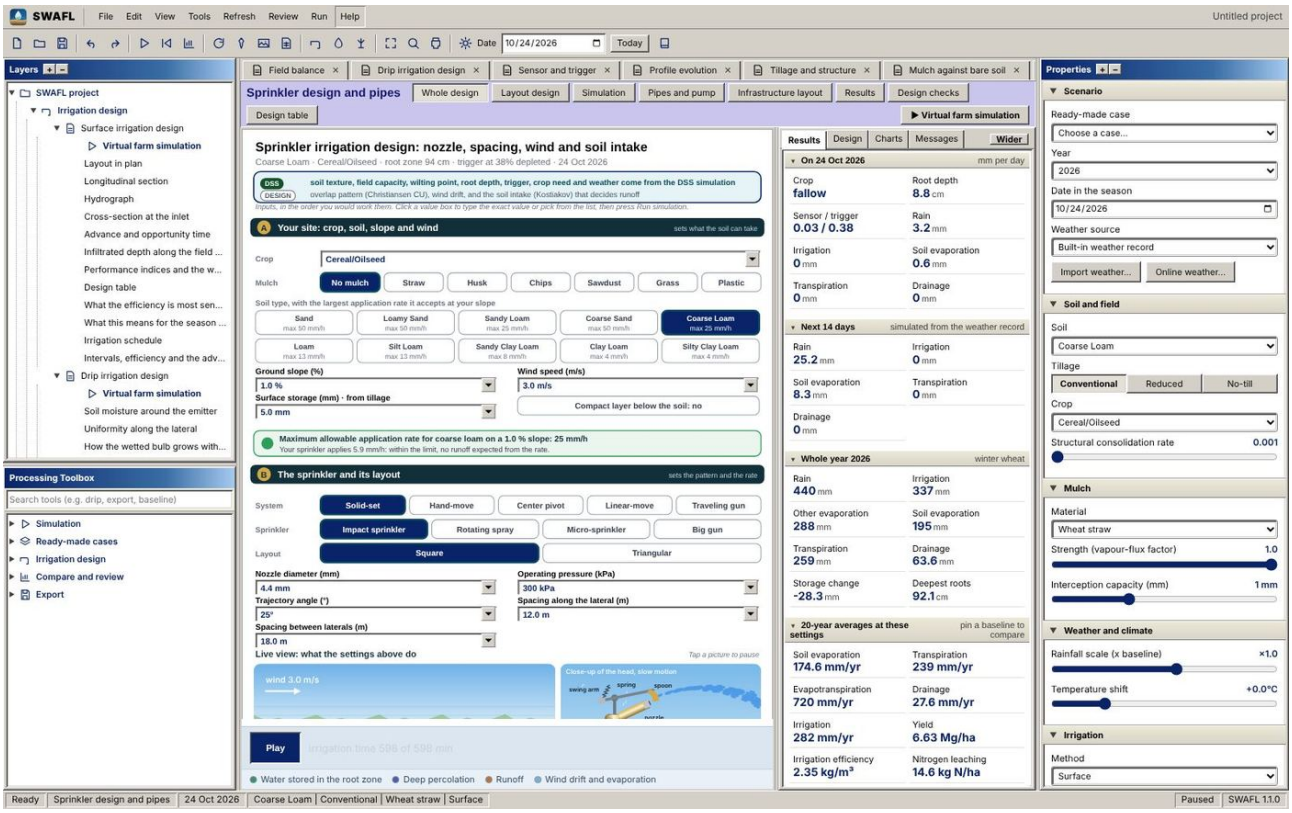


Figure 2.14: Sprinkler design and pipes: the window with the layer open.

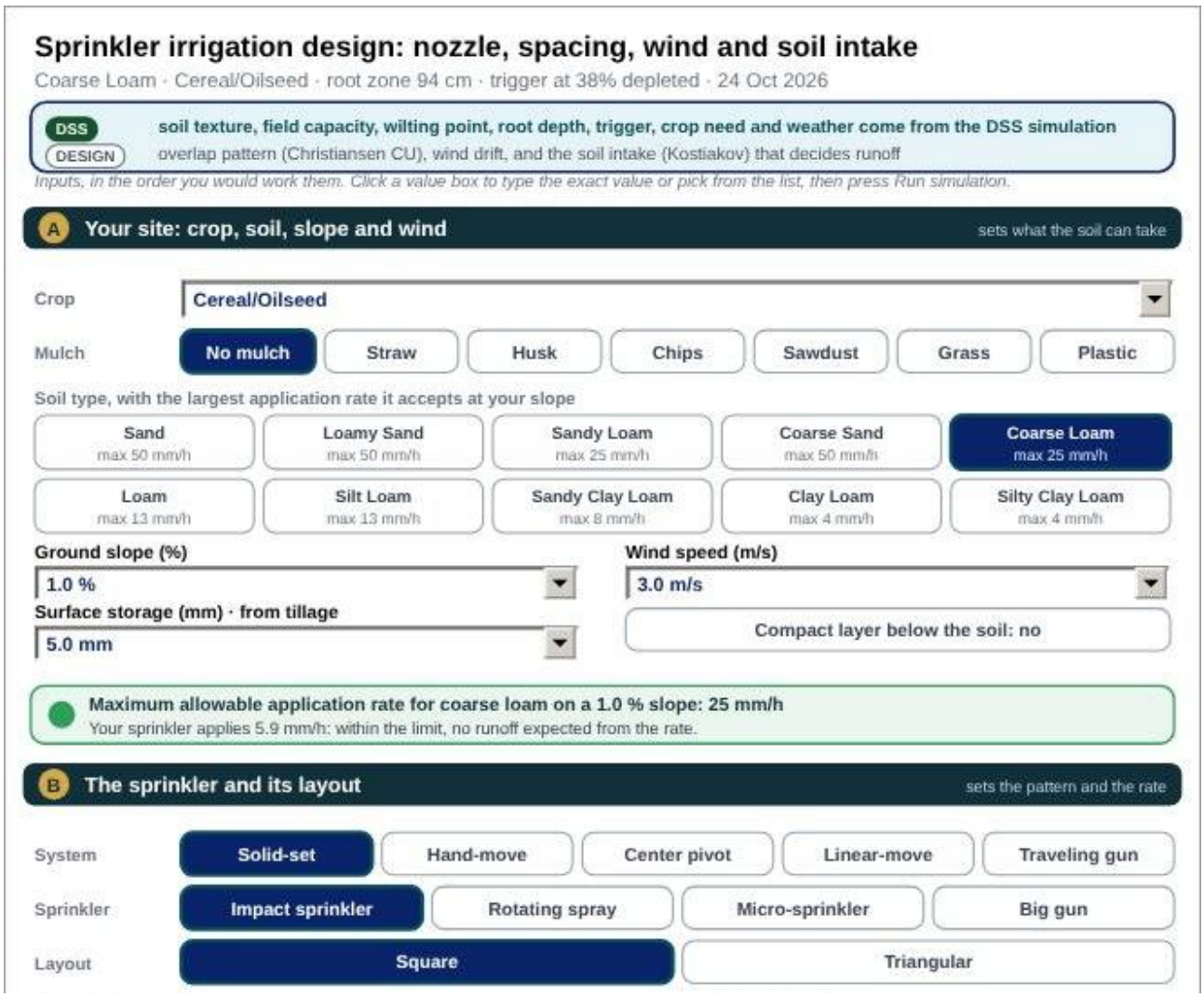


Figure 2.15: Sprinkler design and pipes, part 1 of 12.

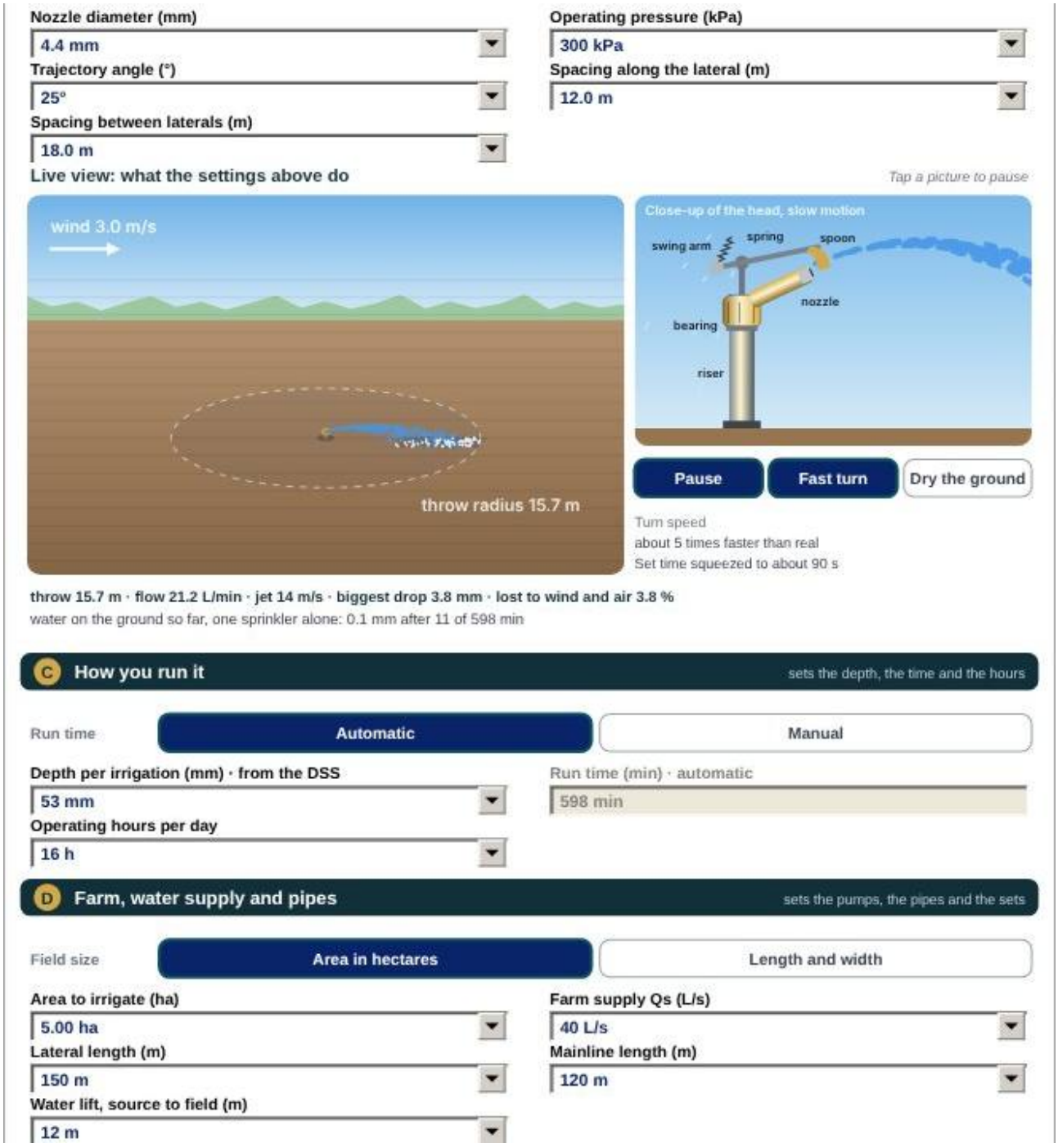


Figure 2.16: Sprinkler design and pipes, part 2 of 12.

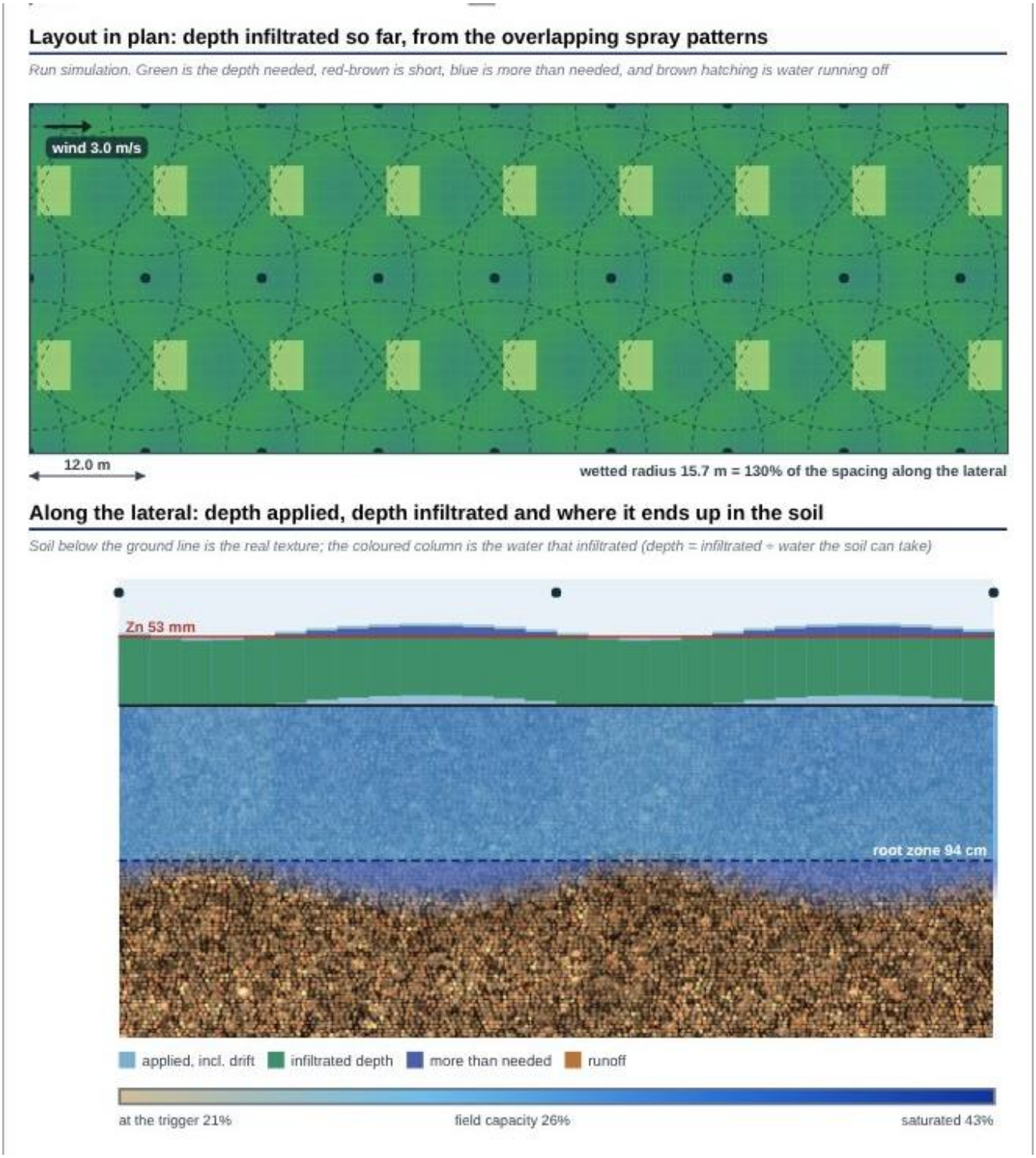


Figure 2.17: Sprinkler design and pipes, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

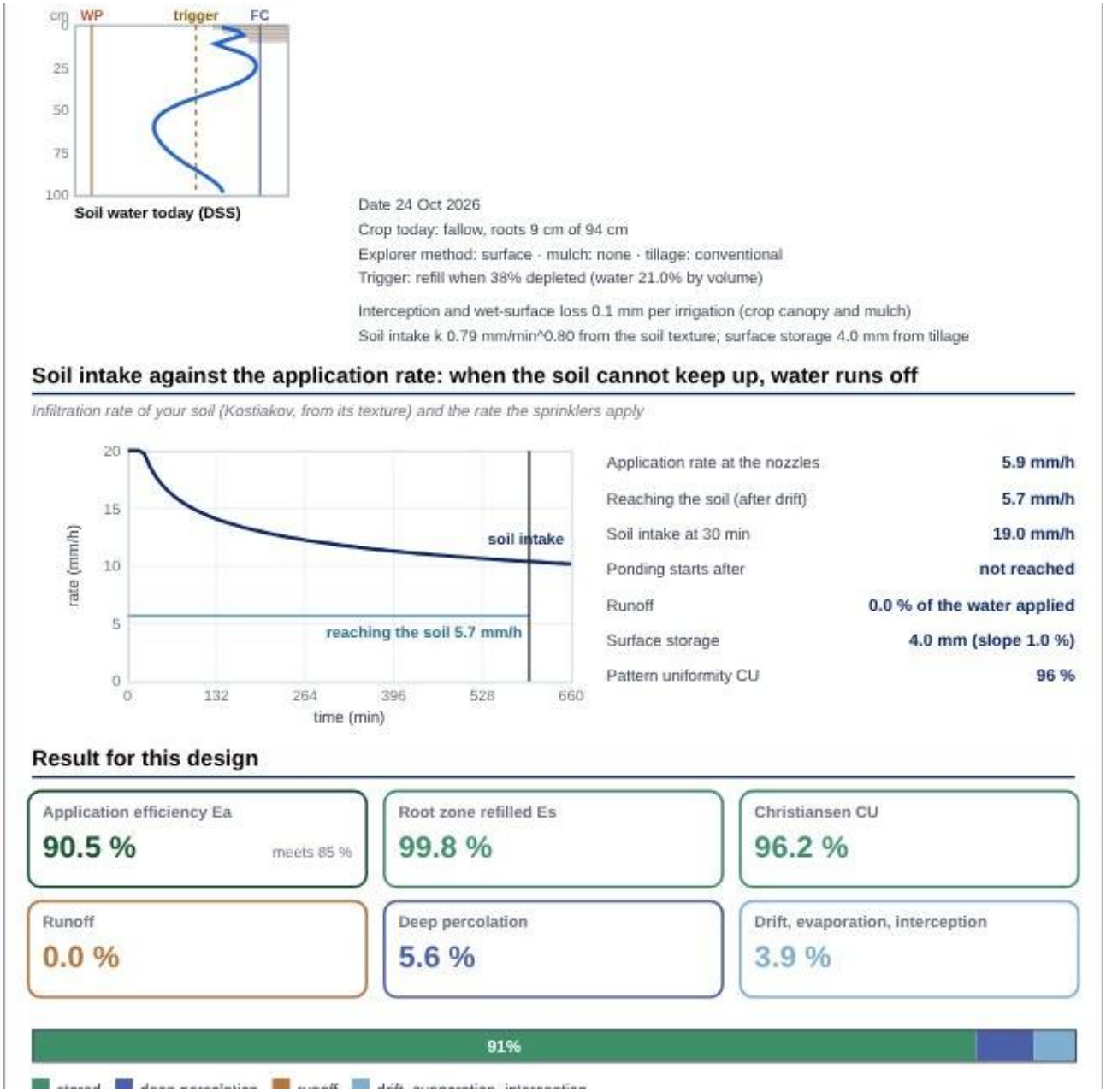


Figure 2.18: Sprinkler design and pipes, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

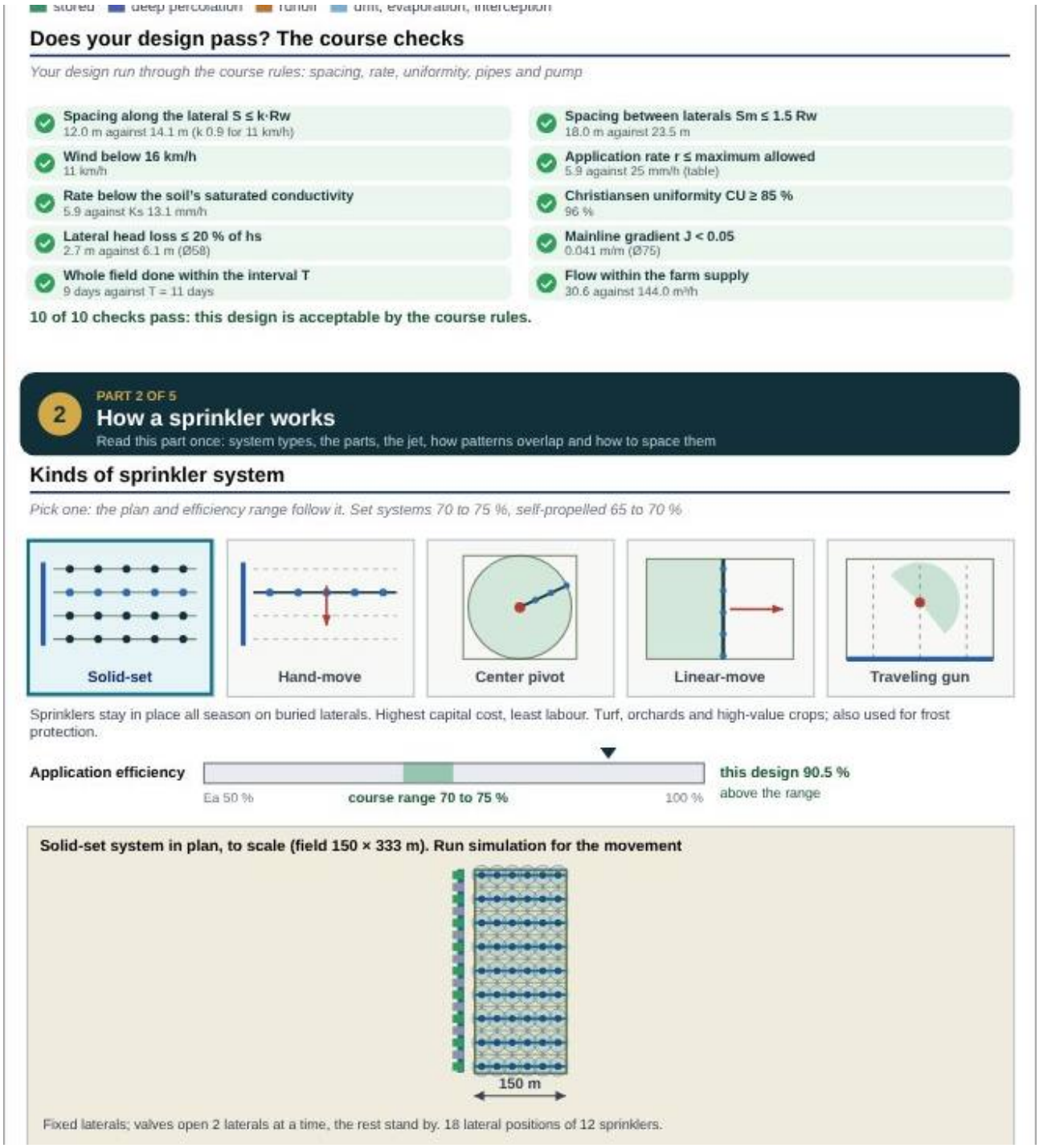


Figure 2.19: Sprinkler design and pipes, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

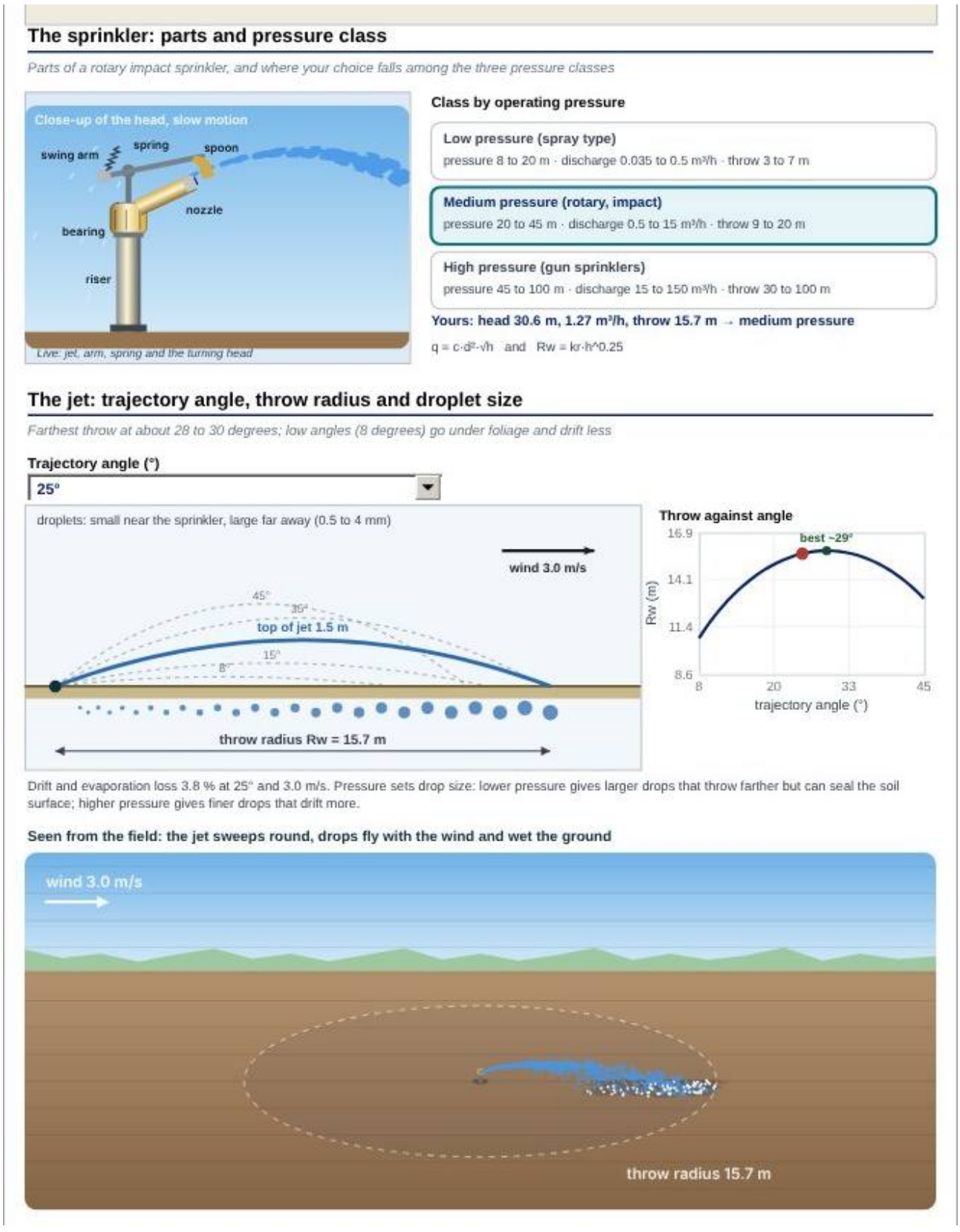


Figure 2.20: Sprinkler design and pipes, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

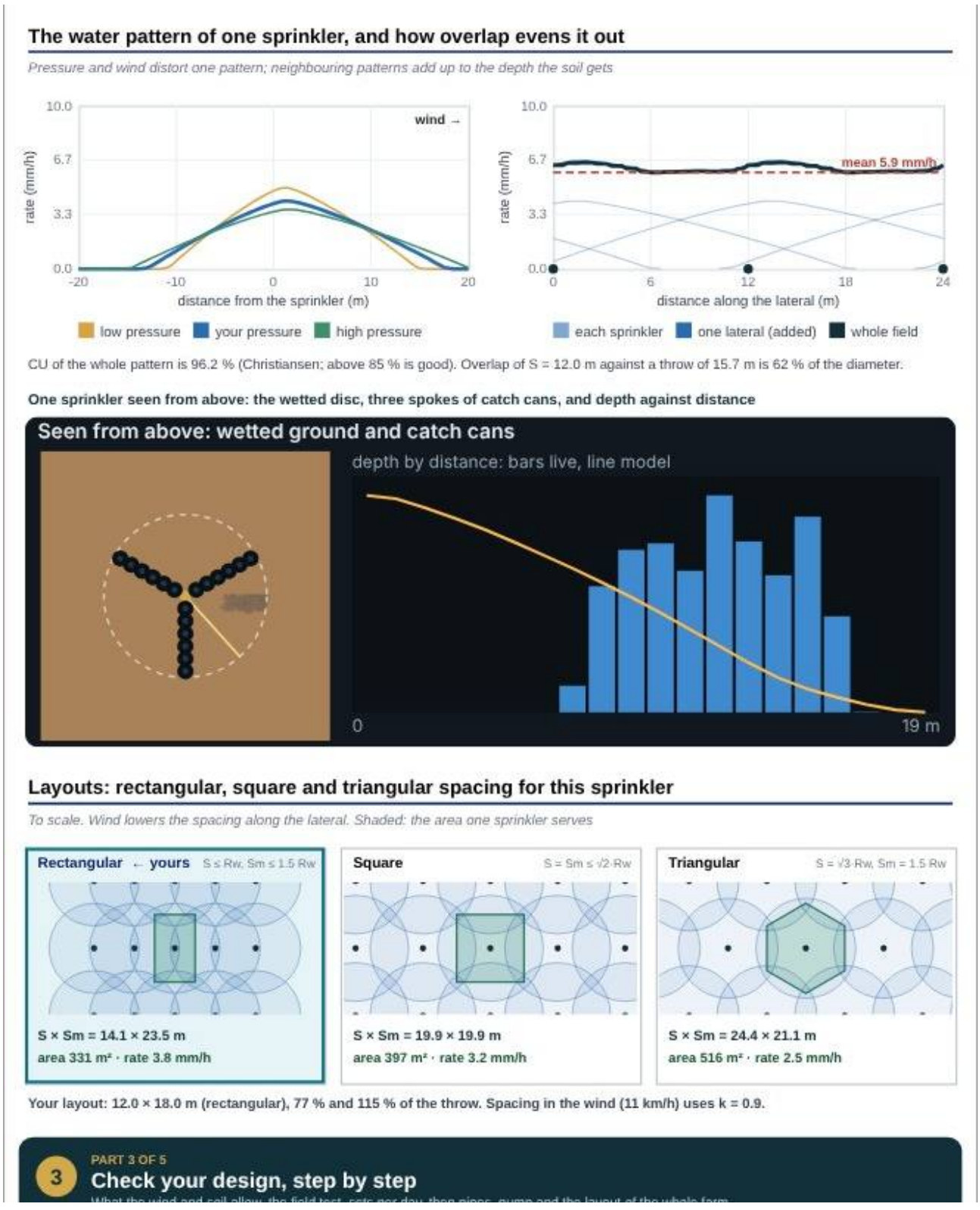


Figure 2.21: Sprinkler design and pipes, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

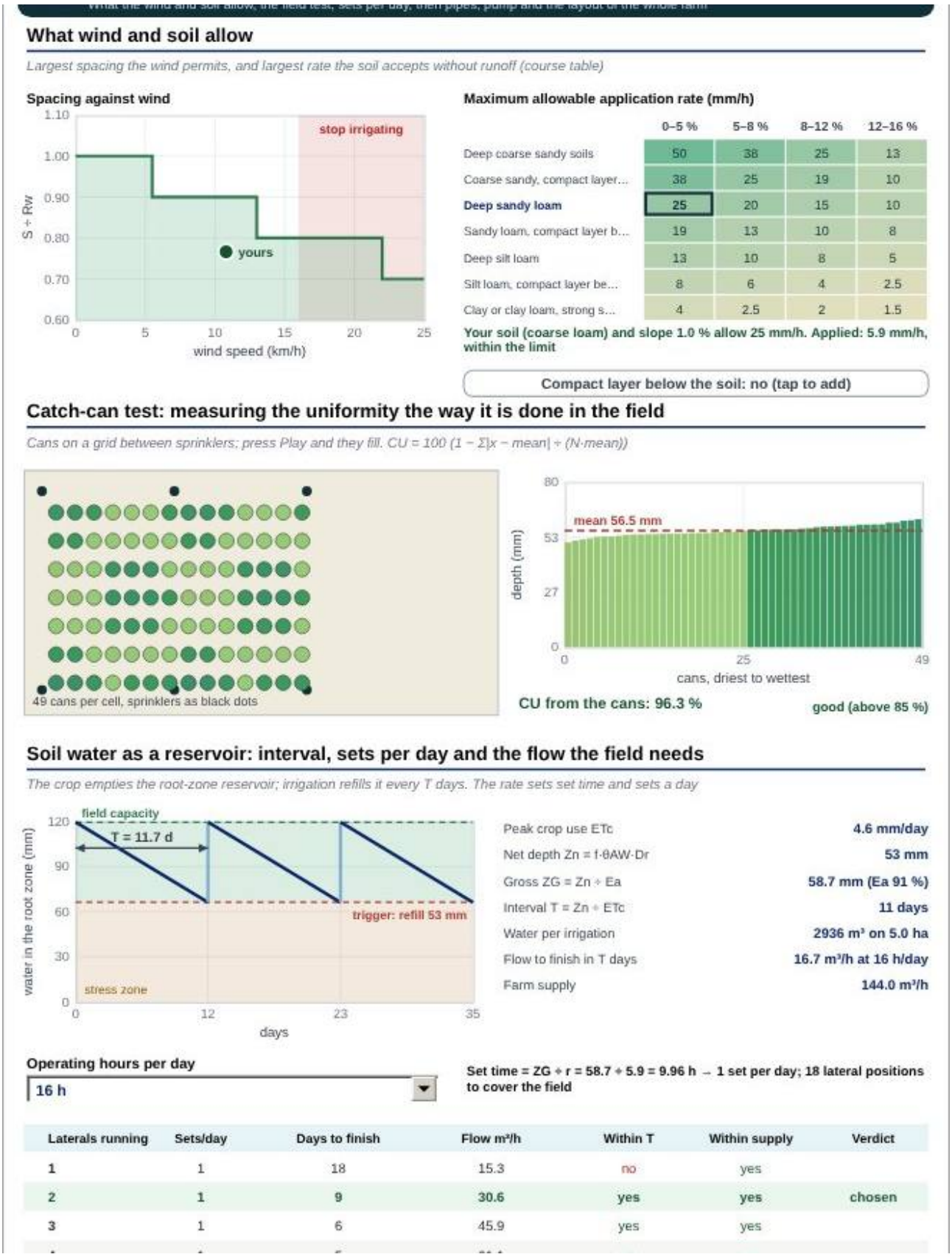


Figure 2.22: Sprinkler design and pipes, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

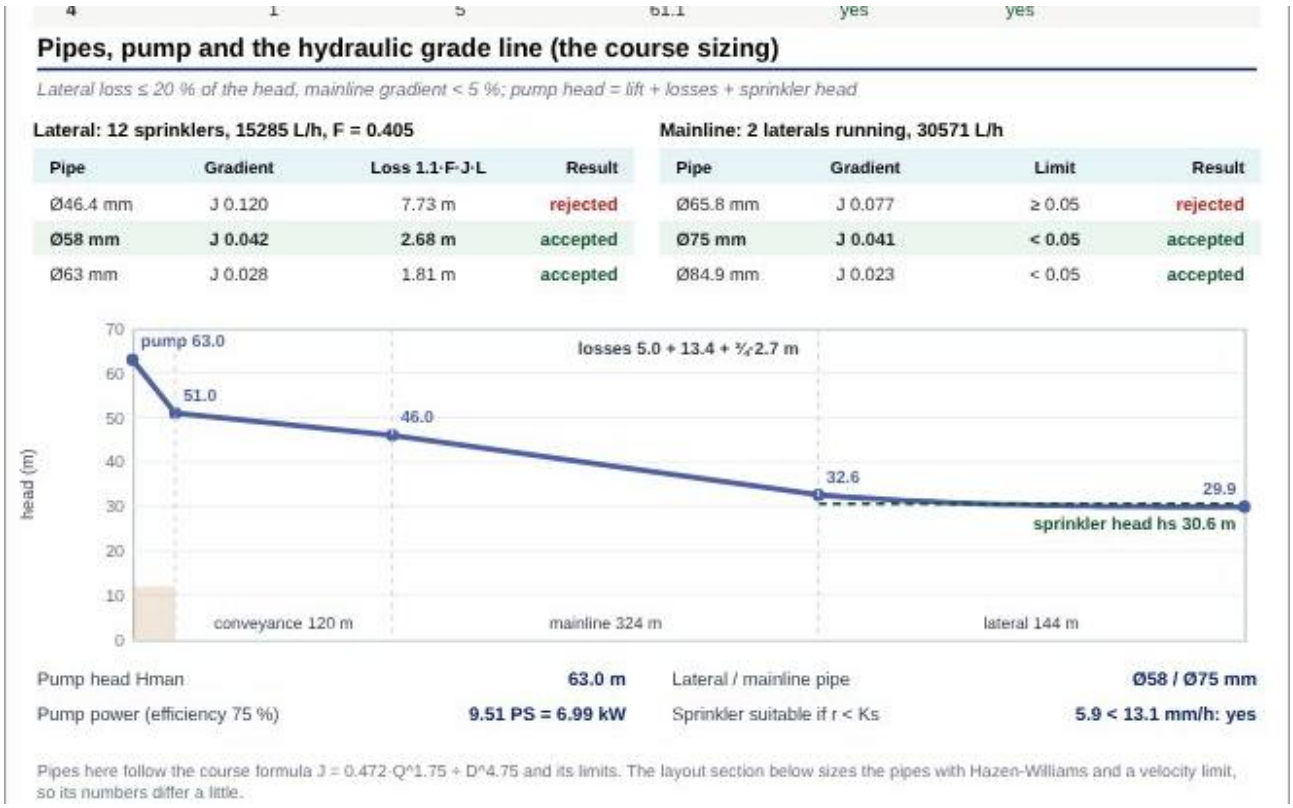
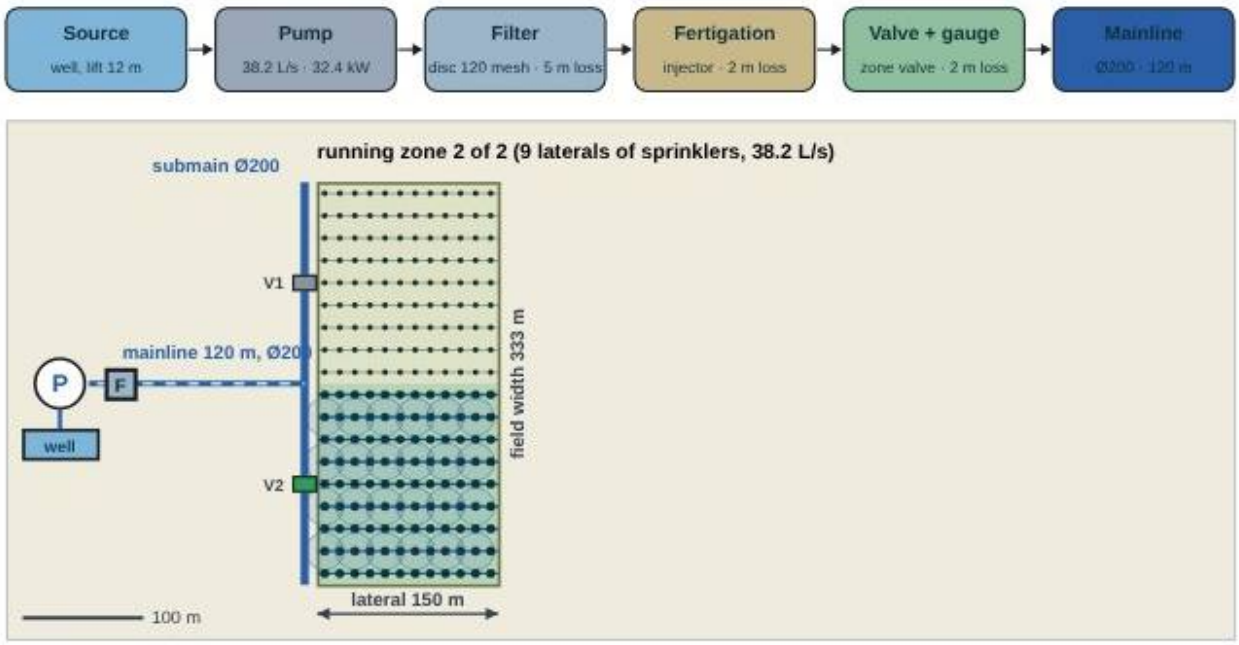
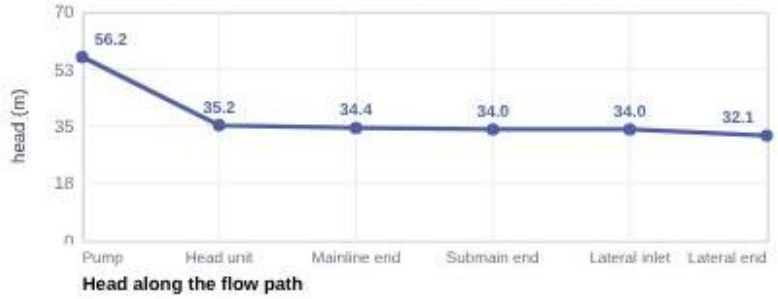


Figure 2.23: Sprinkler design and pipes, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

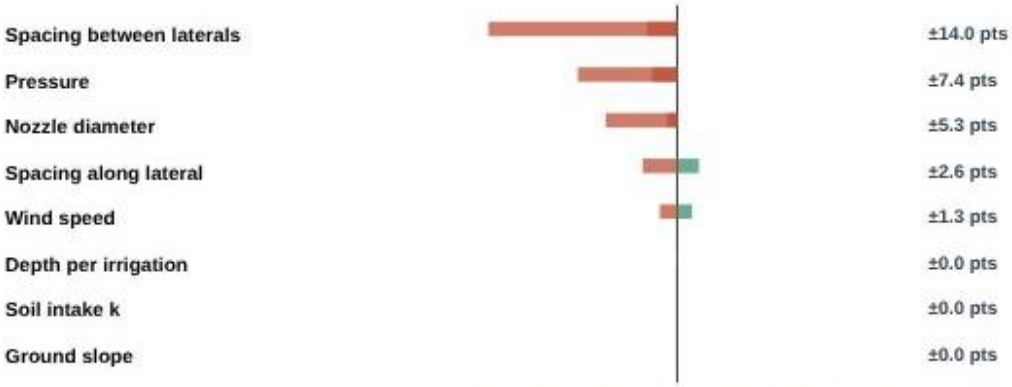
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	598	90.5	99.8	0.0	meets target

Figure 2.24: Sprinkler design and pipes, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

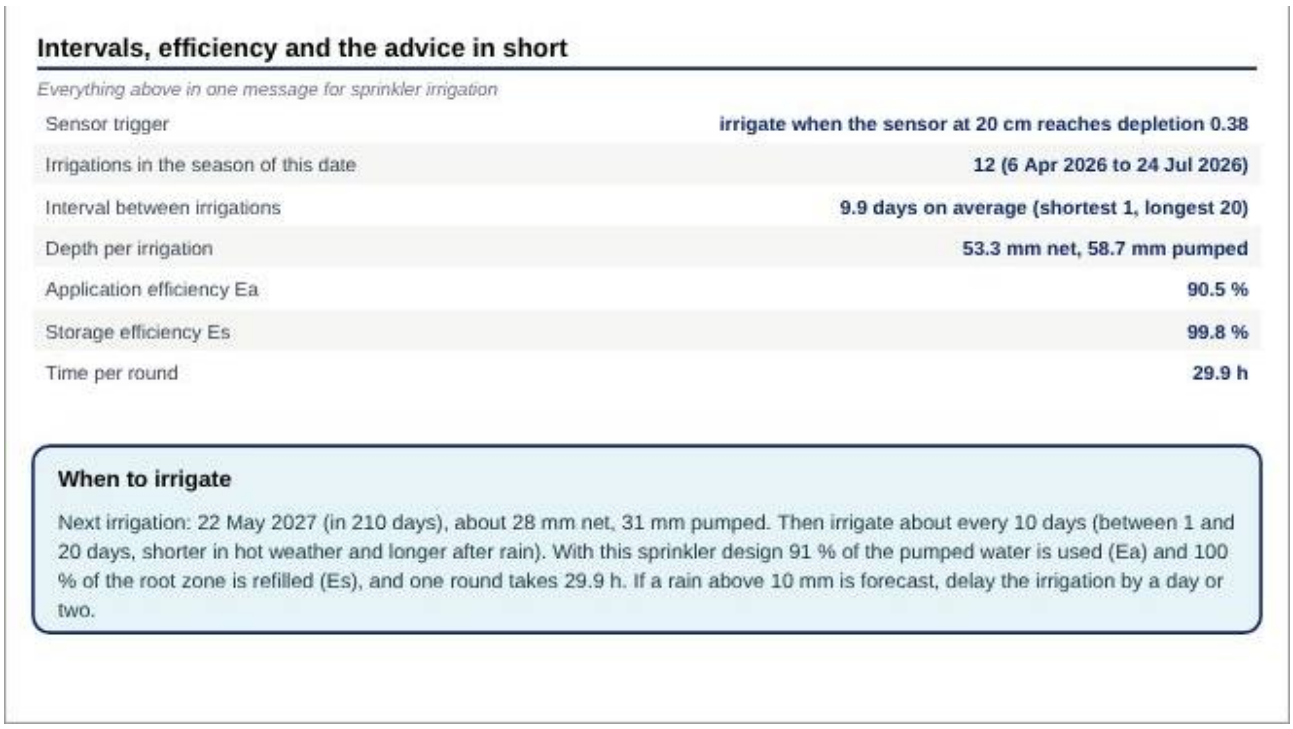
Net irrigation need in a season (DSS)	**339 mm → 6 irrigations of 53 mm**
Gross water per irrigation	**58.7 mm (Ea 91 %)**
Gross water per season	**373 mm = 3734 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 598 min = 29.9 h per round**
Pumping per season	**190 hours · 6162 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**22 May 2027**	210 days	28	31	29.9 h	13 days
2	**4 Jun 2027**	223 days	28	31	29.9 h	11 days
3	**15 Jun 2027**	234 days	28	31	29.9 h	9 days
4	**24 Jun 2027**	243 days	28	31	29.9 h	9 days
5	**3 Jul 2027**	252 days	28	31	29.9 h	10 days
6	**13 Jul 2027**	262 days	28	31	29.9 h	16 days
7	**29 Jul 2027**	278 days	28	31	29.9 h	251 days
8	**5 Apr 2028**	529 days	28	31	29.9 h	-

Figure 2.25: Sprinkler design and pipes, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.38**
Irrigations in the season of this date	**12 (6 Apr 2026 to 24 Jul 2026)**
Interval between irrigations	**9.9 days on average (shortest 1, longest 20)**
Depth per irrigation	**53.3 mm net, 58.7 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**29.9 h**

When to irrigate

Next irrigation: 22 May 2027 (in 210 days), about 28 mm net, 31 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 29.9 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 2.26: Sprinkler design and pipes, part 12 of 12: Intervals, efficiency and the advice in short.

2.3.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.27 (Volume 1, engine)

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\; 25,\; 150\big)\ \text{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.6 (Volume 1, surface)

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 12.3 (Volume 1, drip)

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\text{m}^3\,\text{s}^{-1}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.21 (Volume 1, drip)

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22 (Volume 1, drip)

$$h_{f,sm} = 0.4\,\frac{10.67\, L_{sm}\, Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\, L_m\, Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23 (Volume 1, drip)

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.26 (Volume 1, drip)

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.1 (Volume 1, sprinkler)

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

Eq. 13.2 (Volume 1, sprinkler)

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3 (Volume 1, sprinkler)

$$R = c_R\sqrt{d\,h}\;f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.4 (Volume 1, sprinkler)

$$\hat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5 (Volume 1, sprinkler)

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6 (Volume 1, sprinkler)

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_k a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7 (Volume 1, sprinkler)

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

Eq. 13.8 (Volume 1, sprinkler)

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2{+}0.5u{+}0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}[1{+}0.025\,(T_a{-}20)]\Big[0.6{+}0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.9 (Volume 1, sprinkler)

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \leq t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \text{sup}{-}F, \quad \mathrm{RO} = \max(0, R{-}S_d^{\text{eff}}), \quad I = \text{sup}{-}\mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10 (Volume 1, sprinkler)

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.11 (Volume 1, sprinkler)

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, & \text{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \text{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, & \text{RO} &= \frac{\overline{\text{RO}_k}}{\text{gross}}, & \text{WD} &= w_d + \frac{\overline{I_c}}{\text{gross}}, \end{aligned} \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12 (Volume 1, sprinkler)

$$T = \min\{T :\ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\,2400]\ \text{min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.13 (Volume 1, sprinkler)

$$k = \begin{cases} 1, & W \leq 5.5 \\ 0.9, & 5.5 < W \leq 13 \\ 0.8, & 13 < W \leq 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.14 (Volume 1, sprinkler)

$$S_s \leq k\,R_w, \qquad S_l \leq 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x / S_s \rfloor, \quad N_m = \lfloor L_y / S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.15 (Volume 1, sprinkler)

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}}\ \ (\text{h}), \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right)\ \text{days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.16 (Volume 1, sprinkler)

$$N_d(n) = \left\lceil \frac{N_m}{n\,n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.17 (Volume 1, sprinkler)

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}}\ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.
Eq. 13.18 (Volume 1, sprinkler)

$$h_{f,\mathrm{lat}} = 1.1\,F\,J(Q_L, D)\,(L_x - S_s/2) \le 0.2\,h_s, \qquad Q_L = 1000\,N\,q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).
Eq. 13.19 (Volume 1, sprinkler)

$$H_{\mathrm{man}} = z_{\mathrm{lift}} + J\,L_{\mathrm{conv}} + J\,L_{\mathrm{main}} + h_s + 0.75\,h_{f,\mathrm{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.
Eq. 13.20 (Volume 1, sprinkler)

$$P_{\mathrm{CV}} = \frac{1000\,(Q_m/3.6\times10^{6})\,H_{\mathrm{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\mathrm{kW}} = 0.7355\,P_{\mathrm{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.
Eq. 13.21 (Volume 1, sprinkler)

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\mathrm{lat}}} \Big\rfloor\Big), \quad n_{\mathrm{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\mathrm{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.
Eq. 13.22 (Volume 1, sprinkler)

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\mathrm{lat}} = F_m\,h_{HW}(Q_{\mathrm{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.
Eq. 13.23 (Volume 1, sprinkler)

$$n_{\mathrm{ev}} = \frac{y_I}{Z_n}, \qquad n_{\mathrm{sp}} = \mathrm{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\mathrm{tot}} = \frac{n_{\mathrm{sp}}\,q}{60}, \qquad n_{\mathrm{set}} = \Big\lceil \frac{Q_{\mathrm{tot}}}{Q_s} \Big\rceil, \qquad t_{\mathrm{round}} = \frac{n_{\mathrm{set}}\,T}{60}\ \mathrm{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.
Eq. 18.0 (Volume 1, runs)

$$\kappa(\text{state now}) = \kappa_{\mathrm{run}}, \qquad \kappa = \big(\text{every Properties control except the irrigation method, weather mode}\big), \tag{18.0}$$

Provenance (Eq. 18.0): **[Engineered]**: bookkeeping of the run.
Eq. 18.1 (Volume 1, runs)

$$A_f = \frac{L\,W}{10^4}\ \ [\mathrm{ha}], \qquad L_{\mathrm{lat}} = \mathrm{clip}\Big(\frac{L}{\lceil L/L_{\max}\rceil};\ L_{\min},\ L_{\max}\Big), \tag{18.1}$$

Provenance (Eq. 18.1): **[Engineered]**: geometry; laterals longer than $L_{\max}$ are split into equal runs from a central submain.
Eq. 18.2 (Volume 1, runs)

$$s = E_a - 3\,\max(0,\ 0.95 - E_s) + \begin{cases} 0 & \text{surface} \\ 0.05\,E_u & \text{drip} \\ 0.05\,\mathrm{CU} - 2\,R_o & \text{sprinkler} \end{cases} \tag{18.2}$$

Provenance (Eq. 18.2): **[Engineered]**: the application efficiency, with a penalty when the root zone is refilled to less than 95 % (an efficient design that leaves the crop short is not efficient) and a small reward for uniformity.

Eq. 14.6 (Volume 1, dashboard)

$$Q_{\text{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\; n_z\, t_{\text{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

Provenance (Eq. 14.6): **[Engineered]**: zones of the layout (Eqs. 12.20–12.23 and the sprinkler analogue) run in turn.

2.4 Virtual farm simulation

2.4.1 The farm of the surface design

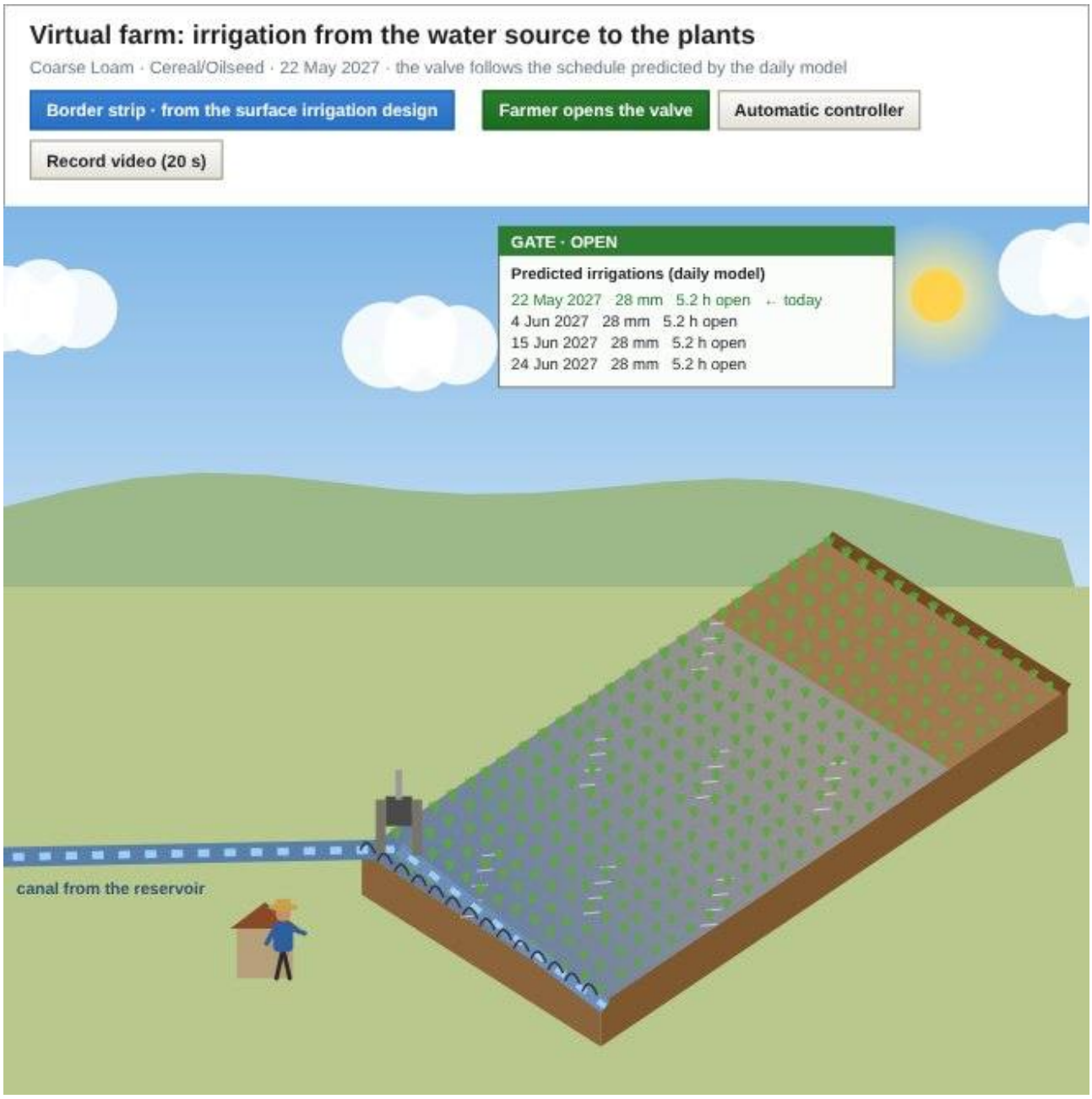


Figure 2.27: Virtual farm simulation, part 1 of 3.



Figure 2.28: Virtual farm simulation, part 2 of 3.

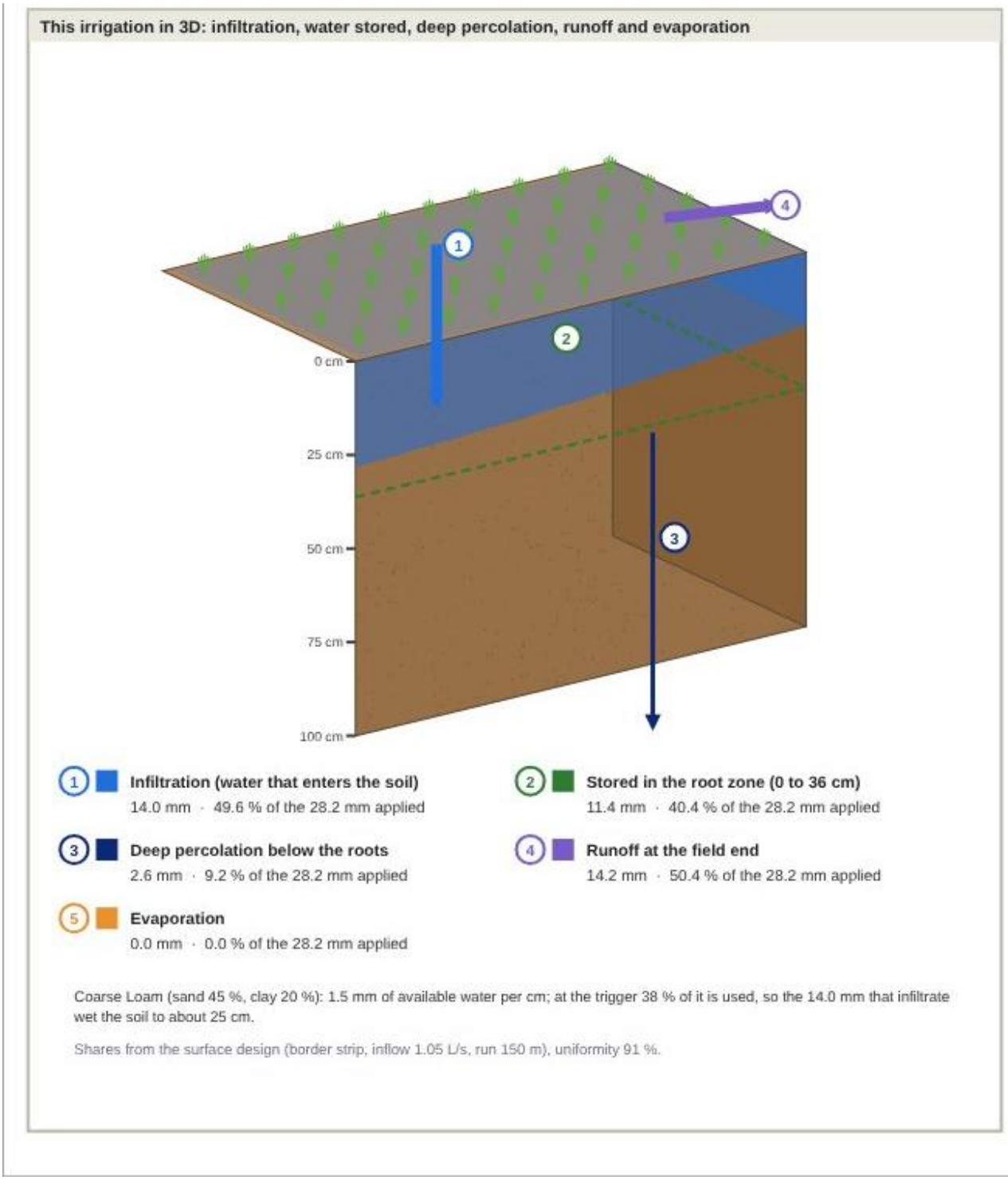


Figure 2.29: Virtual farm simulation, part 3 of 3.

2.4.2 The farm of the drip design

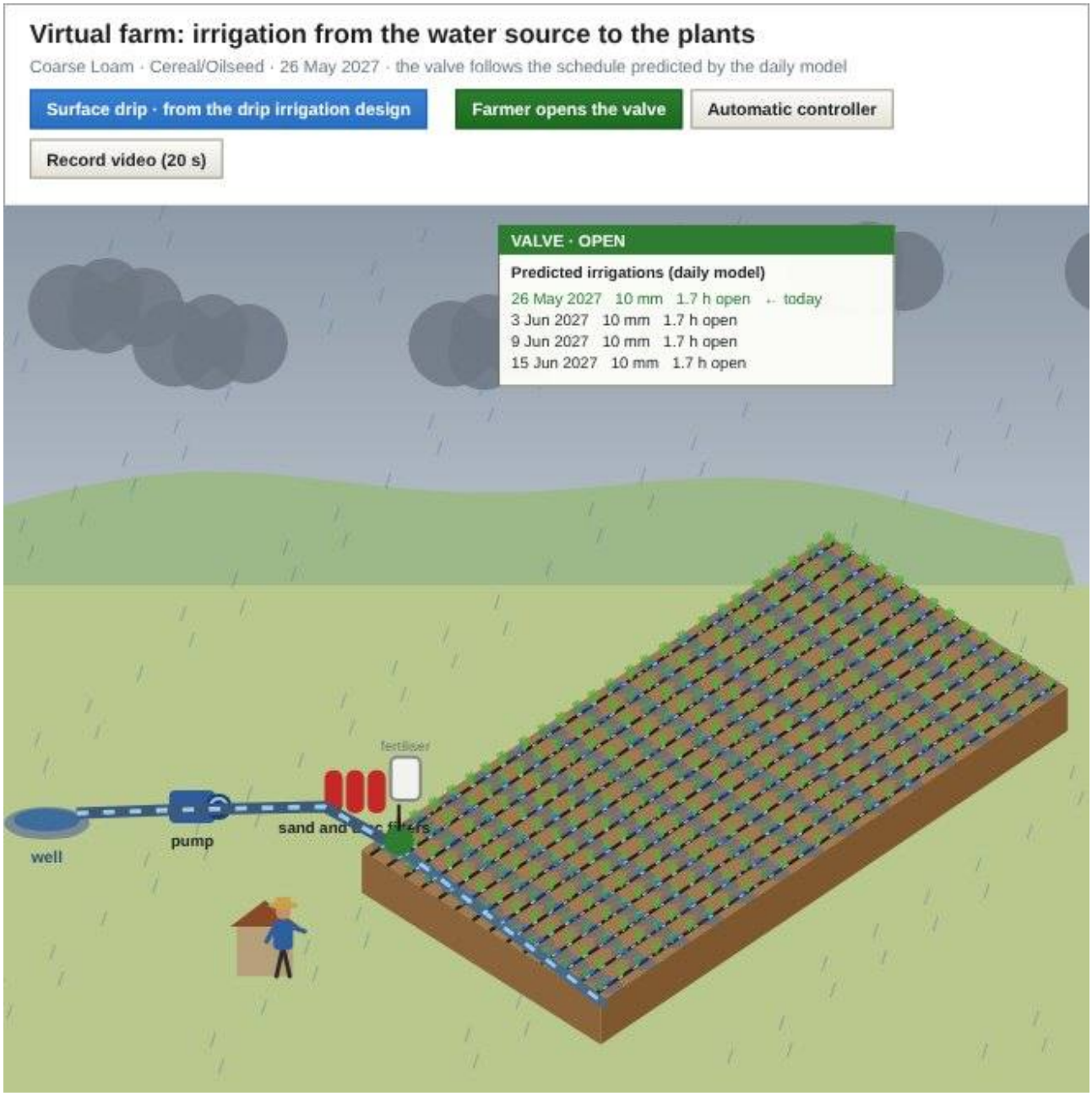


Figure 2.30: Virtual farm simulation, part 1 of 3.

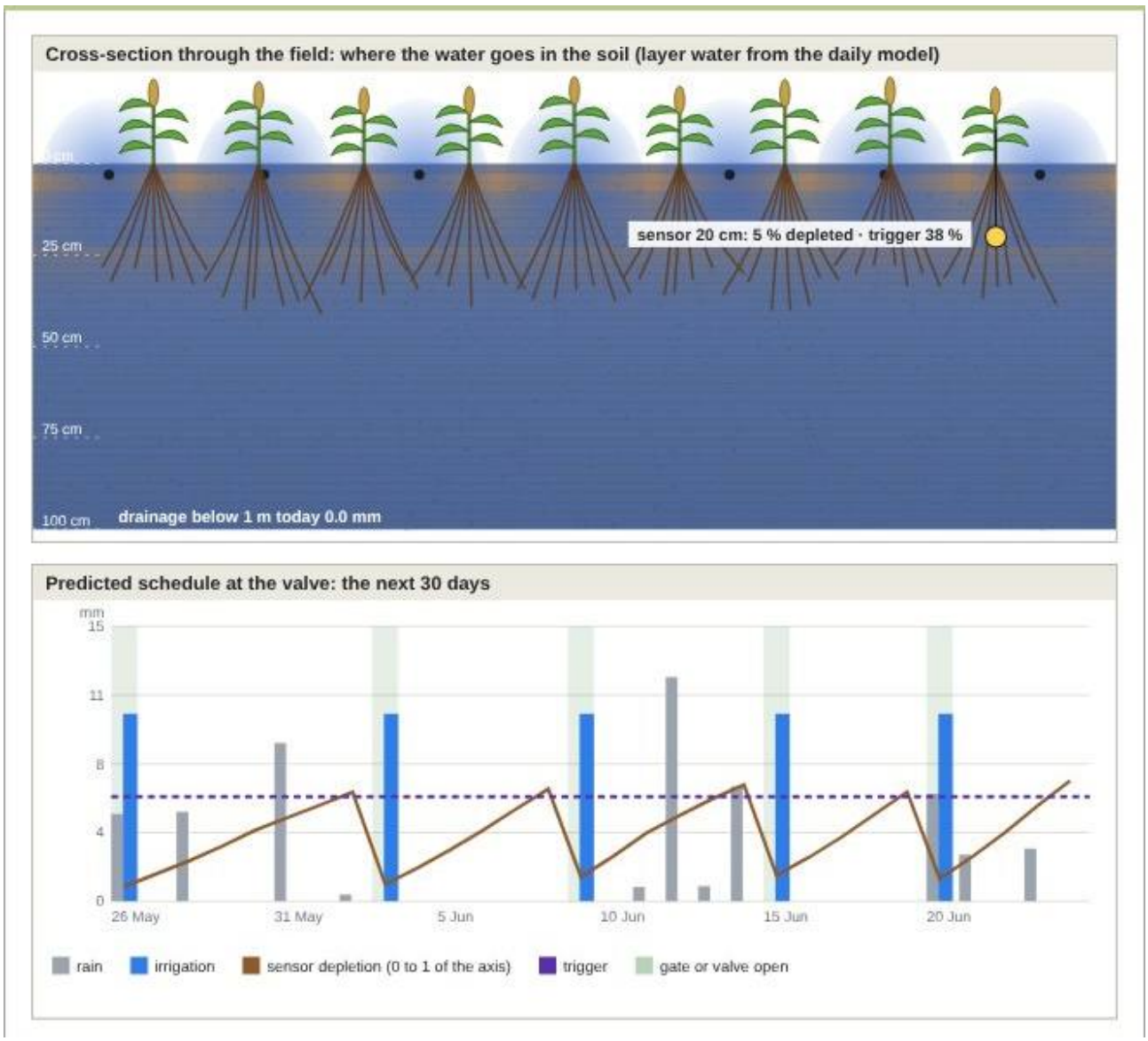


Figure 2.31: Virtual farm simulation, part 2 of 3.

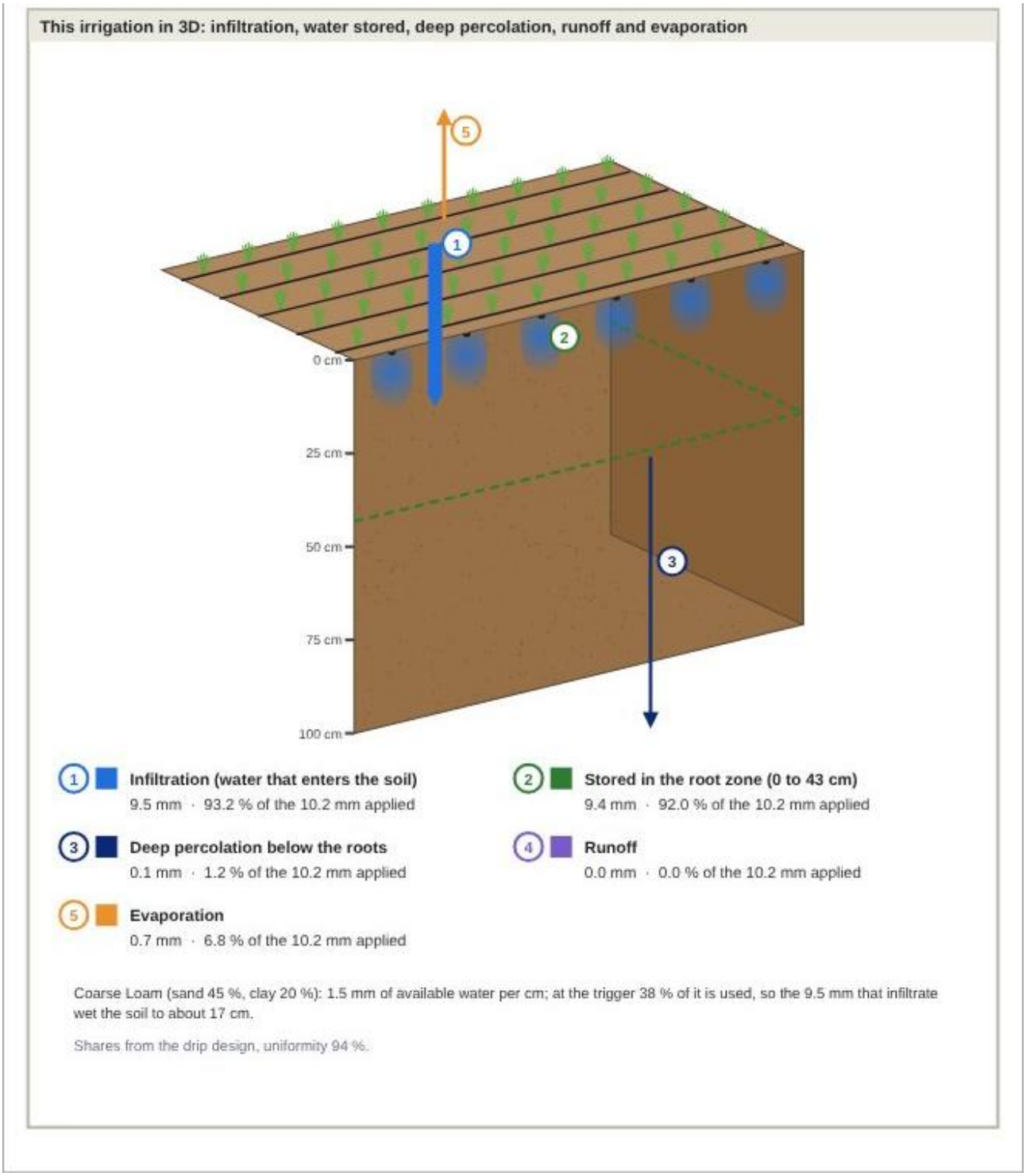


Figure 2.32: Virtual farm simulation, part 3 of 3.

2.4.3 The farm of the sprinkler design

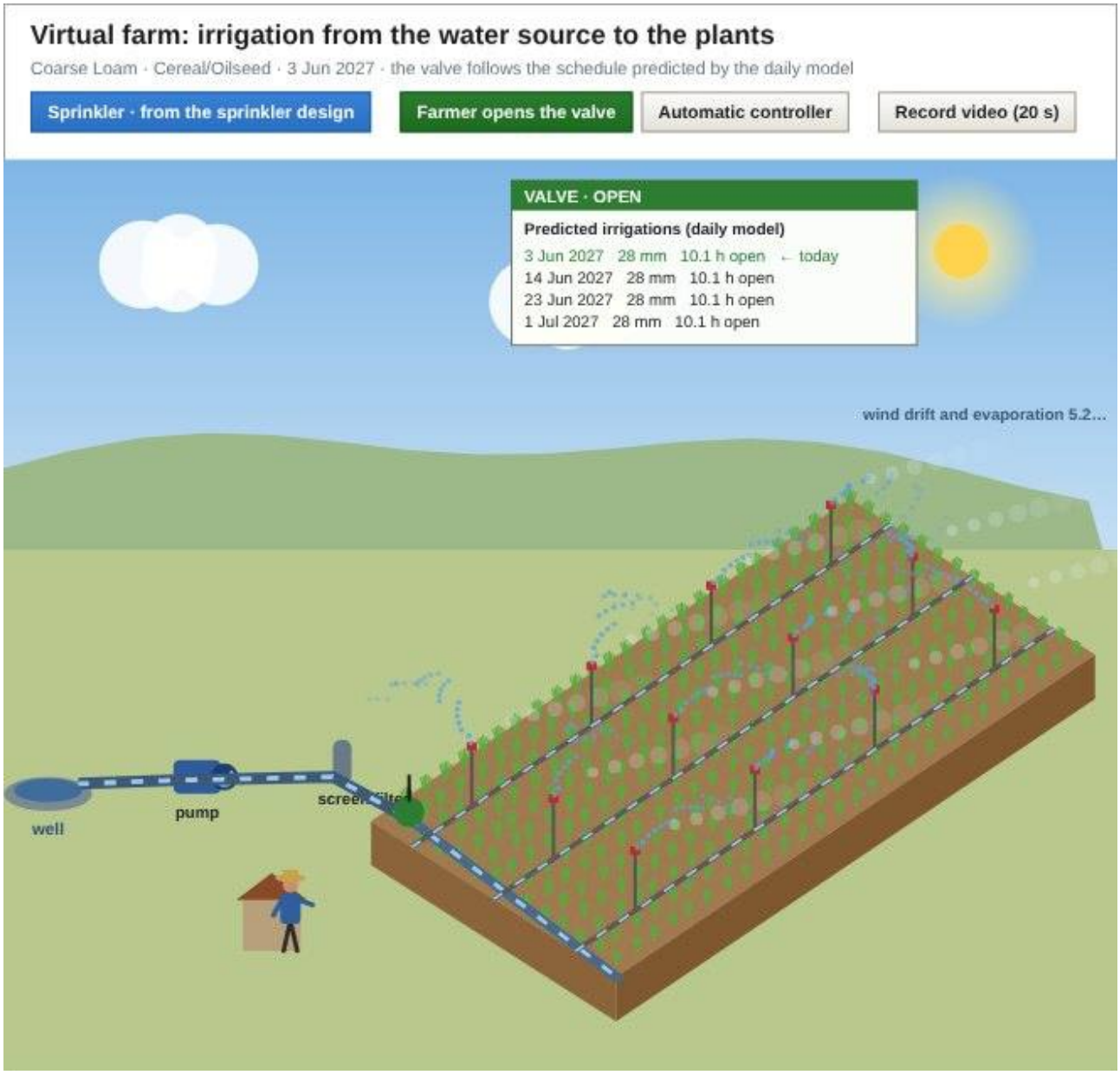


Figure 2.33: Virtual farm simulation, part 1 of 3.

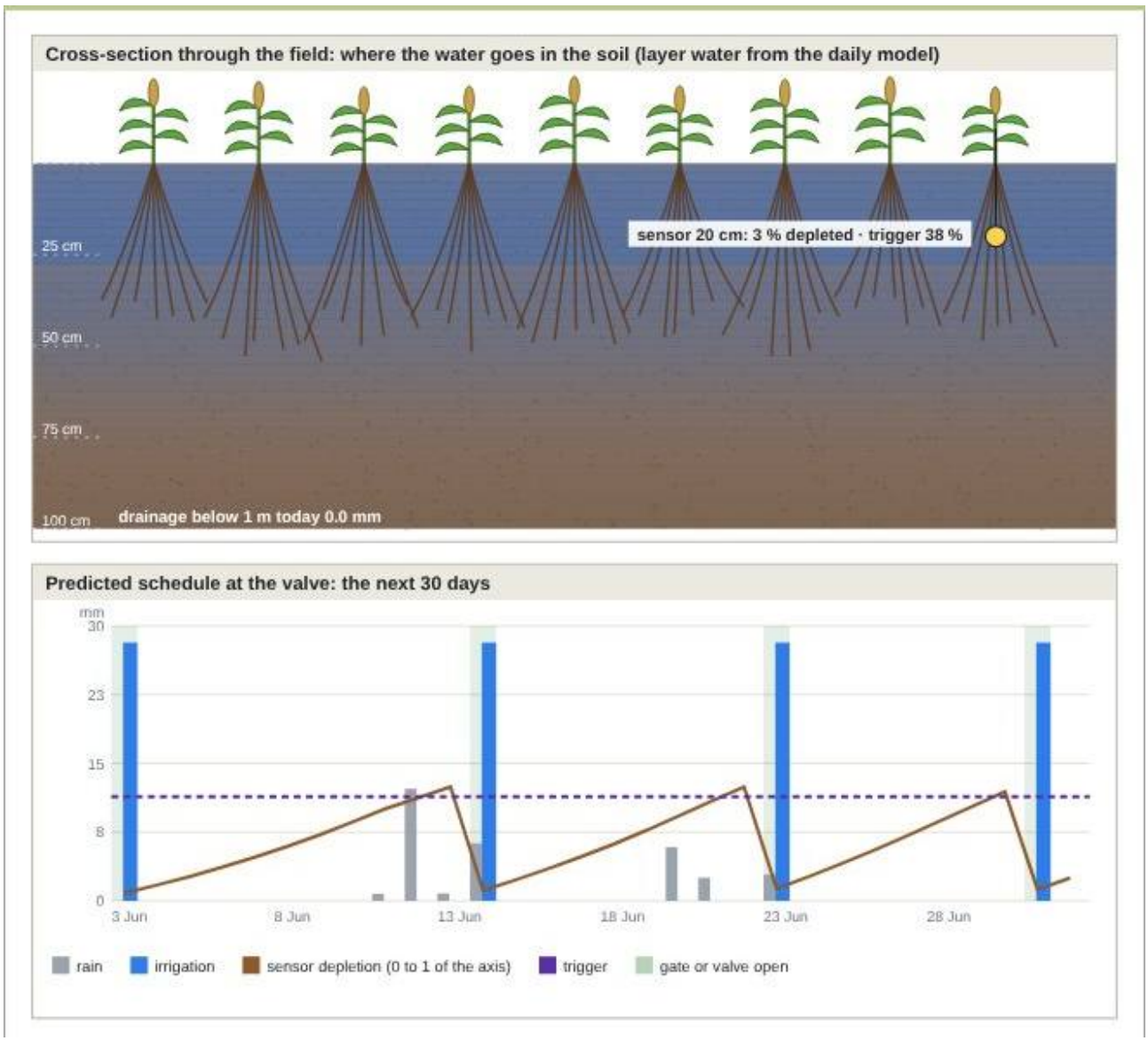


Figure 2.34: Virtual farm simulation, part 2 of 3.

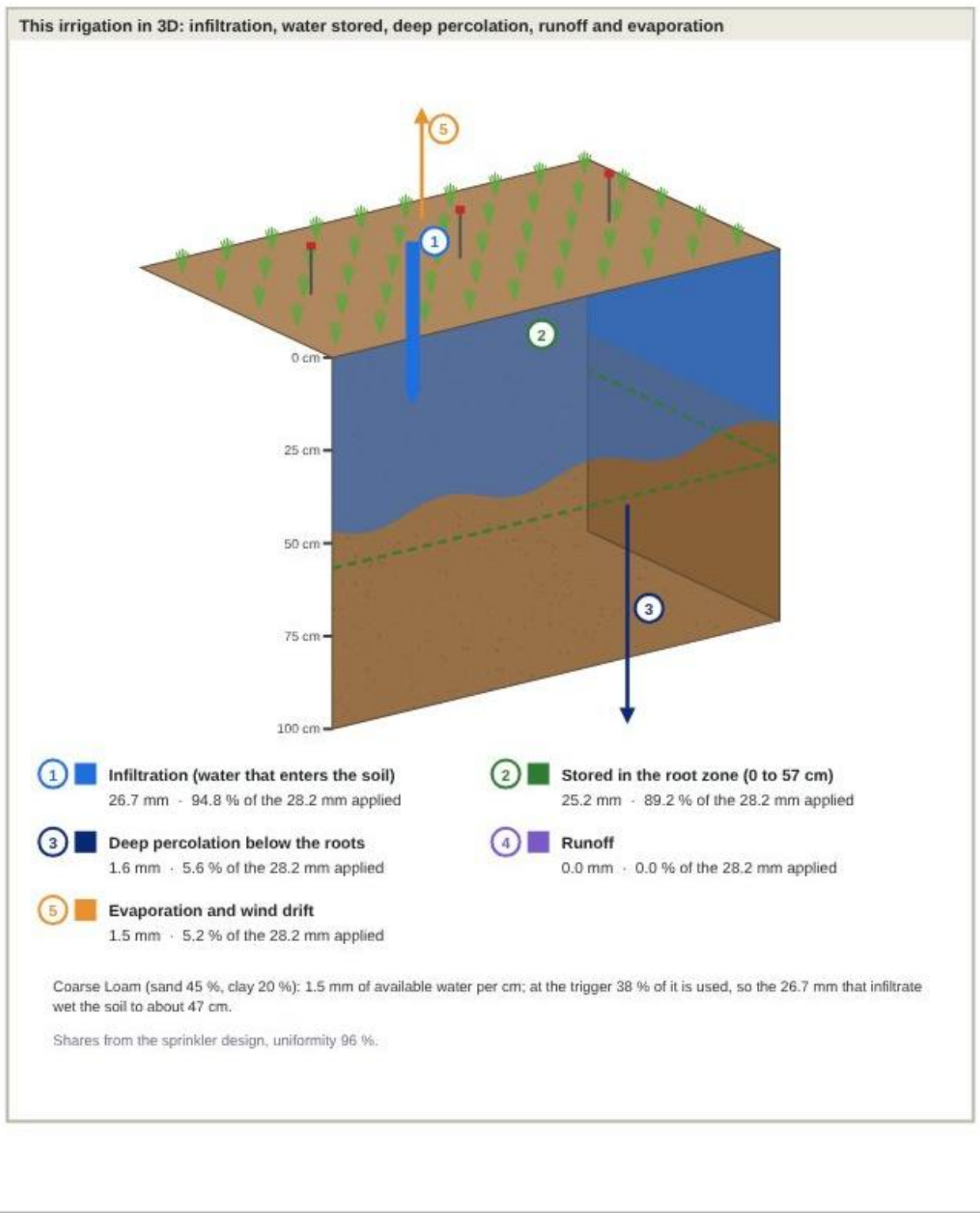


Figure 2.35: Virtual farm simulation, part 3 of 3.

2.4.4 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.27 (Volume 1, engine)

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\; 25,\; 150\big)\ \text{mm} \qquad \big(\text{surface; drip and sprinkler use their own limits}\big) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.16 (Volume 1, surface)

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.18 (Volume 1, surface)

$$\begin{aligned} &V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b < 1:\; Q\,t_L + c_b\,Q\,(t_{co} - t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w, \\ &V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx, \\ &V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx, \\ &V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\; 0). \end{aligned} \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19 (Volume 1, surface)

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 12.19 (Volume 1, drip)

$$E_a = \frac{\langle V_{\text{st}} \rangle}{\langle V \rangle}, \quad E_s = \frac{\langle V_{\text{st}} \rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}} \rangle}{\langle V \rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}} \rangle}{\langle V \rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 13.11 (Volume 1, sprinkler)

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 20.1 (Volume 1, farm)

$$(f_{\text{st}}, f_{\text{dp}}, f_{\text{ro}}, f_{\text{ev}}) = \frac{(E_a,\ DP,\ RO,\ EV)}{E_a + DP + RO + EV}, \qquad d_x = f_x\, d_{\text{gross}}, \tag{20.1}$$

Provenance (Eq. 20.1): **[Engineered]**: the partition of the design run (Eqs. 11.18, 12.19, 13.11) applied to the gross depth of the event predicted by the daily model.

Eq. 20.2 (Volume 1, farm)

$$z_w = \text{clip}\Big(\frac{d_{\text{gross}}\,(f_{\text{st}} + f_{\text{dp}})}{f_T\,\text{AWC}};\ 3,\ 100\Big)\ \ [\text{cm}], \tag{20.2}$$

Provenance (Eq. 20.2): **[Literature]**: the infiltrated water refills the depleted share f_T of the available water AWC (mm per cm) of each centimetre in turn, from the top (the “piston” refill of the FAO-56 root-zone balance).

Eq. 18.0 (Volume 1, runs)

$$\kappa(\text{state now}) = \kappa_{\text{run}}, \qquad \kappa = \big(\text{every Properties control except the irrigation method, weather mode}\big), \tag{18.0}$$

Provenance (Eq. 18.0): **[Engineered]**: bookkeeping of the run.

Eq. 14.6 (Volume 1, dashboard)

$$Q_{\text{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\ n_z\, t_{\text{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

Provenance (Eq. 14.6): **[Engineered]**: zones of the layout (Eqs. 12.20–12.23 and the sprinkler analogue) run in turn.

Chapter 3

Irrigation control

3.1 Sensor and trigger

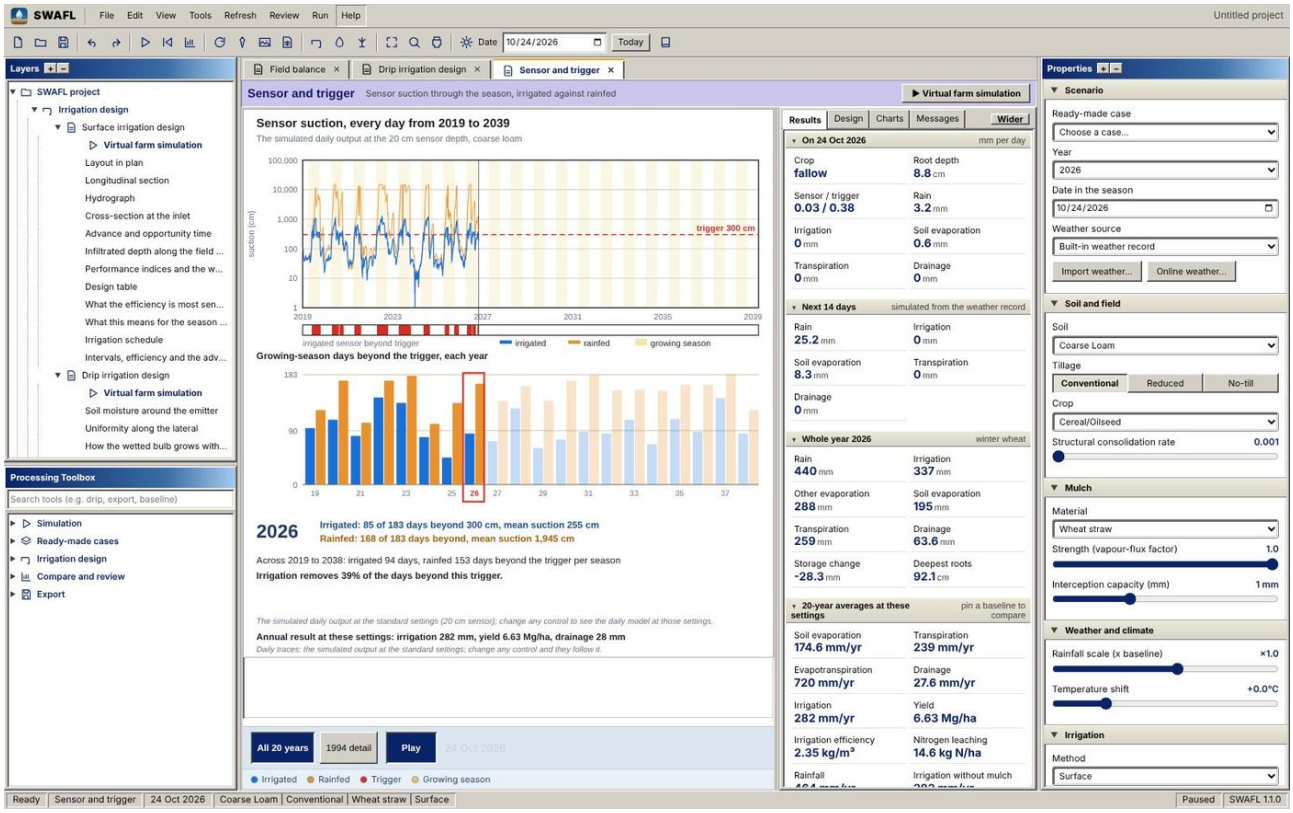


Figure 3.1: Sensor and trigger: the window with the layer open.

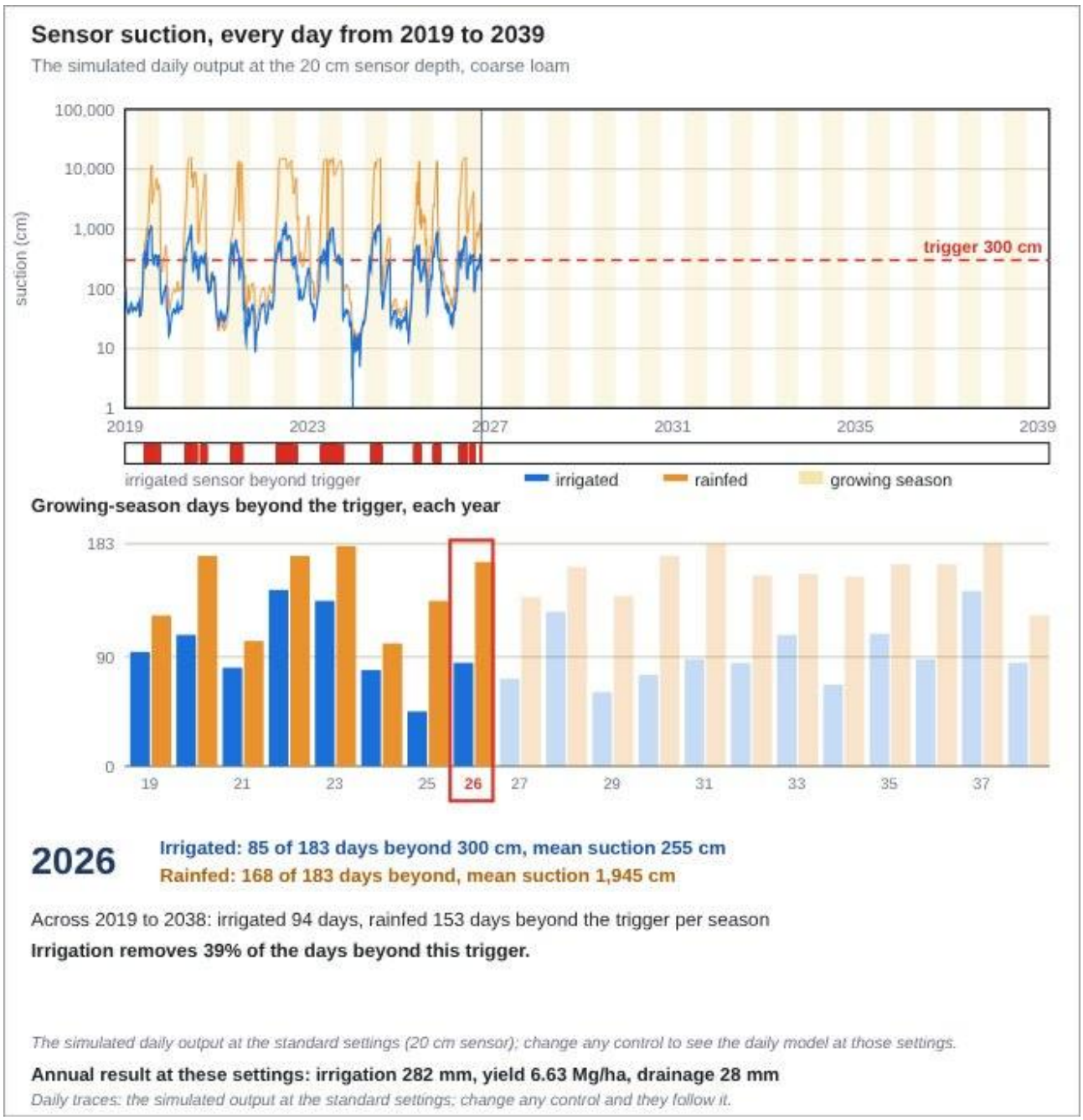


Figure 3.2: Sensor and trigger, part 1 of 1.

3.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 6.1 (Volume 1, control)

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\}\ \ (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.2 (Volume 1, control)

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_y N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.4a (Volume 1, control)

$$h_d^{\text{x}} = h_{\text{soil}}\Big(f_{\text{ref}}(h_d^{\text{rec}}) + \tfrac{1}{7}\sum_{q=-3}^{3}\big[f_{d+q}^{\text{set}} - f_{d+q}^{\text{std}}\big]\Big), \tag{6.4a}$$

Provenance (Eq. 6.4a): **[Data-driven + Engineered]**: the stored simulated record carries the real weather sequence; the change made by your settings is the difference of two runs of the daily model, converted with the van Genuchten curve of your soil (the inverse of Eq. 5.3, tabulated).

3.2 Trigger law

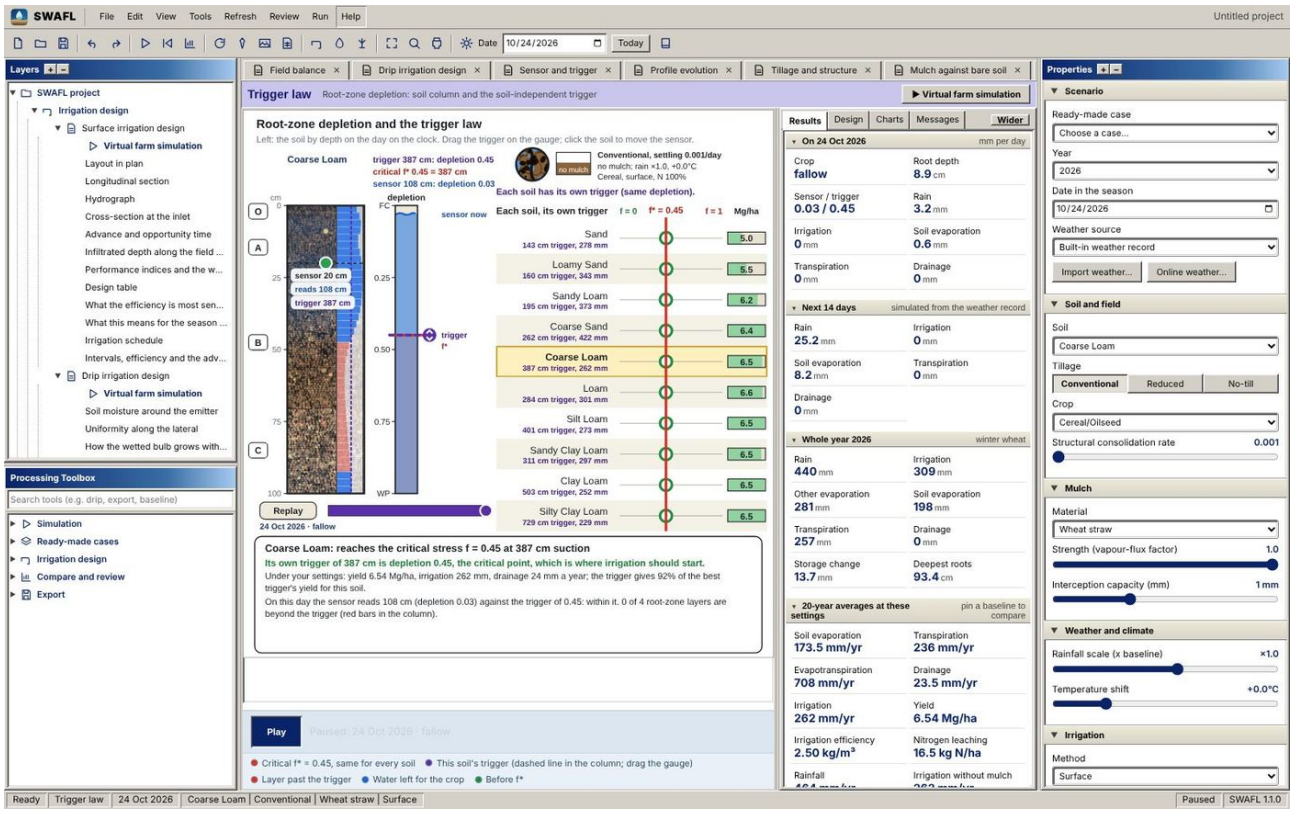


Figure 3.3: Trigger law: the window with the layer open.

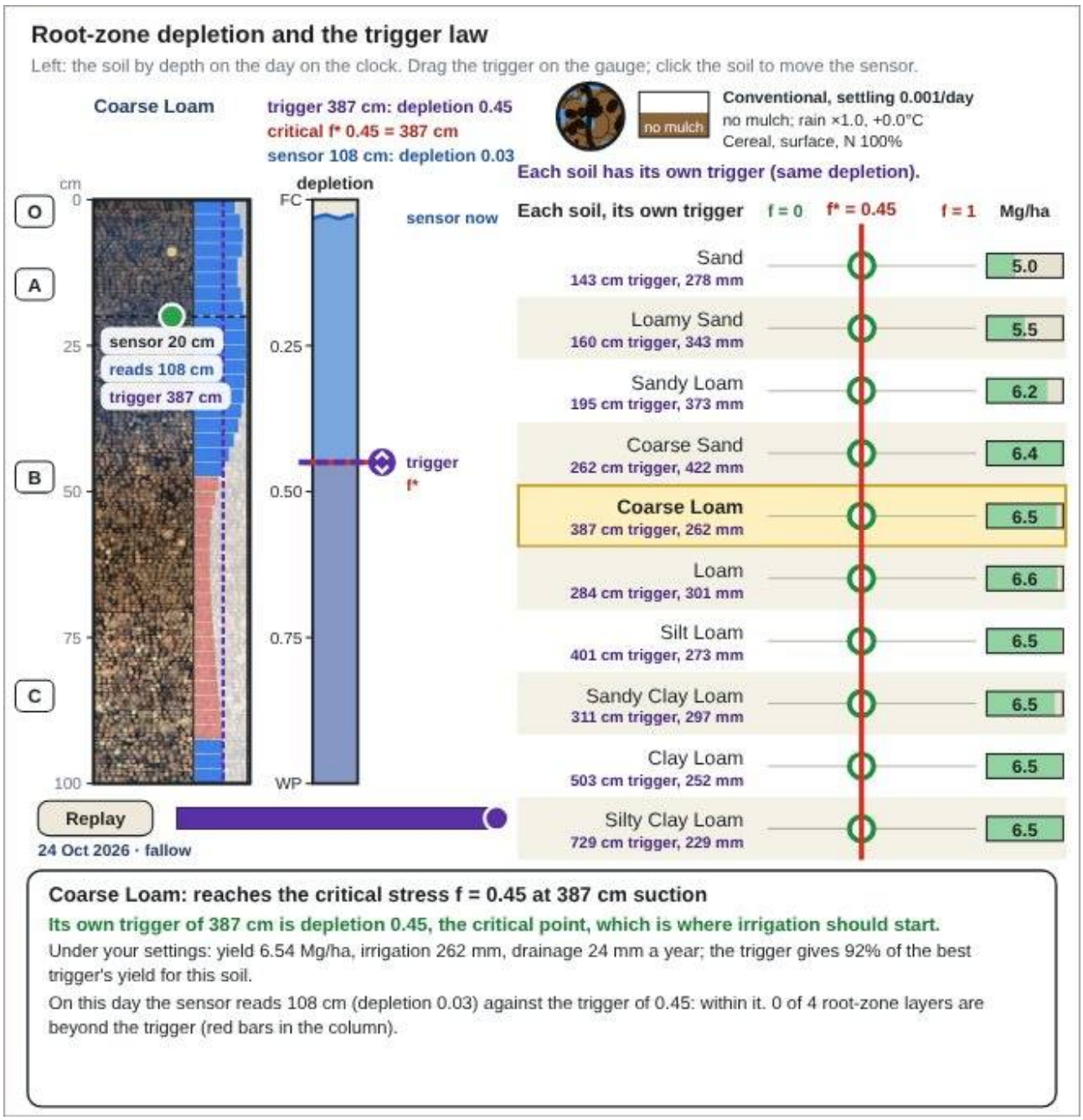


Figure 3.4: Trigger law, part 1 of 1.

3.2.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.27 (Volume 1, engine)

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\text{max}};\; 25,\; 150\big)\ \text{mm} \qquad \big(\text{surface; drip and sprinkler use their own limits}\big) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 6.3 (Volume 1, control)

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

Eq. 6.4 (Volume 1, control)

$$Z_n = f \cdot \text{AWC} \cdot R_{\text{max}} \quad [\text{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.

Eq. 6.5 (Volume 1, control)

$$Y_{\text{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.

Chapter 4

Mulch and evaporation

4.1 Mulch against bare soil

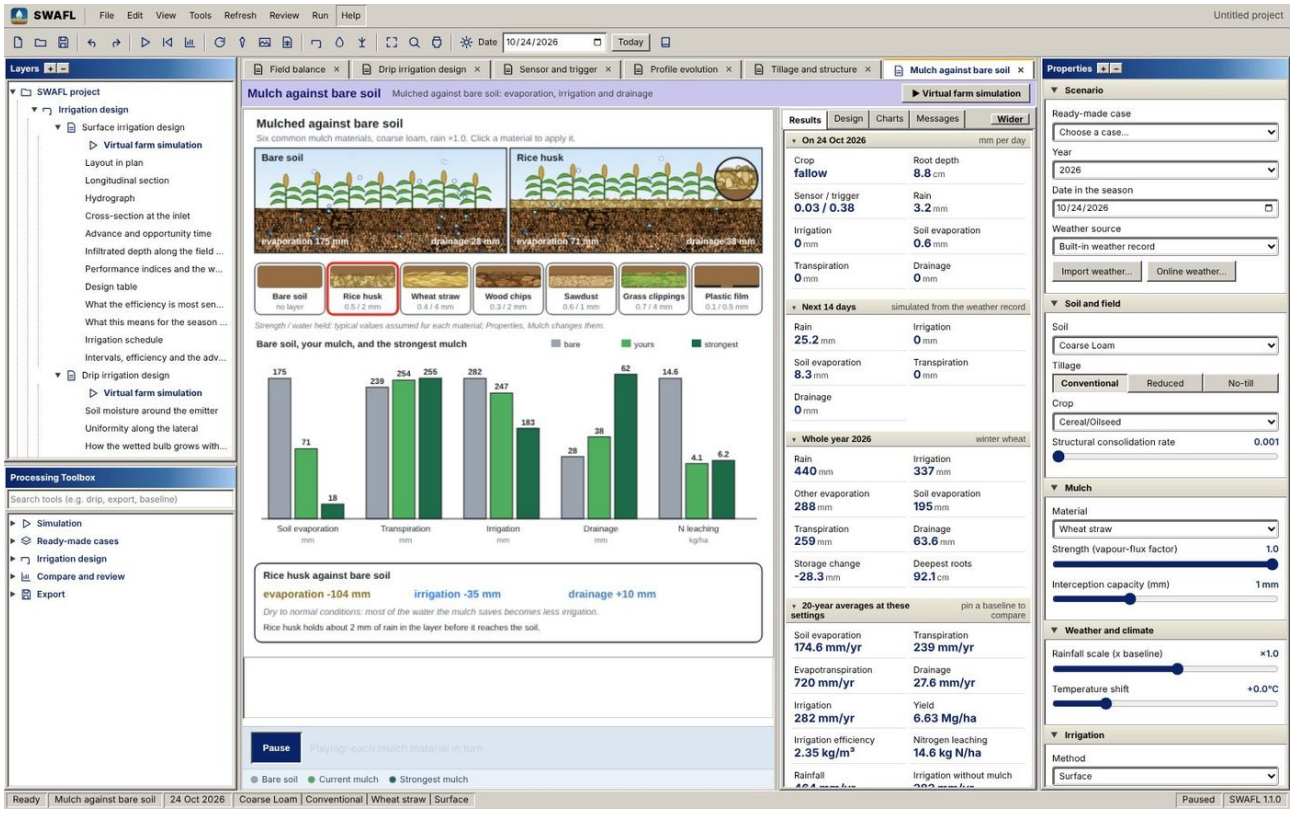


Figure 4.1: Mulch against bare soil: the window with the layer open.

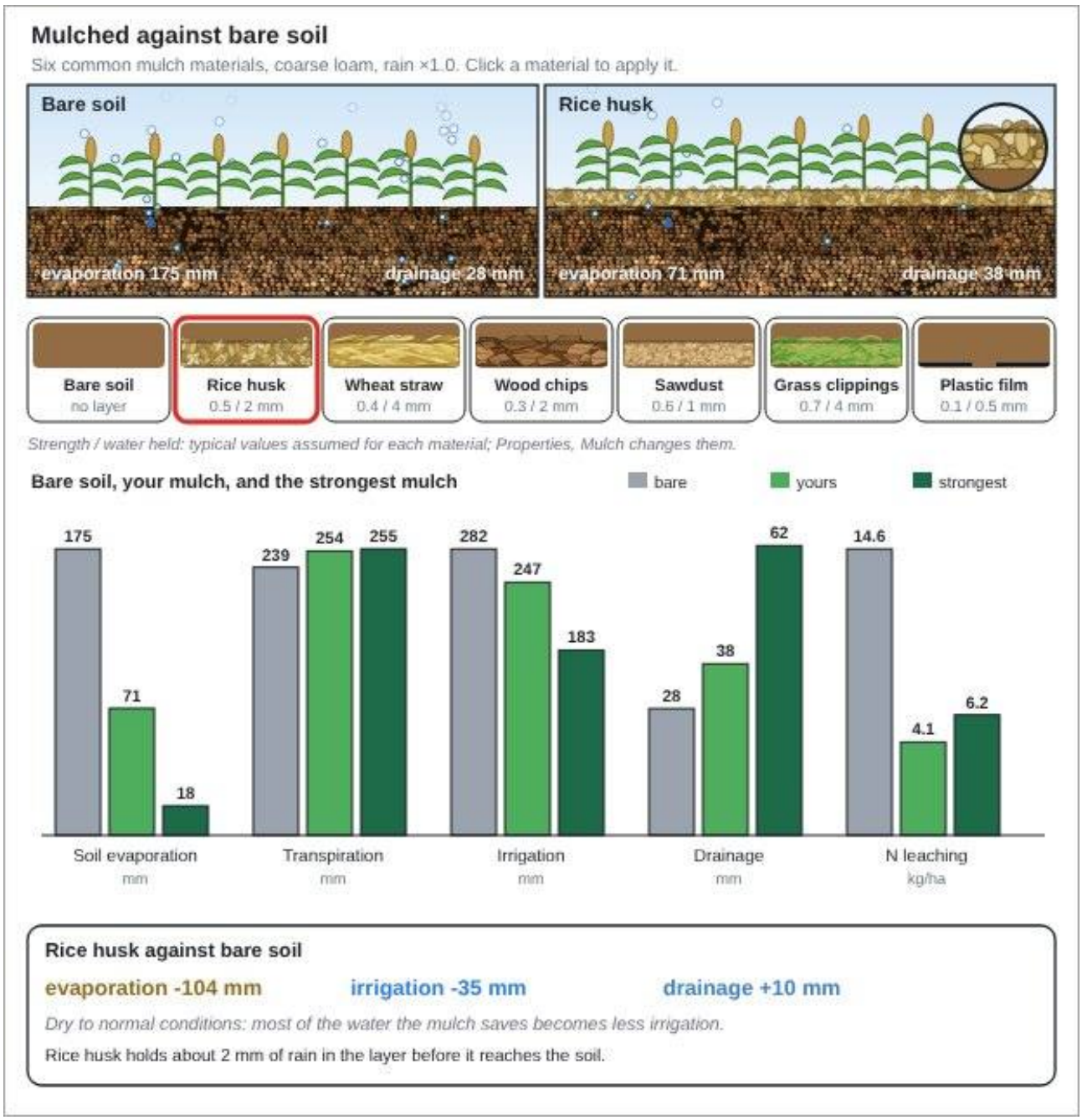


Figure 4.2: Mulch against bare soil, part 1 of 1.

4.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 9.1 (Volume 1, mulch)

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

4.2 Mulch control

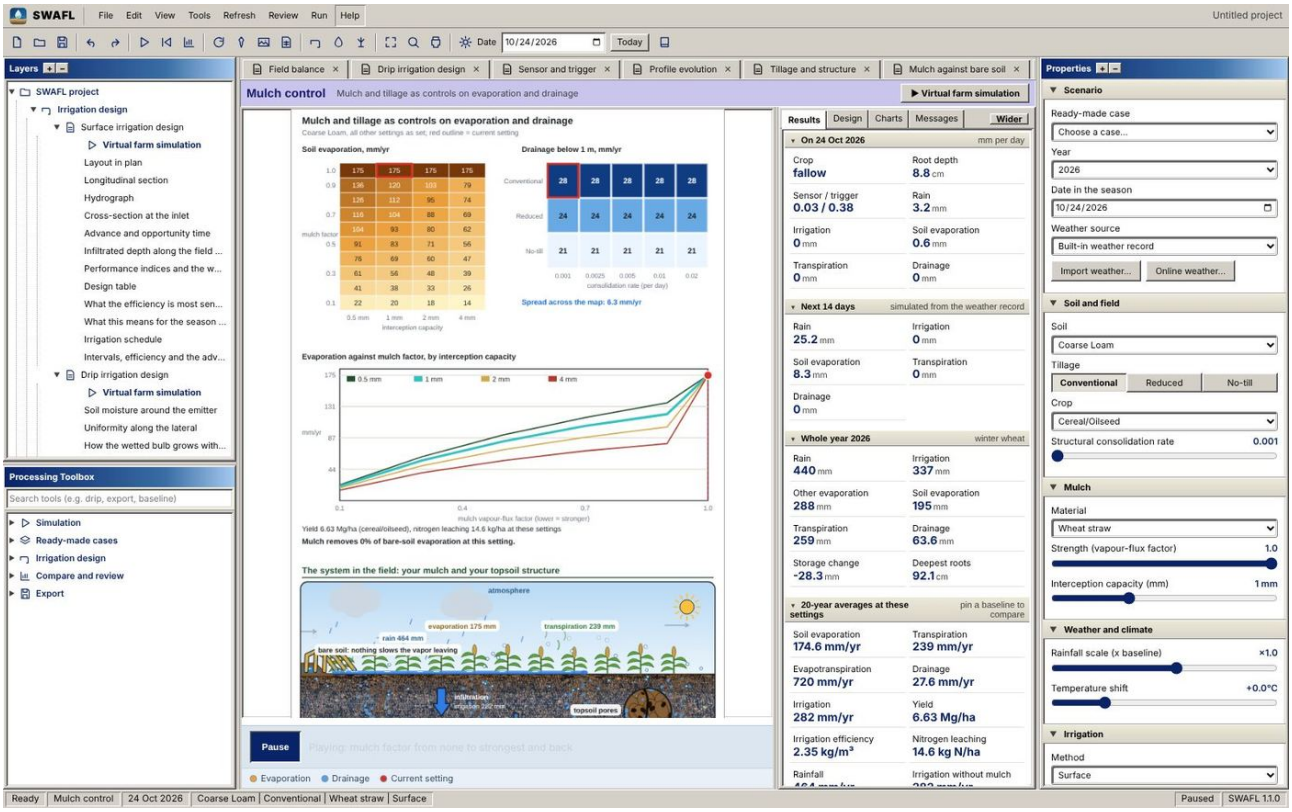


Figure 4.3: Mulch control: the window with the layer open.

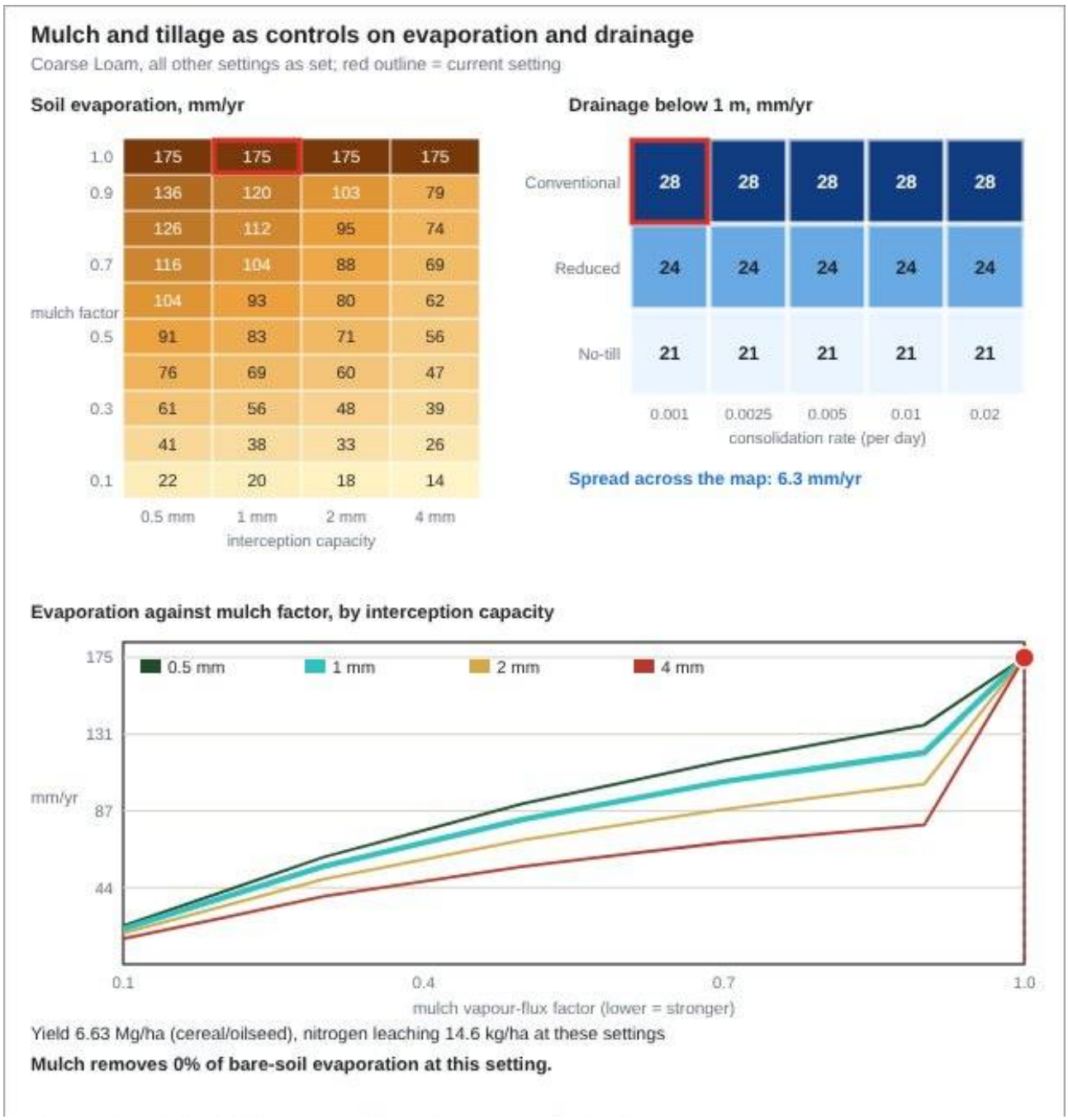


Figure 4.4: Mulch control, part 1 of 2.

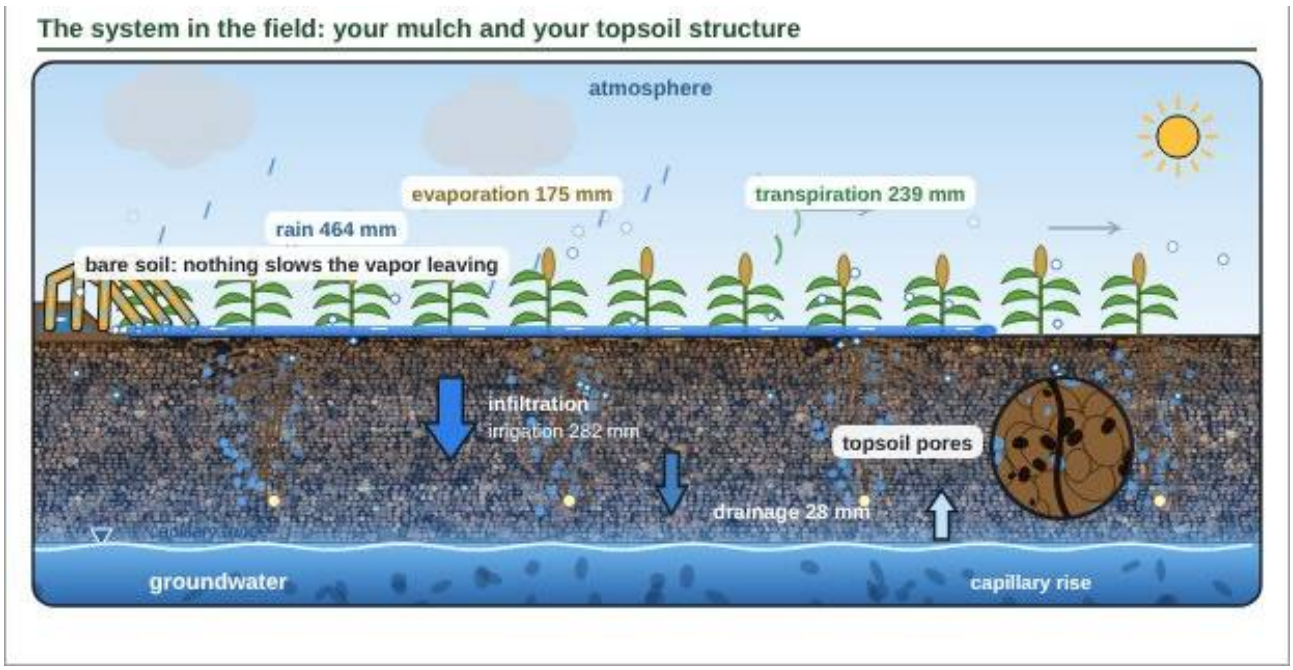


Figure 4.5: Mulch control, part 2 of 2.

4.2.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 9.1 (Volume 1, mulch)

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

4.3 Evaporation physics

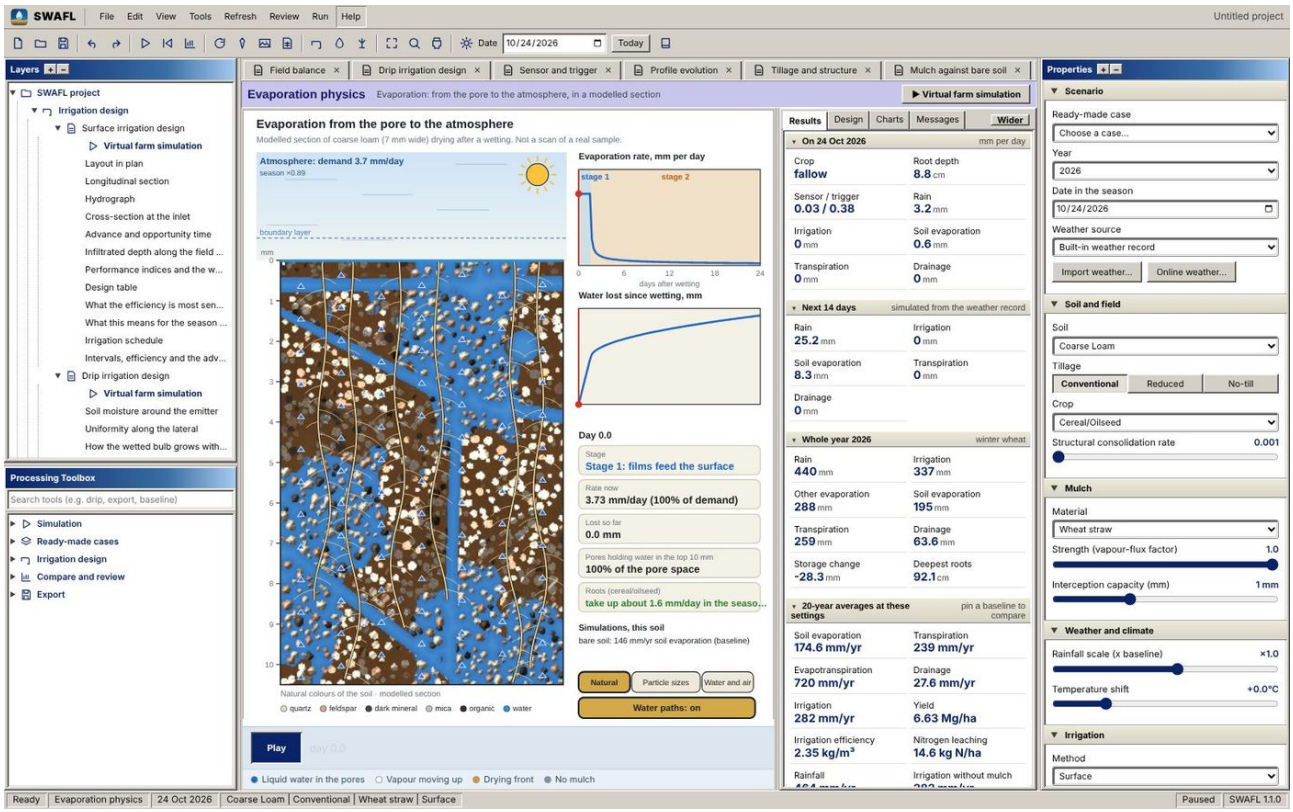


Figure 4.6: Evaporation physics: the window with the layer open.

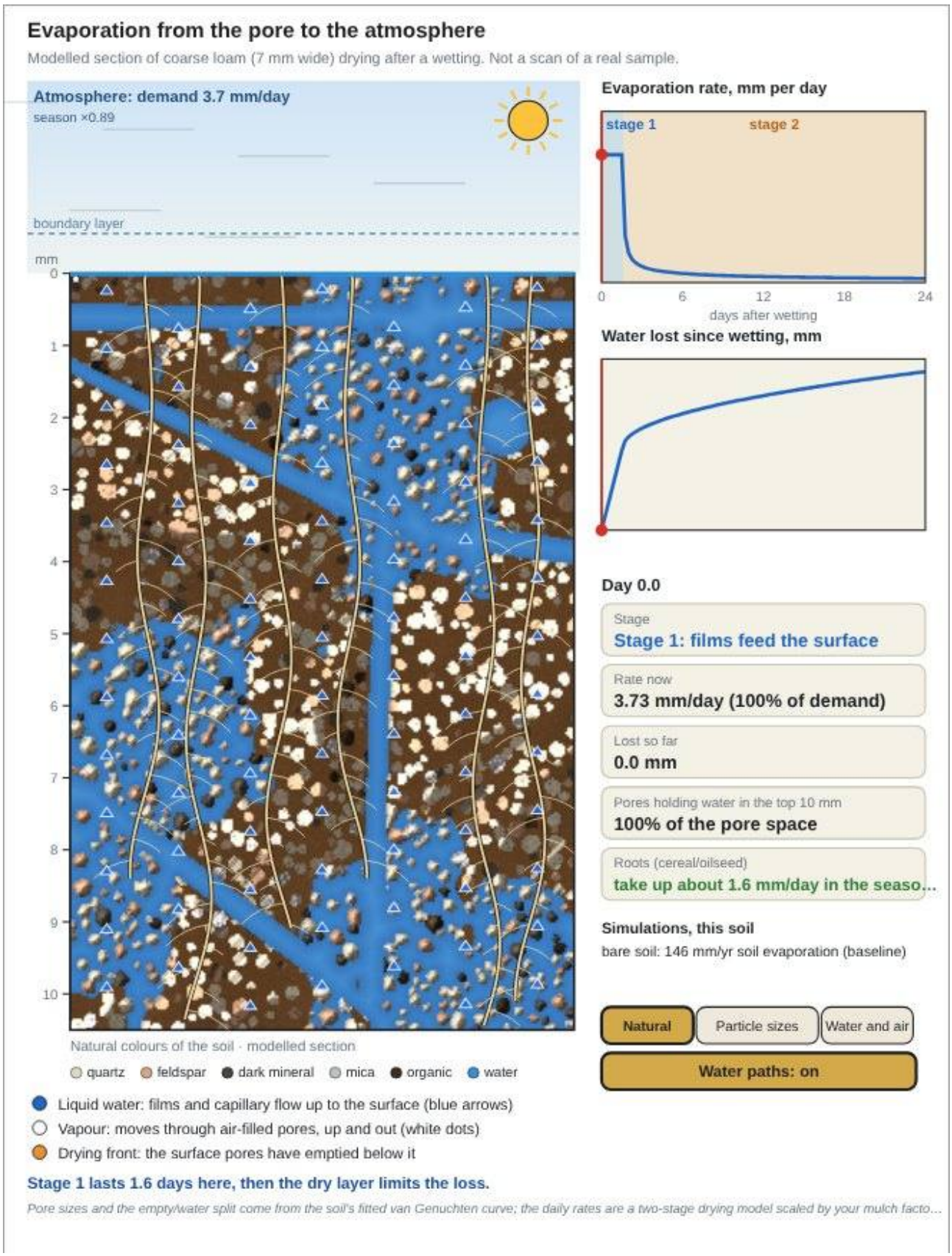


Figure 4.7: Evaporation physics, part 1 of 3.

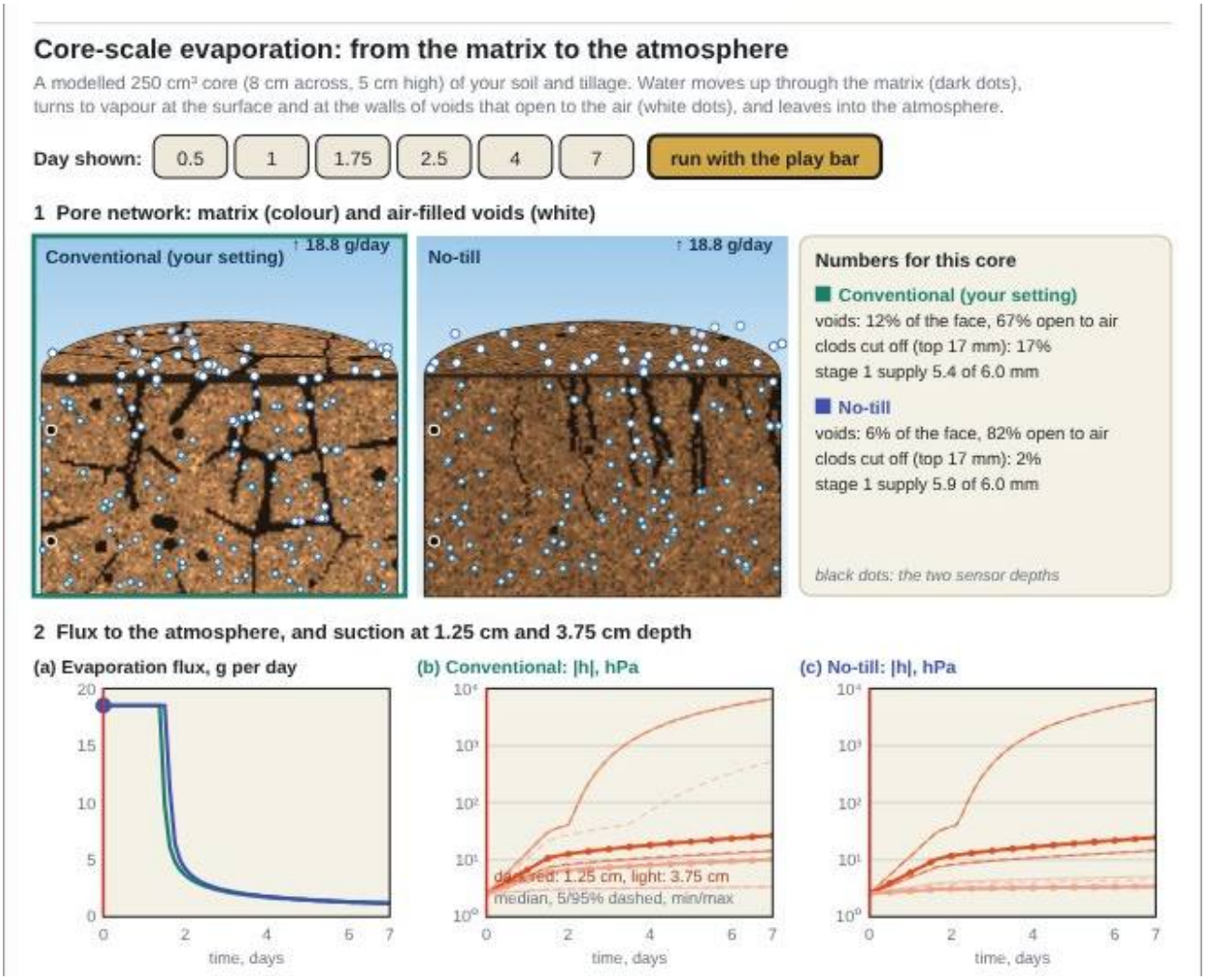


Figure 4.8: Evaporation physics, part 2 of 3.

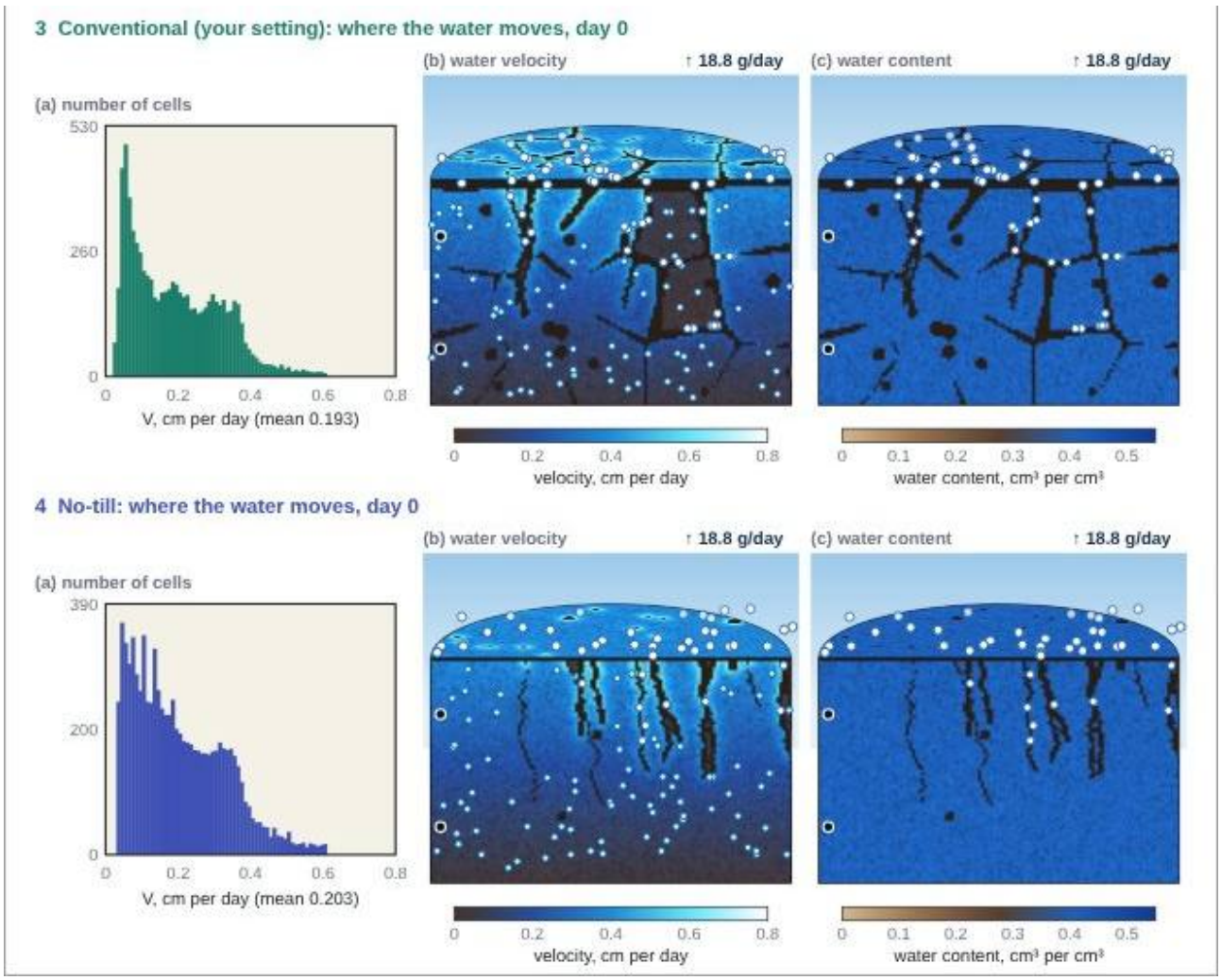


Figure 4.9: Evaporation physics, part 3 of 3.

4.3.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.26 (Volume 1, engine)

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1+\sin(2\pi(\text{doy}-100)/365)]}{0.675}\,\text{clip}(1{-}0.12(\rho{-}1);\,0.85,\,1.12)\sqrt{\gamma_y} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 8.3 (Volume 1, structure)

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4 (Volume 1, structure)

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

Eq. 9.2 (Volume 1, mulch)

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.3 (Volume 1, mulch)

$$Q(t) = q_0 + \begin{cases} e_0\,t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.4 (Volume 1, mulch)

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.5 (Volume 1, mulch)

$$\log_{10} h(z) = \log_{10} h_s + \big(\log_{10} h_i - \log_{10} h_s\big)\big(1 - e^{-(z-\ell)/\lambda}\big) \quad (z \geq \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

4.4 Fate of saved water

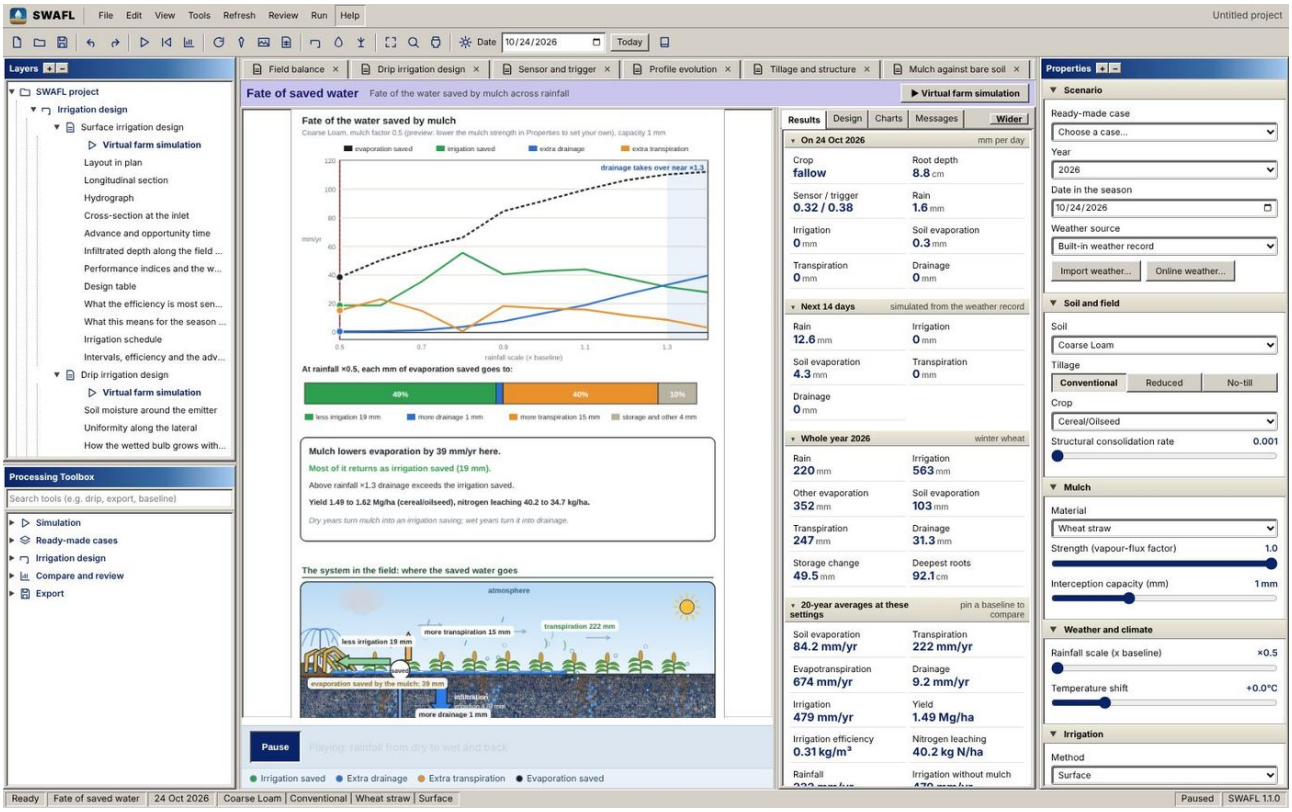


Figure 4.10: Fate of saved water: the window with the layer open.

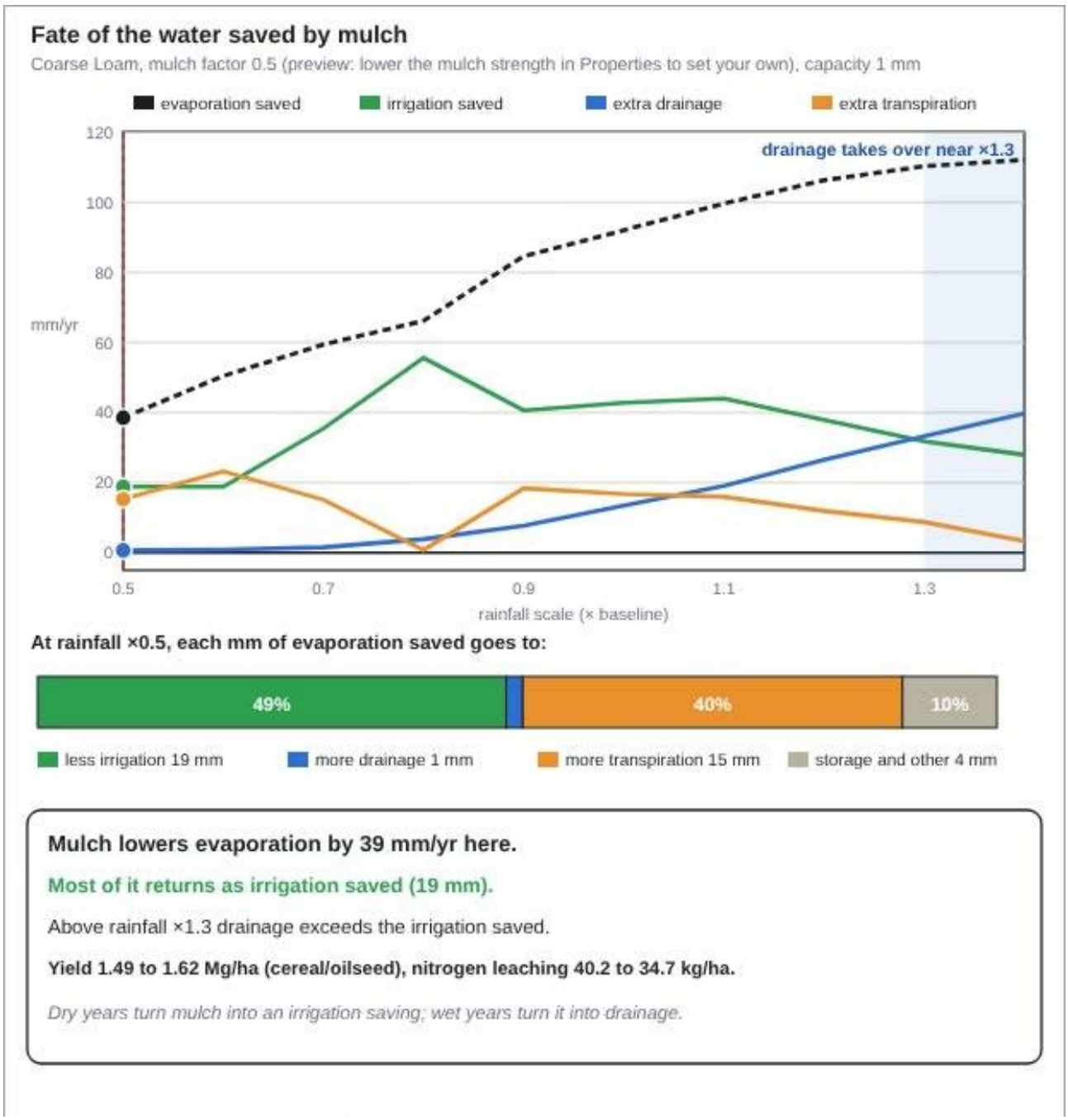


Figure 4.11: Fate of saved water, part 1 of 2.

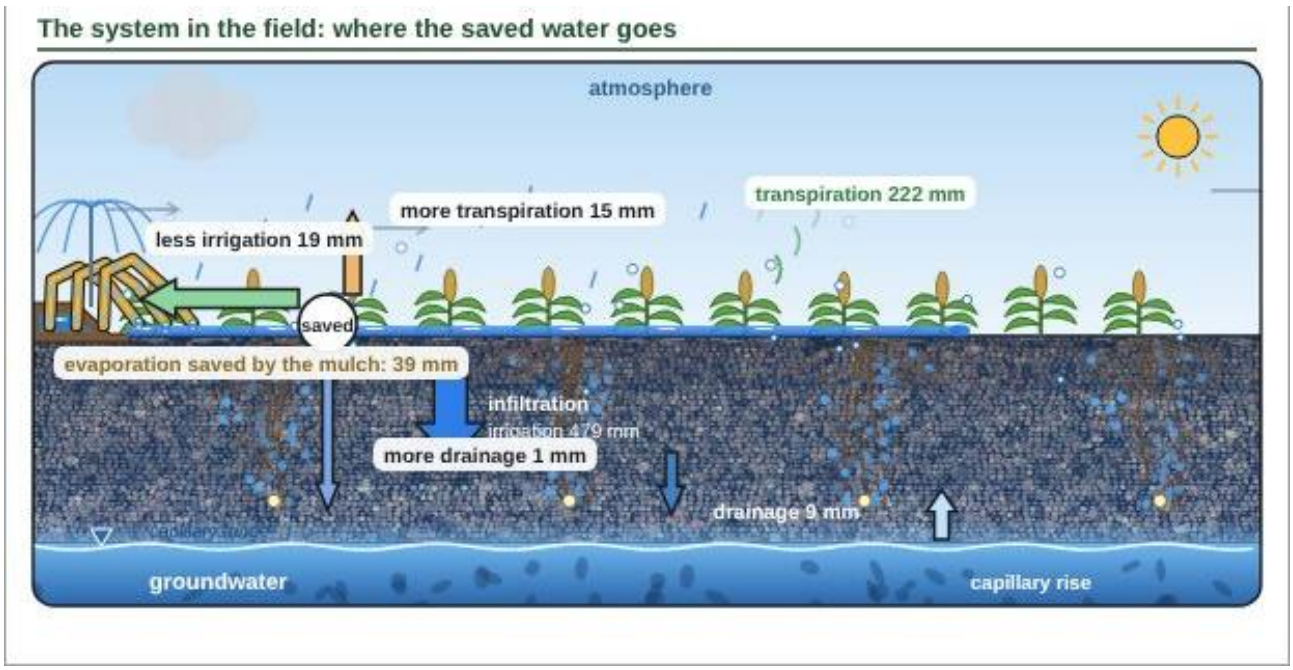


Figure 4.12: Fate of saved water, part 2 of 2.

4.4.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 9.6 (Volume 1, mulch)

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.7 (Volume 1, mulch)

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\; f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\; f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\; f_{\text{rest}} = \max\{0,\, 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

4.5 Regime map

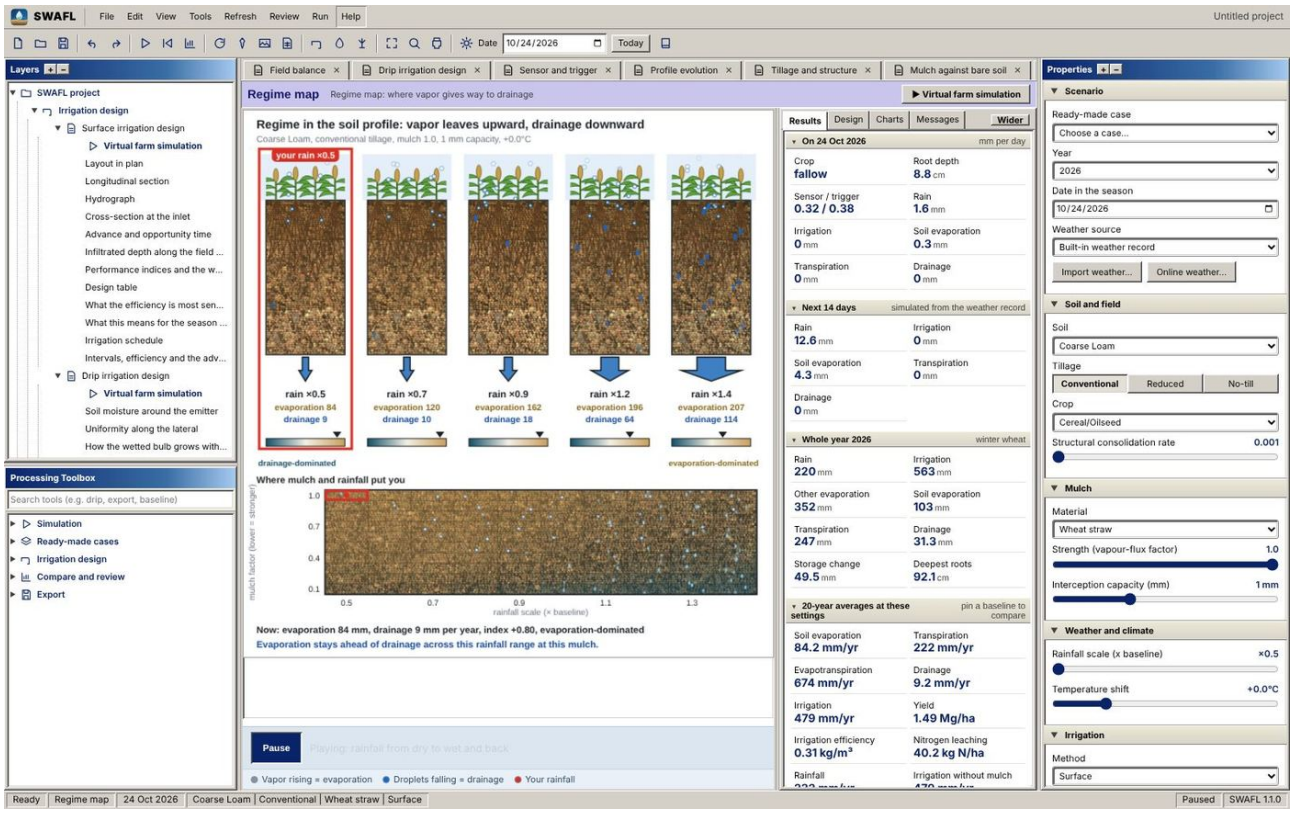


Figure 4.13: Regime map: the window with the layer open.

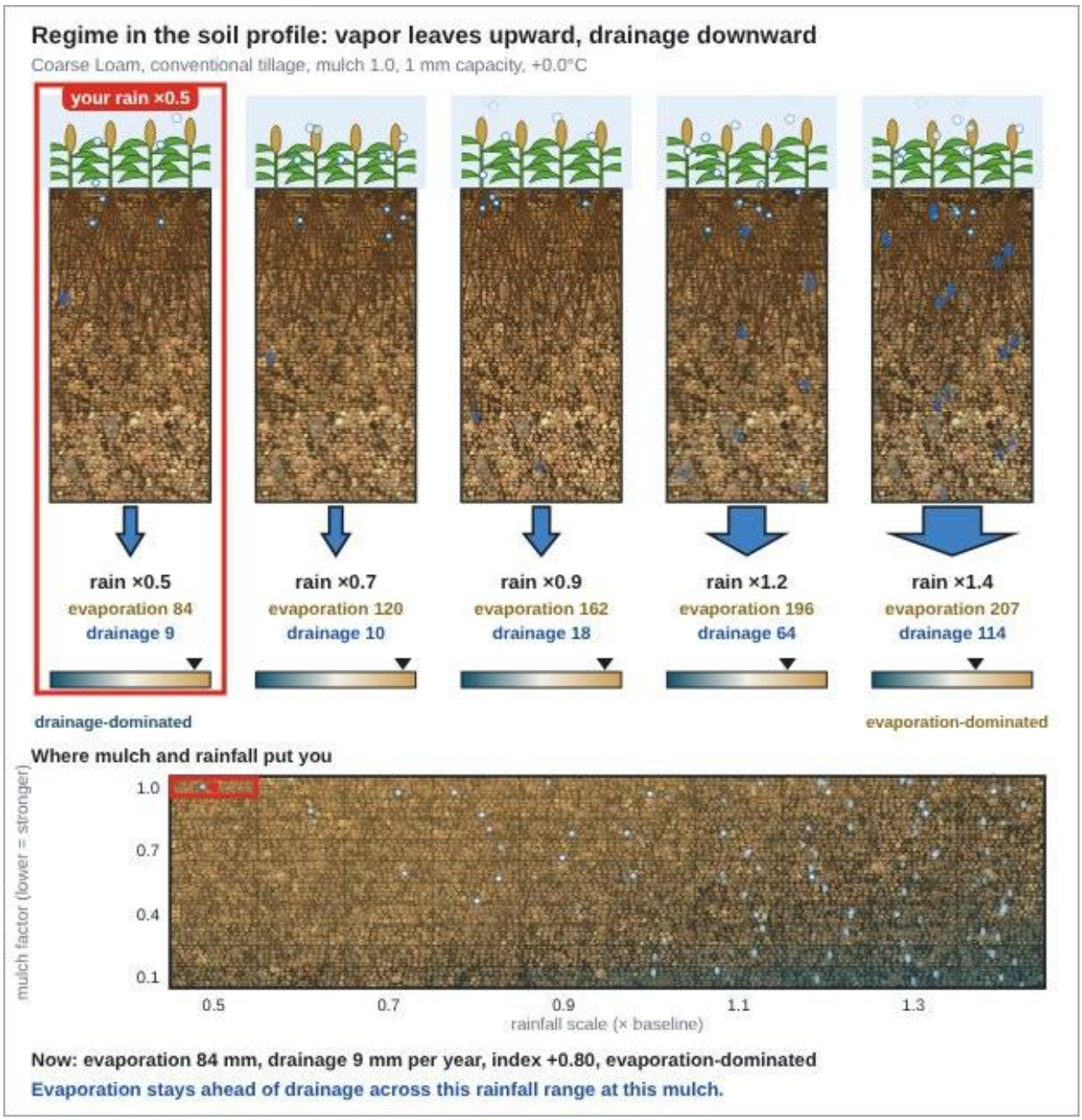


Figure 4.14: Regime map, part 1 of 1.

4.5.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 9.8 (Volume 1, mulch)

$$\mathcal{J} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{J} > +0.05 \\ \text{drainage-dominated} & \mathcal{J} < -0.05 \\ \text{on the boundary} & |\mathcal{J}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

Chapter 5

Soil structure and tillage

5.1 Tillage and structure

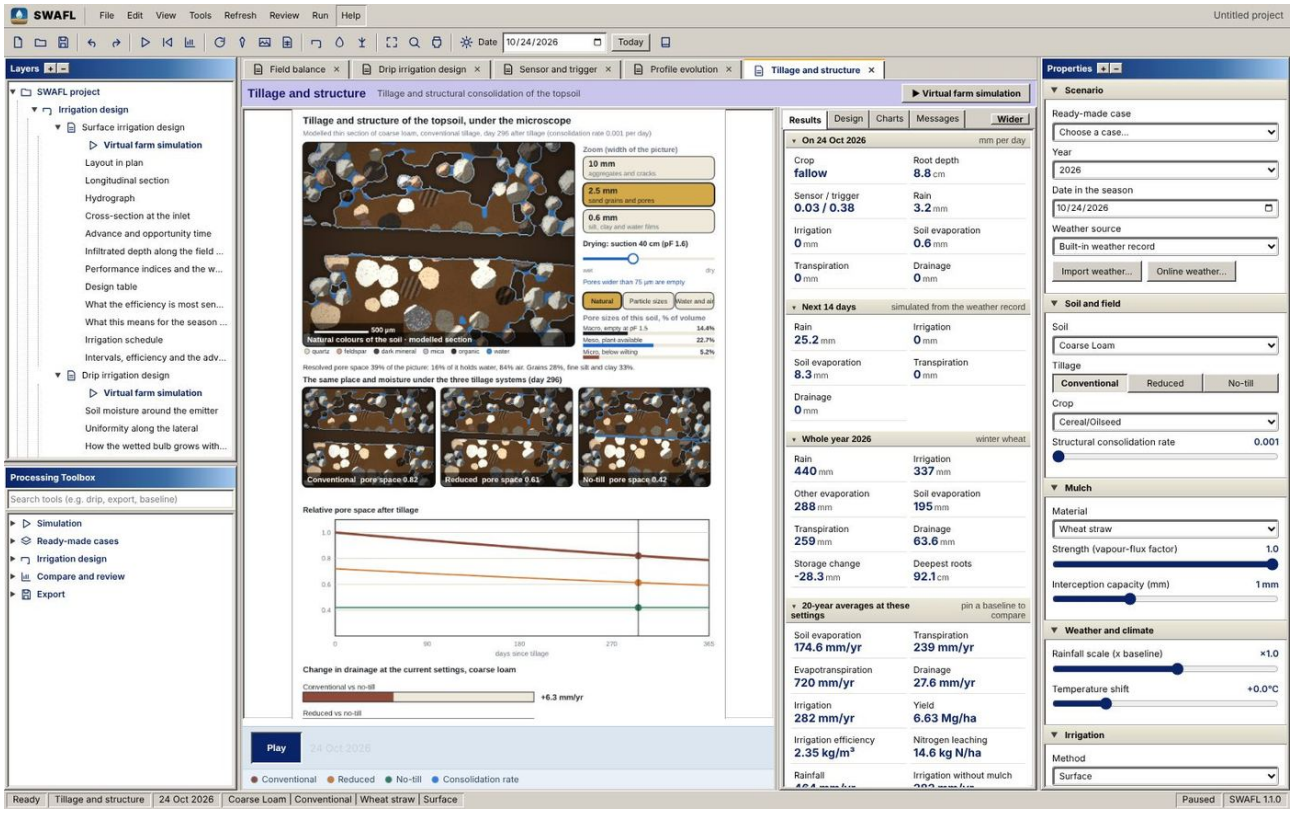


Figure 5.1: Tillage and structure: the window with the layer open.

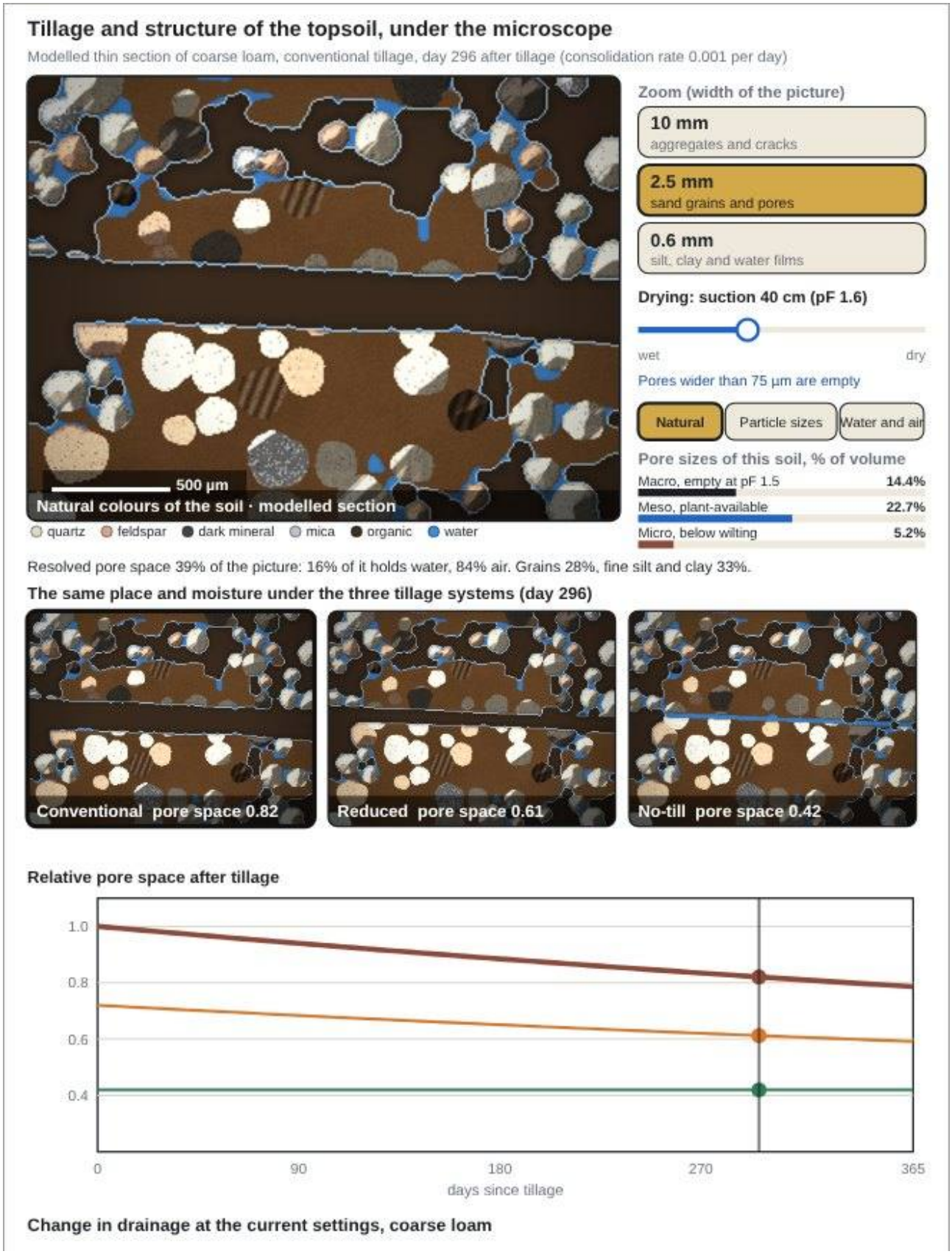


Figure 5.2: Tillage and structure, part 1 of 2.

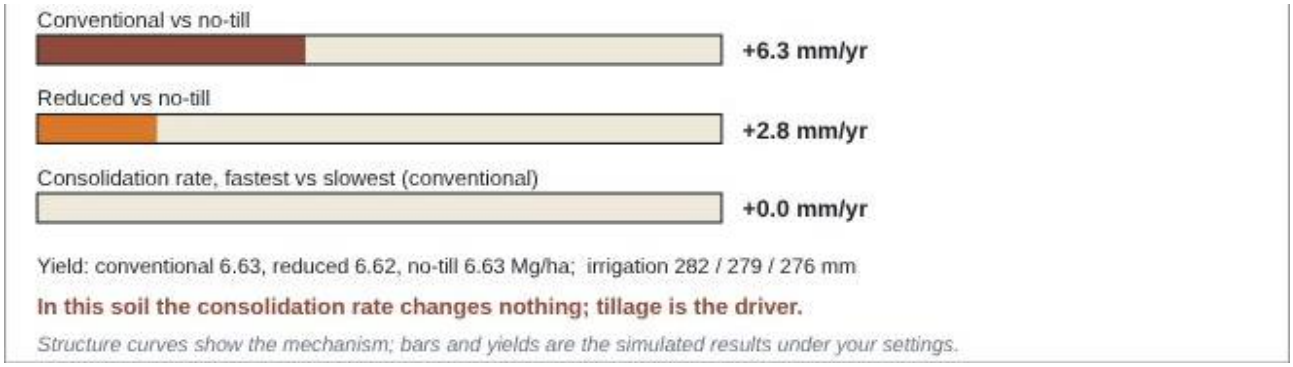


Figure 5.3: Tillage and structure, part 2 of 2.

5.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.1 (Volume 1, engine)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 8.1 (Volume 1, structure)

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.2 (Volume 1, structure)

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

Eq. 8.3 (Volume 1, structure)

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4 (Volume 1, structure)

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

5.2 Soil hydraulic drivers

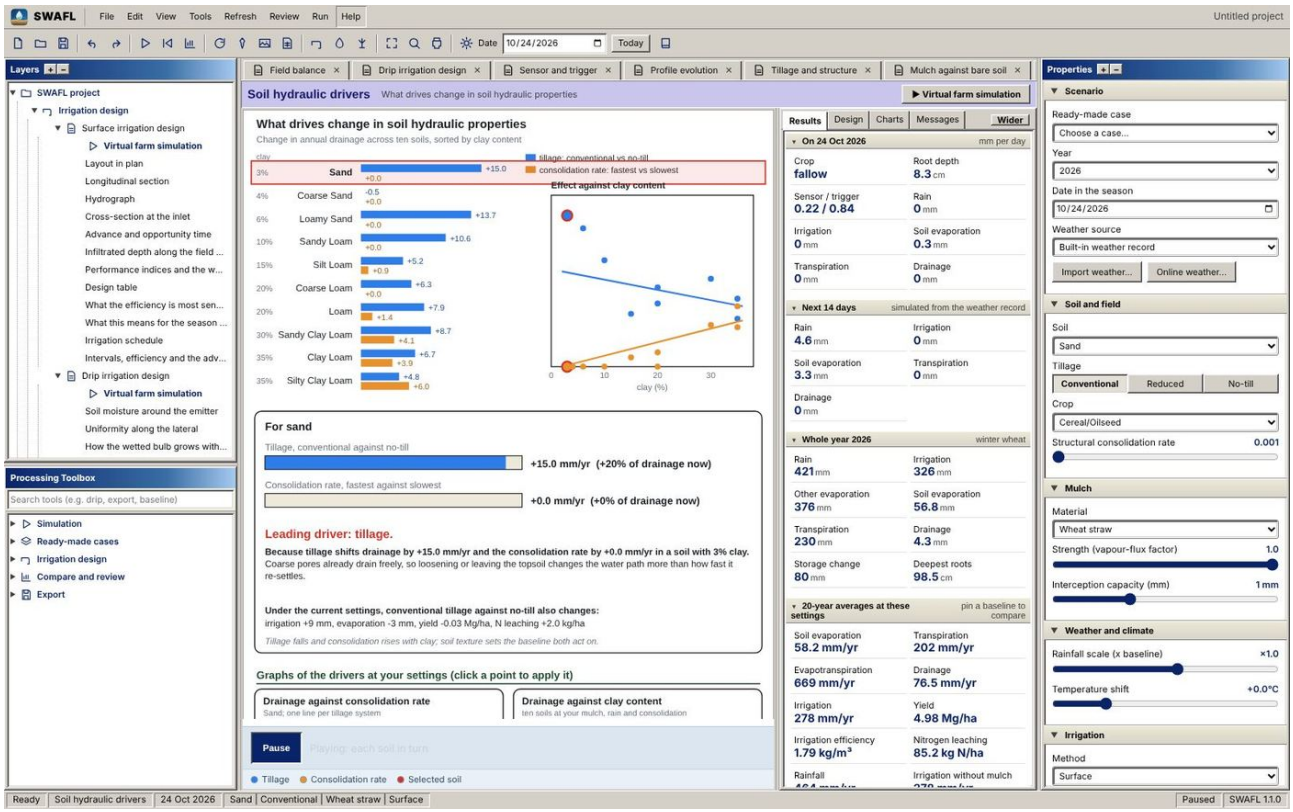


Figure 5.4: Soil hydraulic drivers: the window with the layer open.

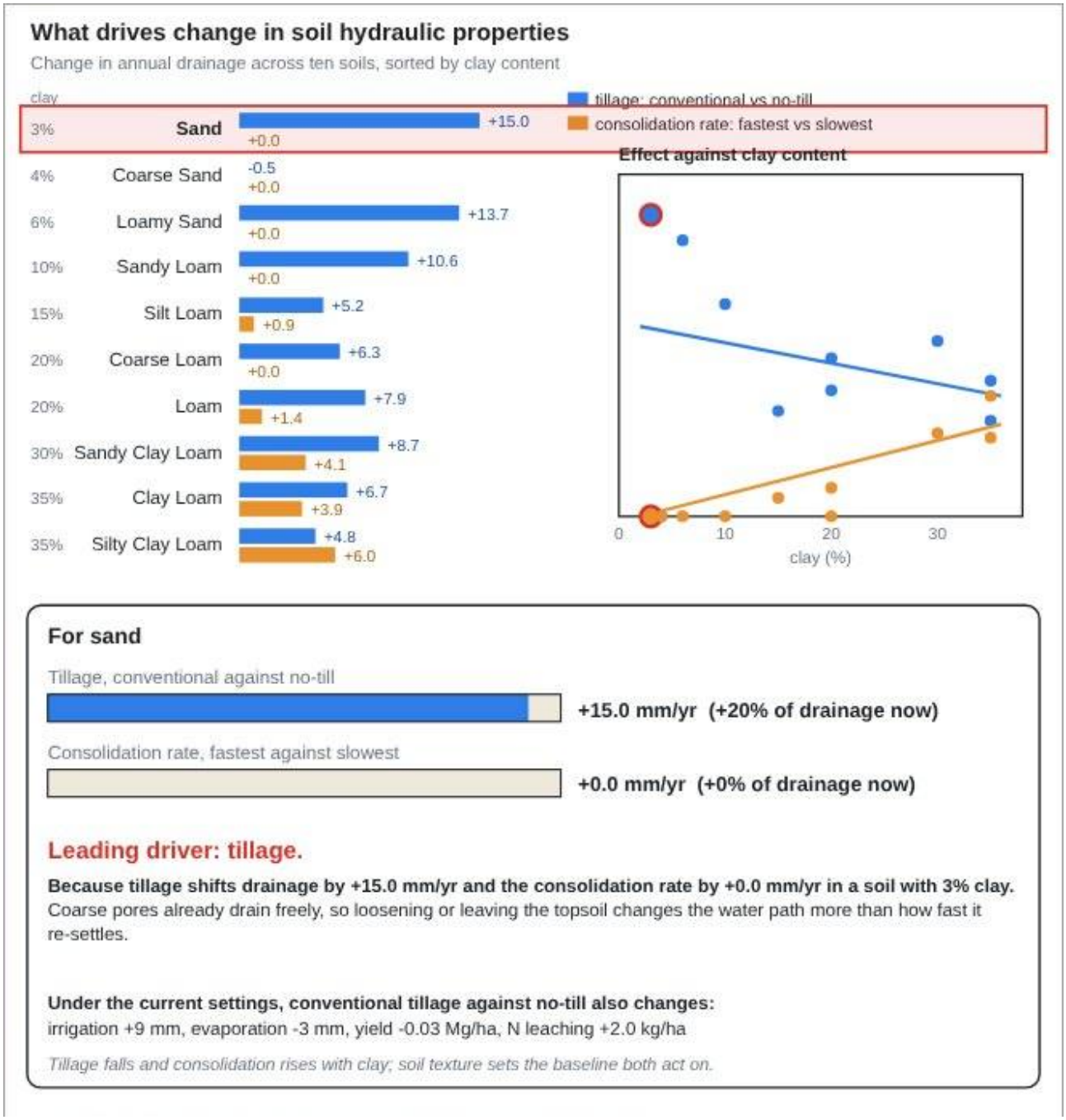


Figure 5.5: Soil hydraulic drivers, part 1 of 3.

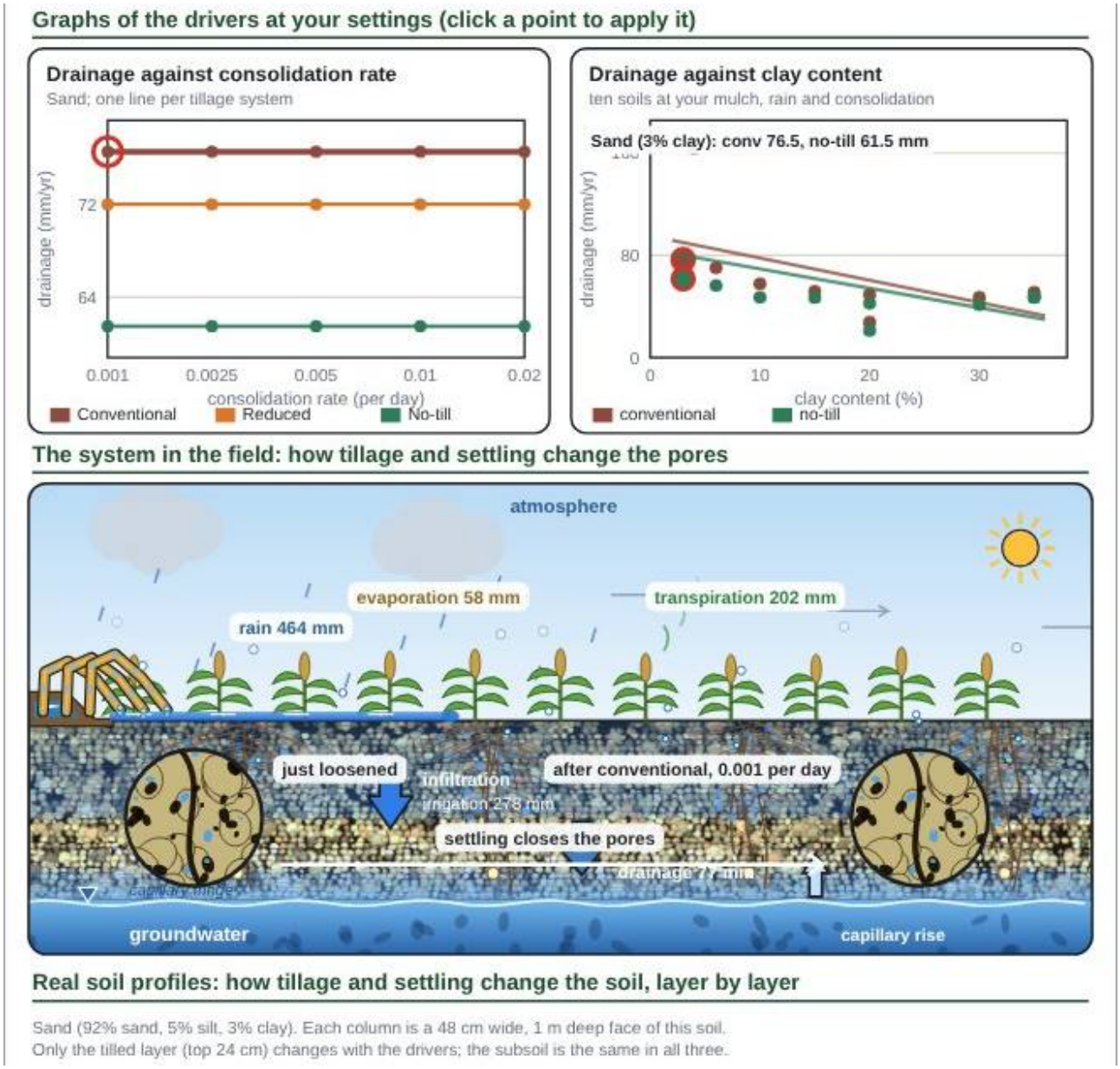


Figure 5.6: Soil hydraulic drivers, part 2 of 3.

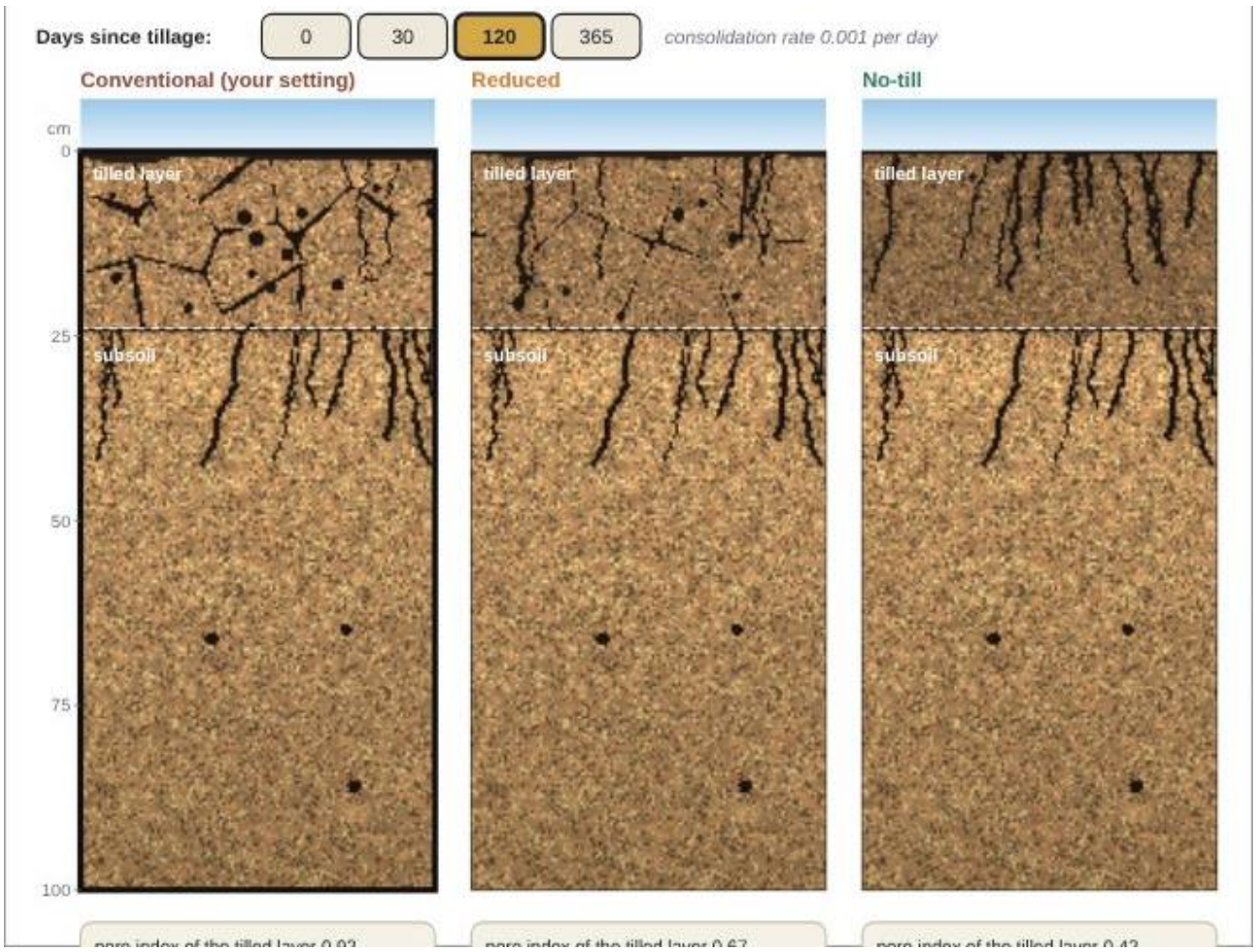


Figure 5.7: Soil hydraulic drivers, part 3 of 3.

5.2.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\,0.05,\,6\Big)\;(Q \in \{E,T,ET,I,Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\; P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\; ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\;\;[\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 8.1 (Volume 1, structure)

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\;(\text{conventional}),\;0.72\;(\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.5 (Volume 1, structure)

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

Eq. 8.6 (Volume 1, structure)

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

5.3 Root structure

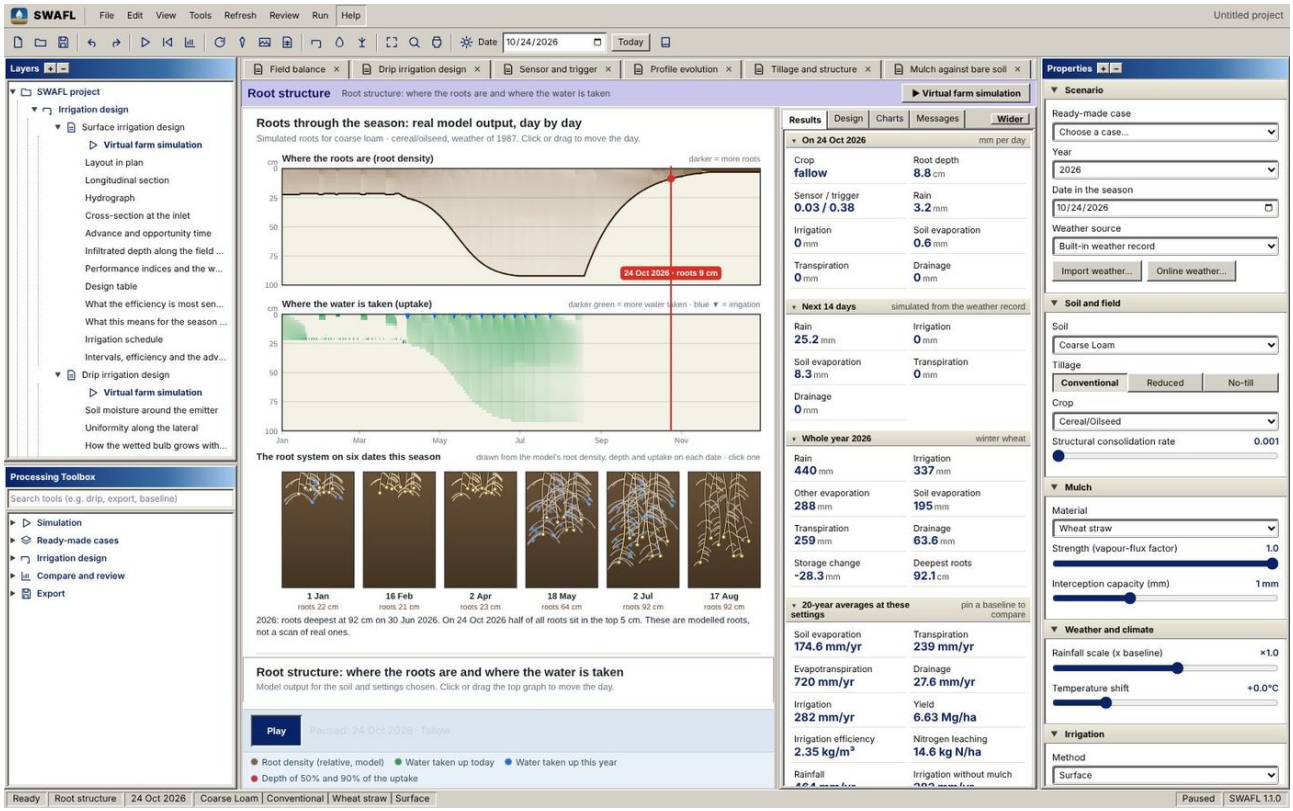


Figure 5.8: Root structure: the window with the layer open.

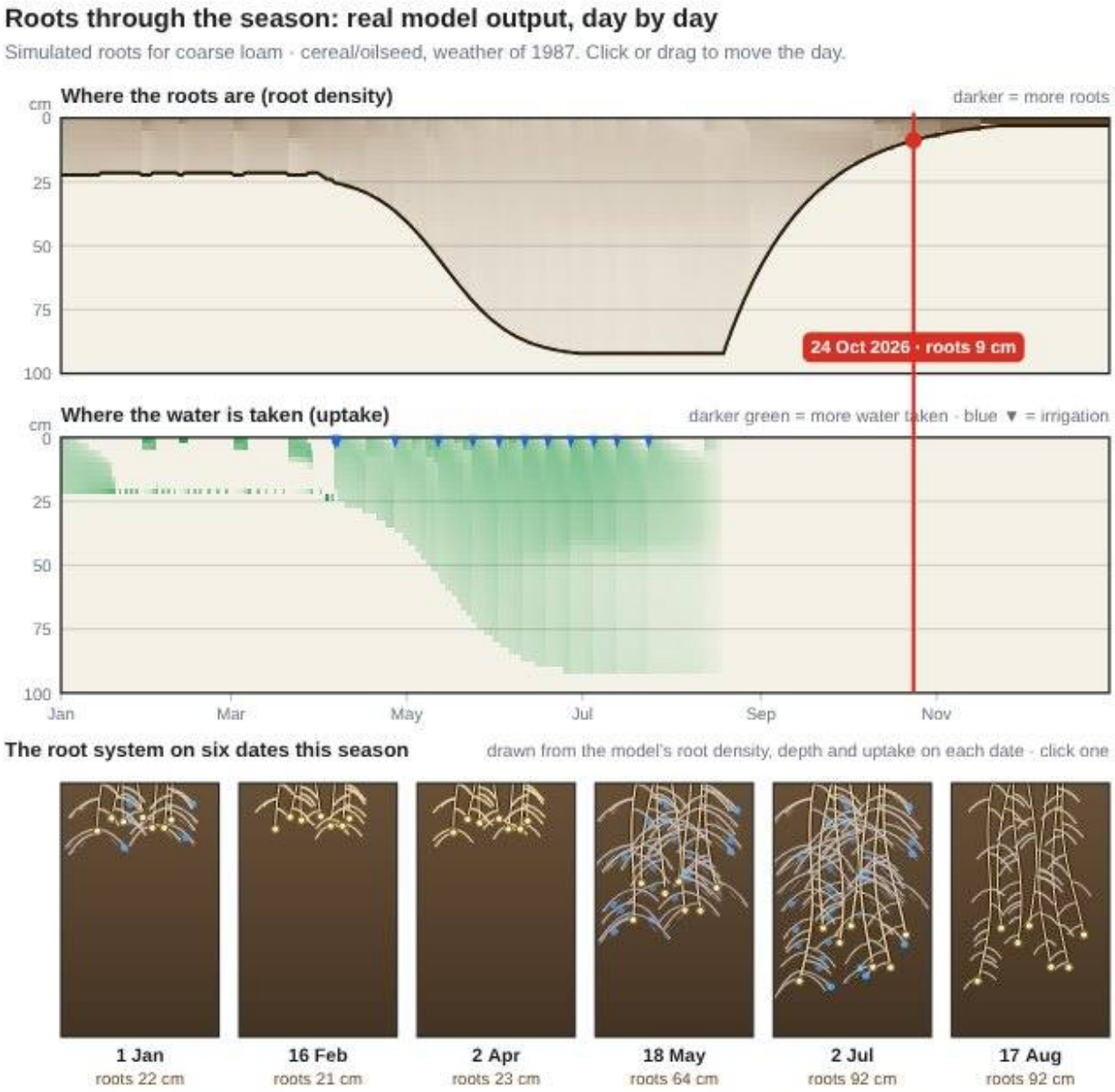


Figure 5.9: Root structure, part 1 of 4.

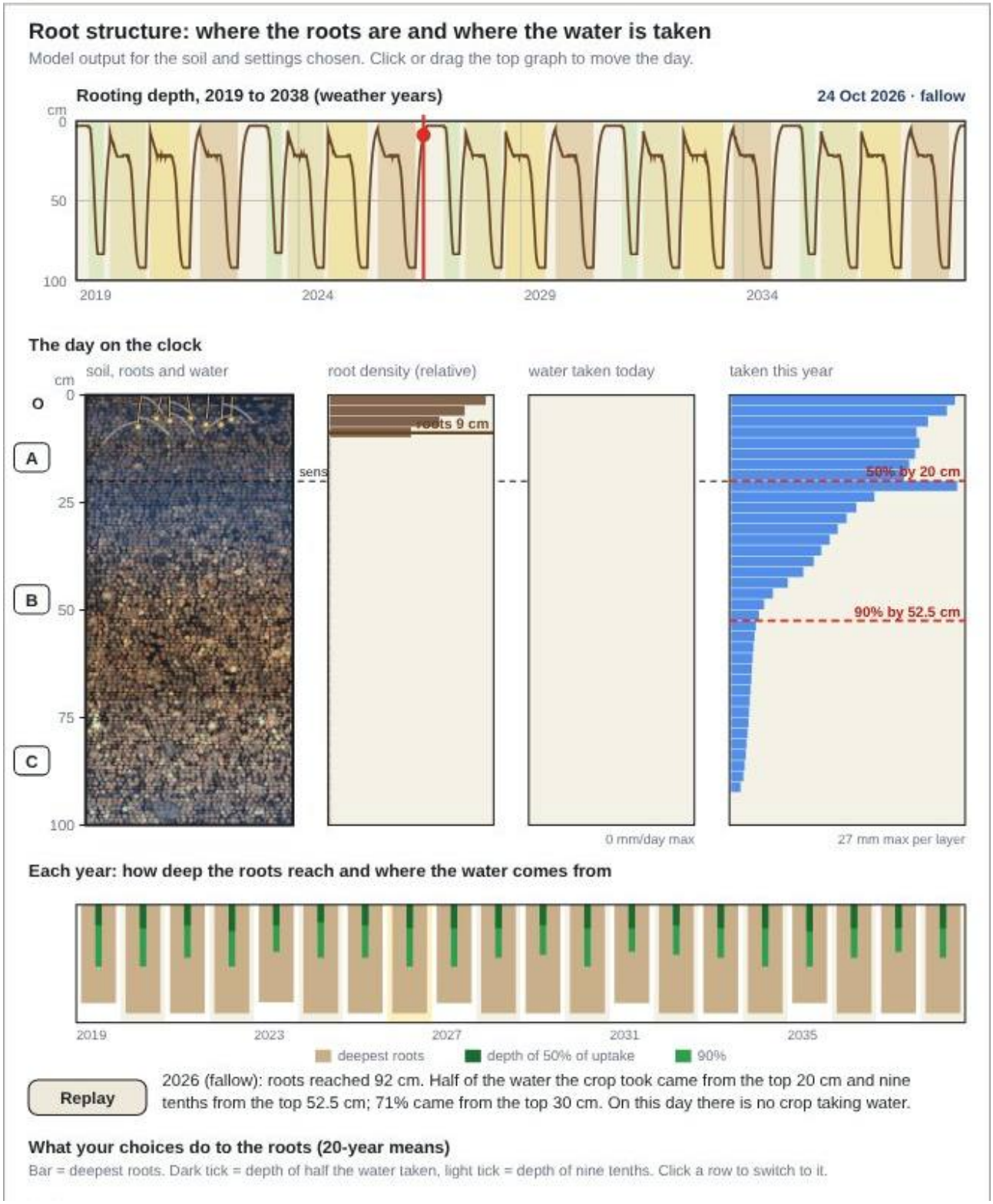


Figure 5.10: Root structure, part 2 of 4.

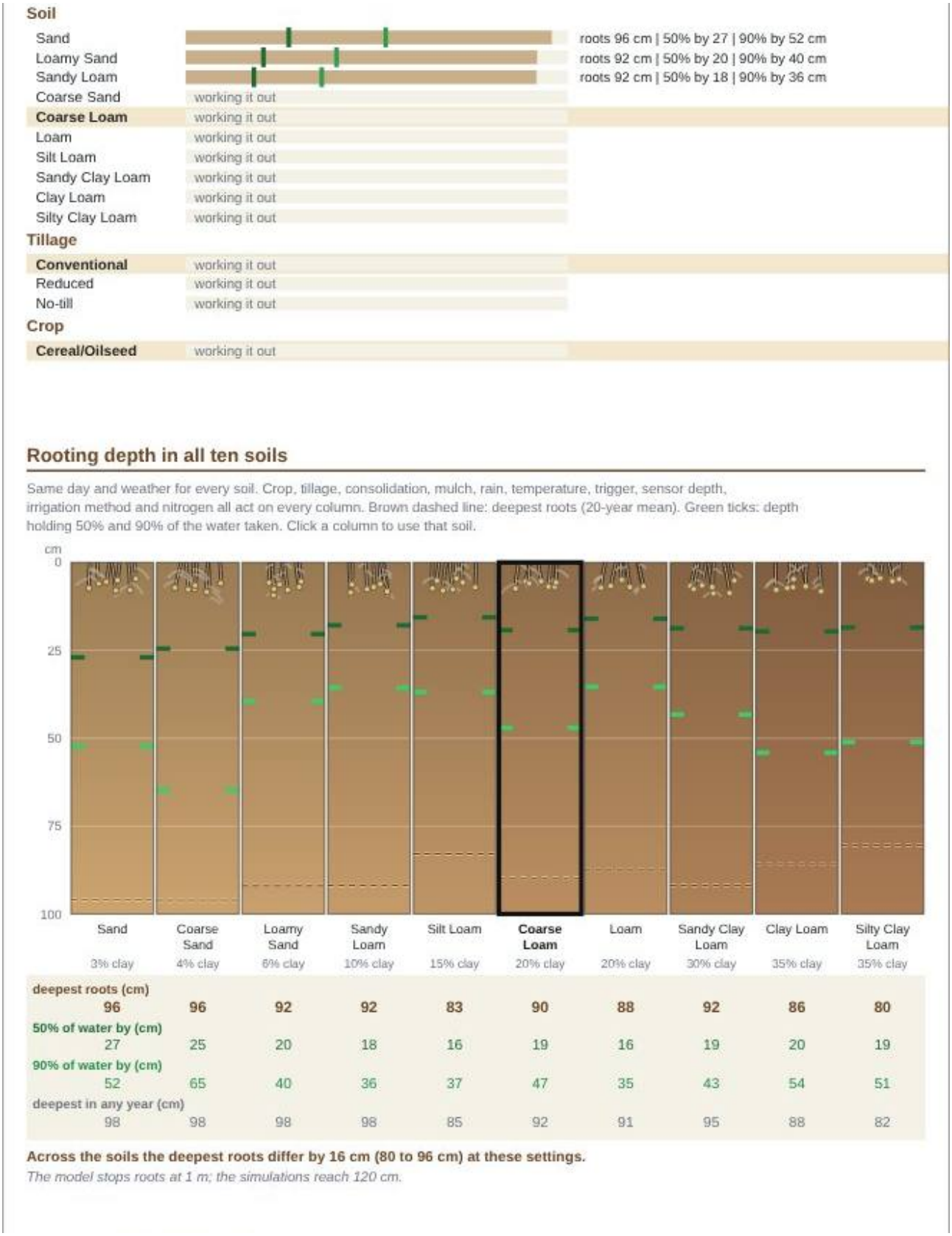


Figure 5.11: Root structure, part 3 of 4.

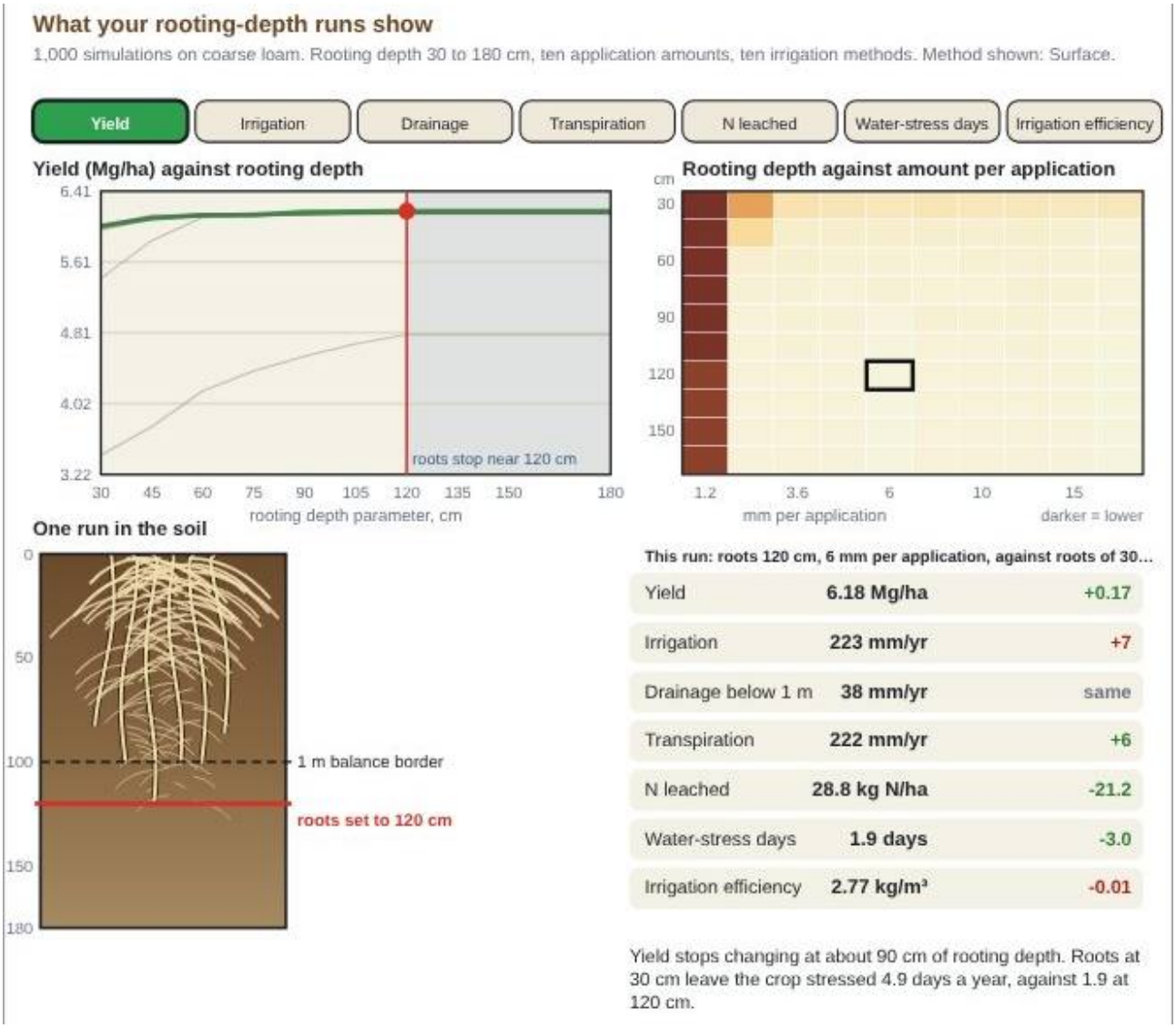


Figure 5.12: Root structure, part 4 of 4.

5.3.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.18 (Volume 1, engine)

$$R_{\max} = \operatorname{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\;25,\;100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19 (Volume 1, engine)

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\operatorname{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35+\bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20 (Volume 1, engine)

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)\,(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21 (Volume 1, engine)

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22 (Volume 1, engine)

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23 (Volume 1, engine)

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \operatorname{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 8.7 (Volume 1, structure)

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k:\ \frac{\sum_{i\le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d\in\text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

Chapter 6

Field water balance

6.1 Field balance

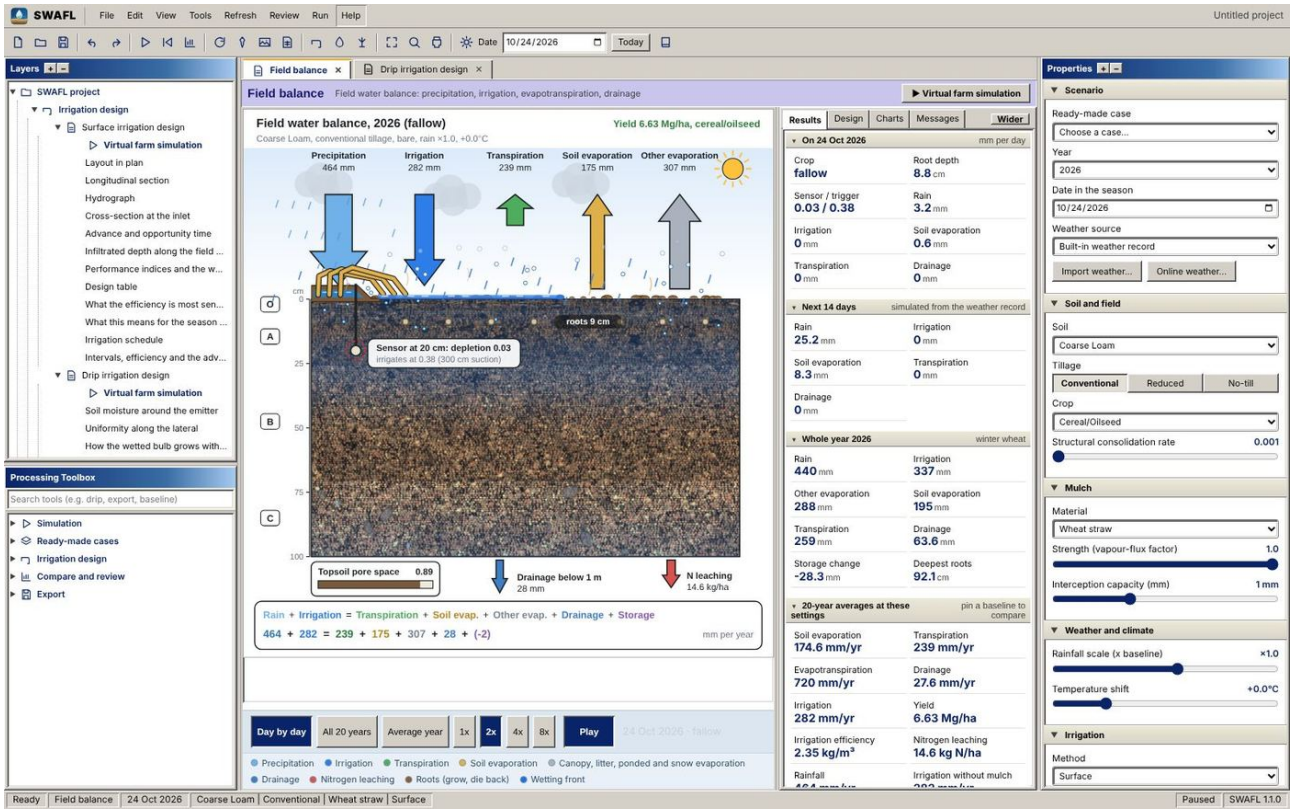


Figure 6.1: Field balance: the window with the layer open.

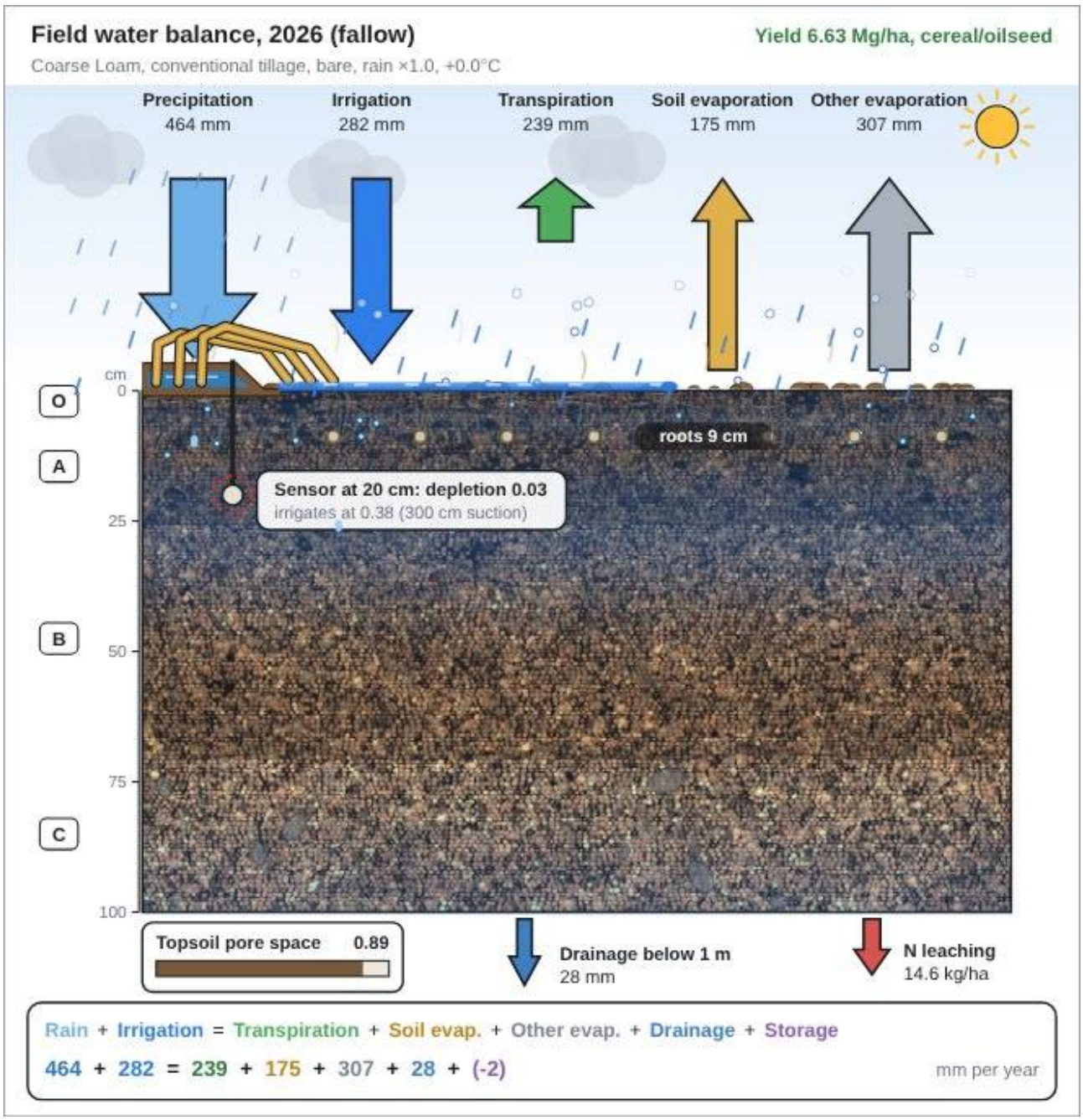


Figure 6.2: Field balance, part 1 of 1.

6.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 5.13 (Volume 1, engine)

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.14 (Volume 1, engine)

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.15 (Volume 1, engine)

$$R = \min\{70,\ \mu_0\, \gamma_y\, (-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.16 (Volume 1, engine)

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17 (Volume 1, engine)

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 5.18 (Volume 1, engine)

$$R_{\max} = \text{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19 (Volume 1, engine)

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\text{clip}\big(\tfrac{c-0.04}{0.96}; 0, 1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20 (Volume 1, engine)

$$\rho_i \propto \exp\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21 (Volume 1, engine)

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22 (Volume 1, engine)

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23 (Volume 1, engine)

$$q_i^T = T_d \, \frac{\rho_i \, a_i}{\sum_j \rho_j a_j}, \qquad a_i = \mathrm{clip}\Big(\frac{s_i - 0.15}{0.6}; \, 0, \, 1 \Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.24 (Volume 1, engine)

$$S_P = \frac{P}{\overline{P_{\mathrm{sim}}}}, \quad S_{\mathrm{thr}} = \frac{P - P_{\mathrm{other}}}{\overline{T_{f,\mathrm{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\mathrm{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\mathrm{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\mathrm{sim}}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

Eq. 5.25 (Volume 1, engine)

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 5.26 (Volume 1, engine)

$$E_{0,\mathrm{bare}} = 4.2 \, (1{+}0.07 \, \Delta T) \, \frac{0.35 + 0.65 \cdot \tfrac{1}{2}[1 + \sin(2\pi(\mathrm{doy} - 100)/365)]}{0.675} \, \mathrm{clip}(1{-}0.12(\rho{-}1); \, 0.85, \, 1.12) \sqrt{\gamma_y} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 8.1 (Volume 1, structure)

$$p(t) = p_c + (p_0 - p_c) \, e^{-\tau_c t}, \qquad p_0 = 1.00 \ (\text{conventional}), \ 0.72 \ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

6.2 Soil, plant and air

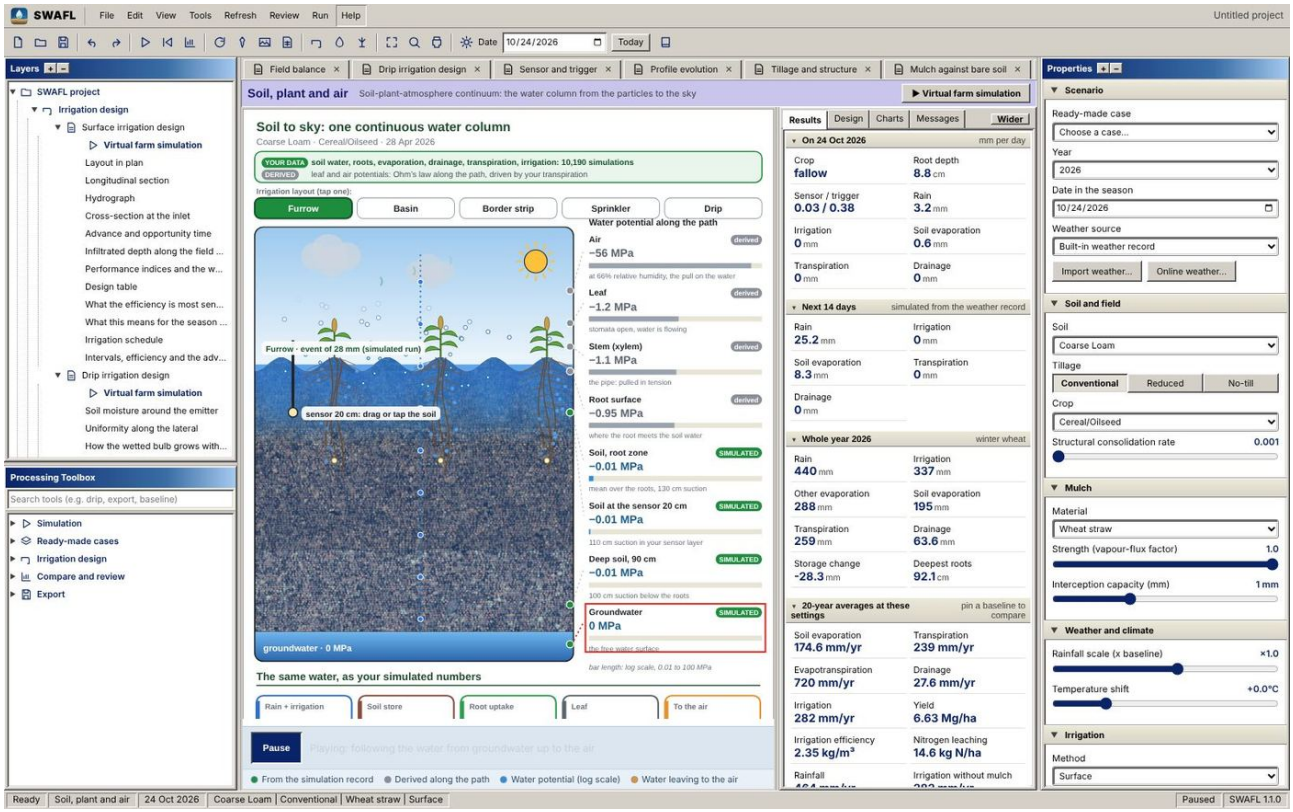


Figure 6.3: Soil, plant and air: the window with the layer open.

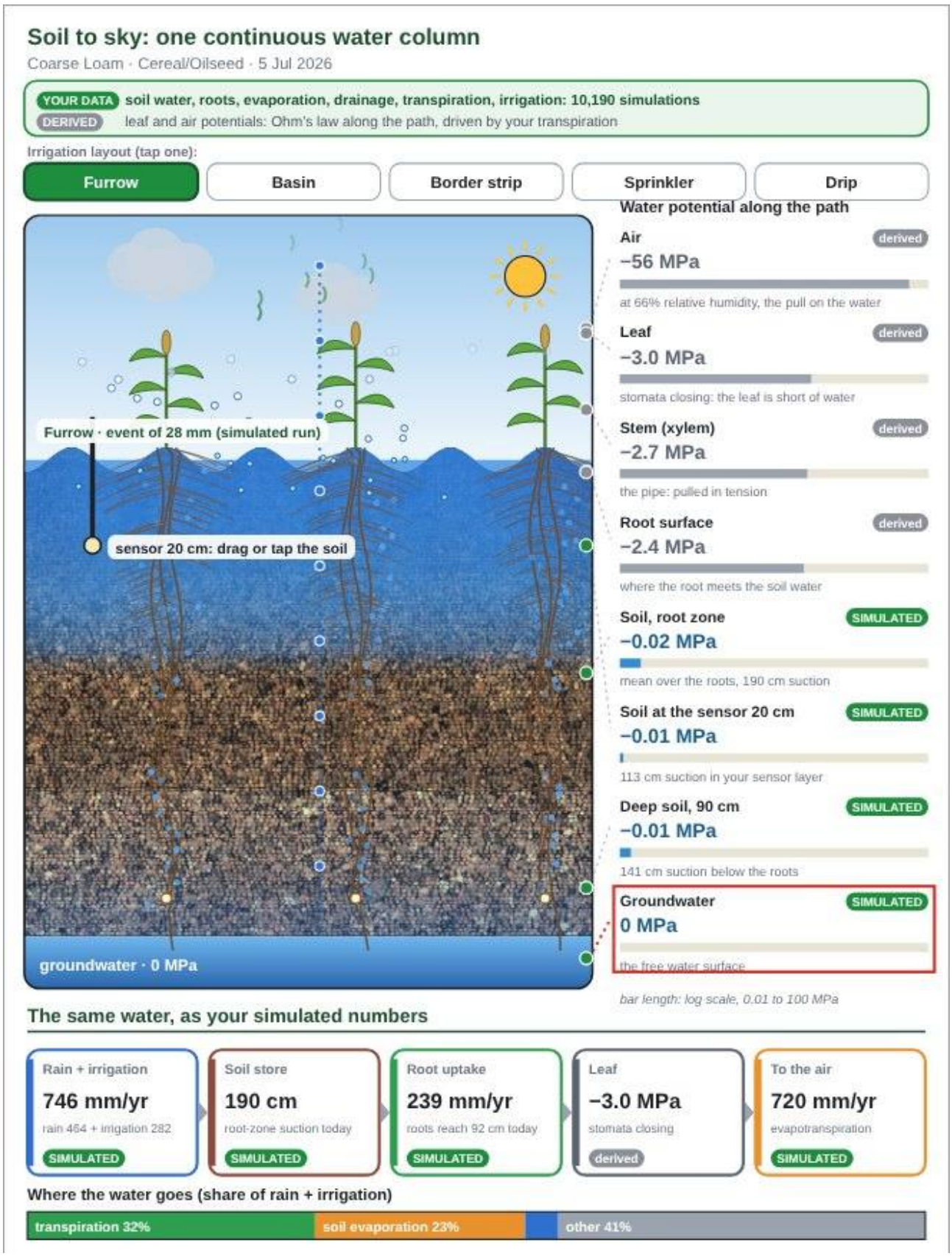


Figure 6.4: Soil, plant and air, part 1 of 3.

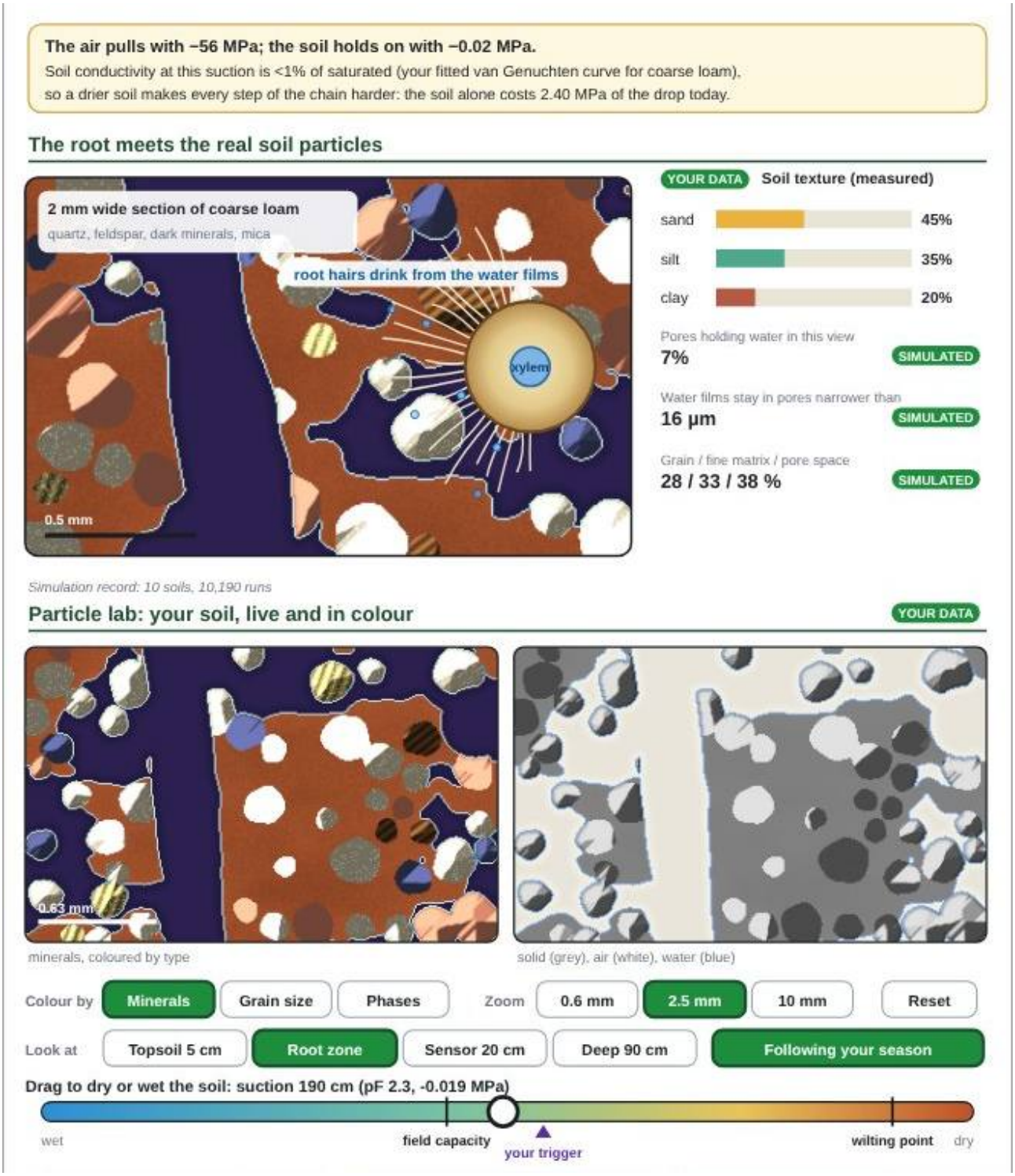


Figure 6.5: Soil, plant and air, part 2 of 3.



Figure 6.6: Soil, plant and air, part 3 of 3.

6.2.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.1 (Volume 1, engine)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 (Volume 1, engine)

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.25 (Volume 1, engine)

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 7.1 (Volume 1, field)

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

Eq. 7.2 (Volume 1, field)

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$
$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$
$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.3 (Volume 1, field)

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$
$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$
$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.4 (Volume 1, field)

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$
$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$
$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.5 (Volume 1, field)

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \;\approx\; 135.5\,\ln \text{RH}\;\;\text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

6.3 Profile evolution

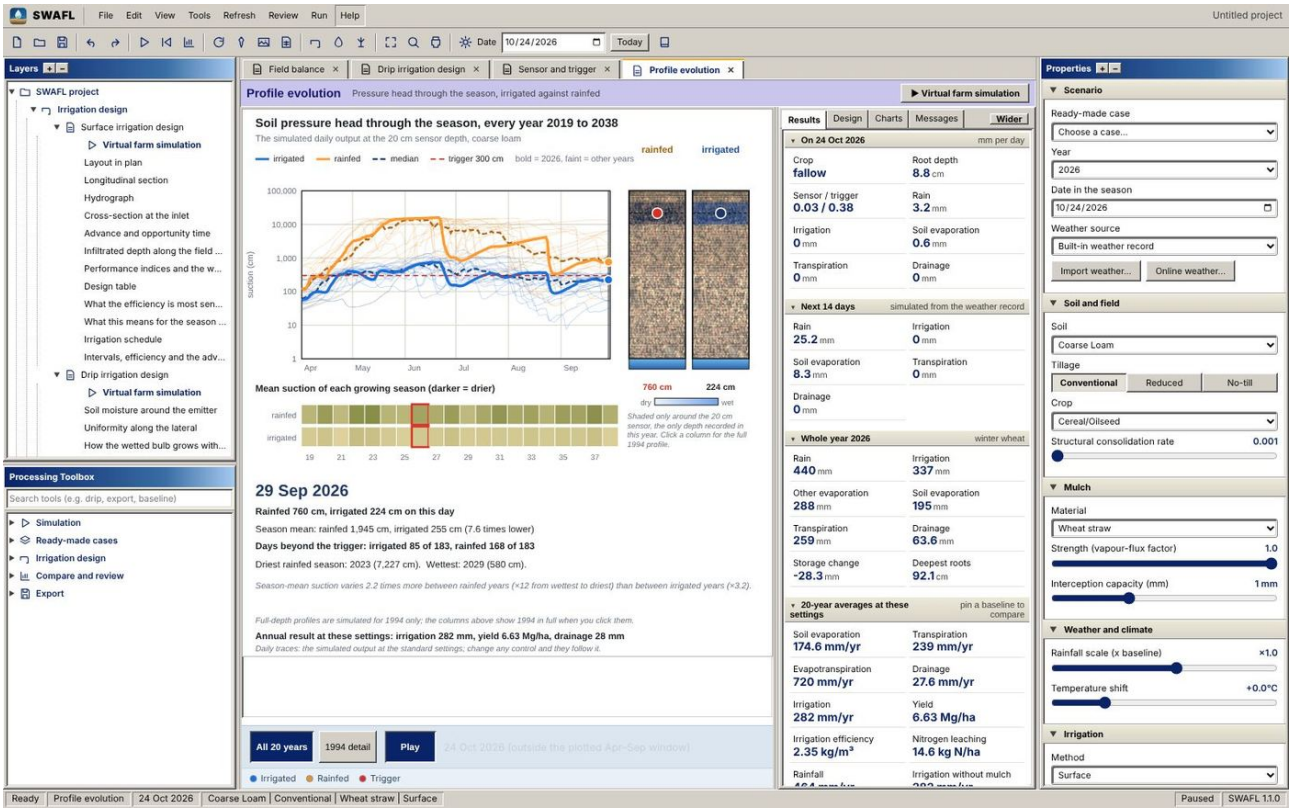


Figure 6.7: Profile evolution: the window with the layer open.

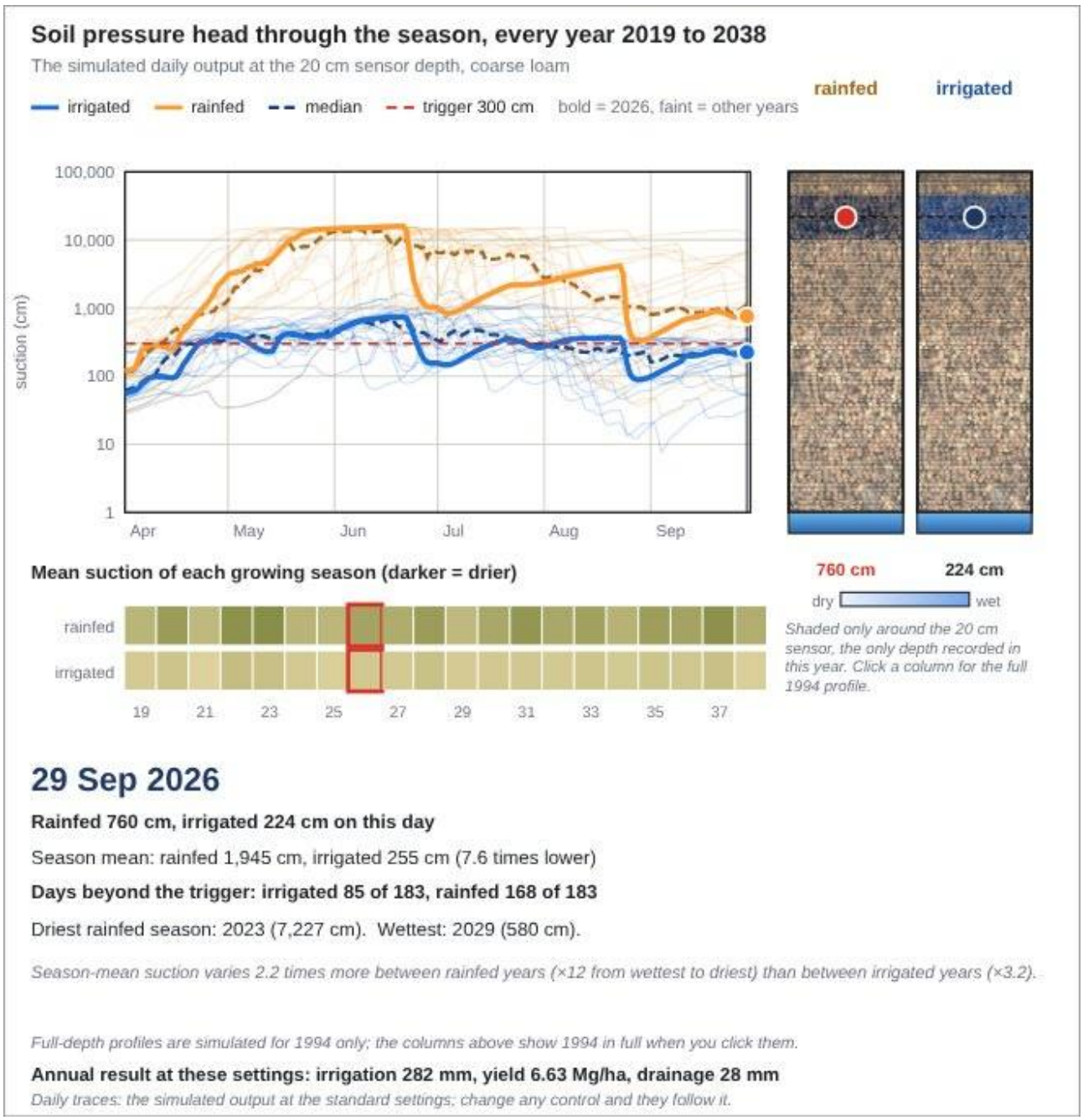


Figure 6.8: Profile evolution, part 1 of 1.

6.3.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.15 (Volume 1, engine)

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17 (Volume 1, engine)

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 7.6 (Volume 1, field)

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

Eq. 6.4a (Volume 1, control)

$$h_d^{\text{x}} = h_{\text{soil}}\Big(f_{\text{ref}}(h_d^{\text{rec}}) + \tfrac{1}{7}\sum_{q=-3}^{3}\big[f_{d+q}^{\text{set}} - f_{d+q}^{\text{std}}\big]\Big), \tag{6.4a}$$

Provenance (Eq. 6.4a): **[Data-driven + Engineered]**: the stored simulated record carries the real weather sequence; the change made by your settings is the difference of two runs of the daily model, converted with the van Genuchten curve of your soil (the inverse of Eq. 5.3, tabulated).

6.4 Field balance: all 20 years

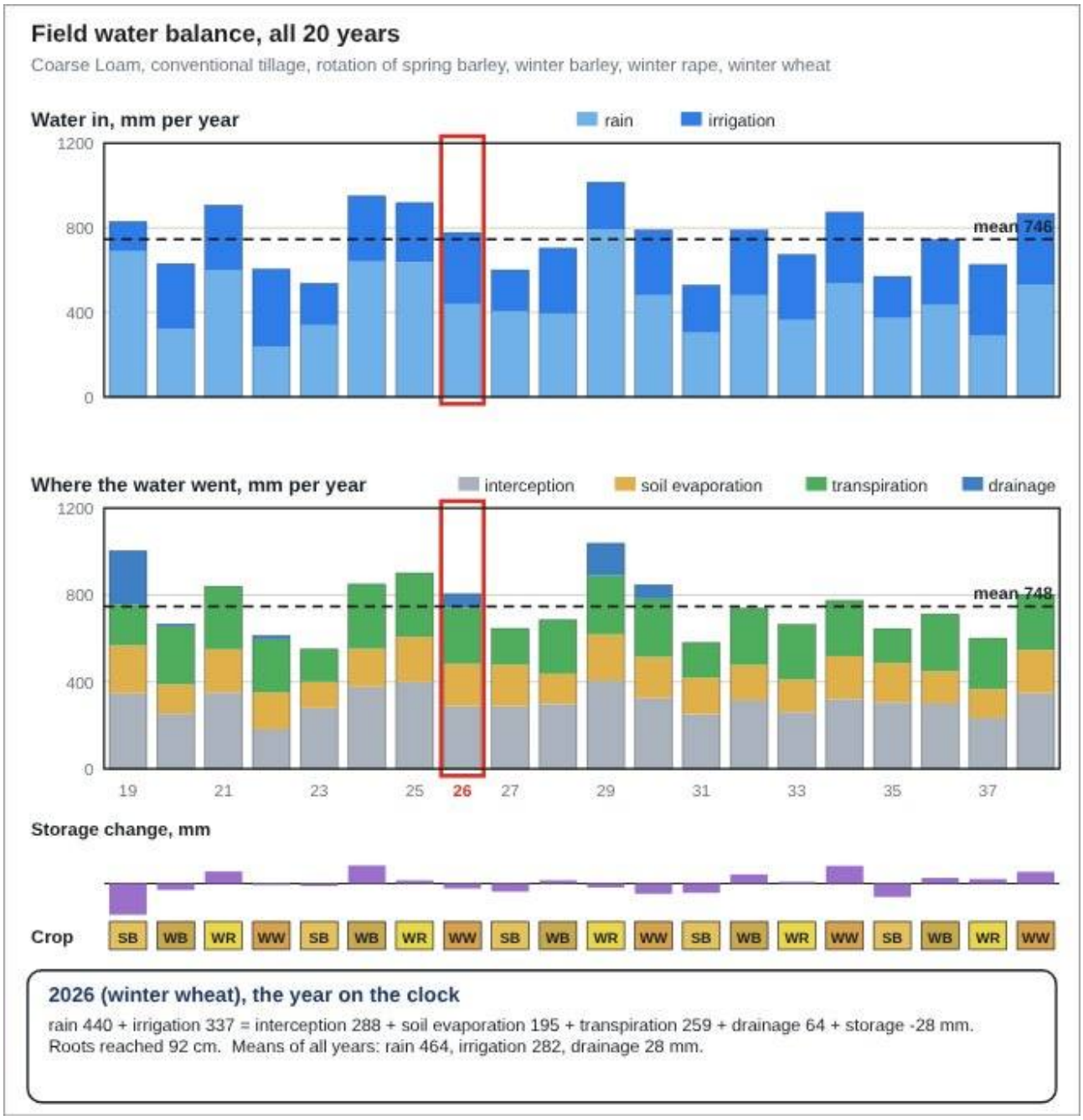


Figure 6.9: Field balance, all 20 years, part 1 of 1.

6.5 Field balance: the average year

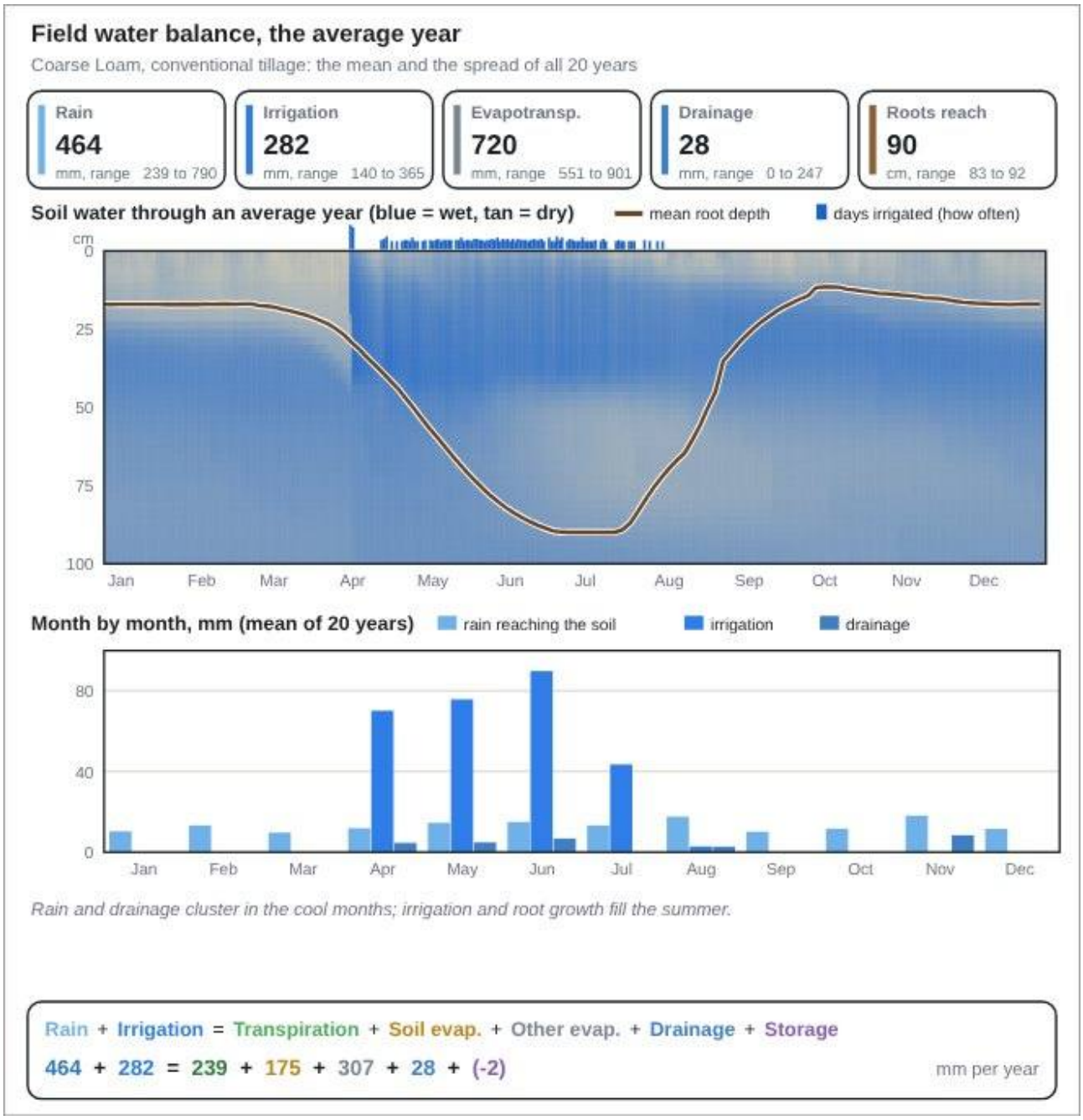


Figure 6.10: Field balance, the average year, part 1 of 1.

Chapter 7

Seasonal data

7.1 Reliability over 20 seasons

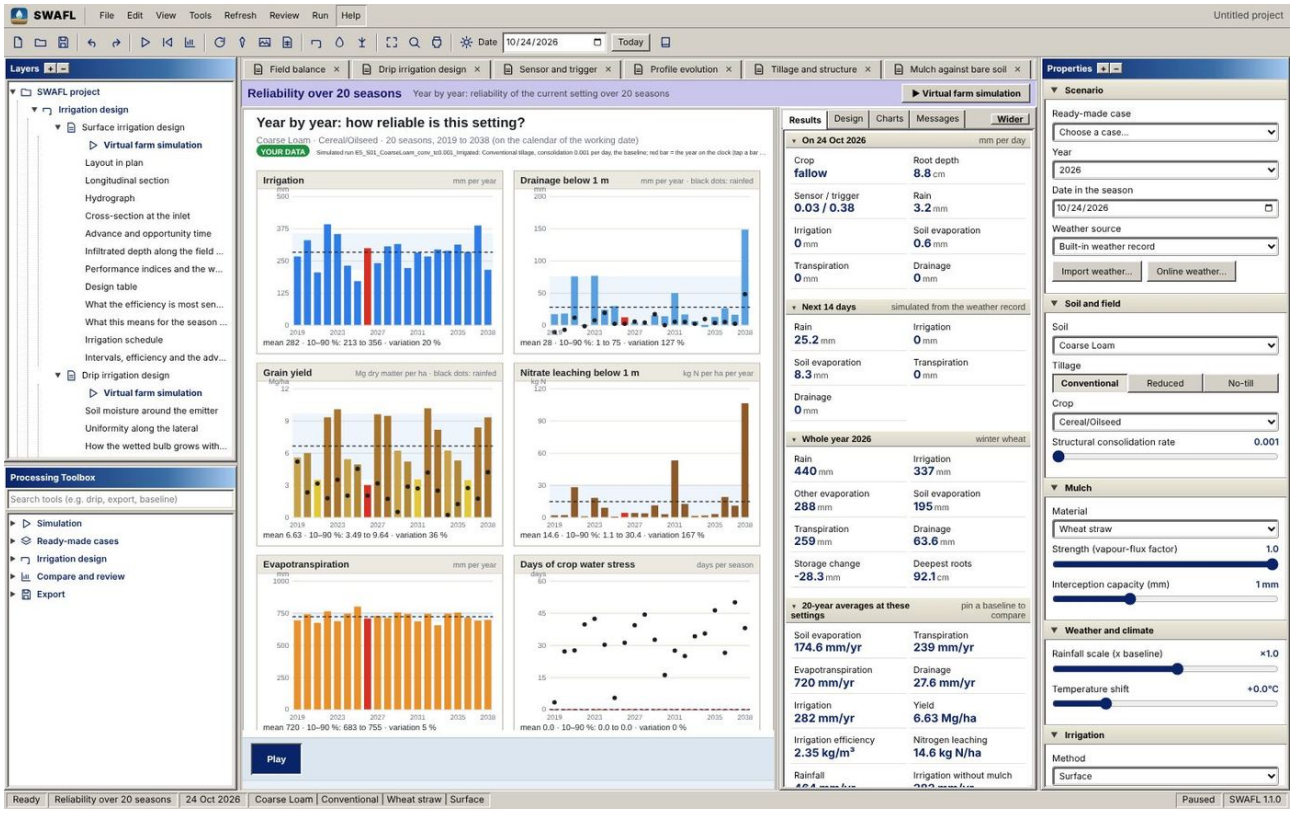


Figure 7.1: Reliability over 20 seasons: the window with the layer open.

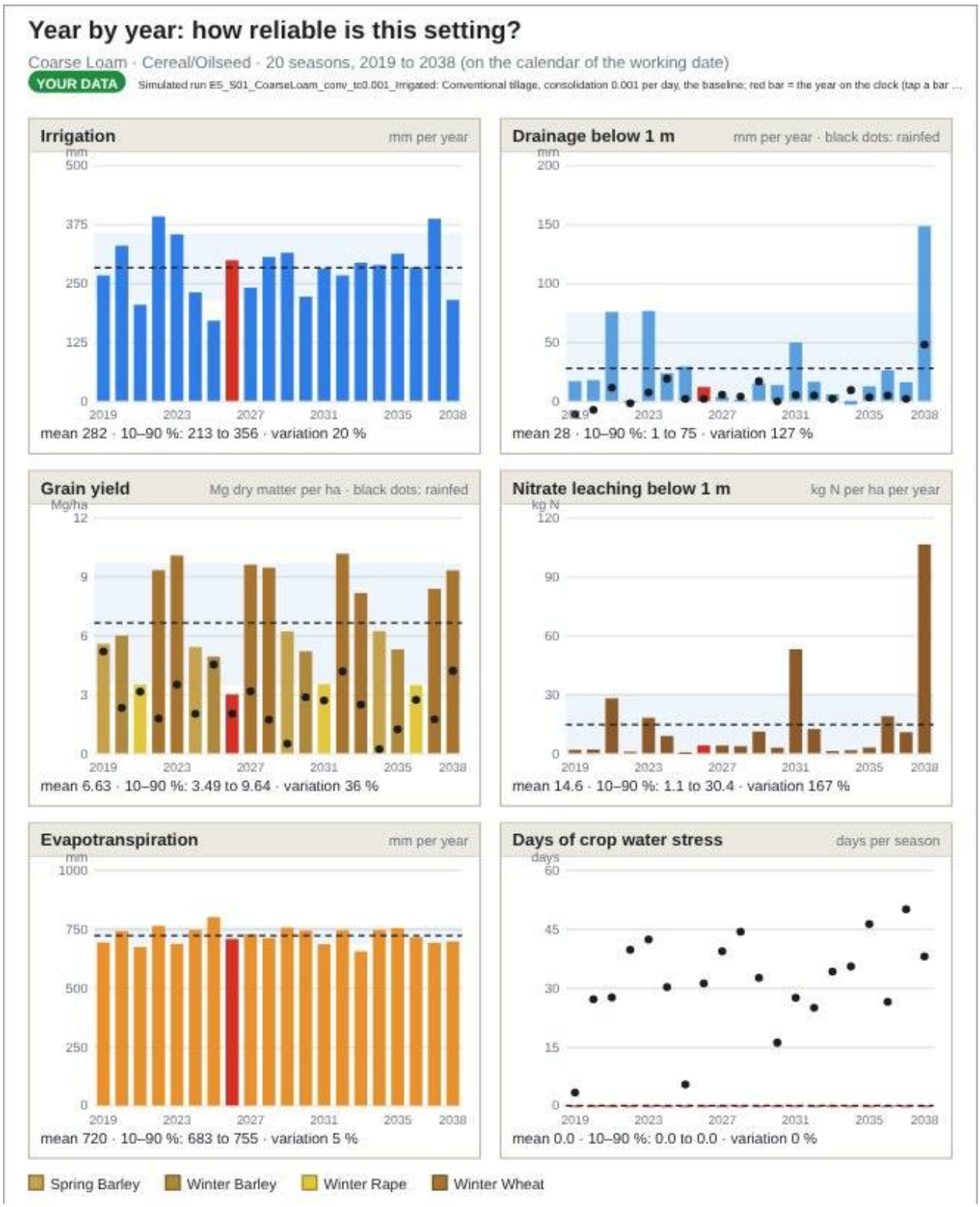


Figure 7.2: Reliability over 20 seasons, part 1 of 2.

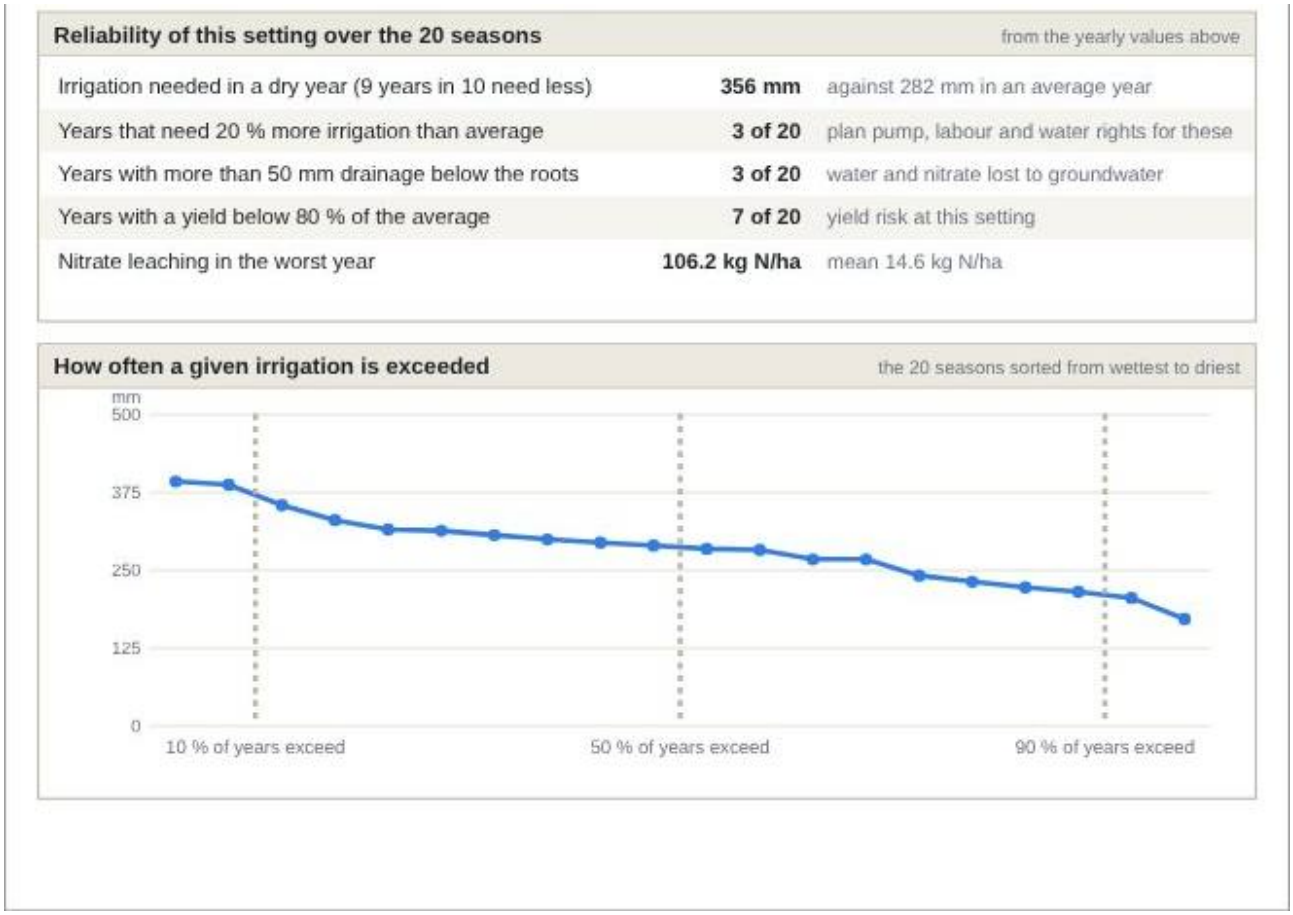


Figure 7.3: Reliability over 20 seasons, part 2 of 2.

7.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.1 (Volume 1, engine)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 (Volume 1, engine)

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 22.1 (Volume 1, seasonal)

$$P'_y = \rho\,P_y, \qquad X'_y = X_y\,\text{clip}\Big(\frac{X^{\text{model}}(\text{your settings})}{X^{\text{model}}(\text{reference})};\ 0,\ 6\Big)\ \ (X \in \{I, T, E, E_{\text{other}}, N_L, Y\}),$$

$$D'_y = \begin{cases} D_y\,\dfrac{D^{\text{model}}}{D^{\text{model}}_{\text{ref}}} & \bar{D} > 5,\ D^{\text{model}}_{\text{ref}} > 5 \\ D_y + \big(D^{\text{model}} - D^{\text{model}}_{\text{ref}}\big) & \text{otherwise} \end{cases}, \qquad ET'_0 = ET_0\,(1 + 0.035\,\Delta T), \tag{22.1}$$

Provenance (Eq. 22.1): **[Data-driven]**: the yearly values of the simulations, moved by the ratio (or, for small drainage, the difference) of Engine 1 between your settings and the reference of the run (your mulch, rain ×1.0, no warming, nitrogen 100 %); for a library crop the yield is the relative yield of Eq. (21.5) year by year and the irrigation follows the ratio of the two crop balances of Section 21.4.

7.2 Full water balance

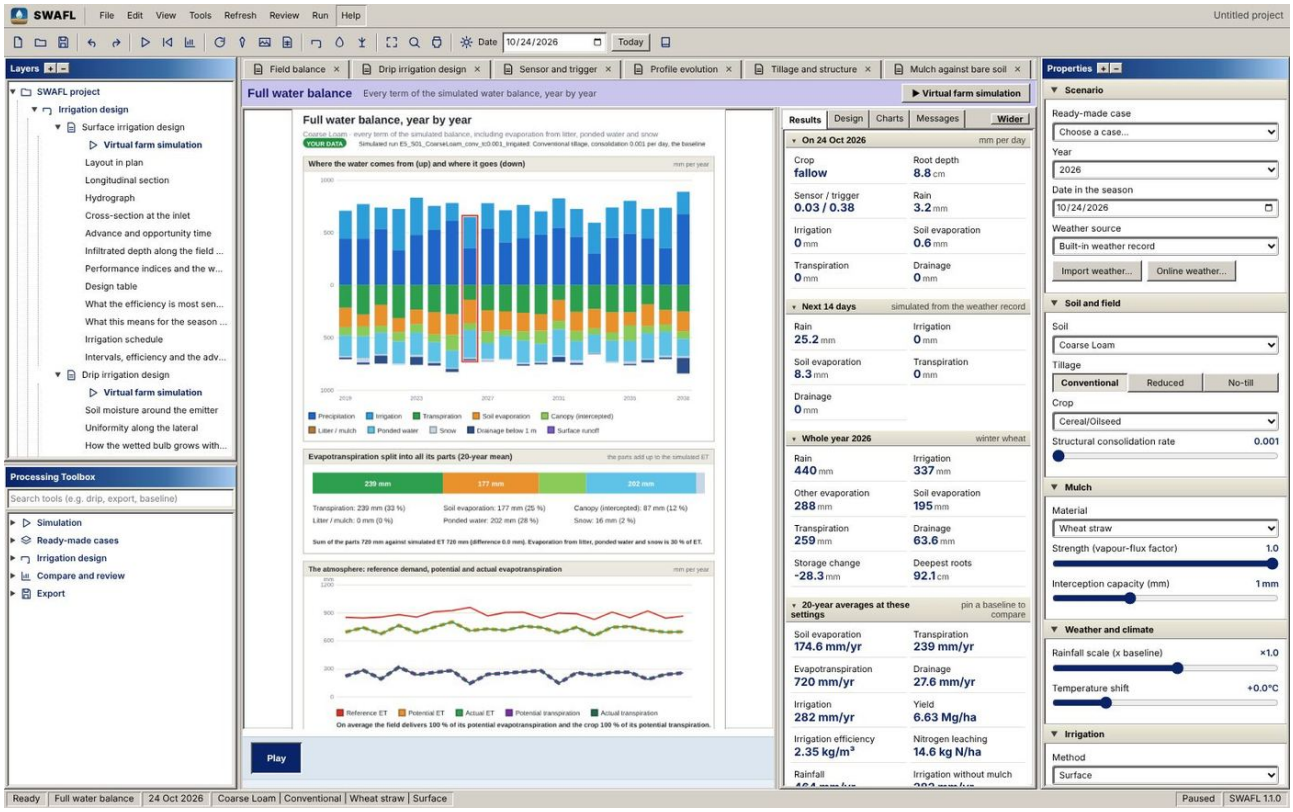


Figure 7.4: Full water balance: the window with the layer open.

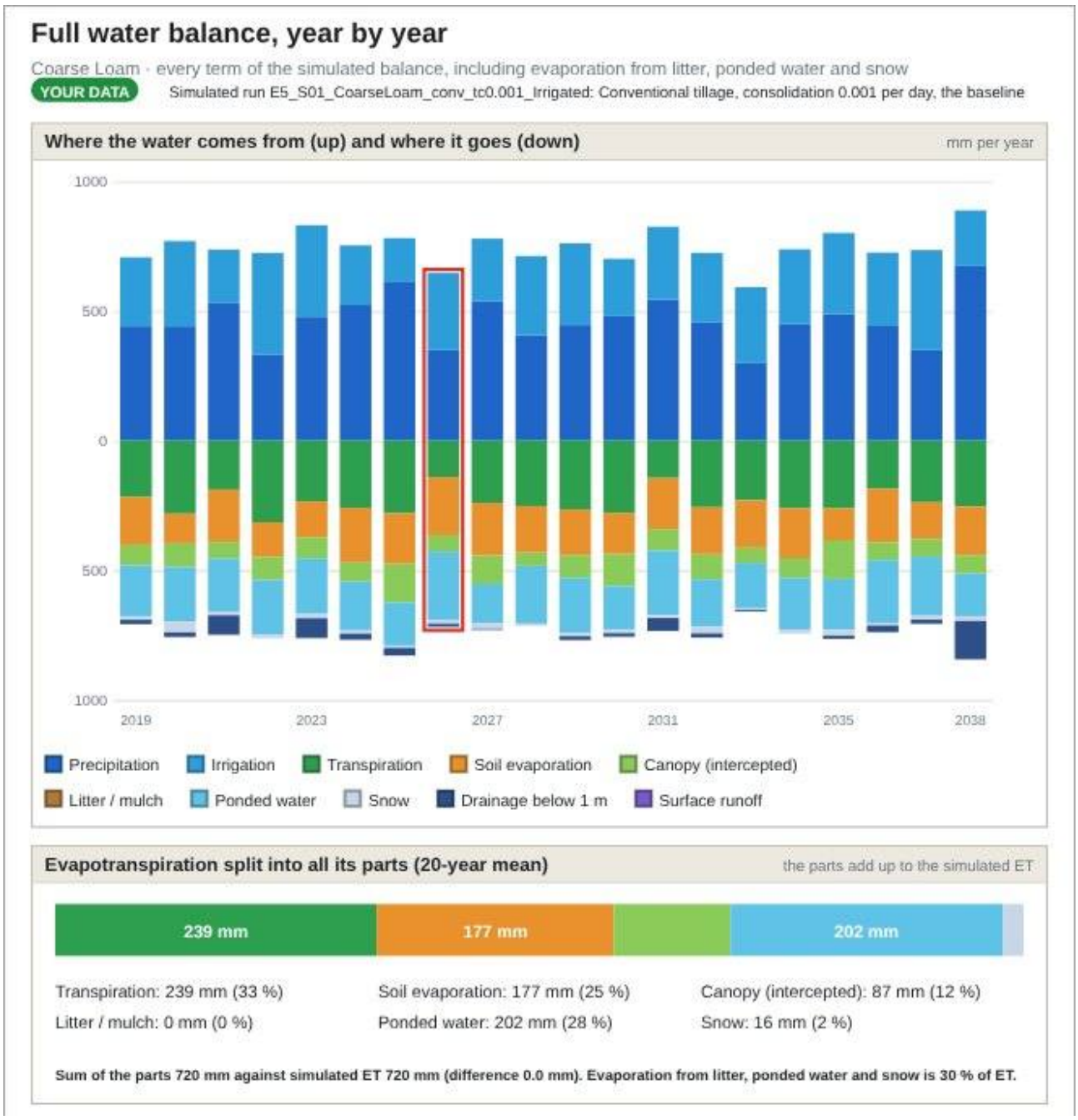


Figure 7.5: Full water balance, part 1 of 2.

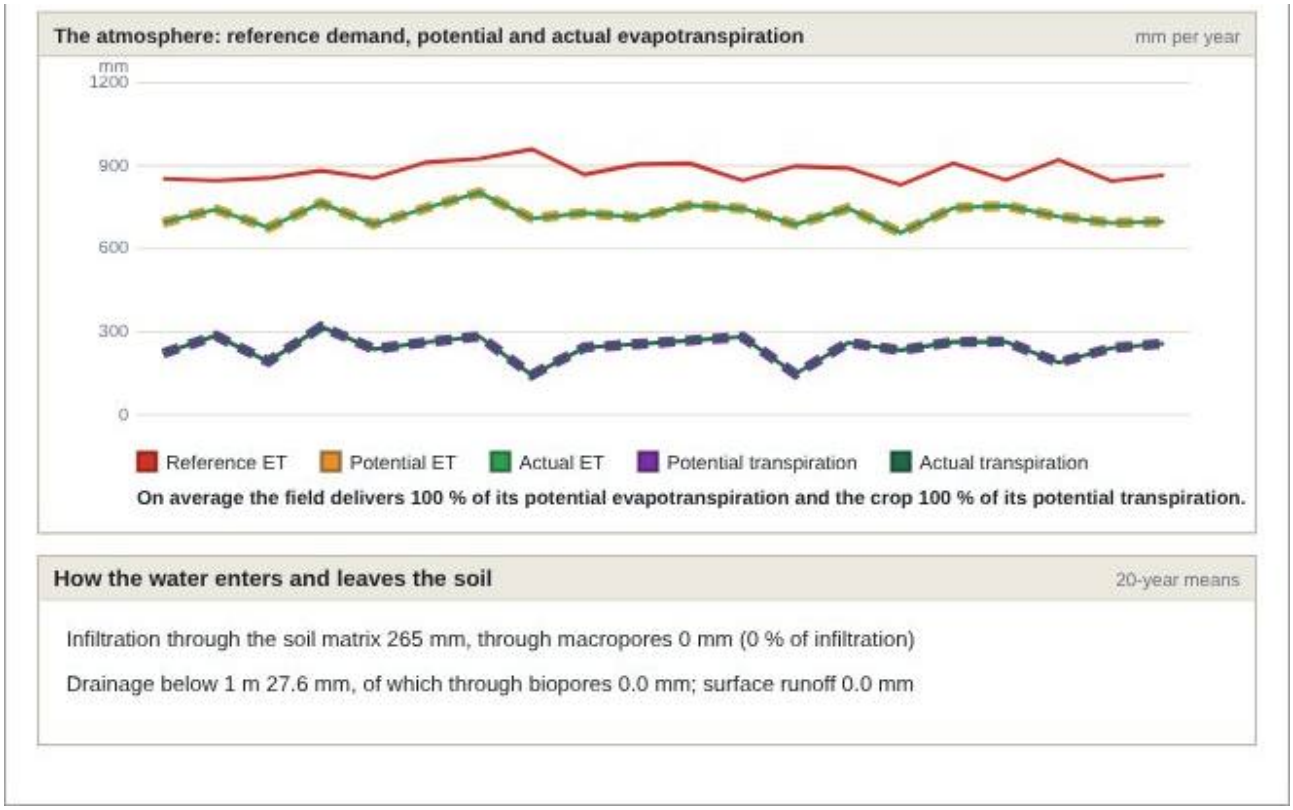


Figure 7.6: Full water balance, part 2 of 2.

7.2.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.1 (Volume 1, engine)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 (Volume 1, engine)

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 22.1 (Volume 1, seasonal)

$$P'_y = \rho\,P_y, \qquad X'_y = X_y\,\text{clip}\Big(\frac{X^{\text{model}}(\text{your settings})}{X^{\text{model}}(\text{reference})};\ 0,\ 6\Big)\ \ (X \in \{I, T, E, E_{\text{other}}, N_L, Y\}),$$

$$D'_y = \begin{cases} D_y\,\dfrac{D^{\text{model}}}{D^{\text{model}}_{\text{ref}}} & \bar{D} > 5,\ D^{\text{model}}_{\text{ref}} > 5 \\ D_y + \big(D^{\text{model}} - D^{\text{model}}_{\text{ref}}\big) & \text{otherwise} \end{cases}, \qquad ET'_0 = ET_0\,(1 + 0.035\,\Delta T), \tag{22.1}$$

Provenance (Eq. 22.1): **[Data-driven]**: the yearly values of the simulations, moved by the ratio (or, for small drainage, the difference) of Engine 1 between your settings and the reference of the run (your mulch, rain ×1.0, no warming, nitrogen 100 %); for a library crop the yield is the relative yield of Eq. (21.5) year by year and the irrigation follows the ratio of the two crop balances of Section 21.4.

7.3 Nitrogen budget

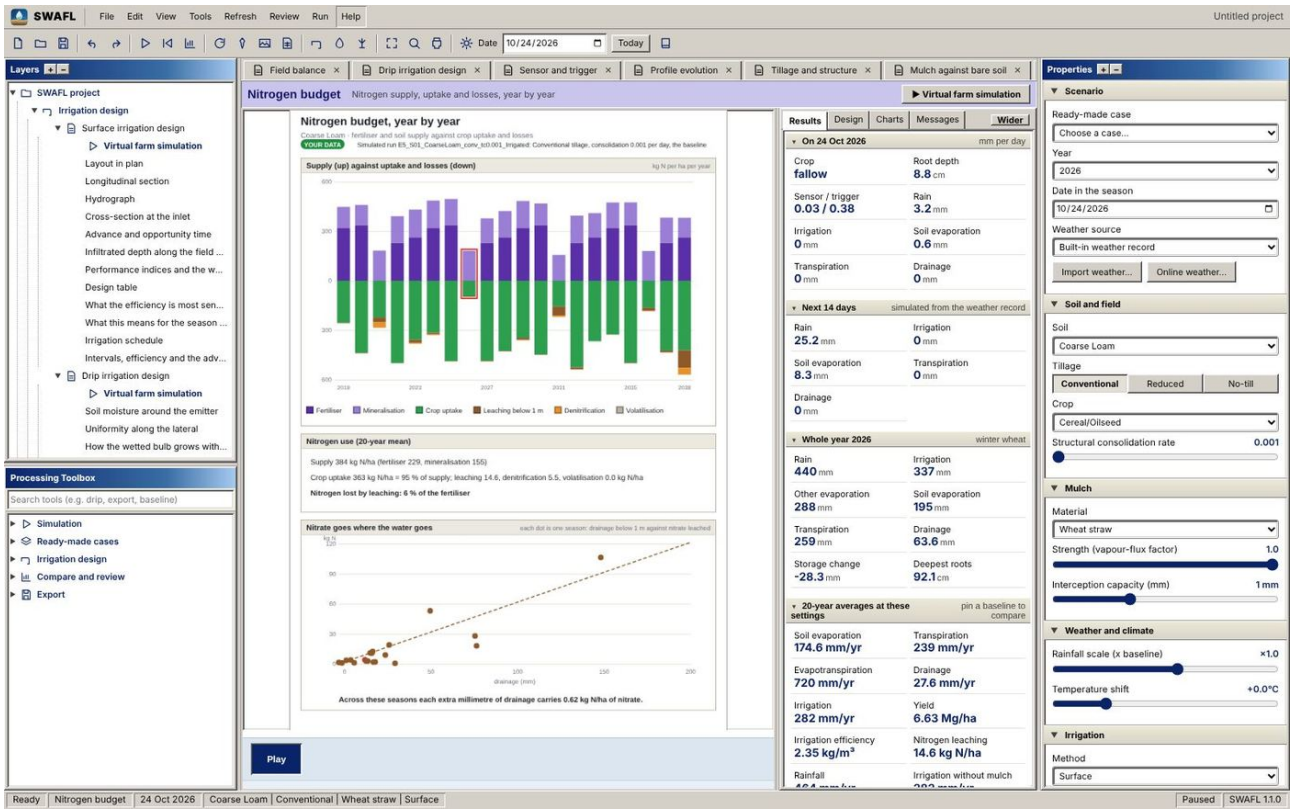


Figure 7.7: Nitrogen budget: the window with the layer open.

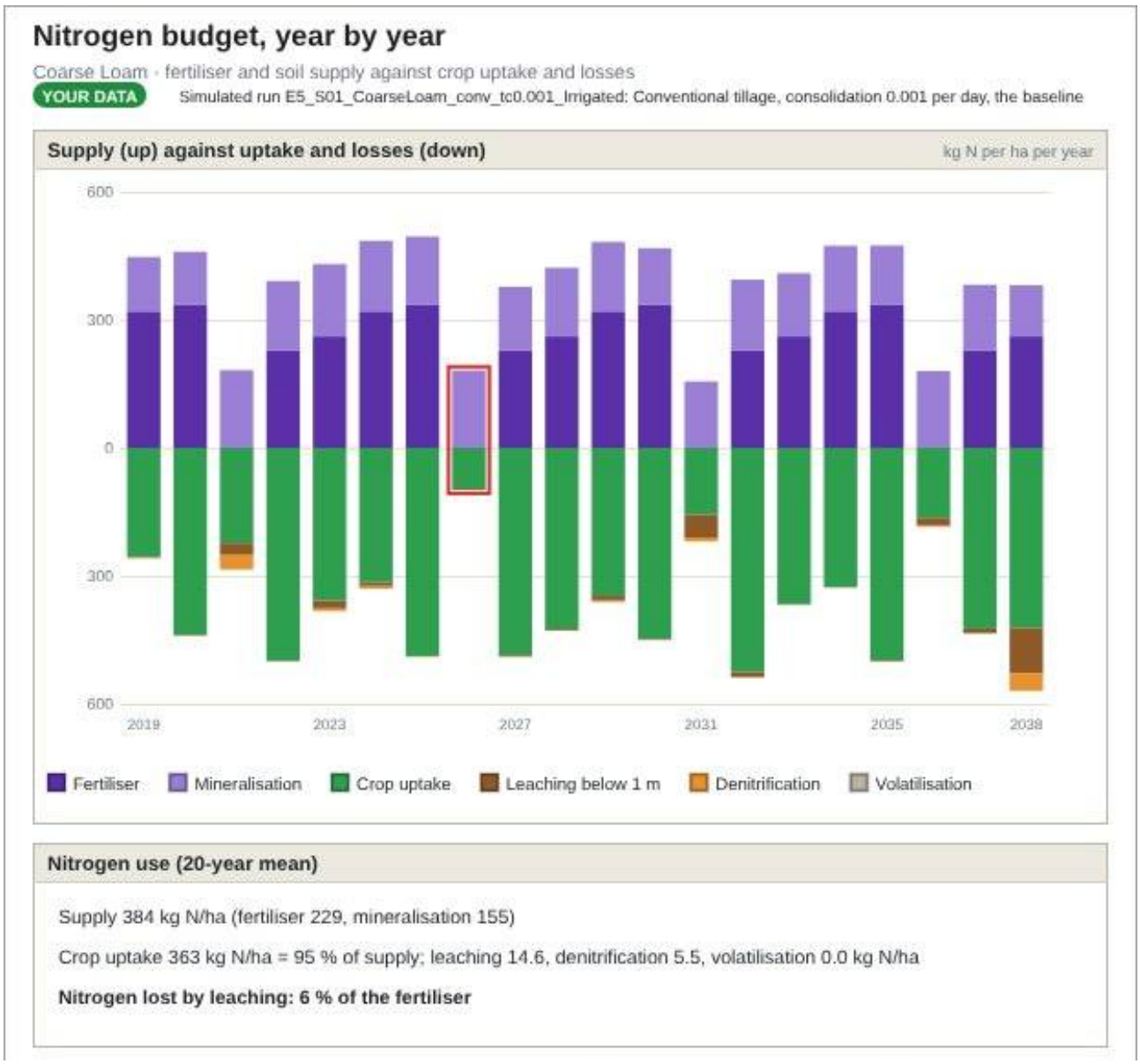


Figure 7.8: Nitrogen budget, part 1 of 2.

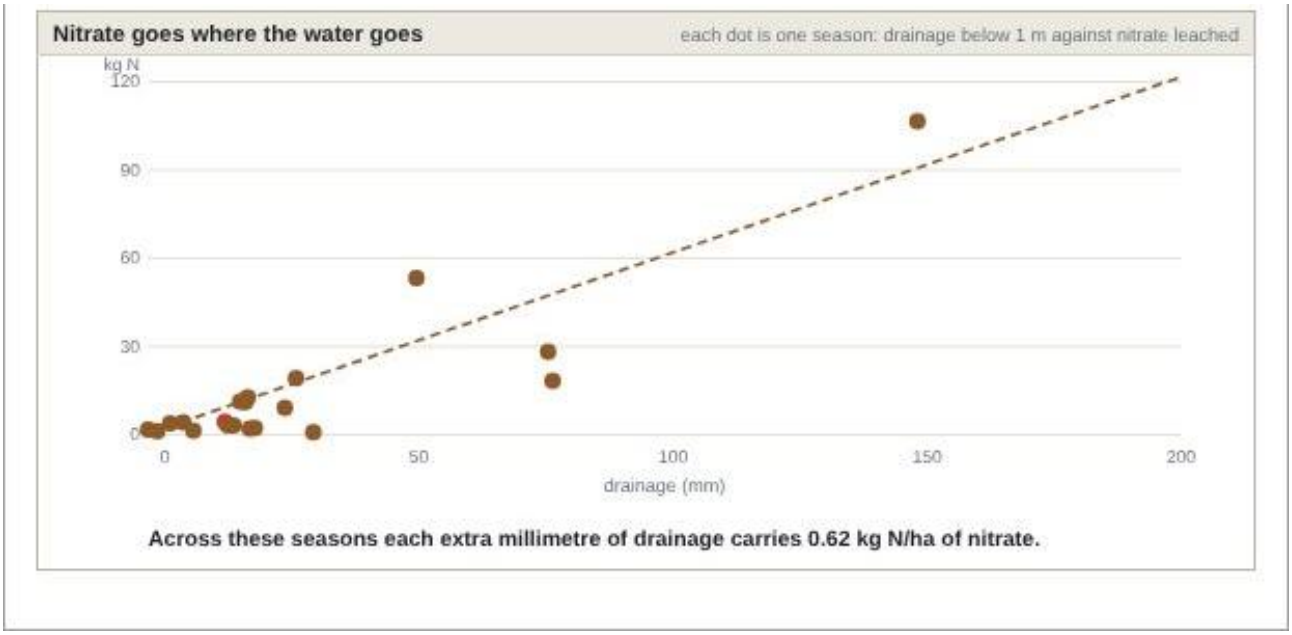


Figure 7.9: Nitrogen budget, part 2 of 2.

7.3.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.1 (Volume 1, engine)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 (Volume 1, engine)

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\,0.05,\,6\Big)\;(Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\; P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\; ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\;\;[\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 22.1 (Volume 1, seasonal)

$$P'_y = \rho\,P_y, \qquad X'_y = X_y\,\text{clip}\Big(\frac{X^{\text{model}}(\text{your settings})}{X^{\text{model}}(\text{reference})};\;0,\;6\Big)\;\;(X \in \{I, T, E, E_{\text{other}}, N_L, Y\}),$$

$$D'_y = \begin{cases} D_y\,\dfrac{D^{\text{model}}}{D^{\text{model}}_{\text{ref}}} & \bar{D} > 5,\; D^{\text{model}}_{\text{ref}} > 5 \\ D_y + \big(D^{\text{model}} - D^{\text{model}}_{\text{ref}}\big) & \text{otherwise} \end{cases}, \qquad ET'_0 = ET_0\,(1 + 0.035\,\Delta T), \tag{22.1}$$

Provenance (Eq. 22.1): **[Data-driven]**: the yearly values of the simulations, moved by the ratio (or, for small drainage, the difference) of Engine 1 between your settings and the reference of the run (your mulch, rain ×1.0, no warming, nitrogen 100 %); for a library crop the yield is the relative yield of Eq. (21.5) year by year and the irrigation follows the ratio of the two crop balances of Section 21.4.

Chapter 8

Crops

8.1 Crop water requirement

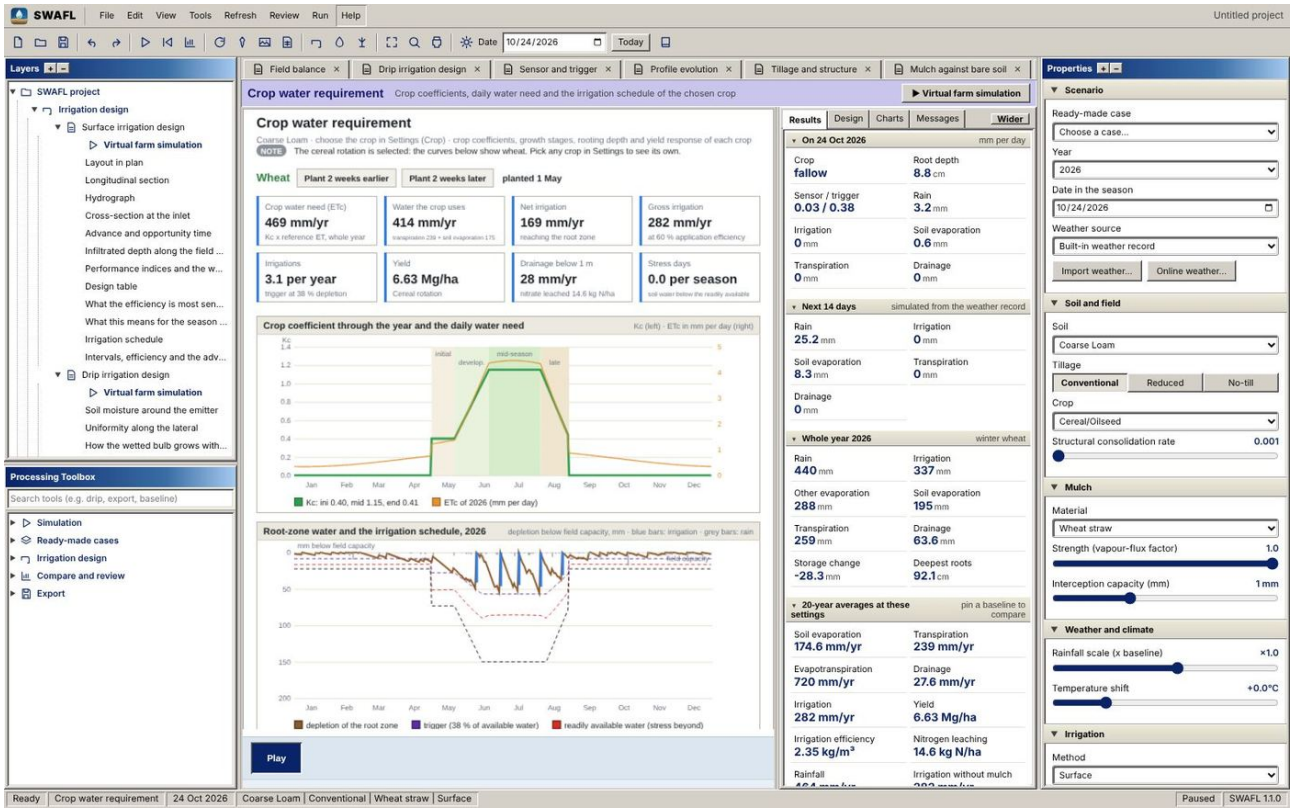


Figure 8.1: Crop water requirement: the window with the layer open.



Figure 8.2: Crop water requirement, part 1 of 3.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %
Chickpea	436	**183**	4.2	100 %
Sorghum (grain)	429	**167**	2.9	100 %
Hazelnut (low)	481	**167**	2.9	100 %
Pecan (low)	481	**167**	2.9	100 %
Cherimoya (medium)	545	**161**	2.8	100 %
Citrus (medium density, short)	552	**152**	2.6	100 %

Figure 8.3: Crop water requirement, part 2 of 3.

Apple (medium)	459	**149**	2.6	100 %
Cowpea (beans)	398	**148**	3.4	100 %
Millet	395	**144**	2.6	100 %
Wine grapes (pergola)	457	**125**	2.2	100 %
Green gram (mung bean)	372	**123**	2.9	100 %
Peach (low)	435	**119**	2.1	100 %
Almond (medium)	435	**119**	2.1	100 %
Apricot (young)	380	**76**	1.3	100 %
Olive (traditional, medium density)	446	**61**	1.1	100 %
Persimmon (young, low)	358	**58**	1.0	100 %
Pomegranate (low)	358	**58**	1.0	100 %
Highbush blueberry (low)	338	**55**	1.1	100 %
Fig tree (young, low)	327	**24**	0.4	100 %

Figure 8.4: Crop water requirement, part 3 of 3.

8.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.3 (Volume 1, engine)

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.26 (Volume 1, engine)

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1+\sin(2\pi(\text{doy}-100)/365)]}{0.675}\,\text{clip}(1{-}0.12(\rho{-}1);\,0.85,\,1.12)\sqrt{\gamma_y} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 5.27 (Volume 1, engine)

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\; 25,\; 150\big)\ \text{mm} \qquad \big(\text{surface; drip and sprinkler use their own limits}\big) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 21.1 (Volume 1, crops)

$$K_c(t) = \begin{cases} K_{c,\text{ini}} & t < L_1 \\ K_{c,\text{ini}} + (K_{c,\text{mid}} - K_{c,\text{ini}})\frac{t-L_1}{L_2} & L_1 \le t < L_1 + L_2 \\ K_{c,\text{mid}} & L_1 + L_2 \le t < L_1 + L_2 + L_3 \\ K_{c,\text{mid}} + (K_{c,\text{end}} - K_{c,\text{mid}})\,\min\big(1, \frac{t-L_1-L_2-L_3}{L_4}\big) & \text{after} \end{cases} \tag{21.1}$$

Provenance (Eq. 21.1): **[Literature]**: FAO-56 crop coefficient curve (Allen et al., 1998, Fig. 25); t days after planting; evergreen crops repeat it every year.

Eq. 21.2 (Volume 1, crops)

$$ET_{c,d} = K_c(d)\,ET_{0,d}, \tag{21.2}$$

Provenance (Eq. 21.2): **[Literature]**: FAO-56 single crop coefficient approach.

Eq. 21.3 (Volume 1, crops)

$$z_r = \text{clip}\Big(15 + (R_{\max} - 15)\,\min\Big(1, \frac{1.15\,K_c}{\max(0.3, K_{c,\text{mid}})}\Big);\; 15,\; R_{\max}\Big)\ \ [\text{cm}], \qquad R_{\max} \le 100\ \text{cm}, \tag{21.3}$$

Provenance (Eq. 21.3): **[Engineered]**: root growth tied to the crop coefficient, limited to the 1 m of the model.

Eq. 21.4 (Volume 1, crops)

$$\begin{aligned} &\text{TAW} = \text{AWC}\,z_r, \qquad p_d = \text{clip}\big(p + 0.04\,(5 - K_c\,ET_0);\,0.1,\,0.8\big), \qquad \text{RAW} = p_d\,\text{TAW}, \\ &K_s = \begin{cases} 1 & D_r \le \text{RAW} \\ \dfrac{\text{TAW} - D_r}{(1-p_d)\,\text{TAW}} & D_r > \text{RAW} \end{cases}, \qquad ET_a = K_s\,ET_c, \\ &D_{r,d} = D_{r,d-1} - (P_d - RO_d) + ET_{a,d}, \qquad RO_d = 0.3\,\max(0, P_d - 35), \end{aligned} \tag{21.4}$$

Provenance (Eq. 21.4): **[Literature]**: FAO-56 Eqs. 82–85 (water stress coefficient, adjustment of p with the demand) and the root-zone balance Eq. 85; the runoff of heavy rain is **[Engineered]**.

Eq. 21.5 (Volume 1, crops)

$$Y_{\text{rel}} = \text{clip}\Big(1 - K_y\Big(1 - \frac{\sum ET_a}{\sum ET_c}\Big);\ 0,\ 1\Big), \tag{21.5}$$

Provenance (Eq. 21.5): **[Literature]**: FAO-33 yield response (Doorenbos and Kassam, 1979); $K_y = 1$ when not tabulated, as the table says.

Eq. 21.9 (Volume 1, crops)

$$I = \begin{cases} I_m\ \text{clip}\big(I_c / I_r;\ 0,\ 4\big) & I_r > 5\ \text{mm} \\ I_c & \text{otherwise} \end{cases} \tag{21.6}$$

$$\begin{aligned} T &= \min\Big(0.9\,ET,\ \ T_m\,\text{clip}\Big(\frac{ET_{a,c}}{ET_{c,c}} \cdot \frac{ET_{c,c}}{ET_{c,r}} \cdot \Big(\frac{f_{c,c}}{f_{c,r}}\Big)^{0.3};\ 0.2,\ 3\Big)\Big), \\ E^* &= E_m\ \text{clip}\Big(\frac{1 - 0.7\,f_{c,c}}{1 - 0.7\,f_{c,r}};\ 0.6,\ 1.6\Big), \end{aligned} \tag{21.7}$$

$$D = \max\big(0.2\,D_m,\ D_m + 0.5\,(DP_c - DP_r)\big), \qquad ET = \text{clip}\big(P + I - D - \Delta S_m;\ 50,\ 5000\big), \tag{21.8}$$

$$(E,\ E_{\text{other}}) = (E^*,\ E^*_{\text{other}})\,\frac{ET - T}{E^* + E^*_{\text{other}}}, \qquad N_L = N_{L,m}\,\frac{D + 5}{D_m + 5}, \qquad Y = 100\,Y_{\text{rel},c}\,[\%], \tag{21.9}$$

Provenance (Eqs. 21.6–21.9): **[Engineered]**: transfer by the ratio of two FAO-56 balances; the clips keep the transfer within the range in which the cereal result is informative; the remainder of the balance is shared between soil evaporation and the other evaporation terms in proportion to their targets, so that the balance closes without inflating either.

Chapter 9

Feedback loops

9.1 Feedback loops

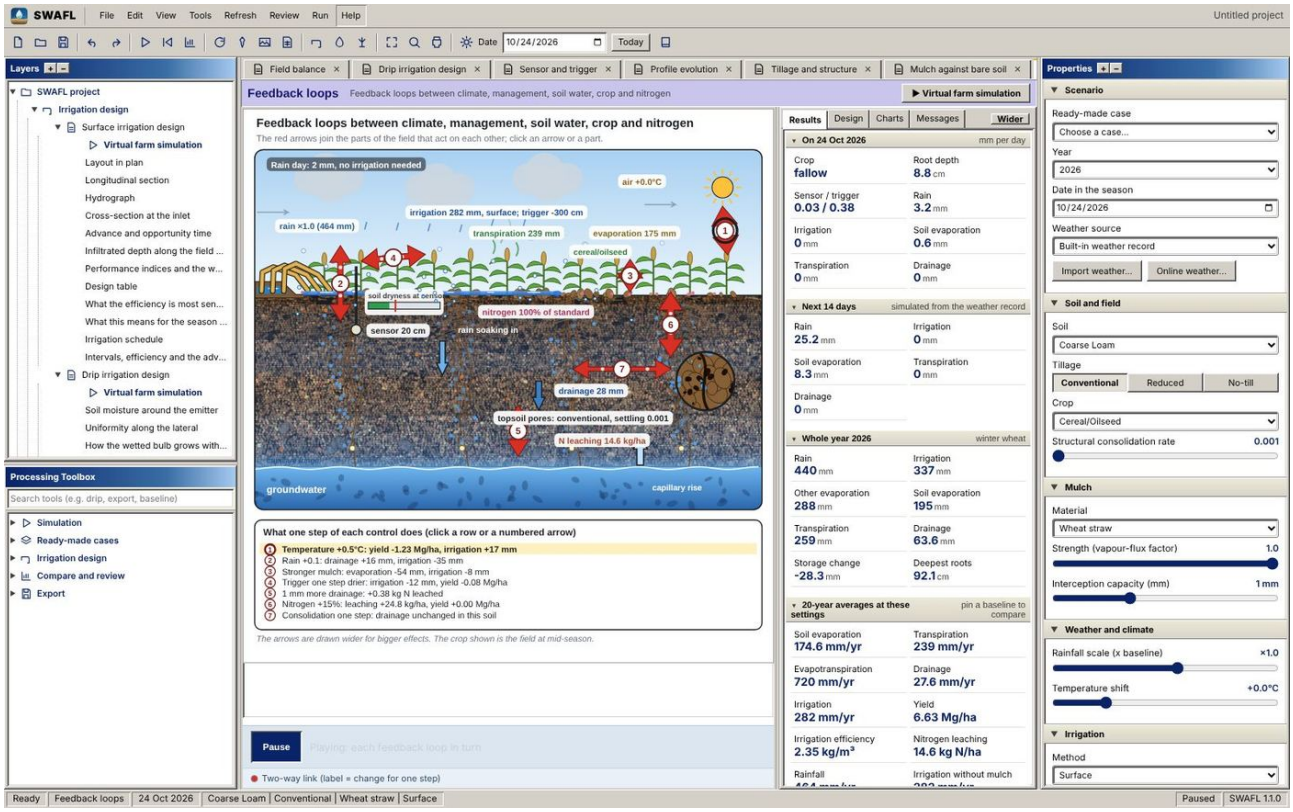


Figure 9.1: Feedback loops: the window with the layer open.

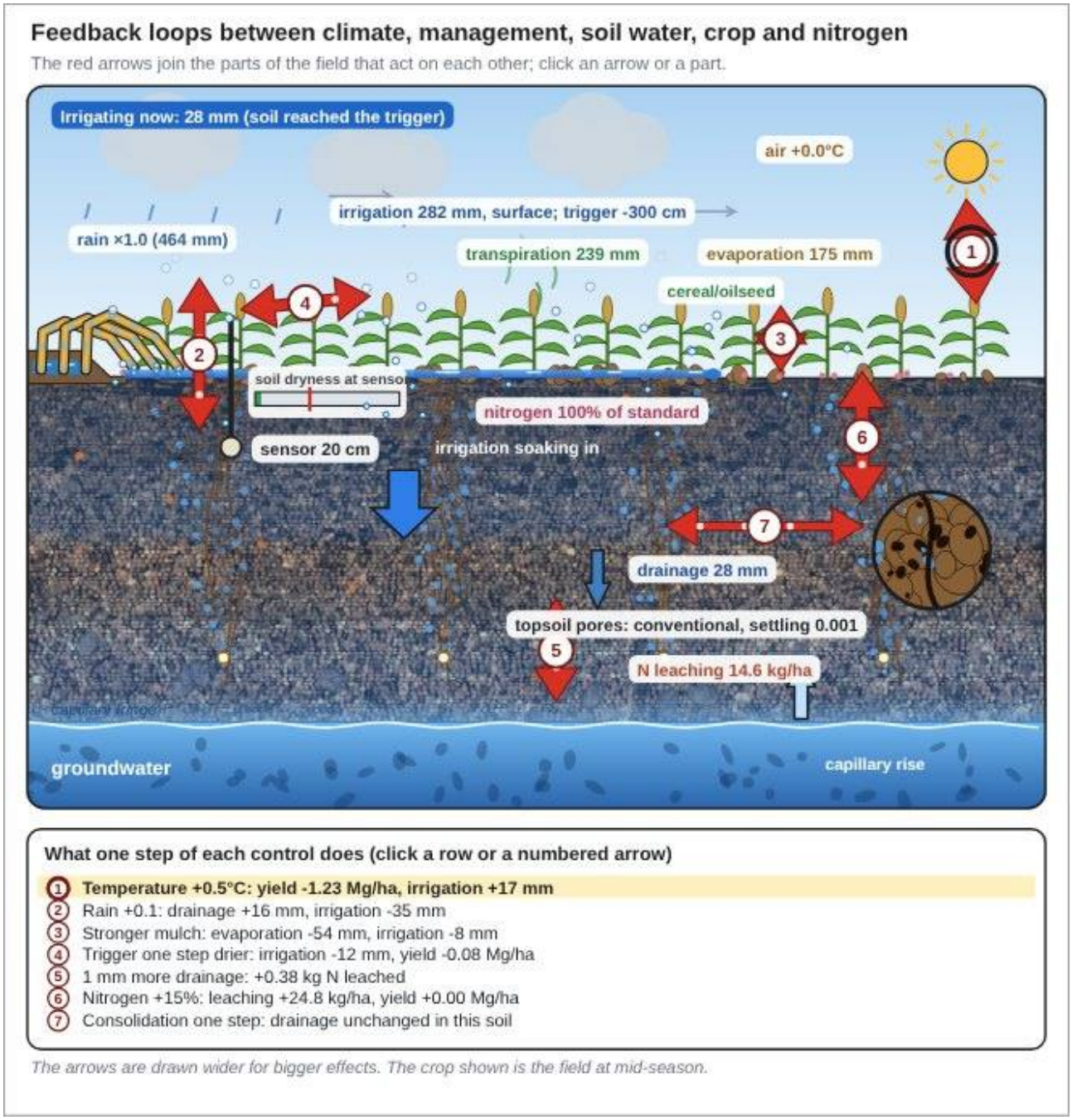


Figure 9.2: Feedback loops, part 1 of 1.

9.1.1 The equations of this layer

Every number and picture of the layer comes from the equations below (numbers as in Volume 1, with the chapter that explains each).

Eq. 5.4 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 (Volume 1, engine)

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 (Volume 1, engine)

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 (Volume 1, engine)

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 10.1 (Volume 1, loops)

$$w_k = 5 + \text{clip}(\mu_k;\ 0,\ 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

Eq. 10.2 (Volume 1, loops)

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D}\quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.